

UMC

2024

Sustainability Report



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Message from the ESG Steering Committee Chairman



While the global economy gradually recovered from the pandemic, 2024 presented different challenges, including geopolitical shifts, inflation, inventory adjustments, and industry overcapacity. Despite these headwinds, our management team demonstrated strong leadership, achieving notable progress in technology development, manufacturing diversification, and positioning for future AI-driven growth. We also made important breakthroughs in ESG, driving sustainable growth and positive environmental and social impact while maintaining stable profitability.

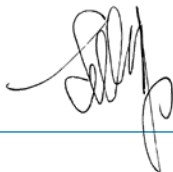
In 2024, we advanced towards our environmental, social, and governance goals. The appointment of our first lead independent director has strengthened communication between the board and our management team, and will play a crucial role in safeguarding shareholder interests and advancing corporate sustainability. We remain committed to enhancing benefits for employees, as demonstrated by the early implementation of an employee stock ownership plan in Taiwan. This initiative enables employees to share in the company's success and fosters a stronger sense of belonging.

Striving towards our net zero greenhouse gas emissions by 2050 and RE100 goals, we have taken decisive climate actions. We introduced internal carbon pricing to incentivize carbon reduction initiatives, and signed an offshore wind purchase agreement for over 30 billion kilowatt-hours (kWh) of power. As of 2024, we have achieved a 31% reduction in Scope 1 and 2 emissions – one-third of the way toward our 2050 net zero target. Group-wide green electricity usage has reached 17%, with subsidiaries HJ and USCXM already fully powered by green energy. Furthermore, we have incorporated reduction targets into the performance indicators for new restricted stock awards, directly linking management compensation to ESG performance.

Our dedication to sustainability has been widely recognized. In 2024, UMC was again selected as a component of the DJSI World Index and received an AA rating in the MSCI ESG assessment. UMC remain the only semiconductor company globally to achieve Double A in CDP disclosures for three consecutive years and are among the few listed companies to maintain a top 5% ranking in the Taiwan Stock Exchange's Corporate Governance Evaluation for ten consecutive years. Additionally, our digital transformation efforts over the years have yielded outstanding results, with our flagship Fab 12A named to the World Economic Forum's Global Lighthouse Network—the first semiconductor foundry worldwide to receive this distinction.

Looking ahead, UMC will continue to invest in technological innovation and develop forward-looking solutions to maintain our competitive edge. We will deepen digital transformation and sustainable governance, strengthen value chain collaboration and climate resilience, and work alongside global customers, employees, and partners to harness the pervasive power of technology for a better world.

Finally, thank you for your interest in and support of UMC's sustainability journey.

Jason Wang and Shan-Chieh Chien
Co-presidents and ESG Steering Committee Chairmen



About This Report

This report is the non-financial information annual report of United Microelectronics Corporation (hereafter "UMC" or "the Company") for the 25th consecutive year since 2001. The report is available in both Chinese and English versions. Based on the principles of pragmatism, integrity, and mutual advancement, through the publication of this report, UMC aims to share the Company's short, medium, and long-term strategies, substantial actions, and performance in various dimensions of the economy, environment, and society (ESG) with stakeholders, in response to the concerns and expectations of the public regarding the development of UMC.

- ❖ Reporting Period: The information presented is from January 1, 2024 to December 31, 2024, which aligns with the period of the Company's annual report.
- ❖ Publication Frequency: Annually
 - Previous edition: Issued in July 2024.
 - Current edition: Issued in July 2025.
 - Next edition: Scheduled to be issued in July 2026.

In support of environmental protection and a paperless policy, an electronic version of this report is posted on the Company website

Reporting Scope and Boundary

The reporting scope of this report covers revenue accounting for over 99% of UMC's consolidated financial statements. The entities in this report include UMC (fabs in Taiwan and Singapore) and its foundry subsidiaries, including Hejian Technology (Suzhou) Co., Ltd. (hereinafter "HJ"), United Semiconductor (Xiamen) Co., Ltd. (hereinafter "USCX"), Wavetek Microelectronics Corporation (hereinafter "Wavetek"), and United Semiconductor Japan Co., Ltd. (hereinafter "USJC"), collectively referred to as the "UMC Group". For chapters in which the boundary scope information was affected on account of differing issues, special notes will be given in corresponding texts.

Note: Please refer to the TWSE MOPS for the Company's complete list of subsidiaries within the consolidated financial statements boundary: https://mops.v.twse.com.tw/mops/web/t57sb01_q10

Reporting Guidelines

The reporting guidelines which UMC was in accordance with while preparing for this report were listed below:

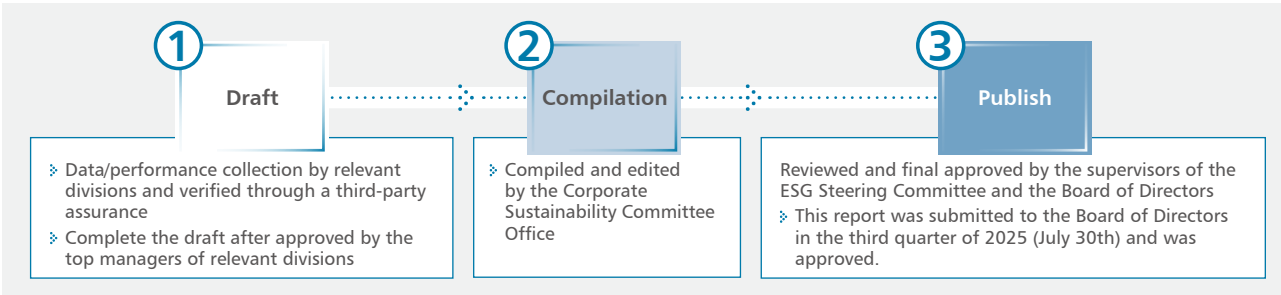
- ❖ GRI Sustainability Reporting Standards, Global Reporting Initiative (GRI)
- ❖ Semiconductors Sustainability Accounting Standard 2018, Sustainability Accounting Standards Board (SASB)
- ❖ Task Force on Climate-related Financial Disclosures (TCFD)
- ❖ Taiwan Stock Exchange Corporation Rules Governing the Preparation and Filing of Sustainability Reports by TWSE Listed Companies
- ❖ AA1000AP AccountAbility Principles
- ❖ UN Global Compact

External Assurance

In June 2025, SGS verified this report in accordance with the GRI Standards and AA1000 Assurance Standard (2008) TYPE II, providing high-level assurance. The assurance items covered not only reporting guidelines such as GRI standards and SASB standards, but also environmental and social indicators from UMC's Sustainable Strategy Blueprint. The related index tables and third-party assurance statements can be found in the appendix of this report. Moreover, financial data in this report is from the Company's financial statement, which has been audited by certified accountants Ernst & Young Accounting. And greenhouse gas emissions inventory and reduction performance conform to ISO 14064-1:2018 standards, and the section on climate-related risks and opportunities follows the TCFD guidelines, and the results have been verified by a third party, DNV Business Assurance Co., Ltd..

Internal Control Process

This report is reviewed and approved by top managers of each division, and verified by a third party. After being compiled by the Corporate Sustainability Committee Office, the draft is submitted to the supervisor of the ESG Steering Committee and the Board of Directors for final approval before publication. Relevant procedures for preparation and verification of the sustainability information are included in the Company's internal control system to ensure reliability, timeliness, transparency and compliance of the reporting.



Contact Information

Thank you for reading this report. For any questions or comment about content in the report, please contact:

United Microelectronics Corp.

❖ GRM & ESH Division ❖ Phone: + 886-3-5782258 / Fax: + 886-3-5782375 ❖ UMC CSR mailbox: csr@umc.com



About UMC

Established in 1980, UMC is Taiwan's first semiconductor company, and has grown to become a leading global semiconductor foundry. The company provides high quality IC production with a focus on both logic and specialty technologies to serve every major sector of the electronics industry. UMC's comprehensive technology and manufacturing solutions include Logic/Mixed-Signal, embedded High-Voltage, embedded Non-Volatile-Memory, RFSOI and BCD.

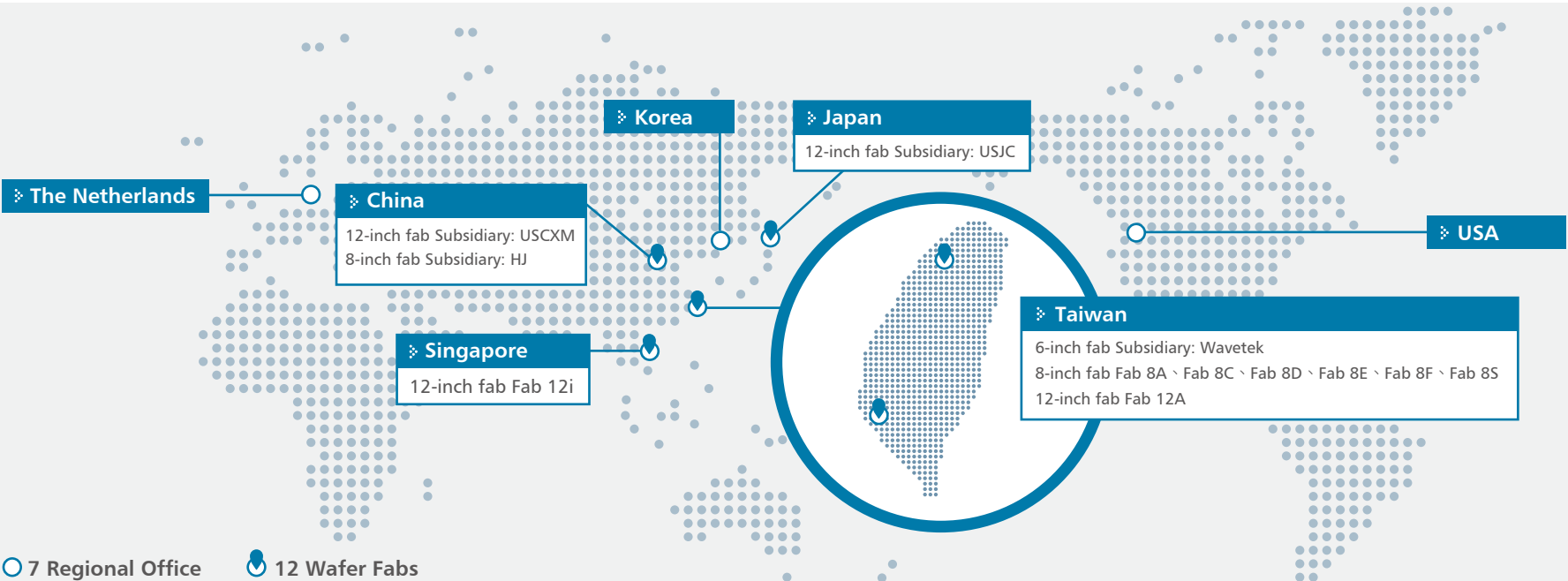
UMC led the development of the semiconductor industry in Taiwan. It was the first domestic company to offer foundry services, and the first semiconductor company to list on the Taiwan Stock Exchange. With approximately 20,000 employees worldwide, UMC offers robust process technologies and comprehensive foundry solutions, enabling customers to strengthen their competitive advantage in today's rapidly changing industry.

UMC has 12 wafer fabs located in Asia, four of which are advanced 12-inch fabs. The locations of our fabs offer customers a geographically diversified and resilient supply chain. In addition, UMC has an extensive network of sales offices in Taiwan, China, Europe, Japan, Korea, Singapore, and the United States to better serve customers.

UMC, with over four decades of manufacturing expertise, ensures industry-leading cycle times and defect densities. Comprehensive process control systems with advanced methodologies and a strong engineering team support fast product ramp. UMC's advanced automation, mature defect density, fast cycle times and more than 400,000 wafers per month (12-inch equivalent) capacity enable UMC to provide the most competitive manufacturing advantages to foundry customers.

Global Presence

Company Name	United Microelectronics Corp.
Date Founded	May 1980
Company Headquarter	No. 3, Li-Hsin Road 2, Hsinchu Science Park
Number of Employees	Approximately 20,000 employees, including those in worldwide affiliated companies
Total Capital Amount	Total capital of NT\$260 billion Paid-in capital amounted to NT\$125.607 billion
Main Operation	Professional integrated foundry services
Product Services	UMC's comprehensive IC processing technologies and manufacturing solutions include Logic/Mixed-Signal, embedded High-Voltage, embedded Non-Volatile-Memory, RFSOI and BCD.
Affiliated Businesse	Affiliated business operations including wafer manufacturing, electronics, optoelectronics, investment, insurance and trading
Consolidated Operating Revenues and Operating Costs	2024 annual revenue was NT\$232.303 billion and 2024 annual operating costs was NT\$156.649 billion



Public Participation

UMC actively builds and maintains productive relationships with governments and relevant organizations to communicate with key stakeholders on issues including industry development, technology, employee welfare, and environmental sustainability. Through membership in industry associations and non-profit organizations, UMC contributes its positions on the development of public issues and in the formulation of relevant industrial policies. In addition, UMC incorporates the sustainability initiatives and policies of governments and international organizations into its long-term planning. UMC established the "UMC Public Affairs Engagement Policy" and regularly reviews associations and organizations in which UMC is a member. UMC also monitors their publicly disclosed positions on climate change and whether they are aligned with the goals of the Paris Agreement.

Association Involvement

Unit: NT\$

Name of Organization	2021	2022	2023	2024	Our Role
Allied Association for Science Parks Industries (ASIP)	1,008,000	1,008,000	1,008,000	1,026,000	Executive Director, Supervisor
Taiwan Semiconductor Industry Association (TSIA)	900,000	900,000	900,000	900,000	Executive Director
Semiconductor Equipment and Materials International (SEMI)	633,490	761,997	871,362	986,333	Member
Association of Taiwan Net Zero Emissions (ATNZE)	-	241,667	150,000	150,000	Director
Responsible Minerals Initiative	260,457	232,013	290,353	383,363	Member
RE100	151,679	177,910	237,313	256,275	Member
Taiwan Climate Partnership (TCP)	-	150,000	50,000	50,000	Group Member
Taiwan IC Industry & Academia Research Alliance (TIARA)	100,000	100,000	100,000	100,000	Alternate Supervisor
Business Council for Sustainable Development of Taiwan (BCSD-Taiwan)	60,000	60,000	60,000	60,000	Permanent Member
Taiwan Electrical and Electronic Manufacturers' Association (TEEMA)	42,000	42,000	42,000	42,000	Member
Chinese Professional Management Association (CPMA)	20,000	20,000	20,000	20,000	Member
Taiwan Carbon Capture Storage and Utilization Association	5,015	5,000	5,000	5,000	Director, Supervisor
Association of Atmosphere Protection in Taiwan (AAPT)	-	-	7,000	6,000	Supervisor
Total Amount	3,275,879	3,698,587	3,741,028	3,984,971	

Costs of Association Participation

Unit: NT\$

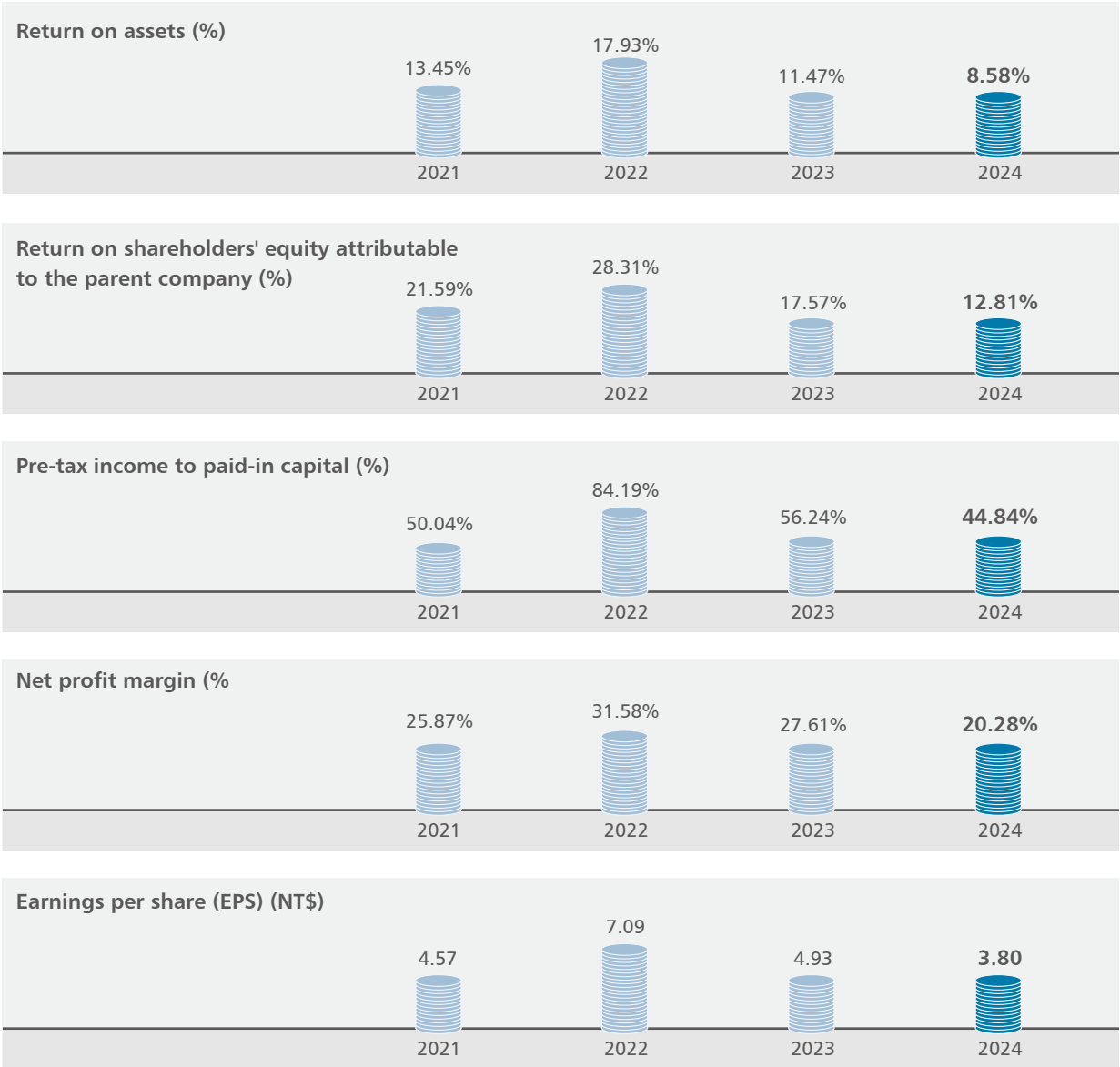
Category	2021	2022	2023	2024
Trade association	3,275,879	3,698,587	3,741,028	3,984,971
Lobbying, interest representatives, or similar	0	0	0	0
Local, regional, or national political campaigns/ candidates	0	0	0	0
Others (e.g., spending related to elections or referendums)	0	0	0	0
Total Amount	3,275,879	3,698,587	3,741,028	3,984,971

Note: 1. The information covers UMC Taiwan and Singapore.
2. Only industry associations that are related to sustainability affairs are listed.

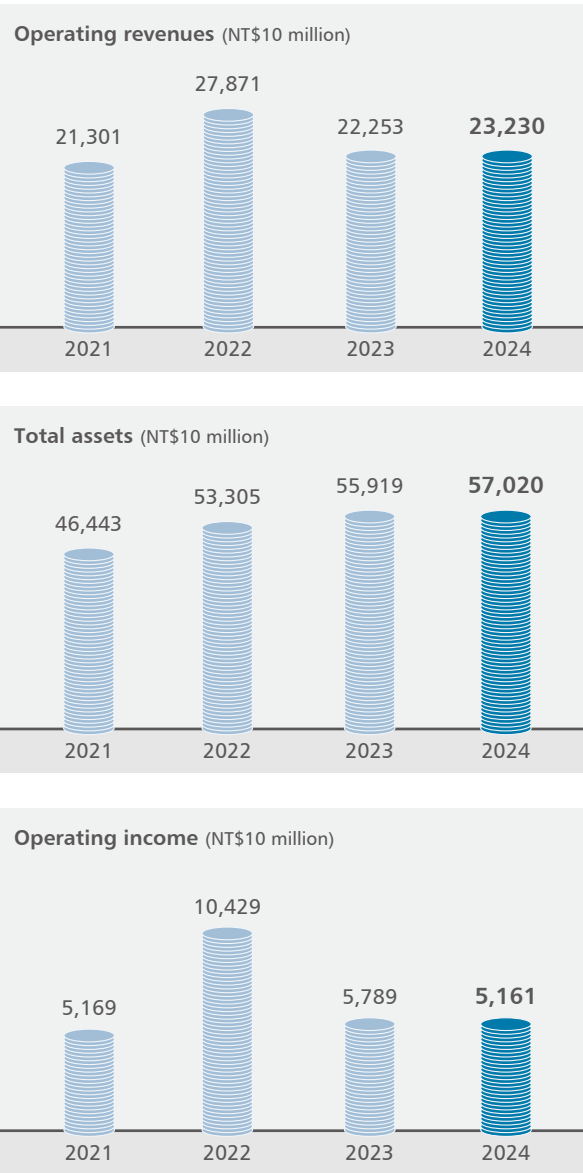


Business Performance

Profitability



Note: The above consolidated financial information is based on Financial Supervisory Commission approved international financial reporting guidelines. Please refer to the TWSE MOPS for parent company information.



Sustainable Governance

UMC's sustainable development is built on the vision of "to strive for a better and more sustainable future that is centered on people, in harmony with the environment, and enables communities to thrive." The Company maintains good interaction with stakeholders to jointly pursue sustainable growth.



Implementation of Sustainable Development Best Practice Principles

To implement corporate social responsibility, promote economic, environmental and social advancement, and achieve the goal of sustainable development, the UMC Corporate Sustainability Committee has formulated "UMC Sustainable Development Best Practice Principles," which has been approved by the Board of Directors for implementation. UMC stays up to date on the development of domestic and international sustainable development standards and changes in the corporate environment, reviews and improves the Company's sustainable development system, and improves the performance of sustainable development activities.

UMC Sustainable Development
Best Practice Principles



Sustainable Governance Structure



The Board of Directors serves as the unit for the highest level of sustainable governance, and also plays a supervisory and guiding role to regularly reviews the progress of the sustainable strategy, and guides the management team to make adjustments when necessary. In addition, the effectiveness of ESG governance is supervised by the Sustainable Development and Nominating Committee, which is composed of all independent directors. The ESG Steering Committee appointed the Co-Presidents as the Chairmen to lead the senior executives (C-Suite) to meet every quarter, and to be in charge of determining UMC's ESG governance strategy and goals. The Chairmen of ESG Steering Committee report to the Sustainable Development and Nominating Committee and the Board of Directors annually on the Company's sustainable performance and plans, as well as key issues and countermeasures that stakeholders are concerned about.

With the Corporate Sustainability (CS) Committee and the Enterprise Risk Management (ERM) Committee under the ESG Steering Committee's supervision in order to ensure that ESG governance and corresponding policies are implemented from the top down. Sustainability affairs are mainly coordinated by the CS Committee and appointed the Chief Sustainability Officer as the Chairman, which is responsible for stipulating the direction and goals of sustainable development. The eight functional groups of the CS Committee appointed the vice presidents as the groups' chairs to review the performance and target achievements quarterly. The Chief Sustainability Officer also holds regular meetings with the Corporate Sustainability Office to oversee and manage the Company's sustainability actions. In 2024, a total of six meetings were held, discussing 21 topics including internal carbon pricing, social impact, renewable energy, greenhouse gas reduction, and the enhancement of the operations of the Corporate Sustainability Committee, among others. The ERM Committee is responsible for coordinating risk management and controls in all aspects of the Company, and promoting relevant countermeasures for major risks. It has six risk groups that horizontally link three existing entity organizations and committees. The Company appointed the vice presidents as the chairs of the groups to review risk changes and management status quarterly, and update risk information on a monthly basis starting from the end of 2021. The representatives of the CS Committee and the ERM Committee report on the evaluation of risks related to corporate governance, environmental, and social issues, as well as plans and their effectiveness to the ESG Steering Committee quarterly.

The main points of the proposal reported to the Sustainable Development and Nominating Committee and the Board of Directors by the Chairman of the ESG Steering Committee and Chief Sustainability Officer on October 30, 2024 include ESG activities and achievements, stakeholder communication and ESG future focus, etc. A total of eleven key ESG issues were reported and communicated, including issues related to corporate governance, integrity management, risk management, information security, supply chain management, diversity, equity, and inclusion (DEI), talent attraction and retention, social participation, climate change, circular economy, and biodiversity. Relevant major issues need to be reported to the Board of Directors for approval. In 2024, the cases related to sustainability approved by the Board of Directors include the achievement of employees' remuneration linked to ESG KPIs, donation to UMC Science and Culture Foundation, "Restricted Stock Awards," which sets GHG reduction and the percentage of female director seats as performance indicators, "the signing of a Corporate Power Purchase Agreement with Fengmiao Wind Power Co., Ltd.," and green investments, including power saving equipment, improvement of pollution prevention facilities and establishment of Circular Economy & Recycling Innovation Center.

Sustainability Strategy and Blueprint

UMC is committed to integrating sustainable development with our business strategy. Based on the Company's core business of wafer manufacturing, UMC has formulated its Sustainability Strategy and Blueprint in 2021 that are aligned with ten UN Sustainable Development Goals (UN SDGs). The Company also set corresponding goals of 2022, 2025 and 2030, and launched action plans to achieve those goals. The 2022 goals have been fully achieved, ensuring the Company's steady operational growth and positively contributing to all ESG aspects, striving to minimize any adverse impact on society and the environment. The Company will continue to develop products with sustainable benefits and create ESG value together with our stakeholders for a better world.

In 2021,
the ESG Steering Committee formulated
four sustainable strategic directions:



Unleash the power
of technology



Integrate UMC's core
capabilities



Minimize environmental
footprint



Create social
value

IMPACT+

Unleash the power of technology

Contribute to a better world by expanding the application of high-performance/low consumption semiconductor products in health-related and eco-friendly fields



Governance

Integrate UMC's core capabilities

Implement sustainable governance, exert UMC's core strengths, and work together with supply chain partners to strive for ESG vision



Environment

Minimize environmental footprint

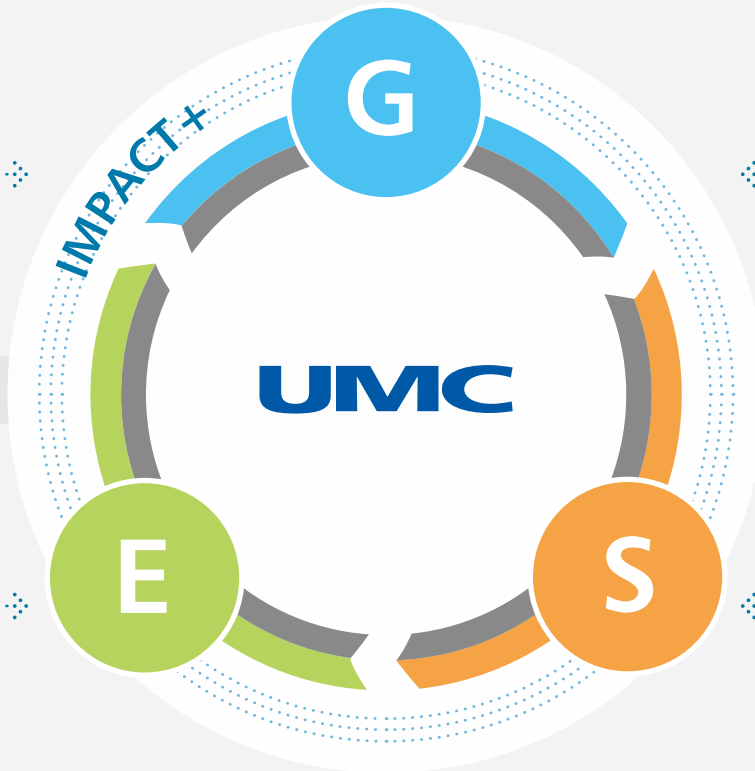
Practice low-carbon sustainability, promote renewable energy and net zero carbon emissions, and reduce the environmental impact of operations and products



Social

Create Social value

Provide opportunities for employees, promote semiconductor education and public welfare



(Continued)		2024	2025	2030		
	Strategy	Item	Achievements	Goals		Location
Unleash the power of technology		Revenue from sustainable products as % of total revenue	82.3% *	≥79%	≥81%	2.1.1 Robust Wafer Manufacturing Services
		Number of innovative energy-saving specialty process platforms (base year: 2020)	4	≥5	≥10	
		Cumulative patents granted worldwide	15,560 * (364 for the year*)	> 16,330	> 17,730	
Integrate UMC's core capabilities		Ratio of satisfied customers	92%	≥90%	≥90%	2.2.1 Strengthen Customer Communication
		Ratio of audit conducted to significant high-risk suppliers	100%	100%	100%	1.3 Building Sustainable Supply Chain
		Ratio of Tier 1 suppliers who have included 'signing Employment Ethics Terms and Conditions Agreement' in their supply chain management requirements	79%	80%	90%	
		Percentage of carbon intensity reduction by suppliers within the project of "low-carbon supply chain"	Annual goal achieved	10%	20%	
		Completion rate of biodiversity survey for Tier 1 suppliers	100%	100%	100%	2.1.3 Smart Manufacturing
		Ratio of products produced by Industry 4.0 Smart Manufacturing Systems	53%	60%	80%	
Minimize environmental footprint		Reduction of water withdrawal per unit product	30.3%	25%	30%	3.3 Water Resources Management
		The total amount of additional water saved (base year: 2015)	6.5 million tonnes	6.1 million tonnes	8.6 million tonnes	
		Reduction of electricity consumption per unit product	28%	25%	30%	3.2 Energy Management
		The total amount of additional electricity saved (base year: 2015)	1,007 GWh	818 GWh	1,000 GWh	
		Reduction of waste generated per unit product	44.2%	35%	50%	3.4 Waste Management
		Cumulative waste reduction (base year: 2015)	Cumulative additional waste reduction: 17,684 tonnes	118,000 tonnes ^{Note 1}	273,000 tonnes ^{Note 1}	
		Reduction of GHG emissions per unit product (with 2010 emissions scenario as the baseline)	56.7%	55%	60%	3.1 Climate Action
		Absolute reduction of GHG emissions (base year: 2020)	31% (Scope 1+2) 23.7% (Scope 3)	-	42% (Scope 1+2) 25% (Scope 3)	3.1 Climate Action
		Ratio of renewable energy in total electricity used	17.1%	25%	50%	3.2 Energy Management
		Reclaimed water and desalinated water usage rate	17.2%	18%	32%	3.3 Water Resources Management

(Continued)		2024		2025	2030		
	Strategy	Item	Achievements	Goals		Location	
 Minimize environmental footprint		Protect nature and promote biodiversity NEW	Completed biodiversity survey of Taiwan Fabs	Conduct TNFD application assessment	Cumulative execution of 5 native species conservation or restoration projects (base year: 2024)	3.6	Supporting Biodiversity
		Develop circular economy solutions/products	2 items in total	4 items in total	6 items in total	3.4	Waste Management
		Ratio of in-fab resource conversion ^{Note 2}	19.7%	22%	35%		
		VOC reduction	97.3% *	>95%	>97% (new fabs) >95% (existing fabs)	3.5	Air Pollution Control
		Key water pollution indicators better than regulated standards	74.3% **	60%	70%	3.3.3	Water Pollution Control
 Create social value		Mid- and high-performance retention rate *	96.1%	≥ 93.5%	≥ 94%	4.2	Diversity, Equity and Inclusion in the Workplace
		Proportion of women among managerial personnel	22.3% *	23%	24%		
		Questionnaire survey coverage rate	98.9% ***	98.9%	99%	4.1.2	Employer-employee Communication
		Work engagement score	82.3 ***	80.2	>80.5		
		Disabling Injury Frequency Rate (FR) and Disabling Injury Severity Rate (SR)	FR: 0.18 and SR: 2, which is better than the 3-year average for the semiconductor manufacturing industry (FR: 0.62 and SR: 36) ^{Note 3}	FR: 0.56 SR: 31	FR: 0.27 SR: 6	4.3.1	Maintain Workplace Safety
		Semiconductor education programs				5.2	Youth Talent Cultivation
		Number of semiconductor promotion events	90 *	62	65		
		Annual number of participants	7,449 *	3,200	3,500		
		Cumulative investment in semiconductor education (base year 2020)	NT\$ 41.65 million *	NT\$ 50 million	NT\$ 87.5 million		
		Semiconductor Science Education for elementary school student and community				5.1	Promote Common-Prosperity Society
		Number of events	25 *	24	33		
		Number of beneficiaries	4,677 *	5,710	7,220		
		Social participation activities					
		Number of beneficiaries	45,028 *	40,448	42,760		
		Employee volunteer hours	9,302.5 *	7,485	9,190		

Scope: *UMC; **UMC (Taiwan); ***UMC, subsidiaries HJ and USCXM; without any * remark is UMC Group (UMC, subsidiaries HJ, USCXM, Wavetek and USJC).

Note: 1. The cumulative reduction benefits for each year from 2015 to the target year.

2. Ratio of in-fab resources conversion = (volume of in-fab resource conversion ÷ volume of total waste generated)

3. Calculation based on OSHA statistical model and the average announced in 2022.

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- 3 Implementing Green Operations
- 4 Providing a Friendly Workplace
- 5 Enhancing Society Mutual-Prosperity

Appendix

Enterprise Risk Management

In the second half of 2018, UMC formed an Enterprise Risk Management Committee, guided by the COSO enterprise risk management framework, to enhance the management and oversight of major risks across all levels of the Company. According to the Company's risk management policy, the Committee serves to manage and prioritize countermeasures for major risks. In 2020, UMC completed the standard operating procedures of "Enterprise Risk Management Manual" for employees. Since 2021, UMC has been periodically cooperating with external consultants to audit the completeness of the Company's enterprise risk management. Additionally, starting from 2021, UMC conducts annual enterprise risk management training to ensure the comprehensiveness of risk management and establish a solid enterprise risk culture.

Enterprise Risk Management Policies

- In order to maintain long-term stable operations, UMC actively conducts company-wide risk management through well-established management organizations and methods:
- Effectively preventing and controlling enterprise risks while simultaneously identifying potential opportunities
 - Comprehensively implementing controls and management procedures in daily operations
 - Shaping and deepening risk culture
 - Establishing complete and transparent risk communication with all stakeholders

Non-Executive Directors' Expertise in Risk Management

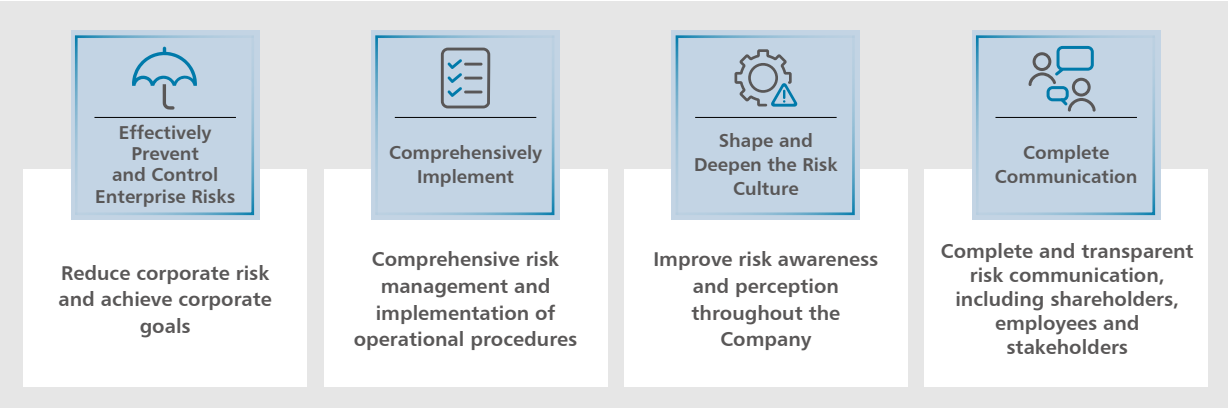
On the UMC board of directors, all non-executive directors possess industry experience in the technology industry or financial institutions before joining the current board. Meanwhile, they have accumulated professional knowledge in Enterprise Risk Management. During their tenure as directors at UMC, they participate in policy planning for enterprise risk management, review the Company's long-term strategies, supervise and guide the execution of the Enterprise Risk Management Committee, and assist the Company in improving its risk management culture and addressing potential crises arising from emerging risks. The Sustainability Development and Nomination Committee is composed of six independent directors and supervises the Sustainability Committee and the Enterprise Risk Management Committee. When the Company formulates important strategies and makes decisions, the Committee leverages its professional expertise to simultaneously consider the potential sustainability risks and opportunities associated with the Company.

For details on the risk management experience of non-executive directors, please refer to Chapter 1.1.1 Operation of the Board of Directors.

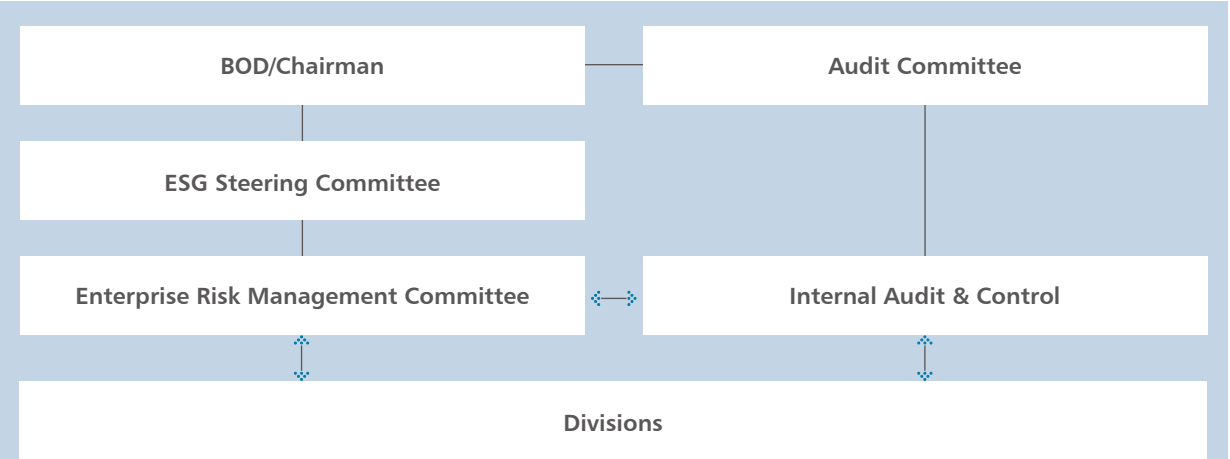
Enterprise Risk Management Structure

The Enterprise Risk Management Committee coordinates relevant divisions to jointly review the internal and external risks that the Company faces, and facilitates risk response measures for company-wide major risks. The Enterprise Risk Management Committee is chaired by the Executive Vice President, who reports to the ESG Steering Committee to ensure alignment with the Company's sustainability strategy. ESG Steering Committee, overseen by the Sustainability Development and Nominating Committee of Board-level (composed of 6 independent directors), reports to the Board of Directors at least once a year to align risk management policies and report on risk results. Additionally, UMC enterprise risk management also incorporates with the Company's internal auditing and control functions, primarily overseen by the Audit Division and reports to the Audit Committee to ensure that the risks associated with operations have been effectively controlled.

Enterprise Risk Management Policies



Enterprise Risk Management Structure



Sustainable Performance Highlights

MSCI ESG Rating

Rated "**AA level**" in MSCI ESG Rating, while also again selected as a component in the MSCI ACWI ESG Leaders Index and MSCI Emerging Markets ESG Leaders Index.

2024 MSCI ESG Leaders Indexes Constituent



Dow Jones Sustainability Indices

Selected for the Dow Jones Sustainability Indices (DJSI)'s World Index, **top ranking among semiconductor peers**, marking the Company's 17th consecutive year of inclusion and was also named a constituent of the Emerging Markets Index.

ISS ESG Corporate Rating

Rated "Prime" by ISS (Institutional Shareholder Services) ESG Corporate Rating for the **8th consecutive year**.



World's Best Employers

Standing out among numerous multinational companies, UMC has been selected for the prestigious **Forbes** list of Best Employers.



CDP Evaluation

Included in A List (Leadership Level) in both "Climate Change" and "Water Security" disclosure projects, **the only semiconductor firm globally to secure the "double A" recognition for 3th consecutive years**.



Corporate Governance Accreditation Top 5%

Ranked in the **top 5%** of companies for the **10th consecutive year** by the Taiwan Stock Exchange and Taipei Exchange.



FTSE4Good Index

Again selected as a component of FTSE4Good All-World Index, FTSE4Good Emerging Index and FTSE4Good TIP Taiwan ESG Index.



CRIF Asia Top 1000

The inaugural Asia Top 1000 survey launched by CRIF recognized UMC for its outstanding financial performance and ESG achievements, ranking the **7th in Asia and the 2nd in Taiwan** among the 68 selected companies in Taiwan.



Taiwan Corporate Sustainability Awards

Awarded "**Top 10 Taiwanese Companies Sustainability Model Award**" and the "Corporate Sustainability Report Award." UMC has won the "Corporate Sustainability Report Awards" for the **17th consecutive year**, and was awarded the highest Platinum award for the 7th time.

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Value Creation: Major Capital Investments and Output

UMC Group refers to the "Integrated Reporting Framework" of the Value Reporting Foundation (VRF) and describes the Company's six categories of operational investments and outputs.

Operational Resources	Capital Investments The Meaning of Each Capital to UMC Group	Key Investment Projects	2024 Corresponding Sustainable Outputs
The below-mentioned resource statistics are reported as of the end of 2023. ❖ Total assets of NT\$559.19 billion ❖ Total market capitalization of NT\$659.09 billion ❖ Approximately 19,833 employees worldwide ❖ 15,196 patents granted in total	 Financial Capital Through operations and investments, the Company's daily operations, production and services are maintained.	Financial structure strengthening, cost-competitive capacity expansion, and product portfolio adjustment.	❖ Total revenue (UMC) of NT\$47.2 billion ❖ Earnings per share of NT\$3.8
	 Manufactured Capital Including the investment and maintenance of the facilities used in wafer fabrication and integration fabs, as well as in production and R&D, in order to produce products that meet customer demands.	Capital expenditures were approximately US\$2.9 billion, primarily allocated for capacity expansions in Tainan, Taiwan; Singapore; and Xiamen, China; and the rest for product mix optimization.	❖ Annual wafer shipments reached 3.446 million pieces of 12-inch equivalent wafers
	 Human Capital By providing a good and positive workplace environment along with proper training, employees' knowledge, experience, expertise, productivity and loyalty can be enhanced.	❖ Education and training costs amounted to NT\$ 77.7 million. ❖ Salary and welfare expenses amounted to NT\$32.8 billion.	❖ Score of employee engagement: 80.9** ❖ Employee turnover rate of 6.1% ❖ Average NT\$4,053 training expense per employee
	 Intellectual Capital Including patents, research and development capabilities, and cooperation agreements that can enhance the company's competitive advantage or generate intangible assets that exceed the company's book value.	The overall number of patents granted has been growing steadily, with emphasis on patent portfolio to protect intellectual property rights of its technologies. ❖ Invested NT\$15.6 billion in R&D.	❖ Obtained 364 domestic and foreign patents, with a total of 15,560 patents granted over the years* ❖ Ratio of satisfied customers: 92%
	 Natural Capital Committed to minimizing the various impacts of fab operations on the environment, including reducing natural resource consumption, decreasing pollution output, and promoting green procurement	Promoted "Green Action Plan" to conserve water and electricity, and reduce waste and F-GHGs emissions. ❖ Environmental protection-related capital expenditures totaled NT\$3.91 billion ❖ Green procurement amounted to NT\$70 million***	❖ 1.02 million tonnes of water savings ❖ 255,530 MWh of energy saving ❖ The volume of waste outsourcing reuse volume reached to over 48,000 tonnes and the re-use rate was 94.2% ❖ F-GHG emissions reduced by 790,000 tonnes of CO₂e*
	 Social Relationship Capital** The sum of resources or capabilities connected through social networks, including the maintenance of long-term partnerships with value chain partners and other stakeholders.	❖ Dedicated a total of NT\$75.89 million to social charity activities ❖ A total of 9,302.5 hours spent in corporate volunteering services.	❖ UMC Eco Echo Award Initiative awarded 3 NGO ecological conservation projects, 8 youth environmental initiatives, and 2 green innovation projects. ❖ UMC Energy Saving Service Team served 11 organizations and reduced about 1,152 tonnes CO₂e. ❖ UMC's public services have reached 45,028 person-times beneficiaries.

Scope: Without any * remark is UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC); *UMC; **UMC, subsidiaries HJ, USCXM and USJC; ***UMC (Taiwan).

Determination of Material Topics

The sustainability report is one of the main channels which UMC Group communicates ESG issues to stakeholders. To ensure that the report focuses on material topics related to external stakeholders and its own operations, **UMC conducts the materiality analysis annually**. In accordance with the GRI Standard: GRI 3 Materiality Topics 2021, material topics with significant impacts are identified through engaging with stakeholders and experts, and the assessing process for the positive and negative impacts. The results of the materiality analysis are then disclosed in the sustainability report to provide a comprehensive overview of UMC Group's management and performance in each material topic.

Process to Determine Material Topics

1 Understand UMC's Organizational Context

Understand links between UMC's activities, business relationships and the sustainability issues as well as identify major stakeholders

7 major categories of stakeholders

Referencing the nature of its businesses as well as the 5 key principles of AA1000 SES (Stakeholder Engagement Standard): Dependency, Responsibility, Tension, Influence and Diverse perspective, UMC identified a total of 7 major categories of stakeholders including customers, employees, investors, suppliers, governmental agencies, community/NGOs, and media.

8 sources of sustainability issues

- ❖ International Sustainability Reporting Guidelines: GRI Standards, SASB Semiconductors Sustainability Accounting Standard, IFRS Sustainability Disclosure Standard
- ❖ ISO 26000 Guidance on Social Responsibility
- ❖ UN Sustainability Development Goals (SDGs)
- ❖ UN Global Compact,
- ❖ Code of Conduct, Responsible Business Alliance (RBA)
- ❖ Industry-related special issues
- ❖ Concerned issues in domestic/international ESG ratings, including DJSI index, MSCI ESG index or FTSE Global index, and etc.,
- ❖ Global Top 10 Risks by World Economic Forum (WEF)

19 sustainability issues

Reviewing the various sources mentioned above, a total of 19 sustainability issues were summarized.

2 Identify Actual and Potential Impacts

To identify the actual and potential impacts of sustainability issues, the materiality assessment is conducted in accordance with the GRI Sustainability Reporting Standards and incorporating the principle of "Double Materiality" from the European Sustainability Reporting Standards (ESRS).

Inventory of positive/negative impacts

❖ Double Materiality

- (1) Sustainable Materiality: identify actual and potential impacts on the economy, environment, and people, including impacts on their human rights, across the Company's activities and business relationship. Define the types and locations of impacts that may occur throughout the value chain.
- (2) Operational Materiality: review whether there are financial, reputational, and operational risks associated with the sustainability issues related to the Company's operations or business relationships.
- ❖ **Sources for impacts inventory:** various management systems, government agencies' laws and regulations, or risk-management-related documents, and channels for collecting stakeholders' opinions, such as questionnaires, grievance mechanisms, etc.

3 Assess the Significance of the Impacts

Quantify the impacts related to each sustainability issue through internal assessment, as well as the involvement of external stakeholder to better understand any concerns.

Double Materiality

❖ Sustainable Materiality

- External Impact: Through inventorying above-mentioned documents by various first-level divisions related to the issues, internal opinions have been consolidated to assess the positive and negative impacts that the Company's operations may have on each sustainable issue through quantifying the severity of impact and the likelihood of occurrence.

❖ Operational Materiality

- Internal Impact: Through reviewing the risk assessment results of sustainability issues via Enterprise Risk Management Mechanism, the significant impacts on the Company's own operations, including the impacts and occurrence probabilities on the Company's financial, operational, personnel, and reputation aspects can be identified.

Involvement of External Stakeholders

- ❖ In order to understand the concerns of stakeholders, a survey was conducted with a total of 388 responses
 - 156 responses: including investors, customers, key suppliers, government agencies, NGOss, and local communities.
 - 232 responses: employees

Opinions from Experts

- ❖ To ensure objectivity and alignment with international trends, opinions from experts and benchmarks from peers and international sustainability ratings have been included in the materiality analysis to improve the contents assessed.

4 Prioritize Significant Impact

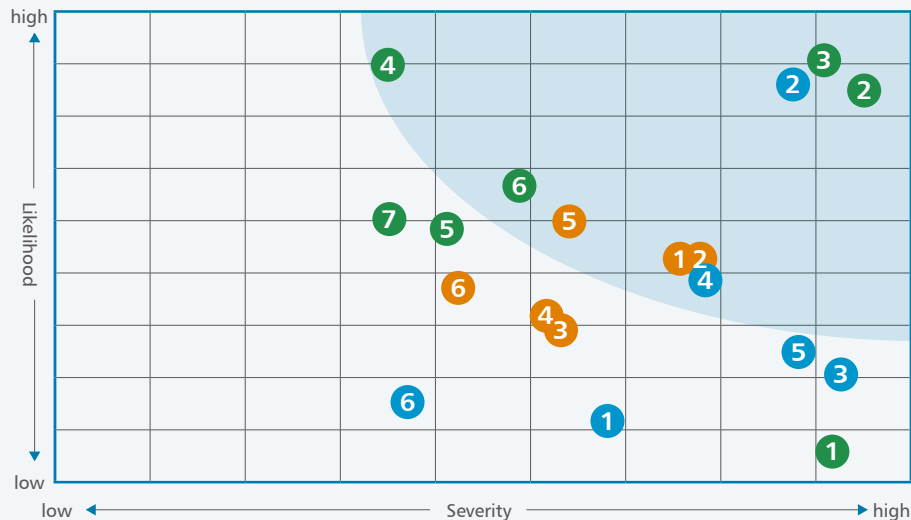
After analyzing collected statistics, material issues are prioritized in a materiality matrix based on the significance of their impacts

❖ In order to prioritize the significant impacts UMC may cause on each sustainability issue, a materiality matrix was created using the X-axis to represent the severity of impact and the Y-axis to represent likelihood of occurrence.

❖ The products of X and Y values of each sustainable issue were used to identify the top 50% of issues with significant positive or negative impacts.

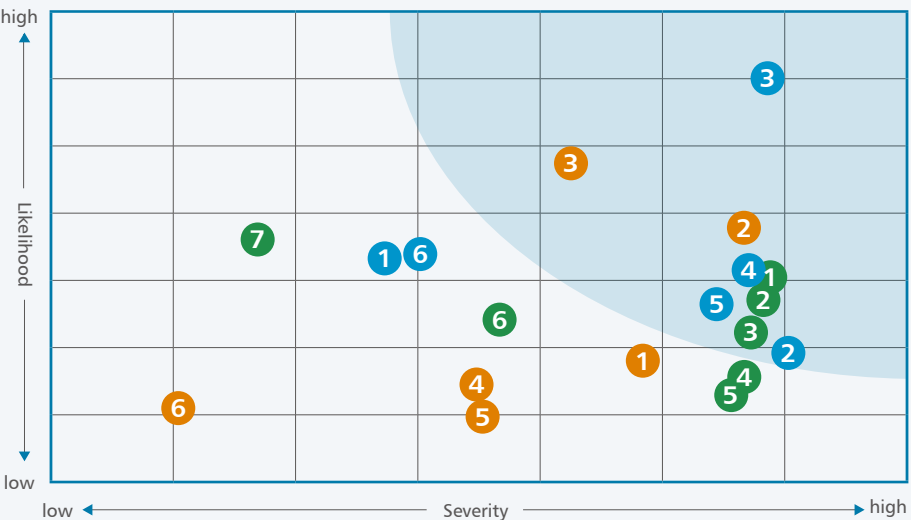
❖ After internal review and discussion, the results of annual materiality analysis identified a total of 13 material topics with the addition of the topic "Information Security and Privacy Protection" compared to 2023.

Positive Impacts



- ❖ Sustainability issues with significant positive impacts
 - Governance and Economic Dimension: 2 Sustainable Supply Chain Management 4 Customer Relationship
 - Environmental Dimension: 2 Climate Strategy and Action 3 Energy Management 4 Water and Wastewater Management 6 Waste Management and Circular Economy
 - Social Dimension: 1 Diversity and Inclusion 2 Talent Attraction and Retention 5 Human Capital Development
- ❖ Secondary topics
 - 1 Risk Management 3 Innovation Management and Intellectual Property Rights Protection Management
 - 5 Information Security 6 Integrity Management
 - 1 Product Responsibility and Lifecycle Assessment 5 Air Pollution Control 7 Biodiversity
 - 3 Occupational Health and Safety 4 Human Capital Development 6 Social Contribution and Participation

Negative Impacts



- ❖ Sustainability issues with significant negative impacts
 - Governance and Economic Dimension: 2 Sustainable Supply Chain Management 3 Innovation Management and Intellectual Property Rights Protection 4 Customer Relationship Management 5 Information Security and Privacy Protection
 - Environmental Dimension: 1 Product Responsibility and Lifecycle Assessment 2 Climate Strategy and Action 3 Energy Management
 - Social Dimension: 2 Talent Attraction and Retention 3 Occupational Health and Safety
- ❖ Secondary topics
 - 1 Risk Management 6 Integrity Management
 - 4 Water and Wastewater Management 5 Air Pollution Control 6 Waste Management and Circular Economy
 - 7 Biodiversity
 - 1 Diversity and Inclusion 4 Human Capital Development 5 Human Rights
 - 6 Social Contribution and Participation

5 Determine Material Topics

The process of materiality assessment was reviewed by the CS Committee before verified by a third-party assurance provider for this report. Furthermore, the materiality assessment results were reported to and signed off by the ESG Steering Committee and the Board of Directors.

❖ The results of materiality analysis will serve as a reference for the Company's Sustainability Strategy Blueprint to determine short-, medium- and long-term goals. This will ensure that relevant strategies and goals encompass the expectations of stakeholders and respond to the identification and management of significant impacts.

❖ In addition to disclosing management guidelines and currently measures for GRI-specific topics in this Report, UMC also discloses the mid- and long-term management goals for each material topic. Based on these goals, effectiveness will be tracked and reviewed.

❖ Moreover, the Company continues to develop and integrate the materiality assessment into its enterprise risk management (ERM). The materiality assessment results will be considered by ERM Committee. Once any topics are identified as having significant impacts on operations, the ERM Committee will regularly review related risks and corresponding managements.

Material Topics

● Direct Impact ○ Indirect Impact

Topic	Operational Significance to UMC	Economic, Environmental, Social and Operation Impacts					GRI Specific Topi	SASB Topic	Corresponding Chapter		
		UMC Group ^{Note1}	Suppliers	Investor	Costumers	Society ^{Note2}					
Governance and Economic Dimension											
Sustainable Supply Chain Management	UMC has numerous upstream and downstream suppliers. Cooperation with global partners can improve sustainability performance in the overall value chain.	●	●				204 Procurement Practices 308 Supplier Environmental Assessment 414 Supplier Social Assessment	Materials Sourcing	1.3	Building Sustainable Supply Chain	
Innovation Management and Intellectual Property Rights Protection	Strengthening UMC core competencies to provide wafer fabrication solutions that meet market trends and customer demands, thereby increasing revenue and enhancing corporate image.	●	●	○	●	○	UMC Specific Topic	Intellectual Property Protection and Competitive Behavior	2.1	Innovation and Technology Development 1.1.3 Legal Compliance	
Information Security and Privacy Protection	Strengthen information security awareness for all employees, and establish an information security framework that complies with regulations and customer needs to provide a safe production environment.	●		●	●	○	UMC Specific Topic	-		1.2.4 Commitment to Information Security Risk Management 1.1.3 Legal Compliance	
Customer Relationship Management	Customer validation can enhance the competitiveness of UMC, and is the key foundation for UMC's long-term operation	●	○	○	●	○	UMC Specific Topic	-		2.2 Customer Satisfaction 1.1.3 Legal Compliance	
Environmental Dimension											
Product Responsibility and Lifecycle Assessment	Provide wafer foundry solutions that meet customer demand, ensure that products are free of hazardous substances, and minimize any impacts may have on the environment	●	●		●		UMC Specific Topic	Product Lifecycle Management	2.3	Product Responsibility	
Climate Strategy and Action	Decent mitigation and adaptation strategies can reduce the direct and indirect impacts of climate change.	●	●	●	●	○	305 Emissions	Greenhouse Gas Emissions	3.1	Climate Action	
Energy Management	Good energy management practices can improve operational performance and also reduce the environmental burdens and carbon footprint	●		○	○	○	302 Energy	Energy management in Manufacturing	3.2	Energy Management	
Water and Wastewater Management	Water is basic to semiconductor production. Effective water resources management can enhance competitiveness.	●	○			○	303 Water and Effluents	Water Management	3.3	Water Resources Management	
Waste Management and Circular Economy	Reducing environmental load at all stages of a product life cycle is one of the important goals in UMC's environmental protection efforts.	●	●			○	306 Waste	Waste Management	3.4	Waste Management	

(continued)



(Continued)

Topic	Operational Significance to UMC	Economic, Environmental, Social and Operation Impacts					GRI Specific Topi	SASB Topic	Corresponding Chapter
		UMC Group ^{Note1}	Suppliers	Investor	Costumers	Society ^{Note2}			
Social Dimension									
Diversity and Inclusion	In terms of personnel selection, employment, education, and retention, there is no differential treatment based on gender, race, religion, political position, marital status, etc., so as to create harmony in the workplace and greater operational performance	●				●	405 Diversity and Equal Opportunity	-	4.2 Diversity, Equity and Inclusion in the Workplace
Talent Attraction and Retention	Attracting and retaining talent can improve UMC's operational performance	●		○		○	401 Employment	Recruiting and Managing a Global and Skilled Workforce	4.2 Diversity, Equity and Inclusion in the Workplace
Occupational Health and Safety	Maintaining employee safety and health and providing a good workplace environment for employees are UMC's basic responsibilities.	●	●				403 Occupational Health and Safety	Employee Health and Safety	4.3 Protecting the Employees and the Work Environment
Human Rights	"Zero Tolerance" for various illegal violations in the workplace, prohibits any tangible or intangible discrimination and harassment in the workplace, and expects subsidiaries, joint ventures, suppliers, and partners to abide by the same policies	●	●				406 Non-discrimination 407 Freedom of Association and Collective Bargaining 408 Child Labor 409 Forced or Compulsory Labor	-	4.1 Enforce Human Rights Management
Note: 1. UMC Group includes the subsidiaries HJ, USCXM, Wavetek and USJC. 2. Society is considered among media, governmental agency, and community/NGOs									
Secondary Topics ●.....									
A total of 6 secondary issues are disclosed in this report with reference to the GRI specific topics.									
Topic		Direction of Response and Management					Corresponding Chapter		
Risk Management		Carry out systematic management for major enterprise-level risks, strengthen the identification of and response to major risks in business continuity, and perform the required drills and adjustments for major risks.					1.2.1	Risk Management System	
							1.2.2	Risk Management and Countermeasures	
							1.2.3	Comprehensive Business Risk Management	
Integrity Management		Establish good business conduct and ethics can ensure the sustainable operation of UMC and avoid operational risks caused by illegal or improper transfer of interests.					1.1.2	Integrity Management	
Human Capital Development		Develop performance improvement plans for underperformers and providing coaching for improvements. Improve employee performance through regular education and training.					4.2.3	Strengthening Talent Cultivation	
Air Pollution Control		Introduce and develop environment-friendly technologies to strengthen emission source reduction and pollutant prevention					3.5	Air Pollution Control	
Biodiversity		Ensure that the impact of company operations on biodiversity ranges from No Net Loss (NNL) to Net Positive Impact (NPI)					3.6	Supporting Biodiversity	
Social Contribution and Participation		Serve the community with core professional skills.					5.4	Enhancing Society Mutual-Prosperity	

Stakeholder Engagement

To maintain effective stakeholder communication, UMC has formulated an engagement mechanism for the identification of and communication with important stakeholders via various engagement channels and regular information disclosure.

Principles for Communication with Stakeholders



Objectives

- ❖ Evaluate and understand the reasonable expectations and requirements of the stakeholders, and provide appropriate responses to key ESG issues that the said stakeholders are concerned with.
- ❖ Consider all concerned issues and analyze the potential impact that each issue may exert upon the environment, society, economy, and business operations.
- ❖ Employ a system-based mechanism to continuously review and enhance corporate sustainability.

Categories of Major Stakeholders

	Customers	❖ Customer products are the main source of revenue. UMC provides a full range of services to promote its growth as well as that of UMC customers, thereby achieving business sustainability and mutual benefit.
	Employees	❖ Employees are the foundation of a company's sustainable development. UMC provides competitive compensation and benefits, and a diverse learning environment for all employees to exercise their potential, in order to enhance the operational performance and efficiency of the Company.
	Investors	❖ Investors provide a source of capital for a company. UMC transparently discloses its operational information and demonstrates the Company's operational effectiveness to increase investor support, stabilize the capital, and create value to attract investment.
	Suppliers	❖ Suppliers are important partners in sustainable development. UMC promotes various economic, social and environmental collaboration programs to enhance the overall value of the supply chain
	Government Agencies	❖ Government agencies set regulations for companies to follow. UMC strives to be a sustainable business by actively participating in communications and providing input that serves as a reference for policy formulation.
	Community/ NGO	❖ Communities and NGOs are important sources of diverse opinions for sustainable development. UMC works together with them to enhance positive social influences through ongoing communication and collaboration.
	Media	❖ The media serves as the communication bridge between the Company and stakeholders. UMC disseminates corporate information through media channels to help stakeholders obtain accurate information about the Company in a timely manner

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Mechanisms for Stakeholder Communication

	Item	Content	Frequency
	Stakeholders communication plan and implementation	✦ Consultation and communication channels with stakeholders are established by respective group under the CS Committee. ✦ Designated personnel to receive, record and reply to messages from stakeholders and to come up with appropriate responding measures	On-time
	Assessment of stakeholder communication outcome	✦ Stakeholder communication outcome reported, and key issues reviewed and responded to the CS Committee.	Quarterly
	Report of major stakeholder concerns	✦ Regularly report major communication issues to the Board of Directors. ✦ Report specific events to the Board of Directors through extraordinary (irregular) meetings	Annually
	Public disclosure	✦ Annual financial reports, sustainability report, etc.	Annually
		✦ Press Releases ✦ UMC official website: The Stakeholder Area was established for Stakeholder engagement ✦	As the occasion demands



Engagement Summary with Major Stakeholder

Stakeholder	Communication Channel/Frequency		Main Focus of Material Topics	Key Communication Outcome in 2024
	Periodic	Aperiodic		
 Employees	✦ Quarterly: Town Hall meetings, Secretary forums, Employee Welfare Committee meetings, employer-employee meetings, communication platforms, UMC ESG Newsletter ✦ Once every 6 months: Fab and Division meetings ✦ Annually: Employee satisfaction survey on benefits measures, service satisfaction survey, HR satisfaction surveys, employee work engagement survey	eUMC information website for employees, BBS message boards, sexual harassment complaint hotline, mailbox for reporting fraud or professional ethics violation, e-suggestion and feedback platform, confidential complaint system, extension 12885 "ER help" hotline, and MyUMC website	✦ Occupational Health and Safety ✦ Talent Attraction and Retention ✦ Information Security and Privacy Protection	✦ Continue to promote a comprehensive personnel health management program, which will be focusing on the three aspects of safe working environment, employee health protection and work-life balance. ✦ Completed employee work engagement surveys to better understand the real needs of employees and their willingness to remain in their employment. ✦ Continue to implement industrial salary surveys and provide competitive performance-based and differentiated remuneration and welfare systems (including bonuses, salary, and stock shares). ✦ Continue to enhance the UMC welfare information platform to improve accessibility to employee benefits and discounts. ✦ Strengthen communication of business strategies and directives and to continue the provision of up-to-date information of corporate performance. A total of 156 communication meetings were held. ✦ The e-suggestion opinion feedback platform received 191 opinions from various employees, and all cases (100%) have been closed.
 Customers	✦ Communication and discussion meetings	✦ Online Service Platform MyUMC ✦ Questionnaire response ✦ On-site audit and discussion ✦ Voice of Customer (VOC) instant customer online complaint system ✦ Customer satisfaction monitoring	✦ Risk Management ✦ Information Security and Privacy Protection ✦ Sustainable Supply Chain Management ✦ Integrity Management	✦ Held 42 quarterly business reviews with 20 customers. ✦ The Enterprise Risk Management Committee coordinates the key departments of risk management and control to jointly review the internal and external risks of UMC, and consolidates the risk response action plans, providing customers with a safe production environment, and reducing the Company's operational risks. ✦ Continually leveraged an international professional security company's resources to help on inspecting the robustness of overall security as a trusted third-party certification, and the results are used as the basis for further improvement. ✦ Continuous customer services: Provided a total of 103 BCM / BCP pieces of risk management information.

(continued)

(Continued)

Stakeholder	Communication Channel/Frequency		Main Focus of Material Topics	Key Communication Outcome in 2024
	Periodic	Aperiodic		
 Investors	General Shareholders: ‣ Annually: General shareholder meeting, Financial reports Institutional shareholders ‣ Quarterly: Investor conferences ‣ Quarterly: Domestic and overseas investor conferences	Institutional shareholders ‣ Domestic and overseas seminars for investing institutions ‣ Written or face-to-face communication	‣ Integrity Management ‣ Information Security and Privacy Protection ‣ Risk Management ‣ Climate Strategy and Action	‣ Continue to hold shareholders meetings and investor conferences in accordance with the annual plan, and communicate with shareholders about the list of director candidates. ‣ Upload multimedia information of the financial and business report in the shareholders' section of the UMC official website. http://www.umc.com/English/investors/e.asp . ‣ Work with the Financial Supervisory Commission (FSC) to complete corporate governance accreditation. ‣ Communicate the Company's ESG planning and performance through written and meeting formats according to the inquires of investment institutions. Investors Overview 
 Suppliers	‣ Annually: Questionnaires and audit visits. Jointly implement ESH and corporate social responsibility program with suppliers	‣ Review reports or meetings ‣ Environmental health and safety, and corporate sustainability related management briefings	‣ Information Security and Privacy Protection ‣ Integrity Management ‣ Occupational Health and Safety	‣ Implemented anti-corruption measures and promoted the signing of "Employment Ethics Terms and Conditions Agreement". ‣ Completed conflict mineral surveys. All 22 suppliers who underwent the conflict mineral audit met UMC's requirements for conflict mineral management. ‣ Held a Supplier Sustainability Seminar and Award Ceremony with the theme of "Sustainability and Prosperity" to promote the concept and policies of sustainable development, as well as to commend outstanding suppliers. ‣ Organized 47 ESH education and training sessions for contractors, with a total of 3,101 person-times trainees. ‣ Provide the latest health and safety regulations and information with contractors through collaboration through the jointly held promotional events. A total of 16 events were held this year, with 1,418 contractors participating.
 Governmental Agencies	‣ Monthly: Participate in the operations of the of the functional organizations of the Science Parks and Science Park Administrations	‣ Participate in public hearings and business seminars organized by governmental authorities	‣ Energy Management ‣ Occupational Health and Safety ‣ Social Contribution and Participation	‣ Played the role of coordinator for The Allied Association for Science Park Industries, facilitating regular discussions on relevant laws and regulations, and offering insights and recommendations based on operational experiences to form policy drafting. ‣ Invited by the Hsinchu Science Park Administration of the National Science and Technology Council to serve as a member of the "Safety and Health Expert Platform," providing professional assistance, consultation, and on-site coaching to enterprises within the Park, with the aim of enhancing occupational safety and health (OSH) management and implementation capacity. In 2024, a total of six on-site visits were conducted in the Park. ‣ Invited by the Occupational Safety and Health Administration, Ministry of Labor, to serve as a guest speaker at the "Northern Region OSHMS Promotion Association," delivering a talk on "Driving Forces Behind Occupational Safety and Health Management from the Perspective of Workplace Sustainability," and fostering knowledge exchange among member companies.
 Community/ NGO	‣ Participate in the operations of outside associations ‣ Participate in community activities or seminars ‣ Sharing activities and exchange visits		‣ Human Rights ‣ Social Contribution and Participation ‣ Integrity Management	‣ Promoted volunteer work culture, serving a total of 45,028 person-times beneficiaries. ‣ Through the diversified semiconductor knowledge promotion program, UMC is committed to building inter-school relationships and fostering high-tech talent by hosting 90 semiconductor knowledge promotion events. A total of 7,449 person-times participants attended these events throughout the year.
 Media	‣ Monthly: Company website updates	‣ Press releases, press conferences	‣ Risk Management ‣ Integrity Management ‣ Sustainable Supply Chain Management ‣ Energy Management ‣ Water and Wastewater Management	‣ Issued 36 press releases on corporate governance and sustainability management ‣ Quarterly newsletters are published to share UMC's sustainability progress on a regular basis. Topics shared in 2024 included "UMC Develops AI Assistant – UGPTs – to Boost Workplace Efficiency," "UMC's Secret Weapon for Combating Water Shortages: U-Water" and etc. Press Center 

Special Feature of the Year

UMC's Fab 12A Leads Supply Chain in Digital Transformation, Becoming the First Semiconductor Foundry in the Global Lighthouse Network

UMC has successfully integrated Industry 4.0 solutions into fab operations, which have enhanced process quality and supply chain efficiency while reducing greenhouse gas emissions. In 2025, UMC's flagship fab was selected for World Economic Forum's Global Lighthouse Network, becoming the first semiconductor foundry globally to earn Lighthouse status.

Located in the Southern Taiwan Science Park, UMC's Fab 12A leverages advanced technologies such as artificial intelligence (AI), the Industrial Internet of Things (IIoT), and big data to boost operating performance. Between 2019 and 2023, product yield increased to 97% and production cycle time reduced by 23%, while scrap rates decreased by 68%. As a result, revenue from specialty processes increased six-fold – driven by strengthened customer loyalty – while enhanced production efficiency

lowered emissions by 23%. UMC also supported our suppliers in introducing related solutions, helping them improve scheduling efficiency and reduce photomask lead time by 53%. The successes at Fab 12A demonstrate that digital transformation benefits not only our own operations, but also extends to the supply chain and environmental sustainability.

UMC's digital transformation started as early as 2010, when individual departments including IT and Manufacturing Technology independently developed solutions. Since then, Industry 4.0-related hardware, software, and network infrastructure have all gradually matured. In 2017, to centralize digital transformation efforts, the Smart Manufacturing division was established to centralize resources and coordinate cross-function collaboration. Given that Fab 12A is UMC's largest production site and R&D center, it was selected as the primary pilot site for Industry 4.0 implementation, from which successful solutions are then fanned out to other sites.

Digital Transformation's Impact on Communication, Quality, and Efficiency

Since the Smart Manufacturing division was created, more than 48 digital solutions have been successfully integrated into UMC's operations and value chain, 95% of which were developed in-house. These solutions can be divided into three key areas: external communication with customers and suppliers, process quality enhancement, and production efficiency optimization.

UMC focuses on specialty processes, which have more stringent specifications and higher complexity. Through machine learning tools, the lead time for customer design kits was by 57%, which has significantly strengthened customer engagement and loyalty. In addition, to ensure accuracy and timely delivery for photomasks, UMC collaborated with suppliers to introduce an AI-powered platform, streamlining data exchange between UMC,



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suppliers, and customers. This upstream and downstream integration reduced photomask lead times by 53%. Notably, these scheduling improvements benefit not only UMC, but also other customers of our suppliers, amplifying the impact beyond just our value chain.

Process quality is critical in earning customers' trust. As a leader in specialty processes including embedded high voltage (eHV), RFSOI, and embedded non-volatile memory (eNVM), UMC targets yield improvements in complex processes through AI-based defect detection, intelligent image recognition, and data-driven predictive models. These tools monitor and adjust abnormal or bottleneck equipment, ensuring timely detection and swift resolution of issues. As a result, product yield has increased to 97%, winning affirmation from customers and significantly contributing to business performance.

To enhance production efficiency, UMC has addressed bottleneck equipment in the manufacturing process through machine learning model prediction and smart gas a switching system. These solutions have reduced setup and queue times caused by gas switching during production, leading to an 87% reduction in cycle times for bottleneck equipment. In addition, UMC has developed an intelligent root cause analysis system that integrates smart image recognition, AI-based anomaly detection, and root cause diagnostics. This system enables engineers to quickly identify root causes and pinpoint equipment responsible for defects. As a result, the production efficiency of engineers has increased by 47%.

Continuous Improvement at All Levels

A long-standing culture of continuous improvement at all levels is a key enabler of UMC's digital transformation. UMC employs a total quality management (TQM) approach that drives employees to constantly seek ways to improve existing processes, products, and services. In 2024 alone, UMC completed 7,298 continuous improvement projects, with digital transformation a central theme,

which are expected to generated NT\$4.2 billion (US\$145 million) in benefits. In the government-hosted Taiwan Continuous Improvement Awards in 2024, UMC won 6 gold and 1 silver awards, and a AI defect identification solution developed by our defect management team was honored with the Best Innovation Award.

To further advance digital transformation, developing and reskilling digital talent has also become a key focus. UMC offers internal training programs such as our newly-established Equipment Academy to ensure that employees can acquire essential technical skills. In addition, UMC is actively working to scale Fab 12A's best practices and operating models to our other operating sites. The goal is to build a data-driven, fully autonomous smart factory, which will further enhance overall corporate competitiveness and drive efficiency in the broader semiconductor industry.



Special Feature of the Year

Inspection by AMR Autonomous Mobile Robot (AMR): AI-Assisted Preventive Management to Enhance Fab Efficiency and Safety



Robots equipped with surveillance cameras, sensors, thermal imaging devices, and ultrasonic acoustic cameras patrol the fab area, reporting any abnormalities immediately. This is not a futuristic scene from a Hollywood movie; it is part of the daily routine at UMC's flagship facility, Fab 12A.

Prevention is Better than Cure: Challenges of Manual Inspections

Although fabs are equipped with surveillance cameras and machines have built-in sensors, they are unable to provide full coverage, meaning traditional inspections still rely heavily on manpower. Inspection personnel must use their hearing, vision, and experience to assess equipment conditions. However, Fab 12A alone is equivalent in size to 10 football fields, making manual inspections inefficient as the fab continues to expand. Inspectors spent several hours each day just to complete routine checks.

In addition, as equipment density increases, it becomes increasingly difficult for inspection personnel to accurately identify abnormalities in loud environments; the sound of gas or liquid leaks is often barely audible, abnormal vibrations may be masked by equipment sounds, and overheating issues may not be noticed until they become visible to the naked eye. These challenges make early detection difficult, increasing the risks of equipment damage and accidents. To enable proactive risk identification and allow fab facility teams to stay ahead of potential hazards, UMC has actively sought effective solutions, one of them being deploying robotics.

Three Technologies Behind the Development of the AMR

Since 2017, UMC has been implementing Industry 4.0 technologies not only to enhance operational efficiency, but also to create a safer working environment for facility operations. With the continued advancement of robotics, UMC launched a robotics project in 2023 through the joint efforts of the Facility Operation Division, Smart Manufacturing Division, and IT Department. The project began with the deployment of larger inspection vehicles, which operated under a basic backend monitoring system to detect abnormal heat sources, gas leaks, water leaks, and vibrations.

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Later, the initiative was upgraded to Autonomous Mobile Robots (AMRs), which serve as the front line of smart inspection and risk detection. Autonomous Mobile Robots integrate three key technologies, featuring autonomous navigation and advanced sensing capabilities. They can monitor equipment conditions in real time, truly becoming the "eyes and ears" of the fab.

Three Key Technologies

1 Ultrasonic acoustic camera

❖ Detects imperceptible abnormal sounds

By detecting high-frequency sound waves, this device can identify subtle noises such as gas or liquid leaks, even in noisy environments, as well as sound wave variations caused by abnormal mechanical vibrations.

2 Thermal imaging camera (IR)

❖ Identifies hidden heat sources and leak indicator

Overheating often signals internal mechanical malfunctions, while fluid leaks are a common safety hazard in factories. Thermal imaging camera can detect residual liquid on the floor that is invisible to the naked eye, enabling the early identification of oil or chemical leaks. It also highlights high-temperature areas to help engineers quickly locate problem zones.

3 Real-time imaging and augmented reality (AR) technology

❖ Smart inspections with reduced human error

Through AI-assisted image analysis, robots can instantly detect work safety violations, such as personnel not wearing helmets or abnormal accumulation of objects in equipment area. More importantly, with AR technology, the system can mark the exact location of abnormal equipment directly on the display interface, allowing operating personnel to instantly understand the situation and dispatch maintenance teams promptly.



Human Expertise + Artificial Intelligence (AI) A Safer and More Efficient Work Environmen

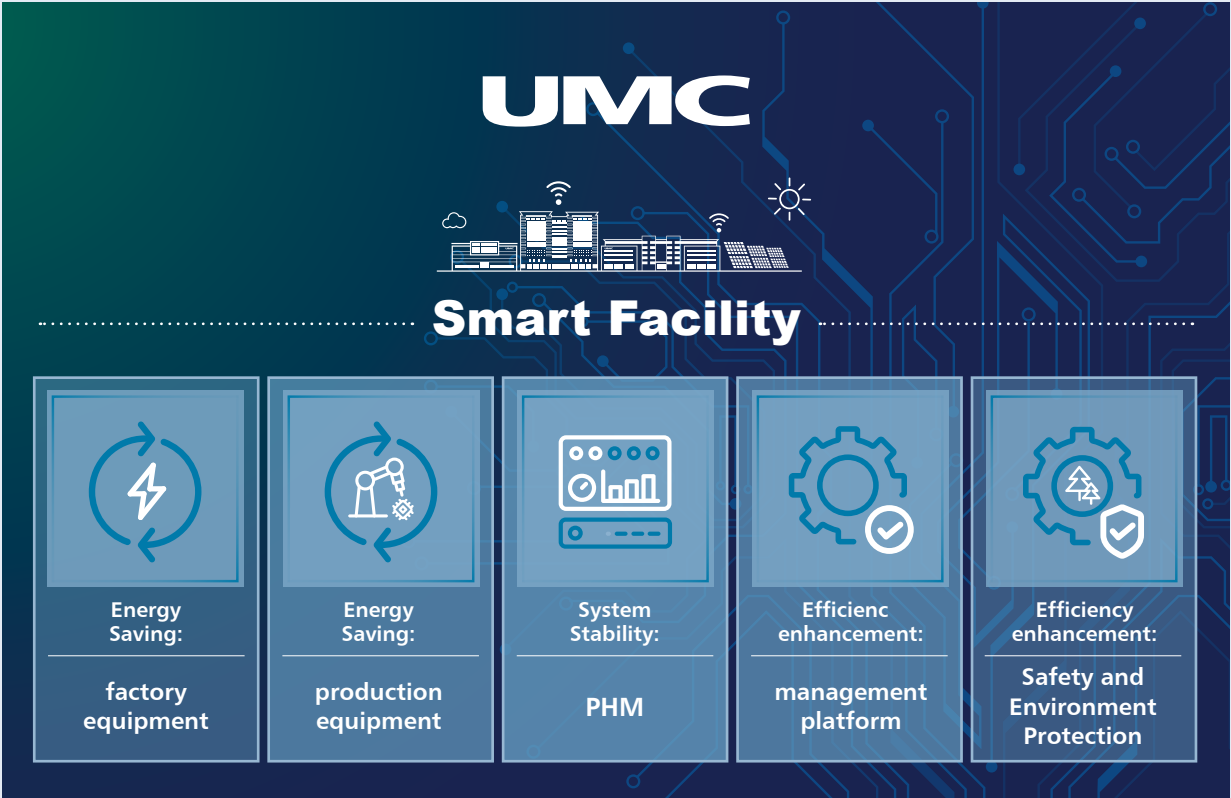
In addition to integrating a range of basic sensors, AMRs are equipped with functions such as wireless network connectivity, path planning, obstacle avoidance, and automatic charging. Problems that were previously difficult to detect through traditional inspection methods can now be analyzed by AI, which accurately identifies potential sources of abnormalities, allowing maintenance teams to intervene early and prevent further equipment damage. This not only significantly reduces the need for manpower and improves operational efficiency, but also minimizes the risk of employees being exposed to potential hazards. Furthermore, with the combination of AI analytics and the robot monitoring backend system (iRobot), AMRs are capable of continuously learning from inspection data. This enables them to refine their decision-making ability over time, improving the accuracy of abnormality detection and enhancing the predictive capabilities of maintenance operations.

AMRs were initially deployed in non-cleanroom areas of Fab 12A P5/6, and is set to enter cleanroom environments in 2025. By 2027, AMRs will be used across all facilities of Fab 12A. In addition, the system will be integrated with the fab's Supervisory Control and Data Acquisition (SCADA) system and Smart Facility Management platform, establishing a more comprehensive framework for smart factory management.

Special Feature of the Year

Smart Facility Industry 4.0 Assists Energy Saving and Carbon Reduction – Three Major Projects Use AI to Cut Carbon Emissions

In a semiconductor fab, facility systems are the largest contributor to electricity consumption, accounting for approximately half of the total use. To effectively reduce carbon emissions from electricity use, it is crucial to identify the biggest energy users. As part of UMC's digital transformation drive, Smart Manufacturing Division, the Facility Operation Division, and the IT Division collaborated on three carbon reduction projects - i-Chiller 2.0, i-CDA, and i-MAU – to leverage Industry 4.0 technologies in reducing electricity use in fab supporting facilities.



i-Chiller 2.0 ✦ Upgrading Chillers from Automation to Intelligence

Precise temperature control is essential for stable yields, making the chiller system, a key component in semiconductor manufacturing. However, it is also one of the most energy-intensive systems in the fab. Through big data and characteristic parameter analysis, the upgraded i-Chiller 2.0 system is able to identify machines requiring the most cooling and dynamically allocates resources to where they are needed the most, optimizing overall system performance and energy efficiency.

The intelligent chiller system has been deployed across three sites at UMC's Fab 12A. When it is implemented across all UMC fabs by 2027, the system will help the company save NT\$50 million (US\$1.7 million) in electricity costs annually.

i-CDA ✦ AI-assisted Decision-making

The semiconductor manufacturing process requires large volumes of clean, dry air to power pneumatic components and valve controls, making the Compressed Dry Air (CDA) system the second-largest energy-consuming facility system in a fab. To reduce its energy consumption, the i-CDA project utilizes AI algorithms to create efficiency prediction models for individual compressors and automatically optimize system configurations, replacing the traditional manual adjustments.

The i-CDA system also features an early warning function that alerts operators of abnormal air supply levels and provides advance notifications for preventive maintenance, ensuring stable operation and achieving intelligent energy savings. Currently implemented at Fab 12A P5/P6, it is expected to save approximately NT\$6 million (US\$206,716) in electricity costs annually.

i-MAU ✦ Smart Ventilation and Filtration Control

The Make-up Air Unit (MAU) is another key facility system that serves as the air exchange and filtration gateway for the

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fab. It draws in fresh air, filters out fine particles, and adjusts temperature and humidity to meet cleanroom standards through a process of heating, humidifying, and cooling before replacing the existing indoor air.

With UMC's expansion of 12-inch fabs and the ramp-up of new equipment, alongside seasonal slowdowns in 8-inch fabs, the MAU system now faces highly variable loads. Traditional manual adjustment methods are insufficient for managing growing system complexity. The i-MAU project uses Industry 4.0's Sensor Model Platform (SMP) techniques, integrating IIoT, AI, and big data algorithms to develop a decision-support system that dynamically optimizes the number of operating units in real time, ensuring efficient resource allocation. Already deployed at Fab 12A, it is expected to save over NT\$10 million (US\$344,554) in electricity costs annually, and will also be implemented in UMC's Singapore and Japan fabs.

Carbon Management Platform and Internal Carbon Pricing

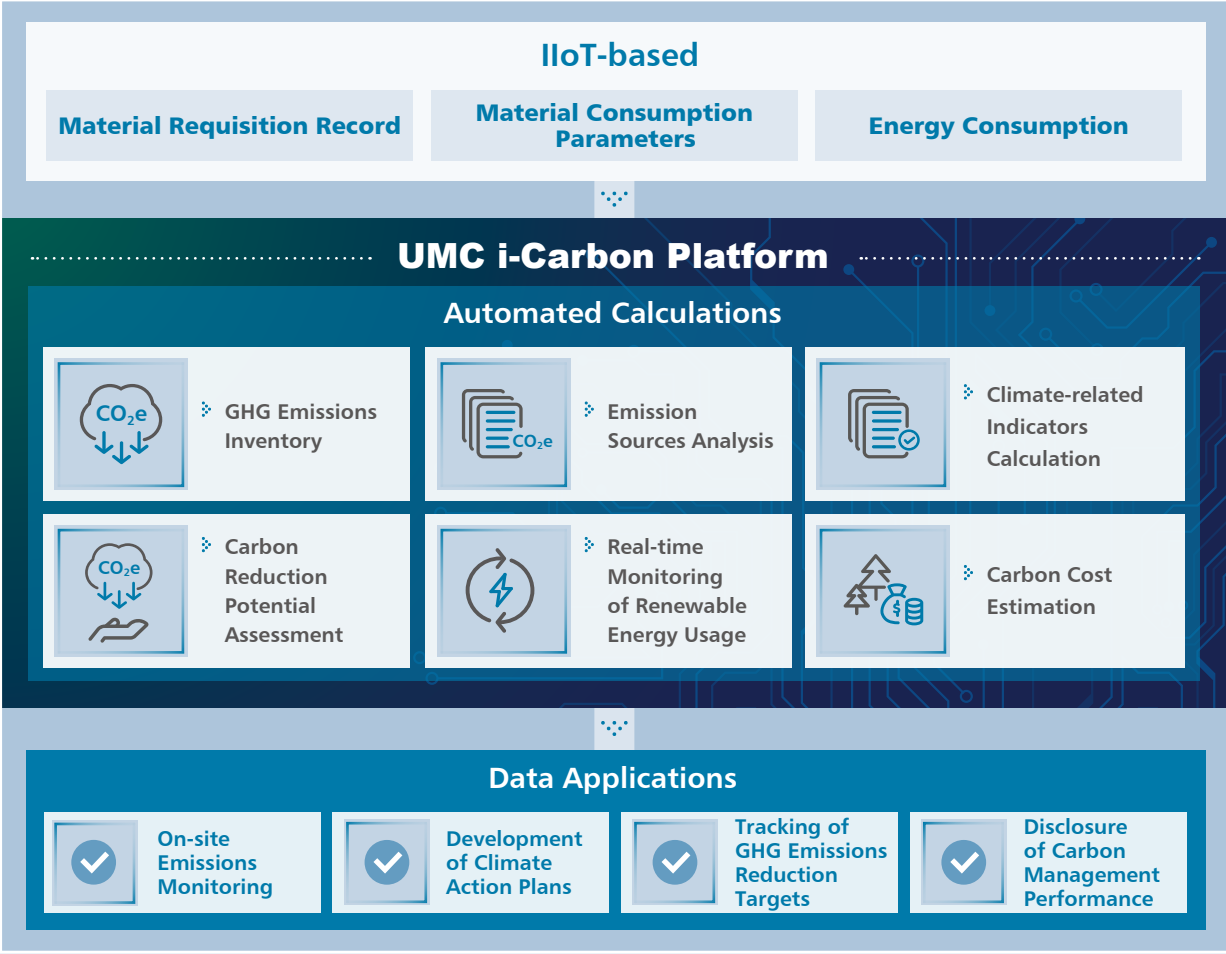
With the launch of more carbon reduction projects, UMC is also advancing systematic carbon management by integrating company-wide carbon emissions data. The carbon management platform connects data from over 1,000 machines, increasing update frequency from annually to monthly. The platform significantly reduces the time needed for manual data collection and calculation, and provides real-time access to essential carbon data for faster responses to production scenarios.

The platform also sorts carbon emissions by processes or individual machines, clearly identifying sources and distribution, which enables the development of more precise and effective carbon reduction strategies. In 2023, the platform drove improvements to the top 30 highest carbon-emitting machines, leading to a reduction of 127,000 metric tons of emissions in 2024.

Internal carbon pricing is also an important tool in driving emissions reduction. By setting an internal carbon price

of US\$100 per metric ton, departments can more accurately quantify the benefits of equipment upgrades. In 2024, several energy-saving improvement projects were approved, including initiatives targeting electric boiler efficiency and cooling tower systems, highlight the positive impact of putting a price on carbon in driving emissions reduction.

Whether employing AI to enhance equipment performance or utilizing internal carbon pricing to incentivize emissions reduction, UMC's actions reflect a strong commitment to environmental sustainability and ensure that the company is on track to achieve the goal of net zero.



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Leading in Corporate Governance



- 1.1 Enhancing Corporate Governance
- 1.2 Implementing Risk Management
- 1.3 Building Sustainable Supply Chain

Leading in Corporate Governance

Important Stakeholders | Shareholders | Suppliers | Government |

UMC continues to refine its corporate governance practices by establishing a rigorous corporate governance structure, strengthening the functions of the Board of Directors, and building a supply chain that emphasizes human rights, social responsibility, labor rights, safety, health, and sustainable development. In order to implement its business strategy, UMC continues to improve its corporate risk management mechanism and strengthen the supervision of all operational levels of the Company. Additionally, UMC is committed to establishing good business practices and ethical standards, living up to its core values of integrity and honesty, and focusing on protecting shareholders' rights and interests to realize UMC's vision.

SDGs



Performance Highlights 2024

★

 Ranked in the top **5%** of the Corporate Governance Evaluation for 10th consecutive year



★

 Designated a **"Lead Independent Director"** to enhance communication efficiency between the management team and the Board of Directors

★

 Being the only company to receive the Taiwan Intellectual Property Management System (TIPS) **AAA** certification

★

 The Nominating Committee has been renamed to the **"Sustainable Development and Nominating Committee"**

★

 **100%** of conflict-mineral suppliers passing third-party independent audits

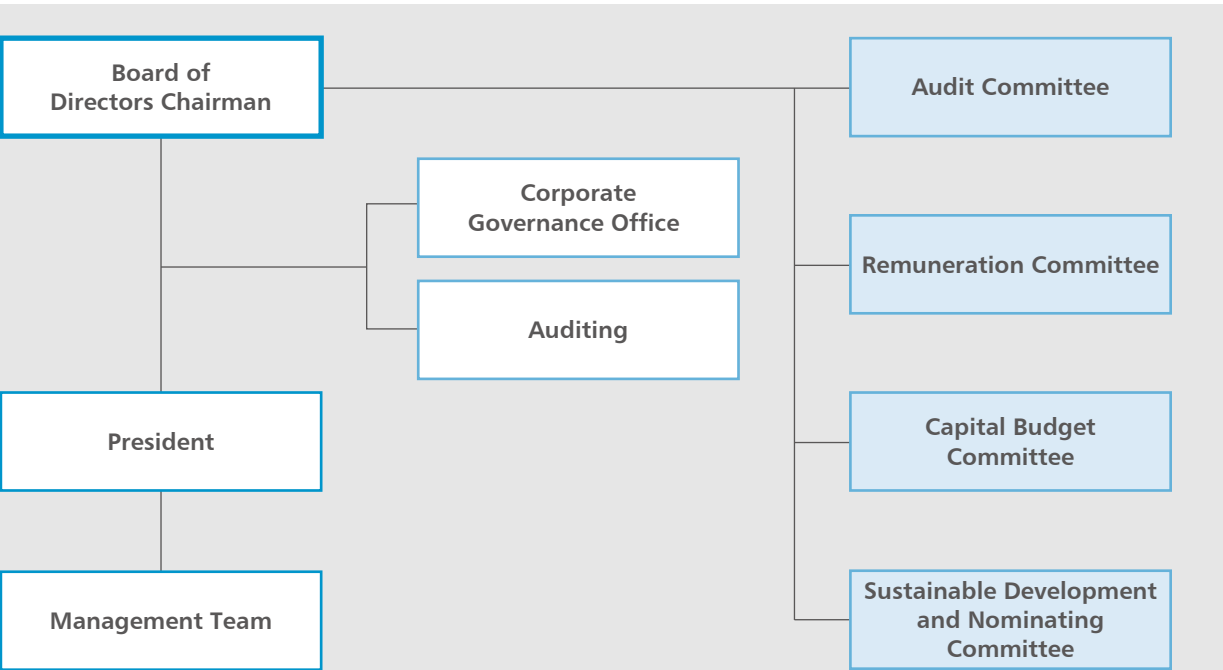
1.1 Enhancing Corporate Governance

- 1.1.1 Operation of the Board of Directors
- 1.1.2 Integrity Management
- 1.1.3 Compliance of Laws and Regulations

1.1.1 Operation of the Board of Directors

The UMC Board of Directors comprises nine members from different professional backgrounds, and is responsible for the Company's operations and supervision. The diverse academic and industry experiences of the board members are an asset to corporate decision-making and long-term strategy planning. Board type of the Company is one-tier system complied with the regulations, and the audit committee composed of all independent directors is in lieu of a supervisor. There are six independent directors in the Board. Currently, two third of the director seats are filled by independent directors. Current board members were elected on May 30, 2024 for a three-year term. There are three female directors in the Board. Each independent director does not serve as an independent director concurrently for more than three listed companies. As of the end of 2024, the average tenure of all directors was 6.2 years.

In 2024, a total of ten board meetings were held, and the average attendance rate was 100%, meeting the 80% requirement for board performance assessment. The ratio of total remuneration for board directors to the Company's after-tax net income was 1.00% in 2024.



Policy for the Nomination and Election of Directors

The composition of the Board should take into consideration the organizational culture, business model, and long-term development of the Company. The criteria established to ensure the diversity of the Board members shall include but not limited to the following three dimensions:



Board of Directors

Board members are elected by shareholders according to director election regulations for during shareholder meetings, and in compliance with the Board of Directors regulations and company constitution. The committees under the Board of Directors are nominated and approved by the Board according to company policy. All six independent directors qualified with professional qualification, work experience, limits of other mandate and independence of Regulations Governing Appointment of Independent Directors and Compliance Matters for Public Companies.

Title	Name	Gender	Member of Functional Committee	Professional and Background	Risk Management Experience	Attendance Rate	Age
Chair of Board, Chief Strategy Office	Stan Hung	Male	None	- Financial accounting and strategic management - Electronics-related industry experience		100%	65
Director, President	Jason S. Wang	Male	None	- Financial accounting and organizational leadership - Electronics-related industry experience		100%	62
	S C Chien	Male	None	- Engineering technology and organizational leadership - Electronics-related industry experience		100%	67
Independent Director	Lih J. Chen	Male	- Lead Independent Director - Convener for Audit Committee - Remuneration Committee member - Capital Budget Committee member - Sustainable Development and Nominating Committee member	- Engineering technology and organizational leadership - Electronics-related industry experience	- Served as the President of National Tsing Hua University and the Deputy Ministers of National Science Council - Serves as the convener of UMC audit committee - Experience in risk management of R&D activities and corporate governance	100%	79
	Jyuo-Min Shyu	Male	- Audit Committee member - Remuneration Committee member - Capital Budget Committee member - Convener for Sustainable Development and Nominating Committee	- Engineering technology and organizational leadership - Electronics-related industry experience	- Serves as independent director of Qisda Corporation, and Far Eastone Telecommunications Co., Ltd. - Was the Minister of the Ministry of Science and Technology and the President of the Industrial Technology Research Institute - Experience in risk management of technology institution and electronic industry	100%	71
	Kuang Si Shiu	Male	- Audit Committee member and financial expert - Convener for Remuneration Committee - Capital Budget Committee member - Sustainable Development and Nominating Committee member	- Financial accounting and organizational leadership - Banking and business decision making	- Serves as independent director of Yuanta Financial Holdings Co., Ltd. and Yuanta Commercial Bank Co., Ltd. - Was formerly Chairman & President of Mega Financial Holding Co., Ltd. and Mega International Commercial Bank Co., Ltd., Chairman of Hua Nan Financial Holdings Co., Ltd., Hua Nan Commercial Bank Ltd. and Land Bank of Taiwan Co., Ltd. - Experience in corporate finance and risk management of finance industry	100%	74
	Wen-Hsin Hsu	Female	- Audit Committee member and financial expert - Remuneration Committee member - Convener for Capital Budget Committee - Sustainable Development and Nominating Committee member	- Financial accounting and corporate sustainability - Electronics-related industry experience	- Serves as the CEO of Global MBA in National Taiwan University, associate dean of international affairs of College of Management School in National Taiwan University, and the independent director of Unitech Printed Circuit Board Corp., and ANT Precision Industry Co., Ltd., the CFO & SEVP of Chunghwa Telecom Co., Ltd. - Was director of Taiwan Insurance Guaranty Fund - Experience in risk management of electronic and insurance industries	100%	48
	Ling-Ling Wu	Female	- Audit Committee member - Remuneration Committee member - Capital Budget Committee member - Sustainable Development and Nominating Committee member	- Knowledge Management and Organizational Behavior - Electronics-related industry experience	- Serves as independent director of Hycon Technology Corp. - Was independent director of Faraday Technology Corp., Unimicron Technology Corp., ITE Technology Inc. and Crystalwise Technology Inc. - Experience in risk management of corporate operations	100%	60
	Su Lin Wang	Female	- Audit Committee member and financial expert - Remuneration Committee member - Capital Budget Committee member - Sustainable Development and Nominating Committee member	- Financial accounting and strategic management - Electronics-related industry experience	- Serves as independent director of Jess-Link products Co., Ltd. and AVerMedia Technologies, Inc. - Was Senior Vice President & Corp. CFO of Delta Electronics., Inc.. - Experience in business administration, risk management of finance and strategy	100%	63

Note: 1. There are three seats occupied by members who also serve as administrative directors, namely the President and the Chairman also serves as Chief Strategy Officer due to his expertise and experience in finance and strategic planning. The current positions of directors at UMC or at other companies are disclosed on page 10-13 of the UMC's 2024 Annual Report.
2. The board of directors have a diverse range of professional experience. Please refer to page 15 of the 2024 Annual Report and official website (https://www.umc.com/en/IR_Director/directors_information) for more details.
3. Training programs on corporate governance, risk management, ESG, and corporate sustainability are planned and arranged by the Chief Corporate Governance Officer for directors and officers annually. Please refer to page 42-43 of the 2024 Annual Report for more details.

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Functional Committee

Committee	Accountability	Achievement in 2024
Capital Budget Committee	The Capital Budget Committee was established in October 2013, comprises independent directors and outside directors to assist in the company's long-term development strategy, financial planning and business performance. The Committee discusses plan implementations, modifies and follows up on the company's capital expenditure budget by auditing its cost-effectiveness and tracking its performance.	The Committee met six times, audited and approved capital budget expenses of NT\$28,418 million.
Sustainable Development and Nominating Committee	The Sustainable Development and Nominating Committee (formerly the Nominating Committee) was established in December 2017, comprises independent directors to enhance management mechanism and to improve corporate governance. The Committee reviews the selection and performance assessment of directors and executives, and supervises issues related to environmental, social and corporate governance.	The Committee met three times, reviewed the Company's achievements in corporate governance, ethical management, risk management, supply chain management, DEI, social participation, climate change, circular economy and biodiversity, approved the Employees' Compensation linked with ESG KPI achievement, enacted the Company's Non-executive Director Shareholding Policy, reviewed and nominated candidates for the 16 th term of Directors of the Company.
Audit Committee	UMC's Audit Committee assists the Board in its oversight responsibilities, and is responsible for tasks dictated by the Company Act, Securities Exchange Act, and other relevant laws. Since UMC is listed on the New York Stock Exchange (NYSE), the Company is also subject to the US laws for foreign issuers. The UMC Audit Committee comprises independent directors, of which three are financial experts. According to the terms and responsibilities stated in the Company's Audit Committee Charter, members shall convene at least four times per year.	A total of seven meetings were held and the actual attendance rate was 100%. The Committee maintains good communication and contact channels with the Company's management, Chief Corporate Governance Officer, internal auditors and independent auditors.
Remuneration Committee	According to Regulations Governing the Appointment and Exercise of Powers by the Remuneration Committee of a Company Whose Stock is Listed on the Stock Exchange or Traded Over the Counter Act, UMC established the Remuneration Committee to strengthen corporate and risk management. In addition, to motivate and retain talent, the Committee reviews and supervises the remuneration system of the Company's directors and managers.	In 2024, the Committee convened five meetings in February, June, July, October, and December. For the actual attendance status of the Committee members, please refer to page 28 of UMC 2024 Annual Report.

Assessment of Board Performance

To implement corporate governance, enhance capability and review performance of the Board, UMC instituted the Rules for Performance Evaluation of the Board of Directors to assess the performance of the Board annually. The self-evaluation result of the 2024 performance of the Board, functional committees (including the Audit Committee, Remuneration Committee, Capital Budget Committee, and Sustainable Development and Nominating Committee), and individual board members was "Excellent", and the result was reported to the Sustainable Development and Nominating Committee and Board of Directors on February 26, 2025.

In addition to self-assessments, the performance of the Board assessed by an external independent professional institution or a panel of external experts and scholars at least once every three years in order to enhance the Board's operation. The Company engaged with an external independent institution, Taiwan Corporate Governance Association to conduct an assessment on board performance, and completed the evaluation on October 25, 2023. The assessment conclusion was reported to the Sustainable Development and Nominating Committee and the Board on December 13, 2023. The report addressed a recommendation on renaming the Nomination Committee to the Sustainability and Nominating Committee, and improving the relevant rules and regulations of the ESG organization, in order to enhance the effectiveness and efficiency of the Board. The Company amended the Articles of Incorporation to rename the Nomination Committee to the Sustainable Development and Nominating Committee in annual general meeting on May 30, 2024, and gradually implement the improvement plan in accordance with the recommendations of the report.



Assessment Method

- ❖ Online self-assessment
- ❖ Interviews with six members, including Chairman, Co-Presidents, Conveners for functional committees, head of corporate governance, and head of internal audit.



Assessment Scopes

- ❖ Composition, Direction, Authorization, Supervision, Communication, Internal Control, Risk Management, and Discipline of the Board



Conclusion

- ❖ The Company paid significant attention to the professionalism and diversity of the composition of the Board, and rigorously selected qualified candidates based on the Company's development needs and strategic goals. The seats of independent directors and female directors has exceeded the regulatory requirements, the Company has made leading arrangements with international perspective and which is praiseworthy.
- ❖ The Board share a good culture with aggressive and effective benefits.
- ❖ The Board focused the importance in sustainable development and risk management issues, which not only integrated sustainable goals with business operations, but also linked employee compensations with the achievement of ESG KPIs, which will strive to enhance the effectiveness of sustainable management.
- ❖ The Company emphasized on the development and cultivation of high-level management, and the succession plan in management team to reserve talents for the company's sustainable development.

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Conflict of Interest Management

UMC has established conflict of interest avoidance regulations for the rules of procedures of the Board of Directors, and the organizational charter of the Audit Committee as well as the Sustainable Development and Nomination Committee. If the matters to be discussed in the Board Meeting are involved with the interests of the directors or the legal person they represented, the directors shall explain the important points of the interests in the meeting. If such interests are in conflict with the interests of UMC, the directors shall not participate in the discussion as well as the voting process, and shall not exercise their voting rights on behalf of other directors. In addition, the names of related directors, contents of important points and circumstances of withdrawal shall all be stated in the meeting minutes.

UMC has established the codes of integrity management, regulations for the management of related party transactions, codes of ethical conduct for directors and managers, and codes of ethics for employees. There are clear provisions on avoidance of conflicts of interests and the implementation of the codes of ethics for employees is reported to the Audit Committee. For the communication with stakeholders, UMC has set up a special section for stakeholders on the website and relevant regulations and information are disclosed in details. There is no shareholder with controlling power in the Company, neither undisclosed conflict of interest between stakeholders, such as directors, suppliers and customers. UMC also has a spokesperson and an e-mail box, and has assigned dedicated personnel to be responsible for handling the questions and suggestions of stakeholders. There is no reported conflict of interest for UMC.

Internal Audit

UMC has established the internal audit function under the direct jurisdiction of the Board for the following purpose and tasks:



Main Purpose

- Examine and evaluate the effectiveness of the internal control system.
- Evaluate the efficiency and effectiveness of business operations.
- Ensure reliability, timeliness, transparency, and legal compliance of reports.
- Provide timely suggestions for improvement to ensure the sustainability and effective implementation of internal control tasks.



Duties and Responsibilities

- Conduct an annual audit in accordance to the provisions of the Taiwan authorities and in the event of risks.
- Issue audit reports and track improvement.
- Regularly revise the internal control system and audit implementation details as well as annual internal control self-assessments.
- Communicate with independent directors and report to the Audit Committee and Board of Directors.

UMC complies with the regulations of the Taiwanese Authority's "Regulations Governing Establishment of Internal Control Systems by Public Companies", completed the 2024 internal control self-assessment of the Company and its subsidiaries, and prepared a statement on the design and implementation effectiveness of the internal control system, which was submitted to the Audit Committee and Board of Directors for approval on February 26, 2025.

As UMC is listed on the New York Stock Exchange (NYSE), the Company is subject to US laws for foreign issuers. Since 2006, UMC has complied with the Sarbanes-Oxley Act Section 404 (SOX Section 404) and has been audited by the independent auditors annually. To date, the independent auditors have issued unqualified audit reports for UMC's internal control design and implementation performance.

Announcement of UMC 2024
Internal Control System Statement



The Organization and Operation of
UMC's Internal Audit



Executive Compensation

The performance evaluation and the policies, system, standards, and structure for salary remuneration of directors and executives are determined and reviewed by the Remuneration Committee. The committee members include independent directors who provide suggestions and guidance from an external perspective. The committee meets at least twice a year to review and ensure the competitiveness of compensation packages.

The remuneration of the president and vice president, including salaries, bonuses, severance payments, and retirement pensions, is disclosed in the Company's annual reports every year. UMC is committed to the advancement of the three aspects of ESG (environmental, social, governance), and has formulated medium- and long-term goals for sustainable development. Therefore, in addition to the Company's competitiveness in the industry, the executives' overall compensation also takes the environmental sustainability developments and talent cultivation into consideration. ESG was added to key performance metrics for variable compensation in order to align the interests of executives with that of the Company, making corporate sustainability a common goal.

Total Compensation	Performance Factor
Fixed Compensation	Based on job duties and relative importance of the position and other factors
Variable Compensation ^{Note}	According to the execution and achievement status of the operating strategy and objectives that the manager is responsible for. - Financial Performance Index: Return on equity (ROE) %, operating margin (OM) %, earning per share (EPS) and total shareholder return(TSR), etc. - Other Sustainability Index: Increasing revenue proportion and market share of sustainable products, integrate UMC's core capabilities, minimize environmental footprint, and create social value.

Note: 1. In consideration of different purposes of reward, diversified compensation will be provided, such as performance bonuses, incentive bonuses, sign-on bonuses, employee compensation (profit sharing) and share option certificates, restricted stocks, etc.
2. The calculation is based on the actual cash compensation received in the current fiscal year.

Country	Ratio of maximum remuneration to median annual remuneration (%)	Ratio of % increase in highest total annual remuneration to % increase in median total remuneration
Taiwan	59.4	3.48
Singapore	18.99	1.85

Note: 1. The calculation is based on those who were employed throughout the year of 2024.
2. Starting in 2025, subsidiaries HJ, USCXM, Wavetek, and USJC will be included in the data collection scope, with plans for public disclosure in the subsequent year after the data has been finalized.

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Long-term Incentives

In order to attract and retain key talents for the achievement of its mid- and long-term objectives, UMC issued restricted stock awards for employees with a four-year performance evaluation period from 2021 to 2024. The performance indicators for the restricted stock awards are return on equity % (ROE%), operating margin % (OM%), earning per share (EPS), total shareholder return (TSR), MSCI ESG ratings, in addition, starting from 2024, ESG performance indicators will include new targets such as reducing greenhouse gas emissions and increasing the proportion of female directors, intending to motivate employees to achieve performance goals and to create greater value for the Company and its stakeholders.

Clawback Policy

As an organization that highly values integrity, UMC believes that every employee is obligated to act in the Company's interests within legal limits and is responsible for preventing damages to or loss of the Company's interests. Violators of the Company's Code of Conduct shall pay back any benefit improperly obtained, and will be penalized according to the severity of the incident, including deduction of performance cash award, year-end bonus and profit-sharing bonus, demotion, removal from post, and, when necessary, legal actions. UMC expects all employees, especially executive level managers, to fulfill their management and supervision responsibilities, and strictly observe the Code of Conduct to ensure UMC's sustainable growth and development.

In addition, in order to enhance the improvement of corporate governance effectiveness, the Company formulated, implemented and disclosed the "Incentive-based Compensation Clawback Policy" in 2023, which clearly defines the circumstances under which the Company needs to claim back specific remuneration in the event of a major restatement of financial statements, ensures to comply with the best interests of the Company and its shareholders.

Executives' Stock Ownership Guidelines (Equity Policy)

In 2022, UMC established its Equity Policy for executives' ownership of the Company's stock, intended to strengthen the link between the interests of executives, company performance, and the interests of stockholders. The chairman and presidents are required to hold stock valued at least ten times their annual base salary. The requirement for other executive officers is at least 5 times their annual base salary. The actual achievement is reviewed every year to ensure the implementation of the policy. As of the end of 2024, the value of the President's shareholding is equivalent to 152.63 times the annual salary, and all executives have achieved the shareholding policy target ahead of schedule.

Office	Title	Shareholding Value (Multiple of Base Salary)	Status ^{Note}
Stan Hung	Chairman & Chief Strategic Officer	10X	✔ Meet
SC Chien	President		
Jason Wang	President		
Ming Hsu	Executive Vice President	5X	✔ Meet
Oliver Chang	Senior Vice President		
Chitung Liu	Senior Vice President & Chief Financial Officer		
TS Wu	Vice President		
C C Hsu	Vice President		
JT Lin	Vice President		
S F Tzou	Vice President		
Osbert Cheng	Vice President		
G C Hung	Vice President		
Steven Hsu	Vice President		
Jerry CJ Hu	Vice President		
Y S Shen	Vice President		
Steven S Liu	Vice President		
Francia Hsu	Vice President		
Mindy Lin	Vice President		
Linwu Kuo	Vice President		
Eric Chen	Vice President & Chief Human Resources Officer		
Olivia Liao	Vice President & General Counsel		

Note: Shareholding value is calculated based on the average stock price during the 30 days prior to and including the evaluation date of December 31, 2024.

1.1.2 Integrity Management

UMC values its credibility. The Company is committed to maintaining the highest ethical standards and professional competence, and strictly requires its employees and business partners to abide by the applicable anti-corruption and anti-bribery laws and regulations no matter where business is conducted. In order to comply with the above-mentioned global anti-corruption and anti-bribery laws and regulations, UMC has adopted three management methods of integrity management, namely to establish relevant regulations, to internal review and to external review for all employees and business partners. By gaining public trust through business integrity and ethics, the Company fulfills its corporate social responsibility and achieves sustainable development.

In view of the high correlation between anti-corruption regulations and conflicts of interest, UMC conducts internal investigations and management to address conflicts of interest, and actively prevents related risks by rotating job responsibilities among personnel with different roles and duties. In 2023, UMC did not receive any penalties for violating anti-corruption laws and regulations, and no proof of corruption or bribery was found in 2024.

Management of conflicts of interest: please refer to Chapter 1.1.1 "Operation of the Board of Directors"

Ways to demonstrate integrity in the workplace

UMC has initiated an analysis of the differences between its internal system and the ISO 37001 since 2023, and will continue to conduct risk assessments. The results of the aforementioned risk assessment will be used to optimize UMC's internal management.

Establish Relevant Regulations

Relevant regulations

UMC has established the "UMC Code of Conduct" to declare the Company's emphasis on integrity and adherence to professional ethical standards in order to gain public trust and ensure the Company's sustainable growth and development. The Company has also developed the "UMC Anti-Corruption Statement" to emphasize its zero-tolerance policy towards corrupt behavior and to clearly define bribery, conflicts of interest, and avoidance of such conflicts. Additionally, the Company provides whistleblower channels for reporting corrupt behavior and protection measures for whistleblowers.

UMC Code of Conduct



UMC Anti-Corruption Statement



Expand education and training

(1) Board of Directors

Directors for every board term receive training on anti-corruption policy and procedure communication. The nine directors of this term have all completed the training on July 31, 2024, with a completion rate of 100%.

(2) All employees

UMC provides integrity-related education and training for new employees and provides relevant training materials on the Company's internal website. UMC requires all IDL employees to complete training and testing on anti-corruption and anti-bribery regulations, which include an introduction to the regulations at UMC's operating locations, emphasizing the importance of relevant laws and regulations.

Internal Review

Internal control self-assessment operation

Through the annual internal control self-assessment operation, UMC requires all fabs, divisions and subsidiaries to conduct self-inspection on their business, including the awareness and assessment of possible risks in legal compliance and business ethics, and adjust the design and implementation of internal control systems to complete the self-monitoring mechanism. In addition, the Audit Division develops an annual audit plan according to legal regulations and risk considerations to conduct related audits. It regularly reports audit results to the Audit Committee and the Board of Directors and monitors improvements.

Reporting system and channels

Employees and third parties can report any misconduct through the whistleblower channel anonymously, and UMC ensures those who report the misconduct in good faith are not subject to any retaliation. If an investigation shows that a reported violation by an employee is true, UMC will take disciplinary action in accordance with internal regulations.

Channels for communication: please refer to Chapter 4.1.2 Employer-employee Communication

UMC Reporting System and Channels



External Review

Validated Assessment Program (VAP)

Performance of VAP: please refer to Chapter 4.1.1 Enforce Human Rights Management

1.1.3 Compliance of Laws and Regulations

UMC's customers are located around the world, and the Company's operations span a number of countries. To ensure UMC's compliance with the laws and regulations of each country, thereby avoiding losses due to legal violations or avoiding profit losses due to large fines, the Company closely monitors all policy or regulatory changes that might impact its business or finances.

In 2022, UMC established the Global Compliance related unit to coordinate cross-functional regulatory compliance matters and provide employees with updates on relevant regulations. All divisions are required to adhere to the applicable legal requirements. UMC also set up a legal service platform to offer consultation and assistance on legal issues to various divisions UMC also arranges training programs and courses on legal compliance, and provides prompt updates of the latest legal trends.

UMC remains in compliance with the laws and regulations by prompt updates, periodical self-assessment, and periodically review and improve internal regulations and confirm the items to be followed.

Training Courses for Legal Compliance

UMC provides regulatory training courses for legal compliance through online courses, on-site courses, and external courses. UMC arranges training for three levels of trainees based on global policy or local legal requirements: senior executives, highly related employees, and all employees. The training and testing include but are not limited to: anti-corruption laws, securities and exchange laws, anti-trust laws, privacy laws, trade secret laws, copyright laws, and export control regulations. New employees must complete courses and testing on intellectual property and trade secrets laws upon arrival. The aforementioned training courses and tests are offered routinely to strengthen compliance with the laws and regulations.

Legal staff also participate in external training courses to update their knowledge of amendments and latest news and details to ensure compliance with latest requirements.

Legal Compliance Status

In 2024, UMC did not receive any penalties of fines exceeding NT\$1 million due to violations of corporate governance, anti-corruption, anti-trust or other related laws and regulations, nor did UMC experience any privacy breaches or violations. There were also no incidents of managers violating insider trading laws. In addition, if there were any major litigation cases, they would be reported to the Audit Committee and the relevant communication and response mechanisms.

Moreover, UMC attaches great importance to the health and safety of employees. Related risks are controlled by various internal standard operating procedures, and regulations are regularly reviewed and improved upon. UMC did not incur any fines related to the health and safety of employees, or financial losses due to related judicial procedures in 2024.

For more details, please refer to Chapter 1.2.4, 1.3.4, 3.3.3, 3.5, 4.1.1, 4.3, 4.3.1.

The legal compliance status of UMC is summarized as follows:

Anti-trust Laws

- UMC formulates and announces the Anti-Trust Policy, which is the guideline for internal divisions to follow when conducting business. The key points are:
 - a. key provisions of anti-trust laws,
 - b. code of conduct for business operations, and
 - c. countermeasures when related risks occur.
- The training materials are provided on the internal website, and training courses and tests are regularly held, which require all IDL employees to complete training and testing on anti-trust laws, which include an introduction to the regulations at UMC's operating locations.

Securities and Exchange Laws – Insider Trading

- UMC not only formulates and announces the "Policies and Procedures for Refraining from Insider Trading", but also keeps emphasizing the importance of not violating the insider trading laws to directors, management team and employees.
- The training materials are provided on the internal website, and training courses and tests are regularly held, which require all IDL employees to complete training and testing on insider trading knowledge in the Securities and Exchange Laws. In addition, UMC strengthens the promotion of local legal knowledge for IDL employees in Singapore, emphasizing the importance of relevant laws and regulations.

Legal Compliance Risk Management

UMC through the Enterprise Risk Management Committee, examines the internal and external risks faced by the Company. After evaluating and analyzing issues such as privacy protection, export control, and trade secrets, appropriate management and preventive measures have been taken, and it is believed that the impact on the Company is low.

For more details, please refer to Chapter 1.2 Implementing Risk Management

Trade Secret Laws and Regulations

Reaction and Precaution Measures Taken by UMC

Regarding the issue of protecting trade secrets, UMC has established several corresponding management measures and regulations, and has implemented internal inspection methods such as training and audits. In addition, UMC's "Key Trade Secret (KTS) System" has been completed and launched in 2022, and the Company will continue to promote the KTS system to all employees to maintain UMC's competitiveness.

UMC signs non-disclosure agreements with all suppliers and customers, and customer information is handled by specialized units to avoid the possibility of improper disclosure. The communication of information between vendors, customers, and the Company must also comply with the Company's management regulations and norms. UMC also provides information security promotion to customers and business partners, and retains complete training records.

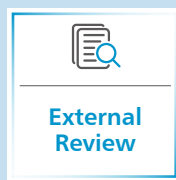


Relevant regulations

UMC formulates and announces "Trade Secret Protection Management Regulations" and the "Evidence Information Protection Procedures." The Company continuously reviews and implements the relevant regulations, or protects necessary evidence for suspicious cases. As of 2024, no trade secret related cases have occurred.

Training Courses

UMC provides intellectual property-related training for new employees and offers relevant training materials on internal website. The Company requires all IDL employees to complete training and testing on trade secret laws, emphasizing the importance of relevant laws and regulations.



Taiwan Intellectual Property Management System" (TIPS) AAA Certificatio

In 2021, UMC officially implemented the "Taiwan Intellectual Property Management System" (TIPS), which includes a system for reviewing the management of trade secrets by companies. In the 2023 verification, UMC passed the AAA-level certification and was awarded a certificate by the Industrial Development Bureau of the Ministry of Economic Affairs. Since the establishment of the TIPS system, only two companies have received this honor, demonstrating UMC's long-term investment and active implementation of the intellectual property management system.

Privacy Laws and Regulations

The Usage of Personal Data by UMC

UMC values the privacy rights of our product or service users, customers, suppliers, contractors, applicants, website visitors, employees of UMC, and other individuals (collectively referred to as "Data Subjects"). UMC only collects and uses the personal data of Data Subjects within the necessary scope and for the purposes specified in the consent form and the Privacy Policy. The percentage of usage for secondary purposes is 0%.

Regarding the collection and use of personal data within the necessary scope, UMC has established clear collection methods, personal data categories, purposes and legal bases, ways to process personal data, cross-border transmission regulations, protection of minors, potential sharing of personal data, cross-border transmission limitations, Data Subject's rights, personal data retention periods and expiration destruction mechanisms, and has established personal data communication channels and related information security management measures.

The Data Subject can view UMC's "Privacy Policy", "Privacy Protection Procedure", and "Cookie Policy" through the following links. Data Subject can also communicate privacy-related needs through the "Data Privacy Request Form" or "Feedback Mailbox".

Privacy Policy

https://www.umc.com/en/Html/privacy_policy



Privacy Protection Procedures

https://www.umc.com/en/Html/privacy_protection_procedure



Cookies Policy

https://www.umc.com/en/Html/cookie_policy



Data Privacy Request Form

https://www.umc.com/en/Privacy_Request/privacy_request



Feedback Mailbox

z_website_privacyadmin@umc.com



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Impact and Implications of Privacy Protection Policy

UMC believes that, by providing open and transparent privacy policy information, Data Subject will be able to enjoy the rights to make independent decisions about their personal data. Data subject's trust in UMC will advance the Company's progress in corporate social responsibility, enhance overall awareness of privacy management in the industry, and will ultimately lead to a positive environment that respects and upholds privacy laws and regulations.

UMC provides regular training to enhance the understanding of global privacy laws and regulations to employees. Different levels of personnel will be made aware of their authority to handle personal data, storage restrictions, and the mechanism for reporting personal data security incidents. Personal data should also be protected in accordance with the "UMC's personal data file security maintenance plan and methods of disposing personal data after business termination".

UMC periodically reviews Data Subjects' consent forms and current retention of personal data in the Company, and establishes appropriate protection and security mechanisms to prevent personal data security incidents, such as accidental or unlawful damage, destruction, alteration, disclosure, access, transfer, storage, or other processing (collectively referred to as "Personal Data Security Incidents").

Region	Applicable Laws and Regulations	Specific Response Measure
European Union	General Data Protection Regulation (GDPR)	✦ Through the operation of the UMC's Personal Data Protection Implementation Task Force, UMC ensures that the personal data of the Data Subject is well protected. ✦ UMC updates the Privacy Policy, Privacy Protection Procedure, and relevant documents such as consent forms and contracts in accordance with current laws or regulations. ✦ Providing relevant training materials on internal website
United States	California Consumer Privacy Act California Privacy Rights Act	
Taiwan	Personal Data Protection Act	
China	Personal Information Protection Law	

UMC's Personal Data Protection Implementation Task Force

Given that UMC's business and services are expanding globally, in order to implement the protection of personal data collected and utilized by UMC, and to avoid situations such as leakage, abuse, and being stolen that may cause harm to the Dara Subject, UMC has referred to the Personal Data Protection Act of Taiwan, the General Data Protection Regulation (GDPR) of European Union, and the California Consumer Privacy Act (CCPA) and California's Privacy Rights Act (CPRA) of United States, and has established and announced the "Privacy Protection Procedure" to explicitly regulate UMC's management methods of Privacy Policy and personal data security incidents handling procedures, and UMC will continue to monitor global privacy laws and regulations to continuously optimize management processes and personal data security incidents reporting mechanisms.

According to the "Privacy Protection Procedure", UMC has established the Personal Data Protection Implementation Task Force, which regularly reports its work results to the level of the board of directors. if UMC discovers or receives any personal data security incidents, UMC's Personal Data Protection Implementation Task Force will address the following matters:

- 1 Conduct an incident analysis within 24 hours of receiving the report. The incident analysis shall confirm the type and severity of the incident, the scope of impact, media exposure risk control, and the cause of the incident;
- 2 Immediately after the incident analysis, UMC shall develop incident response procedures and contingency measures to prevent the incident from expanding and take evidence preservation measures to avoid changes or alterations to the original electromagnetic recording and evidence;
- 3 Once the incident is identified, UMC shall promptly communicate with the Data Subjects about the facts of the infringement and the measures taken in an appropriate manner;
- 4 UMC shall report the incident in a timely manner according to the laws. After the aforementioned procedure is completed, UMC's Personal Data Protection Implementation Task Force will also review the cause of the personal data security incidents and formulate improvement plans to prevent the incident from happening again.

UMC has a zero-tolerance policy against any violation of the Privacy Policy and the Privacy Protection Procedure. UMC may impose disciplinary measures, including but not limited to imposing warnings or demerits on those who violate the Privacy Policy and the Privacy Protection Procedure. If an employee abuses others' personal data that is not within the scope of his or her duties and without obtaining approval of his or her supervisor, the Company may terminate the employee's employment without prior notice in accordance with Article 12, paragraph 4 of the Labor Standards Act. UMC may also file civil claims and/or criminal charges against employees who violate the Privacy Policy and the Privacy Protection Procedure, and seek for provisional remedies.

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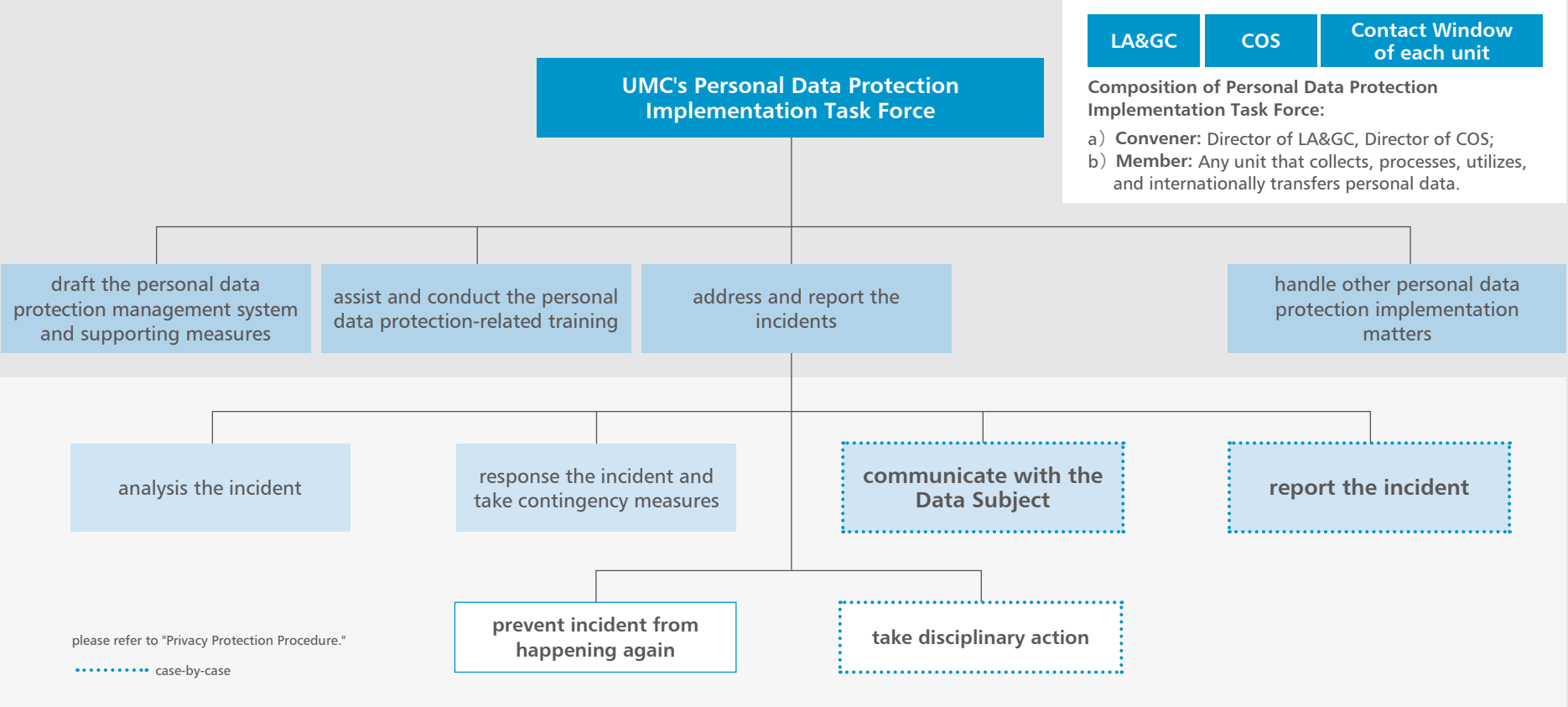
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Risk Reporting Mechanism

UMC has obtained the certification of ISO/IEC 27001, the international standards for information security management, and has set up system maintenance and management standard operating procedures. The prevention, notification, and response mechanisms for incidents that may cause personal data leakage are systematically supervised in accordance with the management regulations of the information security management division. In 2024, the Company received no complaints.

Information security management: please refer to Chapter 1.2.4 Commitment to Information Security Risk Management

Complaint Handling and Resolution

UMC takes actions in accordance with the Privacy Policy and the Privacy Protection Procedure upon receiving notice of personal data security incidents. During 2024, no complaints about UMC's infringement of privacy rights were received by external personal data protection authorities, UMC's internal Data Privacy Request Form, feedback mailboxes, and MyUMC platform, indicating that UMC has successfully achieved the expected results in privacy management internal control mechanisms.

	External	Internal	
	Personal data protection authorities	Data Privacy Request Form and feedback mailboxes	MyUMC platform
Compliant	0	0	0
Penalty	0	not applicable	not applicable

1.2 Implementing Risk Management

- 1.2.1 Risk Management System
- 1.2.2 Risk Management and Countermeasures
- 1.2.3 Comprehensive Business Risk Management
- 1.2.4 Commitment to Information Security Risk Management
- 1.2.5 Managing Climate Risks and Opportunities

1.2.1 Risk Management System

Enterprise Risk Management Mechanism

The Enterprise Risk Management Committee consolidates and evaluates risks in the areas of strategy, operation, finance, and hazard (including climate change risks). According to the risk categories, the Committee sets the risk appetite that the Company can accept, evaluates the possibility of risk occurrence and severity of risk impact, creates a risk map, defines the priority order and risk level of risk items, and executes sensitivity analysis and stress testing of important financial and non-financial risks. Risk management plans are developed based on the level of risk the Company can accept and the costs it can afford. On a monthly basis, each risk group conducts rolling monitoring and reviews the changes in the likelihood and severity of risks exposure. The risks status is subsequently reported to the Risk Management Office, in order to ensure the effectiveness of risk management plans and related control operations. Meanwhile, the Company also seizes business opportunities associated with risks to formulate company development strategies.

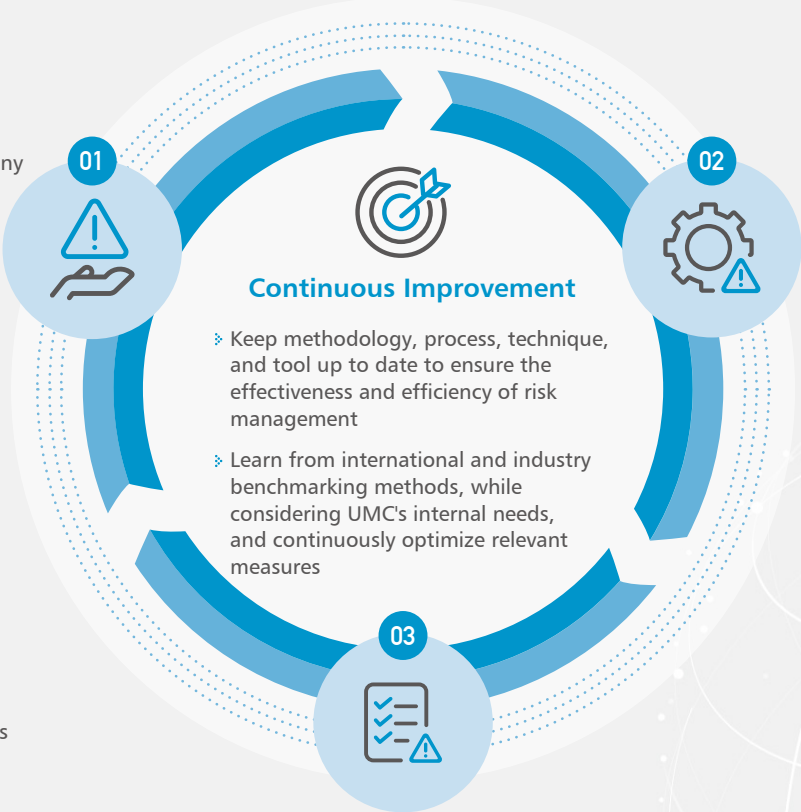
Please refer to the "Risk Management" section for information regarding the risk management organizational structure.

01 Risk Identification and Assessment

- Identify and assess the possibility of occurrence and the severity of their impact on the Company (including finance, operations, personnel, company reputation, etc.)
- Establish UMC risk pool and enterprise risks prioritization

03 Risk Monitoring and Reporting

- Monthly rolling risk assessment to monitor the changes in the likelihood and severity of risks exposure
- Mitigation progress monitoring
- Summarize risk management progress for BOD review
- Annual internal auditing addressed in CSA reports along with regular third-party external audits to ensure effectiveness of internal management



02 Risk Response Strategy and Implementation

- Develop response actions of risks through sensitive analysis, stress testing and considering the Company's risk appetite
- Execute response actions for risks, and embed in internal control activities and management procedures
- Risk response strategy coordination

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Risk Management Three Lines of Defense

UMC has a comprehensive enterprise risk management and internal control mechanism in place. The Company has established the roles, functions, and responsibilities within the three lines of defense model to ensure that UMC's risk management related objectives are achieved and that the risk governance structure operates effectively. The details are as follows:

1 Functional units conduct self-assessment and management

Executing Unit: Each functional unit

- ✧ The first line of defense is to take each functional unit as the operational risk owner. Based on its functions and business scope, and according to its professional knowledge and understanding of its core business, each functional unit designs and supervises the control procedures of divisional business activities from the perspective of identifying where risks are most likely to occur and maintaining core competencies.
- ✧ Each functional unit is responsible for identifying, evaluating, controlling, and reducing risks generated by operational activities through techniques such as Failure Mode and Effects Analysis (FMEA).
- ✧ Each functional unit establishes internal control procedures in a timely manner and implements risk management procedures to ensure that all business executions are in line with business policies and objectives.

2 Robust risk management and compliance mechanisms

Executing Unit: The Enterprise Risk Management Committee

- ✧ Independent from the functional organizational structure, the Company's overall risk policy was formulated and the risk management process was established to assist and supervise the first line of defense in managing risks.
- ✧ The Committee assesses company-level significant risks and coordinates cross-organizational risk management operations to ensure comprehensive coverage of risk management.
- ✧ The Committee monitors risk changes and impacts within the Company's tolerable range, and regularly reports risk management effectiveness to the Company's senior management and board of directors.

3 Independent and impartial internal audit mechanism

Executing Unit: The Audit Division

- ✧ The Division, which is independent of enterprise risk management and daily risk management operations and reports to the Audit Committee in board-level directly, carries out independent auditing and evaluation to ensure the effectiveness of existing first-line internal control mechanisms and second-line risk management systems, and establishes audit plans in accordance with regulations and obtaining approval from the board of directors on an annual basis.
- ✧ The Division reviews the self-assessment operations of internal control systems of each unit and subsidiary, along with improving internal control deficiencies and abnormal issues discovered, as a reference source for the board of directors and the management to evaluate the effectiveness of internal control systems on an annual basis.
- ✧ The Division participates in important meetings of the Enterprise Risk Management Committee, and provides suggestions to ensure that policy compliance and implementation processes are consistent with performance objectives and risk tolerances selected by management.

External Audit of Enterprise Risk Management Process

UMC regularly engages third-party external consulting firms to conduct compliance audits of the Company's risk management processes based on the ISO 31000 Risk Management System Guidelines and the COSO ERM 2017 Enterprise Risk Management Framework. Through questionnaire surveys, interviews and examination of the actual operation of the Company's risk management, including risk governance, risk culture, organizational structure and resources for risk management, the scope and process of risk management, reporting data and indicator tools, and information technology applications, etc., comprehensive evaluations of the completeness of risk management are provided, along with reports containing improvement recommendations. The results of the previous external consultant audit (2024) rated UMC as a company with mature enterprise risk management, and issued a statement confirming that UMC's enterprise risk management processes comply with the COSO ERM Framework. For the detailed statement, please refer to the "Risk Management" section on UMC's official website.

Risk Culture Establishment

UMC is committed to establishing a comprehensive risk culture and continuously improving it to deepen the awareness and capabilities of all employees in risk management. Through measures such as the following, risk awareness and self-management are internalized into employees' DNA.

- ✧ **Ongoing Risk Training:** Providing employees with comprehensive training courses on enterprise risk management and job-related risks to enhance their risk awareness. This includes training and drills related on company enterprise risk management, Business Continuity Management (BCM), the concepts of environmental health and safety risks, intellectual property rights, legal compliance and information security. Some examples of training conducted in 2024 include:
 - "Introduction to UMC Enterprise Risk Management" course: A total of 11,736 employees completed
 - "Business Continuity Management" course: A total of 9,335 employees completed BCM awareness promotion; 10 sessions of BCP drills were conducted
 In addition to offering training programs for its employees, UMC annually provides its board members with various training courses including risk management.
- ✧ **Standardized Implementation of Core Business Risk Management:** Rigorous management processes are implemented from the assessment of new technologies or products to process development and mass production, ensuring standardized risk management to achieve risk prevention and control. During the assessment phase of new technologies or products, comprehensive evaluation and review process are required, considering aspects such as market demand, profitability, technical capability, environmental sustainability, etc., and incorporating considerations of potential risks. The evaluation results are reported to the Chief Technology Officer for approval before development begins. During the process development phase, project teams must follow the standard operating procedure - the "New Process Development Procedure," which includes Failure Mode and Effect Analysis (FMEA) for risk prevention management design, defining strict standards to prevent risks from impacting research and development results. During the mass production phase, adjustments to production methods such as processes, machinery, and materials must undergo the risk assessment management process of FMEA before changes are approved.
- ✧ **Smooth Channels for Employee Risk Feedback:** Encouraging employees to proactively identify surrounding risks and use innovative thinking to uncover potential company risks and propose improvement suggestions. Establishing diversified risk communication platforms where employees can immediately report hazard, operation, strategy, financial, and other related risks 24/7. Using automated systems for management to ensure that relevant risk issues reported by employees are reviewed within the standard timeframe to achieve continuous PDCA management and risk communication effectiveness.

Related Communication Platform Channels:

- The Incident Reporting and Investigating System, The Inspection System, and The ESH Forum: Communicating relevant information and risks according to the Environmental, Safety, and Health Management System (ISO 14001/ISO 45001).
- The Improvement Proposal System
- Information Security and Confidentiality Protection E-mail and Hotline
- Whistleblower Reporting for fraud or violations of professional ethics
- CHO (Chief Human Resources Officer) mailbox
- ✧ **Linking Risk Management Effectiveness with Incentive Measures:** UMC promotes the linkage of risk management with individual rewards through various management competition activities, annual bonuses, and employee restricted stock units. This system aligns rewards of employees and managers at all levels with the effectiveness of risk management and the achievement of performance goals. It aims to motivate individuals to prioritize their focus on accountability and effectively manage risks within their scope of responsibility or influence, thereby contributing to the achievement of both personal and company performance goals. Since 2021, personal performance evaluations have been systematically linked to key company performance indicators, and a portion of employee bonuses is also linked to the results of Environmental, Social, and Governance (ESG) performance indicators to reward effective risk management. This approach aims to collectively achieve the Company's operational performance and ESG-related KPI goals.

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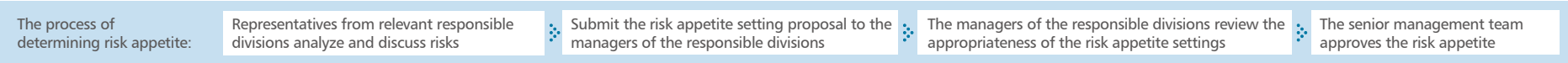
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1.2.2 Risk Management and Countermeasures

UMC identifies key risks and formulates countermeasures in four areas: strategy, operation, finance, and hazard. It also includes important risks related to environmental, social or corporate governance issues in the Company's operations. The growth of Taiwan's semiconductor industry outpaces the global average, and UMC is optimistic that Taiwan will maintain its leading position in the foreseeable future, driven by factors such as continued development of new process technologies and the clustering effect of equipment and material suppliers. However, in recent years, competition among powerful countries has increased geopolitical risks, geopolitical conflicts and inflation have led to higher operating costs, trade and technology wars have impacted the supply of equipment and raw materials, cyberattacks have increased the complexity of information security and maintenance costs, climate change has made green energy and carbon reduction more urgent, and the shortage of international semiconductor talents with international perspectives have impacted the international development of companies. These factors may cause adverse effects on the Company's investments and operations, but they also present some noteworthy opportunities.

As a member of the supply chain, UMC monitors the global risk situation in addition to the emerging risks in the semiconductor industry. The Company refers to the Global Risk Report published by the World Economic Forum (WEF) every year to understand how risk trends are evolving, and pays special attention to potential risks with the following characteristics: newly emerging risks, risks with long-term impacts on the Company's business, risks affecting the Company's strategy and business model, events originating externally and beyond the Company's influence or control, and risks with impacts that can be specifically defined. These are categorized as the Company's emerging risks and the Company should initiate risk response measures as early as possible.

In response to the risks faced by UMC, appropriate risk-mitigation measures are developed within the framework of the Company's enterprise risk management operations. According to the categories of risks under each risk area, they are analyzed and discussed by the relevant dedicated units and then submitted to senior managers for approval to set the risk appetite, which is divided into 5 levels (Low, medium to low, medium, medium to high, high). For example, the risk category of property damage and business interruption, after discussions between the hazard division and the finance division, followed by analysis through financial models, the risk appetite was defined and finally approved by the CFO (risk appetite level: medium to low). For the risk category of investment strategy, after discussions with the business marketing division, the engineering division, the corporate planning division and the finance division, followed by analysis through financial models, the risk appetite was defined and finally approved by the Co-President (risk appetite level: medium).



Additionally, sensitivity analysis and stress testing are conducted for important risks. After evaluation, it is expected that the overall residual risks of the Company and its affiliates should fall within a manageable range under the implementation of various countermeasures.

Sensitivity Analysis and Stress Testing

Item	Risk Scope	Risk Issue	Risk Level (Likelihood, Severity)	Risk Description/ Potential Impact	Sensitivity Analysis and Stress Testing	Countermeasures
Financial Risks	Finance	Currency Exchange and Interest Rate Risk	M-L (M, M-L)	- Fluctuating foreign exchange rates could result in FX losses and unexpected volatility in expenses when revenue or expense generated from operating activities is denominated in a currency different from the functional currency. - Due to large exposure of loans, changes in interest rates could affect future cash flow.	- Based on economic information provided by professional institutions, the Finance Division set economic scenarios and conducts sensitivity analysis & stress testing on a quarterly and demand basis to monitor the impact of exchange rate and interest rate fluctuations on the Company's net profit. The simulated results are reported to the Chief Financial Officer, and the division also formulates countermeasures to minimize risks. - The simulated data for 2024 indicates that, even under the greatest fluctuations in exchange rate and interest rate, after implementing the appropriate measures, the overall net profit will remain controlled within the Company's acceptable range.	- The Company applies natural hedges on the foreign currency risk to balance the assets and liabilities denominated in foreign currencies, and utilizes spot or forward exchange contracts to reduce foreign currency risk. - The Company manages the tenor and borrows at fixed or floating rates to reduce interest rate risk.
Non-Financial Risks	Operation	Factors such as the Pandemic, Geopolitics, and Climate Change Have Pushed Global Inflation Higher, Making it More Difficult to Effectively Control Costs to Maintain Profitability	M-L (H, M-L)	In recent years, countries have implemented loose monetary policies to save their economies from the impact of the COVID-19 pandemic, which has caused inflation and soaring interest rates. In addition, geopolitics, wars, and climate change continue to impact the stability of the supply chain, further increasing the cost of supplies. This would increase the overall operating costs of UMC, including soaring logistics and raw material costs, escalating labor wages, and rising equipment maintenance contract expenditures.	- To establish a financial simulation model and input key financial parameters (such as raw material and component costs, labor costs, patent fees, etc.) to understand the impact of these parameters on the Company's future profitability metrics such as gross margin (GM%) and return on equity (ROE). - Actual simulation data indicates that the percentage increase in the prices of indirect materials, maintenance, and spare parts (IDM+RMP) serves as a sensitive factor affecting the gross margin. UMC sets different levels of countermeasures based on the extent of cost price increases to ensure the achievement of the Company's annual profit targets.	- Actively introducing alternative materials and machines to increase sourcing flexibility and bargaining power. - Promoting continuous improvement and reduction of raw materials through innovative methods. - Improving cross-fab parts management system and benchmark management to exert the effectiveness of integrated management. - Increasing the in-house repair rate of machine parts. - Strengthening the procurement risk assessment process and long-term contract review mechanism to reduce procurement costs with appropriate long-term contracts. - Using digital transformation to increase personnel productivity and human capital utilization efficiency.

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The remaining significant risks and emerging risks for 2024 are outlined in the following table. Additionally, we provide a special chapter to explain the countermeasures for dealing with risks related to information security, natural disaster prevention, and climate change.

Strategy

Risk Issue	Risk Level (Likelihood, Severity)	Risk Description	Potential Impact	Countermeasures
Emerging Risks Risks to the Company's Investment and Operations due to Intensified Competition among Countries	M (M-H, M-H)	Strategic competition among major powers has led to industrial chain decoupling or de-risking measures, and international industries have seen ABC (anywhere but China) or even "Taiwan plus one" trends. The United States, Japan, South Korea, and the Netherlands have imposed restrictions on the export of advanced semiconductor equipment to China. In response, China has restricted the export of specific materials, leading to the gradual emergence of a competitive dynamic between two major camps, led by the United States and China. Major powers have realized that semiconductors are not only essential components of technological products, but also critical strategic resources for their countries. Therefore, they have been actively developing domestic semiconductor manufacturing capabilities with state support. This competitive tension shows no sign of reversal in the short term and will continue to increase the uncertainty of future operations.	The impacts on UMC include costs, revenue, delivery schedule and investment plans, for examples: Relevant regulations and sanction measures will impact the stability of the semiconductor supply chain, which may affect production plans and delay the delivery of customer orders. Prohibiting the export of key equipment or materials to specific countries may affect UMC's global production footprint, such as restricting access to U.S., Japanese, and Dutch equipment or parts for UMC's China fabs. This could even require urgent modifications to investment plans, resulting in the loss of potential business opportunities. Customers may leave China fabs due to ABC (anywhere but China) policy or concerns about product technology continuity, which affects the capacity utilization rate of UMC's China fabs, resulting in a decline in revenue. Major countries' policy subsidies for local production and domestic semiconductor production capacity may further reduce UMC's market share and cost competitiveness in the industry. Rapid changes in policies will increase corporate compliance costs and management complexity, which may affect the Company's operating performance and strategic deployment.	Diversifying the customer base and strengthening technology competitiveness: In addition to its existing business in North America and the Asia-Pacific, UMC also plans to further expand its business in Europe, Japan and other regions. The Company also keeps improving technology competitiveness and differentiation, and analyzes long-term trends, and seizes more opportunities Strengthening the diversification of production locations and the flexibility of production allocation: In addition to existing capacity at UMC's China fabs, UMC continues to expand production capacity in Taiwan, Japan, and Singapore, while collaborating with Intel to establish a manufacturing base in the U.S., enabling UMC to serve global customers with diversified production locations. Furthermore, UMC is increasing the flexibility of cross-regional and cross-fab support, and making timely adjustments to production allocation and capacity expansion plans. Agilely tracking changes in the external environment, including global regulations, politics, and economic conditions: We evaluate the impact of external changes on the acquisition of production resources, the changes in customer demand, and take necessary risk response measures in a timely manner. Some practical measures have been taken: Establishing a tracking and risk monitoring mechanism for raw material supply chain to accurately track the component ratio and risk status of the countries where raw materials come from. Closely monitoring the national policies of important equipment and material exporting countries, and actively establishing second suppliers to diversify risks. Initiating mergers and acquisitions, forming alliances to build new facilities, and collaborating on capacity/technology at the appropriate timing in order to seize opportunities for international cooperation.
Emerging Risks Geopolitical Environment prompts China to Build Domestic Semiconductor Manufacturing Capabilities and the Risks it Poses to Company's Development	M-H (H, M-H)	Due to the influence of the geopolitical environment and restrictions on the import of advanced technologies and equipment from the United States, Japan and Europe, China has been developing semiconductor supply chain autonomy, massively expanding of mature node production capacity and actively subsidizing domestic manufacturers. It is estimated that as China semiconductor capacity continues to grow; the competitive situation will be more exacerbated and have lasting impact.	The impacts on UMC include orders, revenue, and profits, for examples: China's large-scale expansion in mature process capacity overlaps with UMC's production capacity. Moreover, Chinese semiconductor manufacturers may take advantage of government subsidies to trigger aggressive price competition to win customers, especially Chinese customers or customers serving the Chinese market, resulting in a decline in UMC's revenue and profits. Policy subsidies for China's IC design industry may impact the market share of other IC design companies, resulting in the loss of customer orders related to Taiwan's semiconductor foundries. Numerous emerging small and medium-sized semiconductor foundries in China, each focusing on the development of specific technologies, may challenge UMC's leading position in mature process technologies, leading to the loss of business.	Analyzing future market opportunities, improving customer demand management, proactively establishing technological differentiation, and accelerating FinFET technology development to raise the barriers to technology competition. Promoting structural cost improvement and productivity enhancement to maintain cost competitiveness. Fully utilizing the advantages of UMC HJ and USCXM to actively serve Chinese customers, as well as global customers planning to enter the Chinese market, and to create opportunities for collaboration. Continuously monitoring industry policies in China and actively seeking communication with the government to obtain relevant subsidies. To respond to the competition in different market segments, the Company has adopted more diversified and flexible pricing strategies to win customer orders.
Emerging Risks The Ongoing Risk to Stability of the Company's Operations Caused by the Tensions in Cross-Strait Relations	M (M, M)	The recent competition between the United States and China in politics, economy, and technology has become more intense. Because of Taiwan's crucial position in geopolitics and the technology industry chain, it will inevitably be affected by the complex and dynamic interactions among the United States, China, and Taiwan. According to an analysis of cross-strait relations over the past few years, China is likely to continue to increase pressure on Taiwan, including military or paramilitary actions. This would increase the uncertainty of UMC's future business and operations.	The impacts on UMC include business, revenue, and delivery schedule, for examples: This situation may affect UMC customers' willingness to place orders, resulting in a decline in UMC's business and revenue. The supply of semiconductor raw materials may be delayed or interrupted by military or paramilitary actions of China, which may affect UMC's operational stability and customer delivery schedule.	Establishing cross-functional platform mechanisms to deal with geopolitical risks, closely monitoring the dynamics of cross-strait relations, projecting possible scenarios regarding changes in the Taiwan Strait, assessing potential impacts on various aspects of UMC operations, formulating corresponding risk mitigation measures, and conducting rolling reviews and management adjustments. Following the guiding principle of minimizing the impact on Taiwan fabs and maintaining independent operations of overseas fabs, the Company plans and deploys contingency measures to reduce the impact on operations. Enhancing routine cybersecurity management and emergency response measures to safeguard company confidential information.

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Operation

Risk Issue	Risk Level (Likelihood, Severity)	Risk Description	Potential Impact	Countermeasures
! Emerging Risks The Shortage of International Talent Affects the Company's International Development Strategy	M-L (M, M-L)	According to the operational development strategy of UMC, there is a need for international talent to meet requirements such as global expansion, fab expansion, digital transformation, specialized technology development, and market strategy planning. Through alignment with senior management and talent assessment, it has been identified that parts of international talent are lacking within the current organizational structure. This shortage may pose a risk of being unable to rapidly cultivate talents and pass on experience, thereby affecting the Company's long-term competitiveness. This may become increasingly severe as countries actively develop their own semiconductor manufacturing.	The impacts on UMC include the efficiency of organizational operations, development plan, and company image, for examples: ❖ The Company lacks sufficient international leadership talent, making it difficult to enhance the operational performance of overseas fabs (USCJ, 12i) or further expand global footprint according to the UMC's operational plans. ❖ Due to the shortage of international talent with global perspectives and leadership qualities, the development of international innovation technology and global vision face difficulties. Consequently, it is difficult to timely expand or respond to business developments to maintain competitiveness in the semiconductor foundry industry. ❖ Insufficient internationalization in corporate management affects the diversified development of UMC's culture and image.	❖ Establish a competitive talent recruitment strategy, such as establishing global talent search mechanisms and recruitment channels, developing mechanisms for screening international talent and building talent pools, and utilizing innovative recruitment tools and methods. ❖ Develop and nurture international talent, such as implementing mechanisms for succession training and development of international talent, cultivating internal talent with international perspectives and leadership skills (strategic thinking/management abilities/communication/language proficiency). ❖ Foster a diverse organizational culture to help international talent adapt to the Company, such as establishing organizations designed to support international talent, cultivating a diversified organizational culture through training and managerial guidance.

Hazard

Risk Issue	Risk Level (Likelihood, Severity)	Risk Description	Potential Impact	Countermeasures
! Emerging Risks International Green Energy Trend and National Energy Conservation and Carbon Reduction Policy Increase Enterprise Operation and Investment Cost	M (H, M)	To reduce carbon emissions, governments around the world are taking proactive steps to limit or enhance carbon reduction requirements for enterprises. For example: ❖ The Singapore government has imposed carbon tax since 2019 and announced that the carbon emission price per tonne will be raised gradually. In Taiwan, "Climate Change Response Act" was also promulgated in February 2023 including carbon-fee charging from enterprises. ❖ In response to international green energy trend, UMC declared that joined RE100 in 2021 and committed to using 100% renewable energy by 2050. In the meantime, it is estimated that potential customer demand for green energy coverage across the Company will increase gradually in the coming years.	The impacts on UMC include operation and investment cost increasing, and resulting in the erosion of profits. For examples: ❖ The impact caused by carbon fee/tax in 2 years: no more than NT\$20 million of carbon tax per year in Singapore; no more than NT\$400 million of carbon-fee per year in Taiwan. (The calculation is based on NT\$300/per tonne of carbon emission in TW). ❖ Phased goals are set according to RE100 to meet green energy requirements from customers: 25% and 50% of renewable energy are achieved in 2025 and 2030, respectively. It is estimated that an additional expenditure of no more than NT\$0.5 billion per year on purchasing renewable energy in 2025. For more detailed impact information, please refer to Chapter 1.2.5 Managing Climate Risks and Opportunities.	❖ Participate in governmental voluntary greenhouse gas reduction plan, promote greenhouse gas emission reduction plan, apply for carbon offsets projects for carbon credits, join carbon trade market to purchase carbon credits and offsets, evaluate introducing carbon-negative technologies. ❖ Arrange in advance to ensure access to lower-cost contracted renewable energy in external sites. For other response and achievements, please refer to related chapters: 1.2.5 Managing Climate Risks and Opportunities, 3.1 Climate Action, and 3.2 Energy Management.

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1.2.3 Comprehensive Business Risk Management

Business Continuity Management

As global risks continue to rise, UMC recognizes the importance of business continuity, which means providing uninterrupted services to strategic customers and key stakeholders. The ability to provide uninterrupted foundry manufacturing as well as diverse offering and excellent quality is the ultimate purpose of UMC's business continuity management.

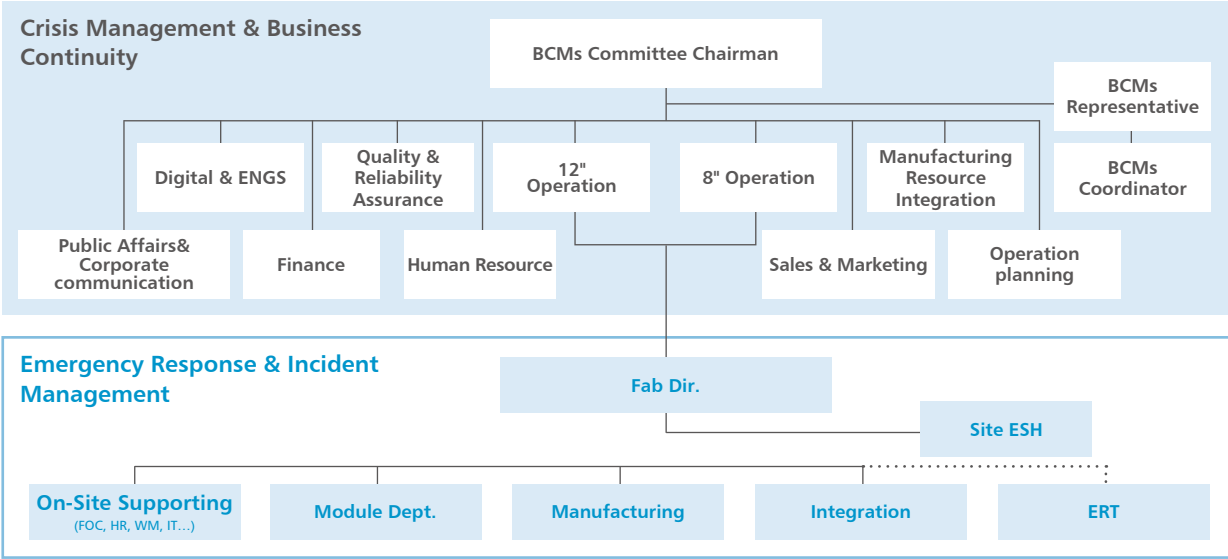
The Business Continuity Management Committee is based on the existing organization and responsibilities. The management team is in charge of resources integration, review of improvement plans, and operational recovery in the case of adverse events in order to protect the rights and interest of customers and stakeholders.

UMC Business Continuity Policy and Objective

UMC strives to provide the competitive wafer foundry services while pursuing business sustainability. The Company aims to provide uninterrupted services to our strategic customers and serve in the interest of our stakeholders through appropriate risk management. Our business continuity policy and objectives are:

 Policy	- UMC aims to provide customers with excellent and uninterrupted service. - UMC ensures the effective of its business continuity system by implement system and integrate resources. - UMC reviews and improves continuously to enhance the resilience of responding to various disasters before/ during/ after the disruption from business impact to ensure the utmost rights and interests of customers and stakeholders.
 Objectives	- Personnel safety is the top business continuity objective. - Base on risk assessment results, the Company aims to develop different various prevention measures and responses to reduce operational impact and shorten recovery time. - UMC follows ISO 22301 management system standard and executes BCP drill on a regular basis to review the effectiveness of the business continuity plan and make continuous improvements.

BCM (Business Continuity Management) Committee Organization



I. Business Continuity Management Framework

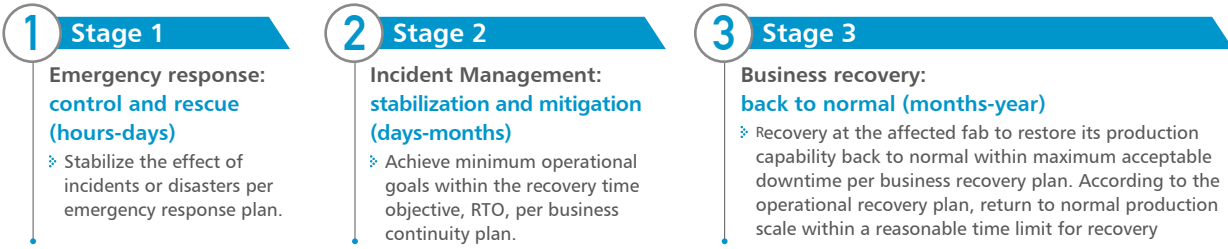
UMC follows the ISO22301 standard to establish, implement, monitor and maintain the business continuity management system. The system's efficacy is regularly revised to ensure that business operations can continue during incidents or disasters without interruption.

All fabs and functional departments have deployed emergency response and recovery plans for specific events such as earthquake, fire, chemical spill, power disruption, water shortage and material shortage. BCP Exercise are conducted every year in order to continually improve emergency response and recovery procedures.

Furthermore, the Company's Hsinchu headquarters, Fab8A, Fab8C, Fab8D, Fab8E, Fab8F, Fab8S, Fab12A in Taiwan, and Fab12i in Singapore, and USCXM in Xiamen, HJ in Suzhou, China and USJC in Japan subsidiaries have passed ISO 22301 certification sequentially from 2013 to 2024.^{Note}

Note: Currently, Wavetek conducts BCP (Business Continuity Plan) and exercise every year to find out opportunity for improvements.

Three steps of emergency response and recovery



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II. BCM Management Achievements

The BCM management review meeting is held annually to review both internal and external risks and the accuracy and effectiveness of the system. For possible risks such as earthquake, fire, chemical leakage, utility outage, drought and raw material shortage etc., UMC conducts risk assessment and improvements in advance. Furthermore, exercises on procedure are held periodically to sharpen response and recovery after incidents.

Achievements of business continuity management

KPI	2024 Achievements	Results
Risk Mitigation in Supply Chain: - Critical material risk identification System establishment - Complete the material inventory for 80% of the total Y24 budget. - Enhance critical material multiple source supply ratio up to 89.9%	- Completed 91% of the material inventory (achieving 113% of the target). - Introduce new suppliers to increase multi-source ratio to 89.9%	✓ Attained
Enhance the Resilience of Utility: - Introduce multiple water resource: reach 16% usage of reclaimed water and desalinated water - Introduce at least one nature gas back-up supplier	- Usage of reclaimed water and desalinated water achieved 17% (Consolidated for the UMC group) - Introduced a nature gas back-up supplier	✓ Attained
Enhance Information Security Protection: - Zero major cyber attack or ransomware event that impact on operation - Maintain score >90 in Security ScoreCard	- Zero in 2024 - Above 90 among all site in UMC Group	✓ Attained

Hazard Risk Control

UMC fully recognizes the impact and influence of natural and man-made disasters on production and operation. Hence, the Company has consistently adopted proactive attitude toward preventive disaster risk management, and seeks to achieve the highest standard of semiconductor industrial safety through rigorous risk engineering controls and implementation of safety regulations and standards.

Disaster Risk Management Topics

 Fire Safety	- UMC incorporated the international standards of the US Factory Mutual Insurance Company (FM), Underwriters Laboratories Inc. (UL), the US National Fire Protection Association (NFPA), the Semiconductor Equipment and Materials International (SEMI), and other international standards into its building construction, equipment, engineering controls and risk assessment, and formulated relevant company regulations for supplementary requirements. - UMC is the only semiconductor company in the industry equipped with a professional fire brigade. In addition to being equipped with government-grade fire trucks and rescue equipment, UMC volunteer firemen regularly receive external simulated fire training from professionals, which enhances their practical disaster relief capabilities.
 Earthquake Protection	- UMC production sites in Taiwan and Japan are located in pacific seismic zone, where the impact caused by earthquake can not be ignored. At the beginning of the construction stage, earthquake-resistant requirements are integrated into the design concept of buildings, facilities, pipelines and production equipment. - UMC continues to introduce new anti-seismic techniques such as seismic isolation platforms for production equipment and Earthquake Early Warning System (EEWS). In addition, for those fabs built before 2000, structural technicians advise on anti-seismic improvements according to regulations and earthquake experience to further ensure personnel safety and reduce the potential loss of process tools and auxiliary equipment.
 Equipment Safety	The Environmental Health and Safety Standards (SEMI-S2) for semiconductor manufacturing equipment, NFPA318, and FM7-7 are the major international standards adopted by UMC. Besides, UMC also adopts the Equipment Purchasing ESH Specifications into its procurement procedures, which requires all equipment to be reviewed and need to comply with the requirements before being brought into the fab and will be checked again during ESO stage to achieve effective equipment safety control.
 Cyber Security	- Construct a multi-layered information security protection mechanism to provide a secure production environment that complies with regulations and meets customer demands. - Establish a network security incident response plan and conduct regular drills. In case of emergencies, implement corresponding escalation and rapid recovery actions to ensure the Company's business continuity. For more details, please refer to Section 1.2.4: Commitment to Information Security Risk Management.
 Supply Chain Management	- Collaborate with suppliers to store materials in UMC warehouses or in supplier facilities located near UMC production sites. This strategy aims to enhance safety stock levels to address urgent demands and reduce response times effectively. Related operational procedures were conducted based on four key topics: Risk Assessment Operations, Business Impact Analysis Operations, Incident Response and Business Continuity Plan, and Emergency Material Procurement Operations. - Continuously develop alternative suppliers and periodically evaluate suppliers' supply capabilities, flexibility, and mechanisms for assessing natural disaster risks for related production locations. To date, no high-risk incidents have been reported. In the event of an emergency, the emergency response mechanism will be activated and handled according to the "Incident Response and Business Continuity Plan" and "Emergency Material Procurement Operations." - Develop and implement a supply chain risk monitoring platform that provides 24/7 monitoring of natural disasters and various geopolitical risks affecting suppliers' production locations. In the event of risk events that affect supply stability, the platform will promptly coordinate inventory management across existing suppliers and production sites to effectively address any supply and demand challenges. Furthermore, risk mitigation mechanisms will be activated to handle the situation effectively.

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1.2.4 Commitment to Information Security Risk Management



Material Topic Information Security Management

Policy

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Information Security Policy



Negative Impact Data leakage or business interrupt caused by major information security virus incidents

Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
If an information security incident causes the leakage of customer and the Company information assets, or if a cyberattack causes operational disruptions, it could lead to damage to both customers and the Company.	- Strengthen information security protection capability: Conduct information security system testing and implement patch regularly. We have business continuity plans in place and test them continually. Establish a network security incident contingency plan and take escalation and recovery actions - Improve information security management procedures: UMC has complied with information security-related certifications such as ISO 15408 and ISO 27001, and carried out continuous improvement through annual recertification - Risk control: UMC has purchased information security insurance as one of the measures to manage information security risks, protecting the Company from cyberattacks and minimizing potential losses - Education and training: Company-wide information security training and social engineering phishing tests are conducted to implant information security awareness in every employee	- To review the information security risk analysis results and the corresponding protective measures and strategies through biannual management review meetings - To report the effectiveness of the information security strategy to the board of directors reviewing annually - To conduct third-party audits annually, such as ISO 27001 and ISO 15408, achieving continuous improvement of the information security system - To align with international requirements and ensure that the information security system effectively responding to current digital environment by participating in ESG evaluations, such as DJSI Dow Jones Sustainability Index, every year

Management goals and indicators	2024		2025	2030
Item	Goal	Achievement	Goal	
Number of major information security incidents	0	⚠️ 1 ^{Note 2}	0	0
Number of major findings by information security related international standard certificatio	0	✅ 0	0	0

Note: 1. Scope: UMC Group (UMC, its subsidiaries HJ, USCXM, Wavetek and USJC)
2. In 2024, there was a DDoS attack on the Company's official website. This incident did not cause system infection, damage, intrusion, or data leakage, and had no significant impact on the Company's operations. For details, please refer to the section "Information Security Risk and Management Plan".

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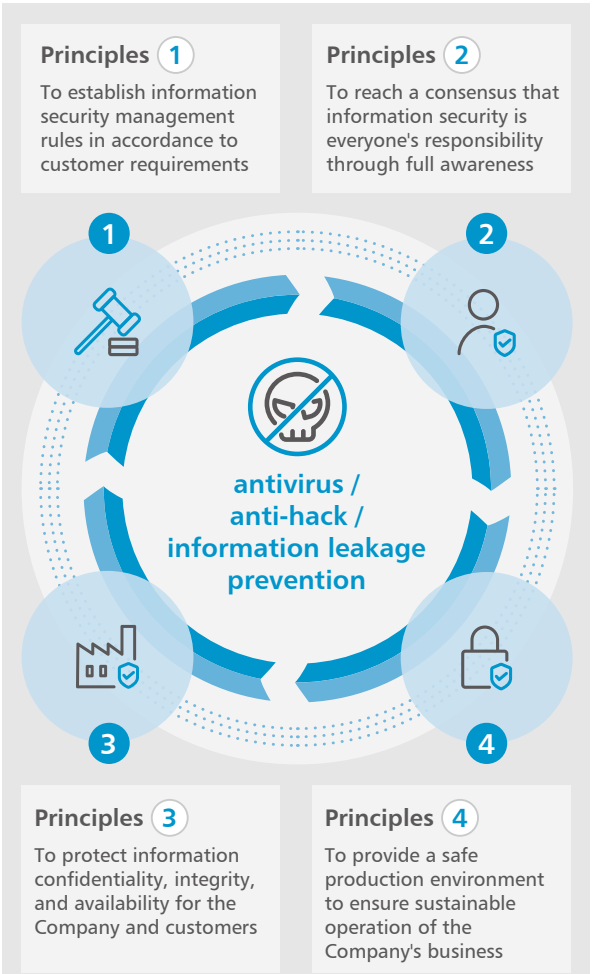
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Information Security Policy
Implementation

UMC's information security policy is based on four guiding principles. The major information security objectives are aimed at antivirus, anti-hack, and information leakage prevention by reinforcing multiple internal controls such as firewall, intrusion detection and antivirus systems to enhance the Company's ability to defend against external attacks and to protect internal confidential information.

Four Guiding Principles



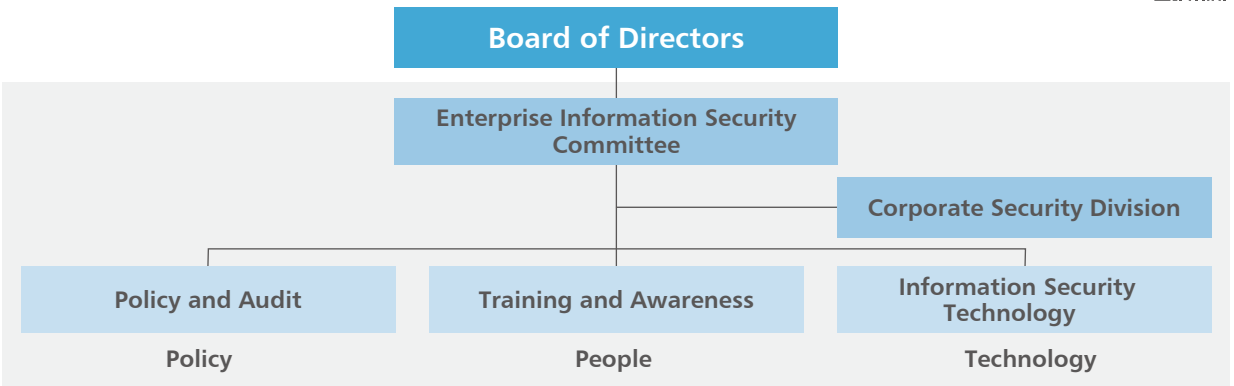
Information Security Committee Organization

The Enterprise Information Security Committee (EISC), established in 2003, is chaired by the Company's president, and Senior Vice President TS Wu of the Digital Function serves as the supervisor/Chief Information Security Officer (CISO), who is responsible for establishing and maintaining the information security strategy and processes that protect information assets. All division directors (including legal, human resources, research and development, engineering, and manufacturing) are members of the committee. In addition, the Corporate Security Division was formed to be responsible for information security, physical security planning, and related audit matters, and to lead EISC affairs.

EISC is responsible for information security management system planning, establishing and maintaining, information security policy formulation and implementation, risk management and compliance verification. The adequacy and effectiveness of the information security management system is reviewed in bi-annual meetings, during which the analysis results for information security risks and the corresponding protective measures are assessed.

The EISC reports on the effectiveness of the Company's information security strategy to the board of directors annually. The "Sustainable Development and Nominating Committee", a board-level committee, is responsible for overseeing information security related matters. All committee members are independent board directors, with Independent Board Director Jyuo-Min Shyu, who has a background in information security, serving as the convener. The committee oversees the information security and cybersecurity strategies, and reviews the strategy regularly.

For more on the background of Mr. Shyu, please refer to the UMC website >



Information Security Risk and Management Plan

According to the Global Risks Report 2024 issued by the World Economic Forum, "Cyberattacks" is selected up to five risks that are believed most likely to present a material crisis on a global scale in 2024, and "Cyber insecurity" is still among the top ten global risks in the future, which indicate that there are numerous threats to cyber security. Insufficient defense against cyberattacks may not only expose the Company to the risks of data leakage and ransom threats, but may also interrupt the production system, causing serious operating losses or even damaging the reputation of the Company.

UMC had no major information security incidents in recent years, except for a Distributed Denial of Service (DDoS) attack on official website in 2024. DDoS is an attack in which hackers use compromised computers on the internet to launch an intensive and massive traffic and packet attack on the target, attempting to paralyze the target's network and servers. This led to slow access to UMC's official website, making it unable to provide normal services. This incident did not cause system infection, damage, intrusion, or data leakage, and had no significant impact on UMC's operations. When the incident occurred, the information security team confirmed it as a DDoS attack and the affected area at once, and they immediately blocked the attacking IP addresses through the security protection system, and activated traffic cleaning services. As a result, the official website was restored back to normal access. UMC has established corresponding DDoS monitoring indicators to detect and respond to potential attacks early.

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Strengthening
Information
Security
Protection

- ❖ Conduct information security systems vulnerability analysis annually by certified third-party, and the vulnerability analysis contains vulnerability scanning, penetration testing (simulated hacker attacks), and application assessing.
- ❖ The analysis results are tracked to ensure that the weaknesses and risks found are appropriately patched and strengthened.



Focusing
Business
Continuity
Mechanisms

- ❖ Establish information security business continuity plans (BCP) to ensure UMC's ability to minimize any impact from incidents that may affect operation, and to provide resilience to the Company's business continuity with minimal productivity loss. The BCP includes the roles and responsibilities of information security response center, procedure to perform impact and loss assessment, and incident response plan from response to recovery.
- ❖ Strengthen backup system with whole server full backups that not only back up data but also the operating system and applications, enabling speedy recovery of business operations in the event of incidents.
- ❖ Conduct information security system business continuity tests at least twice annually, and improve the systems based on the test results to ensure the effectiveness and completeness for all plans.



Improving
Information
Security
Management
Procedures

- ❖ Adopt the National Institute of Standards and Technology standards to establish a corporate information security framework, and develop corresponding metrics to measure the effectiveness.
- ❖ Employees should abide by information security regulations (such as strict control of mobile storage devices), follow the standard operating procedures, and carry out the PDCA (Plan-Do-Check-Act) cycle for continuous improvement.
- ❖ Conduct internal audits of information security management systems twice annually, and all findings and suggestions are corrected and improved to ensure the effectiveness of the related controls and the compliance with the regulations, thereby reducing potential security risks.



Enhancing
Network,
Endpoint and
Application
Security

- ❖ Expand the deployment of Industrial Intrusion Prevention System (IPS) to detect and respond to network traffic, allowing for early detection and blocking of potential cyberattacks.
- ❖ Improve the detection of irregularities and elevate protection capabilities including protocols such as Application Whitelisting and EDR (Endpoint Detection and Response) mechanisms.
- ❖ Improve the efficiency of escalation and processing of information security incidents by establishing SOAR (Security Orchestration, Automation and Response) mechanisms.
- ❖ Deploy the multi-factor authentication protection for host privileged account login for detecting malicious attacks in advance and blocking intrusions.



Expanding the
Application
of Intelligent
Technology

- ❖ Utilize artificial intelligence (AI) to quickly identify malicious attacks, import threat intelligence into the information security defense system, and use robotic process automation (RPA) to continuously block attacks in real-time.
- ❖ At present, employees can use AI Chatbot to obtain instant responses at all times. In the future, it will be expanded to handle repetitive tasks and common issues automatically, further improving work efficiency. Corresponding information security management measures are also established, such as using enterprise-specific computing spaces, encrypting data, and tracing logs, to achieve effective information protection.



Introducing Legal
Compliance and
International
Information
Security
Certification
Standards

- ❖ Comply with information security related certifications and regulations such as ISO 27001, ISO 15408 and Sarbanes-Oxley Act (SOX 404).
- ❖ Corresponding enterprise risk management committee has also been established in the Company for promoting various standardization operations.
- ❖ Conduct independent external audit of information security management systems annually by third-party based on ISO 27001 requirements. All findings and suggestions, that are listed in letters of opinion from external auditors, have been corrected and improved.



Controlling
Potential Risks

- ❖ Leverage an international professional security company's resources to assess the robustness of overall security as a basis for further improvement.
- ❖ Purchase information security insurance to help reduce the losses and operational impact in the event of a security breach.



Deepening
Information
Security
Awareness

- ❖ Conduct information security training twice a year to enhance security awareness.
- ❖ Conduct periodical social engineering phishing tests for all employees to evaluate employees' sensitivity to cyberattacks and the response capabilities, thereby improving the whole information security protection capability.
- ❖ Set up a clear escalation process, that employees notice a suspicious information security or cybersecurity event, they can report it via email or telephone to the information security incident management center. After that, the information security incident management center takes the responsibility to inform related personnel, clarify impact, respond (isolate, block, recover), record and feedback to protection systems.



Deploying
Supply Chain
Information
Security Risk
Management

- ❖ Participate in the Taiwan Chief Information Security Officer (CISO) Alliance to strengthen overall supply chain information security protection by developing industry standards through cross-enterprise alliances.
- ❖ Actively cooperate with the SEMI Taiwan Semiconductor Association's Cybersecurity Committee to promote semiconductor information security risk assessment services, which introduce third-party risk posture and assessment, and tailor common questionnaires for the semiconductor industry, to provide industry benchmarking and increase efficiency in information security assessments.

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1.2.5 Managing Climate Risks and Opportunities

Under the global climate change and low-carbon market transformation, UMC actively pays attention to the potential risks and opportunities of climate change. In order to promote the transparency of information disclosure on climate-related risks and opportunities, the Company independently develops an assessment tool by referring to the Task Force on Climate-Related Financial Disclosures (TCFD) recommendations, and integrates climate scenario data, site-specific environmental characteristics, and historical disaster loss data to scientifically quantify and assess the operational impacts of climate risks. Through relevant responsible divisions to conduct risk and opportunities analysis base on policies and regulations (including current and emerging regulation), market and technology changes, reputation, legal (including litigation) and physical risks to develop adaptation and mitigation strategies, and review the achievements periodically for the purpose of management by connecting to sustainable strategy and blueprint.



TCFD Disclosure Framework and UMC Management

Governance

- Board of Directors and Functional Committees: The Board oversees climate change risks, opportunities, coping strategies, and related promotion plans. The Board also oversees promotion results, including budget implementation, revision and addition; cost-effectiveness review; and following up on implementation results. It also oversees the Company's major fundraising, issuance, and private placement of equity securities that are related to climate change. (Frequency: At least annually, with quarterly reports based on issue development and demand.)
- ESG Steering Committee: Responsible for ESG governance, strategy, objective decision, systems or management guidelines and concrete promotional plans (including climate change). The committee reviews performance and target achievements quarterly and reports to the Board of Directors on a periodic basis.
- Corporate Sustainability Committee: In charge of setting sustainability direction and objective, enforcing systems or related management guidelines (including climate change).It reports on the evaluation of risks related to corporate governance, environmental, and social issues, as well as plans and their effectiveness to the ESG Steering Committee quarterly. If the results of climate change risk assessment have a significant impact on major corporate decisions, they will be escalated to the ESG Steering Committee for approval.
- Enterprise Risk Management Committee: The Committee coordinates related divisions to jointly review the Company's internal and external risks (including climate change risks). For company-wide significant risks, review the changes in risks and management status monthly and update the risk information. Regularly track the changes and report to the ESG Steering Committee quarterly based on the needs for risk improvement and development. Annually, compile the changes in risks, improvement actions, and management implementation status throughout the year and report to the ESG Steering Committee. Additionally, present the overall risk improvement actions to the ESG Steering Committee.

Strategy

- According to UMC Climate Change Risk and Opportunity Assessment Methodology, short-term is defined as 3 years; mid-term is defined as 10 years; and long-term is over 10 years. UMC has formulated a climate change policy as the Company's highest guiding principle in responding to climate change, and has established the "UMC Low-Carbon Commitment" as the guideline for promoting its overall plan.
- Analyze the impact of climate change risks and opportunities on UMC's operations according to the TCFD framework.
- Conduct climate risk analysis for different scenarios, including transition scenarios based on International Energy Agency NZE (Net Zero), Announced Pledges Scenario (IEA APS), Nationally Determined Contributions (NDCs), Stated Policies Scenario (IES STEPS), and physical scenarios according to SSP1-2.6 and SSP5-8.5 in sixth assessment report (AR6) of the Intergovernmental Panel on Climate Change (IPCC).

Risk Management

- Identify possible policies, regulations, markets and technologies transitions, business reputation, and risks that may result in different scenarios for risk and opportunity analysis. Assessment scope includes UMC operation and upstream and downstream of value chain (such as suppliers and customers).
- With the participation of related divisions in the climate change risk and opportunity assessment task force, UMC developed adaptation and mitigation strategies for top three risks. The overall results and related countermeasures will be submitted to CS committee for approval.
- Annual review on risks and opportunities of transitions in policies, regulations, technologies, markets and business reputation; every three to five years, when the IPCC and the Taiwan Climate Change Projection Information and Adaptation Knowledge Platform (TCCIP) updates climate scenarios, re-examinations and on-going corrections on the assessment results of physical risks, transition risks, and opportunities will be carried out.

Metrics and Targets

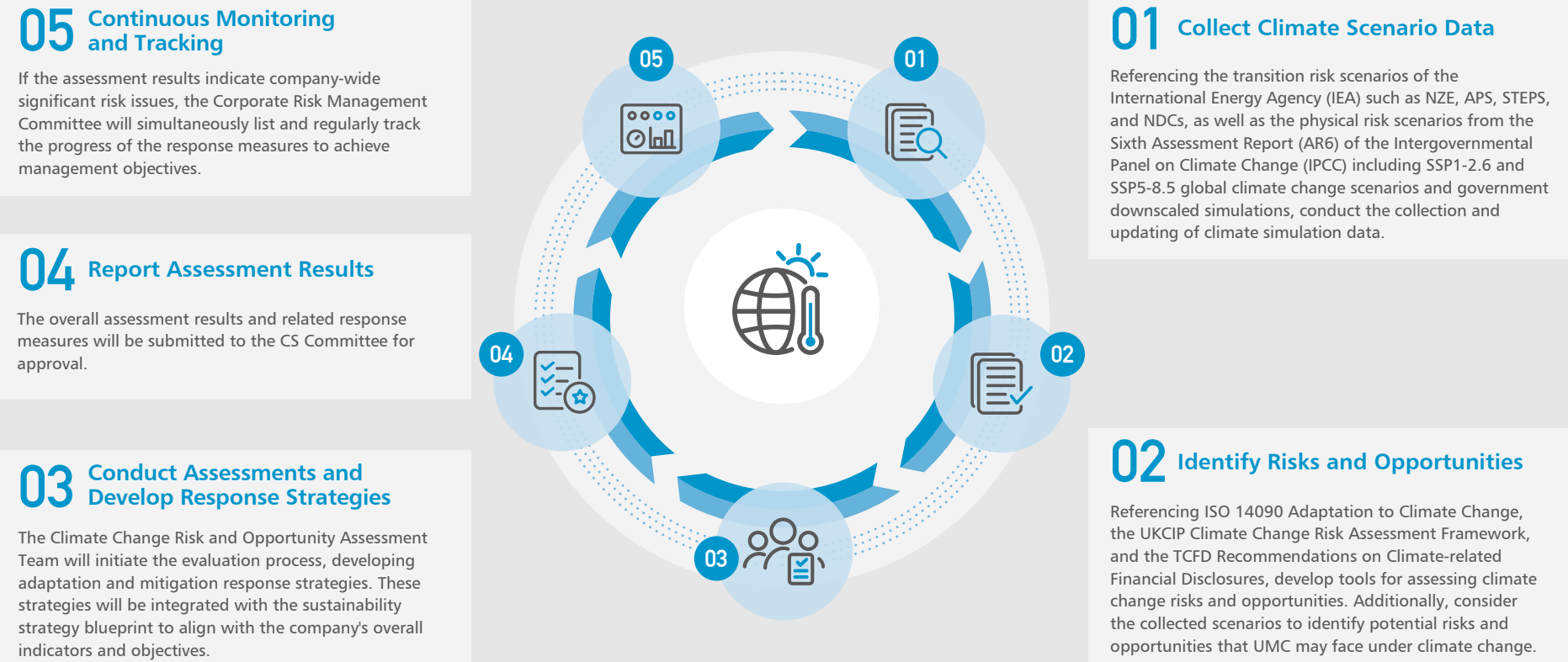
- UMC has established climate-related risk and opportunity assessment management indicators such as water consumption, energy use, greenhouse gas emissions, and the technology development of low energy consumption products.
- Relevant emission information has been checked in accordance with the Greenhouse Gas (GHG) Protocol and the ISO 14064 Standard, and data verification by a third party has been commissioned. In addition to complying with regulations, the current status also complies with IEEE's standards for fluorinated greenhouse gas.
- Established standards for new fabs and new equipment to respond to climate change and greenhouse gas reduction regulations.
- Set year 2025 and year 2030 KPI targets for indicators including greenhouse gas, water, electricity reduction, and product design. For emissions reduction, UMC has also set its net zero by 2050 goal and its 2030 science-based target.

Note: The target has been approved by Science Based Targets initiative (SBTi).

Important Issues Related to Climate Risk and Opportunity Management

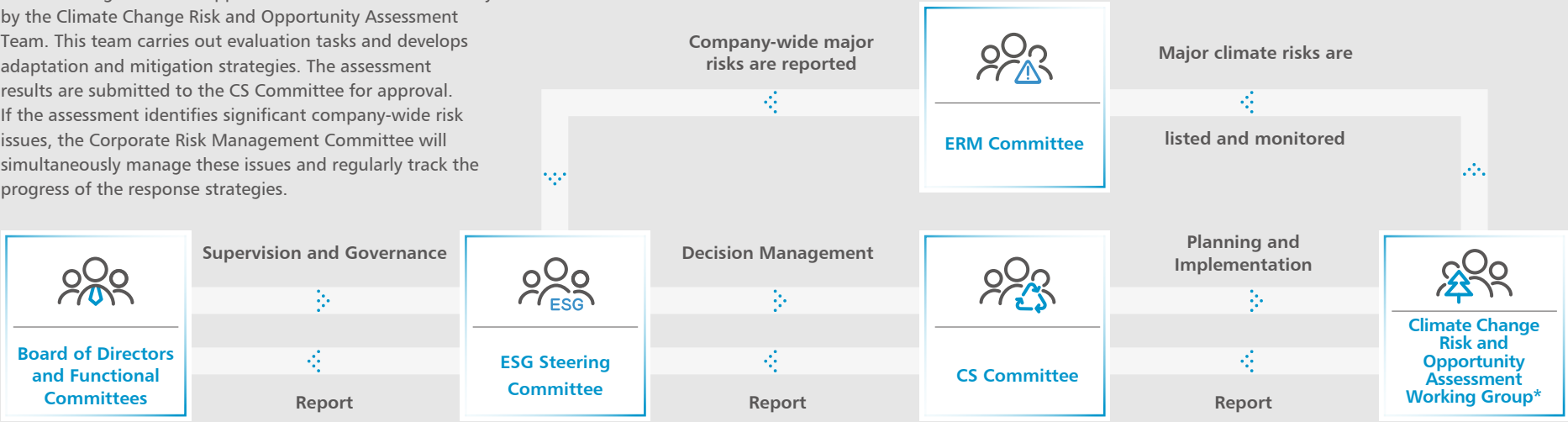
- ❖ The Board of Directors approved the implementation of the capital budget including green investment such as energy saving, improvements for pollution prevention facilities and the Circular Economy & Recycling Innovation Center.
- ❖ The Board of Directors approved the issuance of new restricted employee stock in 2024. One of the conditions for meeting the standards includes the ESG: annual reduction of greenhouse gas emissions targets.
- ❖ The ESG Steering Committee approved an internal carbon price of \$100 per ton in 2024.
- ❖ The Board of Directors approved" the signing of a Corporate Power Purchase Agreement with Fengmiao Wind Power Co., Ltd.," in 2024.

Climate Change Risk and Opportunity Assessment Process



Reporting Mechanism of Climate Change Risk and Opportunity Assessment Results

The reporting mechanism for the assessment results of climate change risks and opportunities is conducted annually by the Climate Change Risk and Opportunity Assessment Team. This team carries out evaluation tasks and develops adaptation and mitigation strategies. The assessment results are submitted to the CS Committee for approval. If the assessment identifies significant company-wide risk issues, the Corporate Risk Management Committee will simultaneously manage these issues and regularly track the progress of the response strategies.



* Climate Change Risk and Opportunity Assessment Working Group members include: procurement, Material Planning & Warehouse Management, Finance, Marketing, Technology Development, Human Resource, Facility Operation, Group Risk Management & Environmental Safety and Health, etc.

Climate Change Risk and Opportunity Assessment Scenario

Risk Type	Scenario	Scenario Abstract Description
Physical Risk	IPCC-AR6 SSP1-2.6	Low Emission Scenario, emphasizing a low-carbon economy, environmental protection, and social equity, with a projected global average temperature increase of less than 2°C by 2100.
	IPCC-AR6 SSP5-8.5	High Emission Scenario, emphasizing economic growth and the continued use of fossil fuels, with greenhouse gas emissions continuing to increase, projecting a global temperature rise of more than 4°C by 2100.

The physical risks brought about by climate change not only directly affect temperature but also cause extreme rainfall events such as heavy rain or drought, which can impact fab operations and lead to operational disruption risks. UMC, in collaboration with its partners, established a "Climate Risk Database" to analyze all UMC fabs, including 14 sites in Hsinchu Science Park, Southern Taiwan Science Park, Suzhou and Xiamen in China, Japan, and Singapore. Based on geographical and environmental characteristics, these sites are grouped into eight locations for analysis, conducting scientific assessments of the physical risks at each location. The database integrates meteorological observation data (Note 1), model estimation of climate scenarios (Note 2), and definition of disaster risk to quantify the "likelihood" and "impact" of disasters. Using a risk matrix to identify the baseline of short-term, mid-term, and long-term risk levels for each fab, the original climate risks of each site can be assessed and recorded. The table below presents the preliminary short-term analysis results for each site.

Note: 1. Historical meteorological observation data released by the Central Weather Bureau or other domestic and international meteorological agencies.

2. Downscaled climate simulation data from official agencies such as the IPCC and TCIP.

The Preliminary Short-term Analysis Results for Each Site

Site/ Location		Scenario	Drought	Heavy Rainfall/Flooding	Warming
Hsinchu science Park	Fab 8A, Fab 8C, Fab 8D, Fab 8E, Fab 8F	SSP1-2.6	Low	Medium	Low
		SSP5-8.5	Low	Medium	Low
	Fab 8S, WTK	SSP1-2.6	Low	Medium	Low
		SSP5-8.5	Low	Medium	Low
Tainan science Park	Fab 12A I	SSP1-2.6	Medium	Medium	Low
		SSP5-8.5	Low	Medium	Low
	Fab 12A II	SSP1-2.6	Medium	Medium	Low
		SSP5-8.5	Medium	Medium	Low
Singapore	Fab 12i	SSP1-2.6	Low	High	Low
		SSP5-8.5	Low	High	Low
Xiamen, China	Fab 12X	SSP1-2.6	Low	Medium	Low
		SSP5-8.5	Low	High	Low
Suzhou, China	Fab 8N	SSP1-2.6	Low	Medium	Low
		SSP5-8.5	Low	High	Low
Mie, Japan	Fab 12M	SSP1-2.6	Low	High	Low
		SSP5-8.5	Low	High	Low

After completing the risk assessment of the fab sites, UMC further quantified the risk levels by incorporating climate science analysis. This approach considers the disaster response capabilities of each fab site and quantifies the climate risks accordingly. The results serve as a reference for financial evaluation and the formulation of adaptation measures. For detailed evaluation results, please refer to the "Climate Change Risk and Opportunity Matrix" provided down below. Regarding the medium-risk drought, detailed impact analysis and response measures can be found in the "Quantitative Financial Impact of Risks and Opportunities" section. For heavy rainfall/flooding, simulations were conducted to assess rainfall intensity against the local drainage return period in days, and the results was less than one day.

Rainfall Intensity Reaching Local Drainage Recurrence Period Days in Various Regions for Short, Medium, and Long Term

Location-Site/Days		Heavy Rainfall Recurrence Period for Sites	Short term	Medium term	Long term
Xiamen, China	Fab 12X	50 years	0.07	0.06	0.07
Suzhou, China	Fab 8N	50 years	0.14	0.05	0.10
Mie, Japan	Fab 12M	10 years	0.21	0.19	0.17
Singapore	Fab 12i	10 years	0.81	0.50	0.49
Hsinchu science Park	Fab 8A, Fab 8C, Fab 8D, Fab 8E, Fab 8F	100 years	0.00	0.01	0.02
	Fab 8S, WTK	100 years	0.00	0.01	0.03
Tainan science Park	Fab 12A I	100 years	0.06	0.02	0.02
	Fab 12A II	100 years	0.04	0.02	0.02

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- Sustainability Strategy and Blueprint
- Enterprise Risk Management
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1 Leading in Corporate Governance

- 1.1 Enhancing Corporate Governance
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Appendix

As global attention to climate change intensifies, the transformation risks faced by enterprises have become an important issue that cannot be ignored. UMC is operating in a rapidly changing market environment and must proactively address challenges from policies, technology, and market demand. According to the framework of the Task Force on Climate-related Financial Disclosures (TCFD), companies should assess and disclose climate-related risks and opportunities to promote transparency and sustainable development.

Explanation of the potential impact of transition risks on UMC:
An analysis is conducted for different scenarios, and the potential impacts are described as follows

Risk Type	Scenario	Scenario Abstract Description
Transition Risk	International Energy Agency Net Zero (IEA NZE)	- Reduce carbon emission gradually through the widespread deployment of clean energy technologies around the world, and achieve zero energy-related carbon dioxide emissions by 2050. - UMC sets net zero emission target for 2050 and conducts reduction and adaptation assessments
	International Energy Agency Announced Pledges Scenario (IEA APS)	- IEA APS scenario takes account of all climate commitments made by governments around the world including Nationally Determined Contributions (NDCs) as well as longer-term net-zero targets and assumes they will be met in full and on time. - UMC conducts reduction and response assessment by referring to climate commitments in the operation region, including Taiwan, Singapore, China, and Japan.
	International Energy Agency Stated Policies Scenario (IEA STEPS)	- Scenario is based on current and proposed policies and measures, and in the absence of additional policies in the future. - UMC conducts assessment under business as usual (BAU) without dramatic reduction plan.

Climate Change Risk and Opportunity Matrix



Transition Risk

- 1. Carbon tax/fee for greenhousegas emission control
- 2. Increasing green energy demand in production line of customers
- 3. Increasing attention and pressure from investors
- 4. Electricity prices rising leads to increasing in operation cost
- 5. Market requires adjustment towards low energy consumption and low carbon emissien products
- 6. Carbon-fee /carbon tax was transferred to raw material cost
- 7. Customers pay more attention to UMC's climate resilience and business continuity
- 8. Govt plans to impose water surcharge during drought season

Physical Risk

- 9. Carbon tax/fee for greenhousegas emission control
- 10. Strong typhoon/extreme rainfall lead to increased frequency of work suspension and voltage dips
- 11. Drought worsens and affects production
- 12. Extreme weather increases supply chain scheduling cost

Opportunity

- 13. Carbon trade and offset opportunities
- 14. Enhance energy efficiency
- 15. Promote low-carbon sustainable supply chain
- 16. Increasing market /customer demand for sustainable products
- 17. ESG performance helps financing costs

△ Short-term (within 3 years) ○ Mid-term (within 10 years) □ Long-term (over 10years)

Climate Risks and Response Actions

Category	Risk	Risk Description	Region	Aspect	UMC's Response	Connection with sustainable strategy and blueprint
Risk Type ❖ Transition						
Policy and Regulation (including current and emerging regulation)	Carbon tax/fee for greenhouse gas emission control*	❖ Carbon tax/fee and rate increase lead to increase in operation cost	❖ Taiwan, Singapore, China, and Japan	❖ Increase operating expense	❖ Promote GHG reduction plans, such as F-GHG's reduction and energy saving plan ❖ Participate in government plans for voluntary GHG reduction ❖ Apply for carbon offset projects for carbon credits ❖ Build UMC's own solar power generation facilities ❖ Purchase green energy/ green energy certificates ❖ Evaluate carbon negative technology and promote net zero ❖ Participate in carbon trade market and purchase carbon credits	✔
	Electricity prices rising leads to increasing in operation cost*	❖ Electricity price rising	❖ Taiwan, Singapore	❖ Increase operating expense	❖ Reduction of electricity consumption per unit product ❖ Adopt high-efficiency energy-saving equipment when installing new systems such as LED lights and pumps. ❖ Use high-efficiency energy-saving devices when replacing old equipment, such as chillers and cooling towers. ❖ Optimize operating conditions, such as adjusting supply pressure and temperature.	✔
	Government plans to impose water surcharge during drought season as climate change causes water shortages	❖ Taiwan imposes water surcharge	❖ Taiwan	❖ Increase operating expense	❖ Magnify usage rate of diversified water resource such as reclaimed water and desalination to reduce the dependency of municipal water ❖ Improve the production efficiency of manufacturing process, reduce water consumption, and increase water recovery rate	✔
	Carbon-fee / carbon tax was transferred to raw material cost	❖ Increase in raw material and transportation prices due to carbon fees/taxes	❖ Taiwan, Singapore, China, and Japan	❖ Increase operating expense	❖ Evaluate the introduction of alternative material or order allocation ❖ Sign contract to secure the supply and protect against price increases	
Market	Increased green energy demand in production line of customers	❖ Increasing the procurement of green energy upon customers' requests	❖ Taiwan, Singapore, China, and Japan	❖ Increase operating expense	❖ Promote energy-saving measures and improve energy efficiency. ❖ Establish a renewable energy task force to regularly track customer green energy demands and formulate renewable energy implementation strategies. ❖ Actively participate in the renewable energy market and evaluate diverse green energy procurement channels.	✔
Reputation	Increased attention and pressure from investors	❖ The Company fails to meet policy regulations or targets of international climate change initiatives, leading to negative news about the Company ❖ The Company is unable to positively respond to ESG related questions raised by shareholders in shareholders' meetings	❖ Taiwan, Singapore, China, and Japan	❖ Rising cost of capital	❖ Clarify and correct news reports if there is any error; take concrete actions to comply with policy regulations or international goals related to climate change	
	Customers pay more attention to UMC's climate resilience and business continuity	❖ UMC is asked to have business continuity capability to withstand natural disaster (extreme weather) and increase the frequency of audit by customers.	❖ Taiwan, Singapore, China, and Japan	❖ Increase operating expense	❖ The coverage rate of "ISO 22301 Business Continuity Management System" in 12-inch Fabs and 8-inch Fabs reaches 100% and the certificate remains 100% effective.	
Technology	Market requires adjustment towards low energy consumption and low carbon emission products	❖ Customer requires product with low energy consumption and low carbon emissions characteristics, which may result in necessary adjustment of UMC's R&D priorities and goals	❖ Taiwan, Singapore, China, and Japan	❖ Increase operating expense	❖ Develop process technology that is superior and has lower energy consumption ❖ Monitor the amount of R&D wafer in production, and and aim to use a method that requires fewer wafers to develop the process. ❖ Look for a faster or streamlined process to manufacture IC	

(continued)

(Continued)

Category	Risk	Risk Description	Region	Aspect	UMC's Response	Connection with sustainable strategy and blueprint
Risk Type ⚡ Physical						
Acute	Strong typhoon/ extreme rainfall lead to increased frequency of work suspension and voltage dips	⚡ Increases in intensity and frequency of typhoons, storms, and natural disasters result in employee inability to work ⚡ Voltage drop caused by typhoon affects production	⚡ Taiwan	⚡ Increase operating expense ⚡ Decrease in revenue	⚡ Build Industry 4.0 to reduce the impact of manpower shortage ⚡ Invest in voltage drop resistance equipment, equip important manufacturing machinery with uninterrupted power supplies (UPS) to reduce risk of voltage drops	✓
	Extreme weather increases supply chain scheduling cost.	⚡ Extreme weather affects transportation and increases the cost of raw material allocation and scheduling fee.	⚡ Taiwan, Singapore, China, and Japan	⚡ Increase operating expense	⚡ Establish risk detection mechanism, with a focus on prevention and post-event management ⚡ Monitor supplier's production areas and the location of major transportation hubs ⚡ Manage production information from suppliers to stay abreast of supply and demand information ⚡ Conduct supplier risk assessment to avoid or reduce placing order in high-risk production areas ⚡ Establish multiple transport routes and avoid risky routes ⚡ Establish second sources to provide additional backup energy ⚡ Increase the storage capacity of local manufacturers and internal storage capacity	
Chronic	Increased electricity consumption due to global warming	⚡ Rising temperatures increase the load of air conditioning, which increases operating costs	⚡ Taiwan, Singapore, China, and Japan	⚡ Increase operating expense	⚡ Reduction of electricity consumption per unit product ⚡ Continuously replace old air conditioning equipment, and promote energy-saving measures for air conditioning ⚡ Implement energy-saving measures for building shell and air-conditioning systems to reduce internal temperature of building ⚡ Introduce green-building design ⚡ Grow plants to increase the amount of greenery and reduce urban heat island effect	✓
	Drought worsens and affects production*	⚡ When the status of water rationing turns to orange light, the daily water supply is reduced. Measures such as using water trucks and Uwater are required to cope, which increases operational costs.	⚡ Taiwan	⚡ Decrease in revenue	⚡ Expand usage rate of reclaimed water and desalinated seawater. ⚡ Enhance water recycle and reuse rate to reduce the pressure of water demands. ⚡ Establish a water contingency plan ⚡ Storage tanks are installed to improve the flexibility of water storage capacity control. ⚡ Contract water tankers and private water sources to make up for water shortages ⚡ Continue to monitor the government's ability to strengthen cross-regional dispatching of water resources	✓
		⚡ Delivery delays from suppliers due to water constraints	⚡ Taiwan	⚡ Increase operating expense	⚡ Conduct supplier water shortage risk assessment, and ask high-risk vendor (such as parts cleaning vendors) to develop business continuity plan (BCP) ⚡ Investigate and understand the impact of water demand in fabs, supplier capacity, and government water restriction policy. ⚡ Request suppliers to provide water storage plans and cooperation models with private water resource providers ⚡ Adopt new products first to reduce the impact ⚡ Monitor the status of suppliers, increase safety stock, and establish relationships with second sources ⚡ Develop new water resources continuously to increase supplement flexibility and mitigate risk	

Note: 1. The items marked with * indicate the top three risk.
2. For detailed index, achievements in 2024 and mid-term or long-term target, please refer to sustainable strategy and blueprint. Items not linked to the sustainable strategy and blueprint are to be incorporated into the daily management of each division by the respective risk identifiers. All identified risks should have corresponding response measures developed and be regularly reviewed by the respective committees.
3. There's no litigation-related risk after this assessment.

Climate Opportunities and Response Actions

Type	Climate-related Opportunities	Region	Aspect	UMC's Response	Connection with sustainable strategy and blueprint
Product/ Service	❖ Promote low-carbon sustainable supply chain	❖ Taiwan	❖ Decrease operating expense	❖ Increase the carbon reduction rate of vendors with high electricity consumption. ❖ Deepen carbon reduction cooperation with suppliers and create a low-carbon supply chain	✔
Market	❖ Increasing market /customer demand for sustainable products.*	❖ Taiwan, Singapore, China, and Japan	❖ Increase revenue	❖ Increase the percentage of sustainable product revenue	✔
	❖ ESG performance helps financing costs	❖ Taiwan, Singapore, China, and Japan	❖ Reducing Financing Costs through the Green Bond MarketLower financing rates through the green bond market	❖ Continue to prioritize the investment of resources to promote ESG, and through the "2030 Sustainable Strategy Blueprint," comprehensively explore the three major aspects of ESG.	
Resource Efficiency	❖ Carbon trade and offset opportunities	❖ Taiwan	❖ Increase revenue ❖ Decrease operating expense	❖ Support government policies to reduce greenhouse gas emissions, improve overall equipment and fab operation efficiency to secure carbon credits.	
	❖ Enhance energy efficiency	❖ Taiwan, Singapore, China, and Japan	❖ Decrease operating expense	❖ Reduction of electricity consumption per unit product ❖ Adopt high-efficiency energy-saving equipment when installing new systems such as LED lights and pumps. ❖ Use high-efficiency energy-saving devices when replacing old equipment, such as chillers and cooling towers. ❖ Optimize operating conditions, such as adjusting supply pressure and temperature.	✔

Note: 1. The items marked with * indicate the main opportunity.
2. For detailed index, achievements in 2024 and mid-term or long-term target, please refer to sustainable strategy and blueprint. Items not linked to the sustainable strategy and blueprint are to be incorporated into the daily management of each department by the respective opportunities identifiers. All identified opportunities should have corresponding response measures developed and be regularly reviewed by the respective committees.

Risks and Opportunities in UMC's Value Chain under Climate Change

In addition to assessing the impact of climate change on operations, UMC also takes the impact on the value chain (including upstream suppliers and downstream customer markets) into consideration. The climate risks and opportunities of the UMC value chain are listed as follows:

		Transition Risk	Physical Risk	Opportunity
	Upstream Purchase	❖ Carbon tax/fee passed on to raw material costs	❖ Extreme weather increases supply chain logistics costs	❖ Promote a Low-Carbon Sustainable Supply Chain
	Operations	❖ Carbon tax/fee for greenhouse gas emission control risk ❖ The increase in electricity prices leads to higher operating costs. ❖ In response to the scarcity of natural water resources due to climate change, the government plans to implement a policy to increase water surcharge during drought seasons.	❖ Drought worsen and affects production ❖ Global warming leads to increased electricity consumption. ❖ Severe typhoons/extreme rainfall result in higher frequency of work stoppages and voltage drops.	❖ Improve energy use efficiency ❖ Opportunities for carbon trading and offsetting ❖ ESG performance benefits financing costs
	Customer Market	❖ Market demands shift towards low energy consumption/low carbon emission products ❖ Increased demand for green electricity in customer production lines ❖ Growing investor attention and pressure ❖ Customers are increasingly focused on UMC's climate resilience and operational sustainability	❖ Heavy rainfall caused operational disruptions and delayed shipments.	❖ There is a growing demand for sustainable products in the market/customers.

Quantification of Financial Impact of Risks and Opportunitie

The following discloses one transition risk, one physical risk, and one opportunity for each production site of the UMC Group, along with the financial impact after considering the corresponding mitigation measures:

Transition Risk Carbon tax/fee for greenhouse gas emission control risk

Scenario

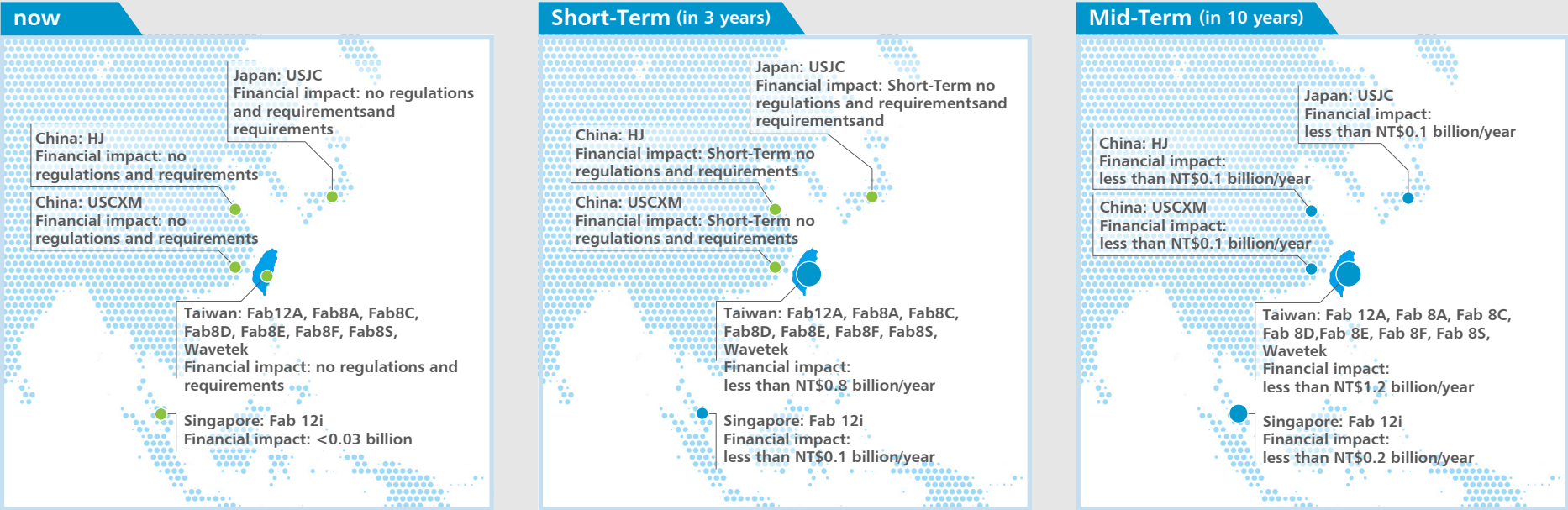
IEA APS including NDCs

Assumptions:

- Singapore's carbon fee rate will increases to SGD 45 per ton in 2026, and with an assumed rate of SGD 80 per ton for the period from 2030 to 2034.
- Taiwan imposes a carbon fee. From 2025 to 2026, the rate is NT\$300 per ton; from 2027 to 2028, the rate is assumed to be NT\$600 per ton; and from 2033 to 2034, the rate is assumed to be NT\$1,500 per ton.
- Carbon fee are levied in Japan and mainland China, assuming a rate of NT\$300-1,200 per ton in the mid-term, while there are no regulations and requirements in the short-term.

Financial shocks and countermeasures:

Low Risk ≤0.1 billion ≤0.5 billion <1.5 billion Unit: NTD



UMC continues to implement measures to reduce greenhouse gas emissions, the current cost of purchasing green electricity is projected to exceed NT\$200 million in 2024 and over NT\$5 billion in 2025 and 2030. The cost of replacing existing equipment with new L/S setups is estimated to exceed NT\$100 million in 2024 and over NT\$900 million in 2025 and 2030. The benefits include a reduction of approximately NT\$9 million in carbon tax in 2024, with an estimated reduction of over NT\$1 billion in carbon fees/carbon tax from 2025 to 2030.

Physical Risk

➤ Drought worsen and affects production

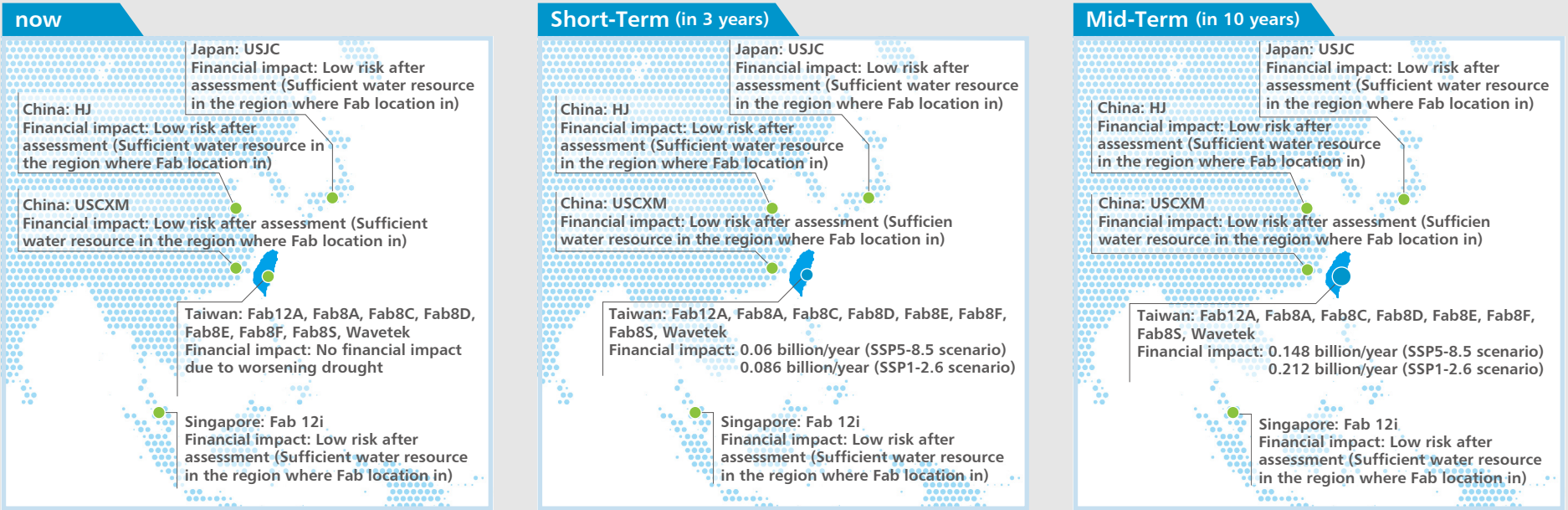
Scenario

- IPCC AR6 SSP1-2.6 and SSP5-8.5
- Risk Analysis Results under the Scenario of Long-term Drought: Significant drought risks are observed in the inland regions of East Asia and the central and southern parts of Taiwan.

Assumptions:

- Evaluation Parameter: The assessment is conducted based on the changes in the Standardized Precipitation Index (SPI). The higher the SPI value, the more severe the drought.
- Baseline Value: The SPI values from years with severe droughts in recent history are used as the baseline. For Hsinchu, the SPI from 2020-2021 is used, and for Tainan, the SPI from 2022-2023 is used.
- Using downscaling models, the annual rate of change in SPI severity under SSP1-2.6 and SSP5-8.5 scenarios is projected to estimate the financial loss over the assessment period.

Financial shocks and countermeasures



UMC will continue to invest in water trucks and self-implemented water-saving measures (including U water), as well as the purchase of recycled water and desalinated water. The investment cost for 2024 is approximately NT\$14.7 million per year, with estimated annual costs exceeding NT\$20 million per year within three years and exceeding NT\$80 million per year within ten years. This financial calculation only accounts for the increased costs of recycled water and the contract fees for water trucks.

Note: Fixed costs such as tap water are excluded.

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Appendix

Opportunity ▶ Increasing Revenue Share of Sustainable Products

UMC is committed to investing in the development of high-efficiency product technologies, providing semiconductor manufacturing services in clean technology fields such as automotive electronics, smart energy management, industrial automation, and system optimization, to reduce carbon emissions across the value chain, i.e., the end products of customers. To align its technology development with global carbon reduction goals, UMC has set phased targets to increase the revenue share of sustainable products by 2030. The Company continues to develop relevant technologies based on customer and market demands, such as the recently developed 28nm ultra-low power embedded high-voltage platform 28eHV-LP process technology. This technology uses the industry's smallest SRAM cells, reducing energy consumption by 15% without compromising image quality or data rate, thereby meeting the demand for battery power savings.

Financial shocks and countermeasures:

now

Based on the consolidated revenue of NT\$232.3 billion in 2024, there is an increase of NT\$9.8 billion in annual revenue.

short term (in 3 years)

The proportion of sustainable products remains at 80% to 82%. With demand for products below 40nm accounting for over 60%, revenue is expected to increase by 2%, approximately NT\$3.8 billion.

Mid-term (in 10 years)

The proportion of sustainable products is expected to remain at 80% to 82%. With demand for products below 40nm accounting for over 60%, revenue is projected to increase by 6%, approximately NT\$11.3 billion.

UMC continues to invest in the development of more sustainable product process platforms to align with the goals outlined in the sustainable strategy blueprint and to increase revenue from sustainable products. This includes products that reduce energy consumption, such as low-power, low-leakage, and power management solutions, as well as durable products like microprocessors and automotive chips, and those related to epidemic prevention and medical care. In 2024, the revenue from energy-saving products, including low-power, low-leakage, and power management solutions, is expected to account for 76.8% of total revenue, contributing approximately NT\$138 billion. The overall investment in research and development expenses is NT\$13.15 billion.

Note: This calculation is based on UMC's projected individual revenue of NT\$179.681 billion for 2024.

Climate-related Response Measures of UMC

UMC attaches great importance to the risks and opportunities that climate change poses to the Company. The response measures taken by the Company are described as follows:

Item	Implementation Status	Location
1. Describe the supervision and governance of the Board and the management team regarding climate-related risks and opportunities.	UMC's Board of Directors and functional Committees oversees climate change risks, opportunities, coping strategies, and related promotion plans. ESG Steering Committee: Throws meeting quarterly, responsible for ESG governance, strategy, objective decision, systems or management guidelines and concrete promotional plans (including climate change), and report to the Board of Directors on a periodic basis. Corporate Sustainability Committee: Throws meeting quarterly, in charge of setting sustainability direction and objective, enforcing systems or related management guidelines (including climate change). If the results of climate change risk assessment have significant influence on corporate major decision, they will be escalated to ESG Steering committee for approval. Enterprise Risk Management Committee coordinates related divisions within the Company to jointly review the Company's internal and external risks (including climate change risks), regularly traces the changes for listed as major company-wide risks, and reports the overall improvement actions to ESG Steering Committee.	1.1.1 Operation of the Board of Directors 1.2.5 Managing Climate Risks and Opportunities
2. Describe how the identified climate risks and opportunities affect the business, strategy, and finances of the company (in the short, medium, and long term).	According to UMC Climate Change Risk and Opportunity Assessment Methodology, short-term is defined as within the next 3 years, medium-term is defined as within the next 10 years, and long-term is defined as more than 10 years in the future. UMC formulates response plans for the identified major risks and opportunities. Taking carbon tax/fee as an example, it is expected that operating costs will gradually increase due to rising rates, resulting in a decline in profits. Therefore, the Company has set a net zero emission target in 2050, and actively promotes greenhouse gas reduction plans to reduce potential financial impacts.	1.2.5 Managing Climate Risks and Opportunities

(Continued)

Item	Implementation Status	Location
3. Describe the impact of extreme climate events and transitional actions on company finances.	UMC has taken into account the potential financial impacts such as revenue, costs, and assets when assessing identified climate-related risks and opportunities.	1.2.5 Managing Climate Risks and Opportunities
4. Describe how climate risk identification, assessment and management processes are integrated into the overall risk management system.	The Climate Change Risk and Opportunity Assessment Working Group conducts assessment annually, and develops adaptation and mitigation countermeasures. The assessment results are submitted to the Corporate Sustainability Committee for approval. If the assessment results are listed as major company-wide risks, ERM committee will simultaneously manage it and trace the implementation progress of countermeasures on a regular basis, and summarizes the overall risk improvements and reports to the ESG Steering Committee to ensure alignment with the Company's sustainability strategy.	1.2.1 Risk Management System 1.2.3 Comprehensive Business Risk Management 1.2.5 Managing Climate Risks and Opportunities
5. If a scenario analysis is adopted for the assessment of resilience to climate change risks, then the scenarios, parameters, assumptions, and analysis factors used, as well as the major financial impacts should be explained.	UMC conducts climate-related scenario analysis, including transitional scenarios based on National Determined Contributions (NDCs) and World Energy Outlook (WEO) published by International Energy Agency (IEA), and physical scenarios in Assessment Report published by Intergovernmental Panel on Climate Change (IPCC). UMC also conducts risk and opportunity analysis for policies, regulations, markets and technologies transitions, business reputation and physical risks. For relevant parameters, assumptions, analysis factors and major financial impact information, please refer to this chapter.	1.2.5 Managing Climate Risks and Opportunities
6. If there is a transition plan for managing climate-related risks, please describe the content of the plan, as well as the indicators and goals used to identify and manage physical risks and transition risks	In order to achieve net zero emissions by 2050, UMC has drawn up a low-carbon transition plan to cover direct emissions from fab operations (Scope 1), indirect emissions from energy consumption (Scope 2), and indirect emissions from value chain (Scope 3).The implementation includes: (1) Persistent and proactive carbon emissions reduction: Through continually developing advanced foundry process technologies, enhancing productivity and improving energy efficiency, UMC is able to minimize carbon emissions in both manufacturing stage of wafers and using stage of IC end products; (2) 100% renewable energy: A 3-stage objective of 25% by 2025, 50% by 2030, and 100% by 2050 is set to substantially increase the proportion of consuming renewable energy, UMC will also invite its value chain to support the low-carbon energy transition; (3) Investment in net-zero technologies: UMC commits to invest in net zero emissions technologies and participate in Carbon Offset Projects to remove and offset carbon emissions that are inevitable or still subject to technical restrictions. UMC has considered the international carbon market prices, the carbon prices of GHG relevant regulations and the costs of reducing GHG emissions of the Company to set an internal carbon price, and used it as a reference for carbon reduction management and planning.	Sustainable Strategy Blueprint Sustainable Governance 1.2.5 Managing Climate Risks and Opportunities 2.1 Innovation & Technology Development 3.1 Climate Action 3.2 Energy Management 3.3 Water Management
7. If internal carbon pricing is used as a planning tool, the basis for setting the price should be stated.	UMC has considered the international carbon market prices, the carbon prices of GHG relevant regulations and the costs of reducing GHG emissions of the Company to set an internal carbon price, and used it as a reference for carbon reduction management and planning.	3.1.2 GHG Emissions Management
8. If climate-related goals are set, information such as the activities involved, the scope of greenhouse gas emissions, planning schedule, and annual progress should be disclosed. If carbon offsets or renewable energy certificates (RECs) are used to achieve the relevant goals, the source and quantity of carbon reductions for carbon offsets or the quantity of renewable energy certificates (RECs) should be explained.	UMC has announced to achieve RE100 and net zero emissions by 2050, and established climate-related targets such as water consumption, energy use, GHG emissions, and the technology development of low energy consuming products.	2.1 Innovation & Technology Development 3.1 Climate Action 3.2 Energy Management 3.3 Water Management
9. Greenhouse gas inventory and assurance results, along with reduction targets, strategies, and specific action plans	In 2021, UMC announced its pledge to reach net zero emissions by 2050 as the first semiconductor foundry globally. The 2030 GHG emissions reduction target set by UMC was approved by the Science-Based Targets Initiative (SBTi) in 2022, and the Company continues to drive further reductions by three resolutions. In 2024, in order to actively implement UMC's low-carbon commitments and take actions to speed up emission reduction, UMC has further signed the SBTi net-zero target commitment and raised the 2030 GHG emissions reduction target in accordance with the SBTi Net-Zero Standard to a 42% reduction of Scope 1 and Scope 2 emissions with 2020 as the base year.The GHG emissions of UMC and its foundry subsidiaries reduced by 31% in 2024 compared to the base year emissions of 2.883 million tonnes.	3.1 Climate Action

Note: The information listed in this table contains the climate-related indicators that should be disclosed as required by Appendix 2 under Article 4 of TWSE's "Rules Governing the Preparation and Filing of Sustainability Reports by TWSE Listed Companies".
Appendix 2 1-1: For greenhouse gas inventory and relevant data verification, please refer to this report's appendix.

1.3 Building Sustainable Supply Chain

- 1.3.1 Localization of Supply Chain and Procurement
- 1.3.2 Supply Chain Management
- 1.3.3 Sustainable Raw Materials Management
- 1.3.4 Conflict Minerals Management

Material Topic Sustainable Supply Chain Management

Related Policy

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

UMC's Supplier Code of Conduct

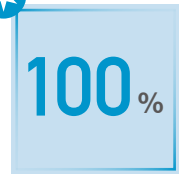


Positive Impact Building sustainability supply chain Negative Impact Insufficient supply chain management

Impact	Management Actions	Processes Used to Track the Effectiveness
Positive: Building sustainability supply chain—Implement sustainable management and ESG evaluation mechanisms for suppliers to enhance their ESG awareness and management practices. Negative: Insufficient supply chain management—Without comprehensive economic, environmental, and social standards for supplier selection and management mechanisms, improper supply chain management may occur.	- Promote "Low-Carbon Supply Chain" Project, inventorying greenhouse gas emissions of suppliers and strengthening supply chain resilience. - Conduct third-party risk assessment to lower potential risks of supply chain. - Formulate a "Code of Conduct" for suppliers to comply in accordance with the requirements of the Responsible Business Alliance (RBA). - Hold annual sessions with suppliers to share sustainability-related information. - Introduce Sustainability Suppliers Assessment. - Due diligence on conflict minerals through questionnaires and on-site audits every year.	The Responsible Supply Chain Committee has launched the Supply Chain Sustainability Blueprint and supply chain self-set goals and projects, with the results being reported quarterly to the Corporate Sustainability Committee and the ESG Steering Committee, and ultimately reported to the Board of Directors, which oversees the responsible supply chain projects.



Maintaining a supply chain with zero conflict mineral procurement



- 100% of new suppliers complied with UMC's Supplier Code of Conduct.
- Achieve 100% of products passing stringent third-party mineral source certification
- 100% of conflict-mineral suppliers passing third-party independent audit
- 100% completion rate of ESG & RBA self-assessment questionnaire for Tier 1 suppliers

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Percentage of carbon intensity reduction by suppliers within the project of "low-carbon supply chain" ^{Note2}	8.3%	✓ Achieved	10%	20%
Ratio of Tier 1 suppliers who have included 'signing Employment Ethics Terms and Conditions Agreement' in their supply chain management requirements	60%	✓ 79%	80%	80%
Ratio of audit conducted to significant high-risk suppliers	100%	✓ 100%	100%	100%
Completion rate of biodiversity survey for Tier 1 suppliers	100%	✓ 100%	100%	100%

Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC)
2. Low Carbon Supply Chain Suppliers: those who are Tier 1 suppliers and willing to participate in the project; Carbon reduction ratio calculation: based on the carbon reduction amount in the year before the sustainability report.

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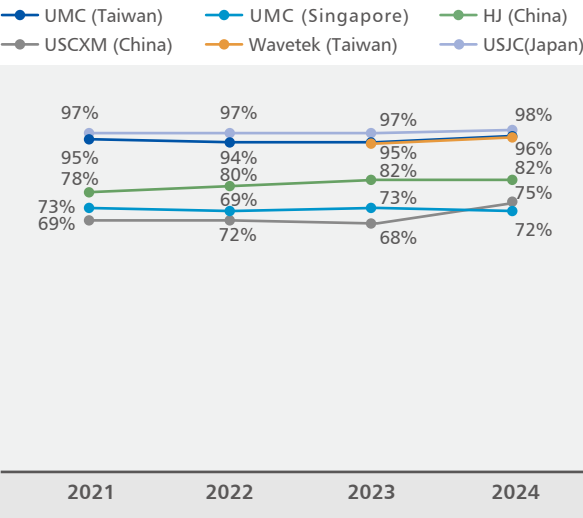
5 Enhancing Society Mutual-Prosperity

Appendix

1.3.1 Localization of Supply Chain and Procurement

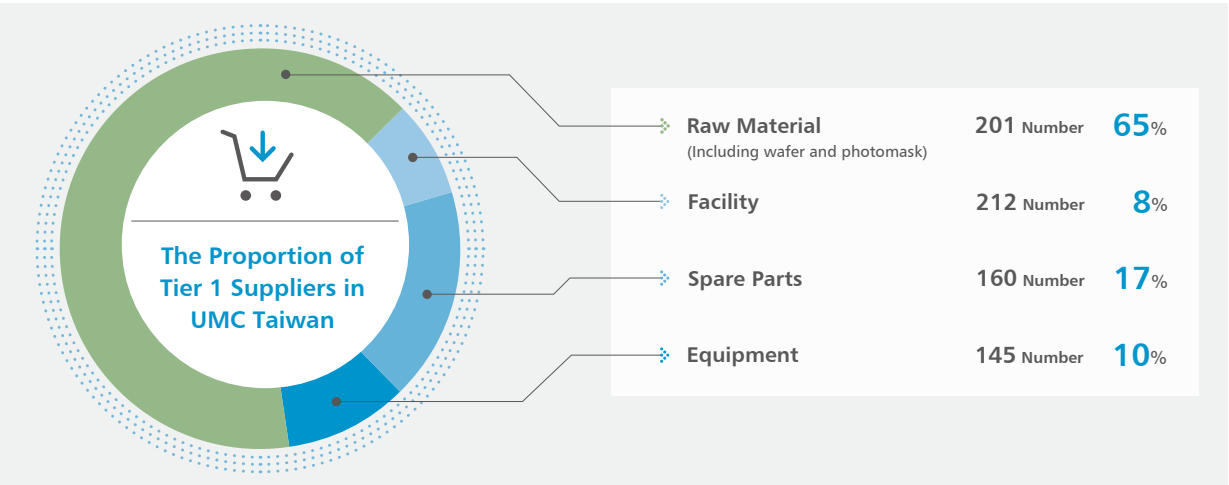
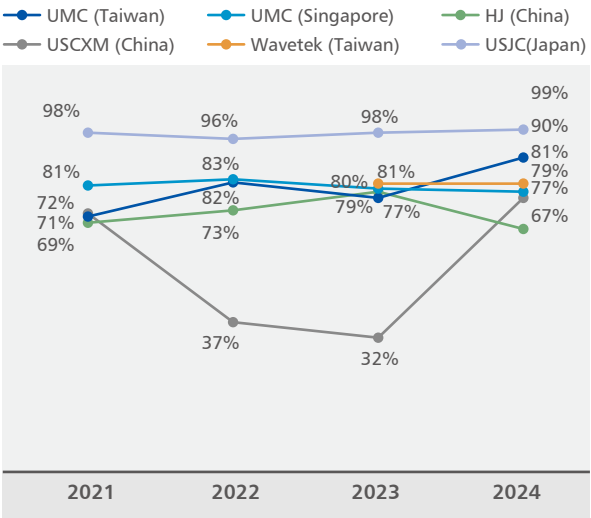
In 2024, UMC, including subsidiaries HJ, USCXM, and UJSC, cooperated with more than 4,000 suppliers worldwide. Procurement localization became a key guideline adopted by UMC when looking for potential partners and suppliers. In addition to enhancing close partnerships with local suppliers, UMC expects suppliers to provide services and products with higher efficiency, shorter delivery time, and lower carbon footprint of raw material required in production. In doing so, UMC also indirectly provides job opportunities and promotes socio-economic development. The number of local suppliers including foundry subsidiaries is generally greater than 50%.

Percentage of domestic suppliers



Note: Wavetek is included in the calculation scope starting from 2023.

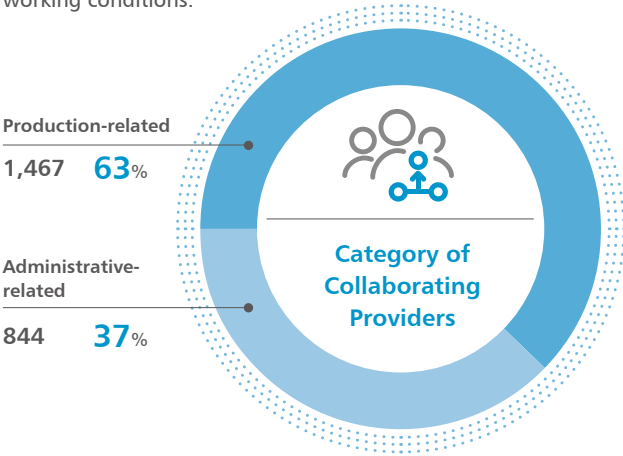
Percentage of domestic procurement amount



Note: The definition of local/non-local suppliers is based on whether the supplier's registered location is in the same country as UMC's main operational production sites. If they are in the same country, it is considered domestic procurement.

Collaborating Providers

Due to the nature of our industry, and in response to operating plan requirements and to maintain the Company's normal operations, UMC requires a lot of labor. Such labor demands come in two major categories: Production related and administrative-related, totaling 2311 people. UMC Group, including our Singapore branch and other subsidiaries, tend to engage third-party providers from nearby regions. This creates long-term, large-scale employment opportunities for nearby regions, while simultaneously building a responsible supply chain. Collaborating providers engaged in UMC fabs have all passed the Supplier Sustainability Assessment, and met the regulatory requirements stipulated in local laws/regulations and UMC's Supplier Code of Conduct. UMC also conducts sustainability assessments for collaborating providers every year, and grades the providers in accordance with assessment results. Those failing to meet required standards for two consecutive years will, in accordance with the Company's regulations, be subject to lower procurement/sourcing or termination of collaboration. The 2024 on-site audit results indicated that all suppliers were in human rights compliance with national laws and regulations, as well as the human rights norms required by the Responsible Business Alliance (the RBA). No suppliers had use of child labor, forced labor, or any violations of regulations regarding working conditions.



Note: 1. The production-related category includes collaborating providers in equipment maintenance, service, etc. in production and technical departments; the administrative related category includes collaborating providers in non-production departments' administrative operations, such as cleaning, staff canteen, security duties, etc.

2. The number of collaborating providers will be surveyed once every two years, with the next survey scheduled for 2026.

1.3.2 Supply Chain Management

Supplier Management Policy

01

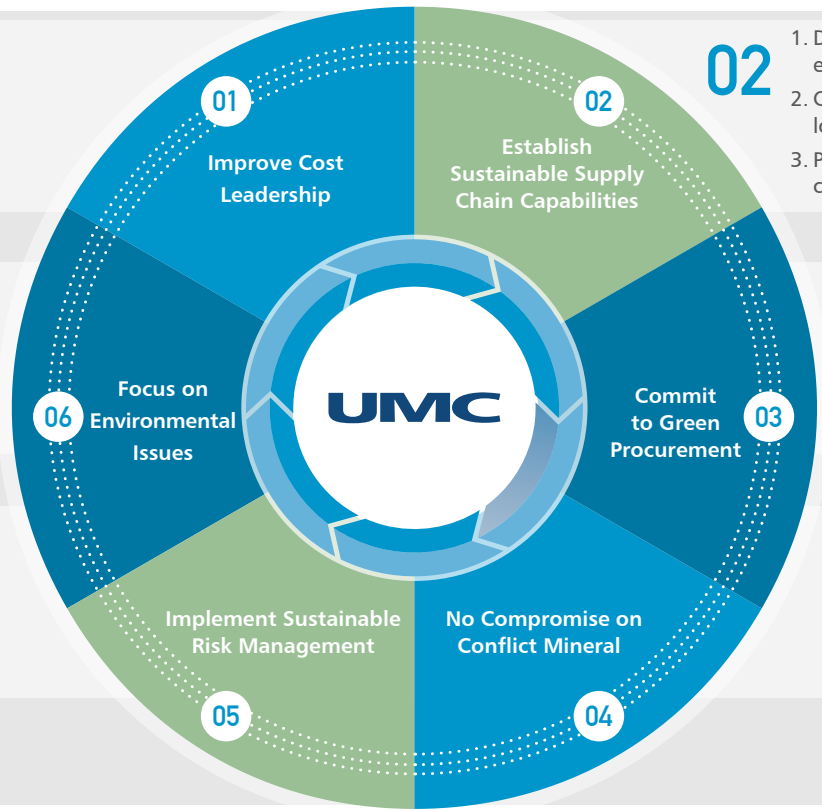
Integrate UMC Group resources to achieve the most competitive supply chain value through strategic cooperation.

06

Strengthen recycling of waste resources and strive to reduce the impact of environmental pollution.

05

Focus on suppliers' energy resource use and water resources management to respond to the impact of extreme climate change on the supply chain.



02

1. Drive suppliers to improve economic, social and environmental performance.
2. Continue to guide suppliers to achieve diversified and localized supply capabilities.
3. Promote suppliers' moving from energy savings and carbon reduction toward the circular economy.

03

Proactively implement green procurement while pursuing economic benefits.

04

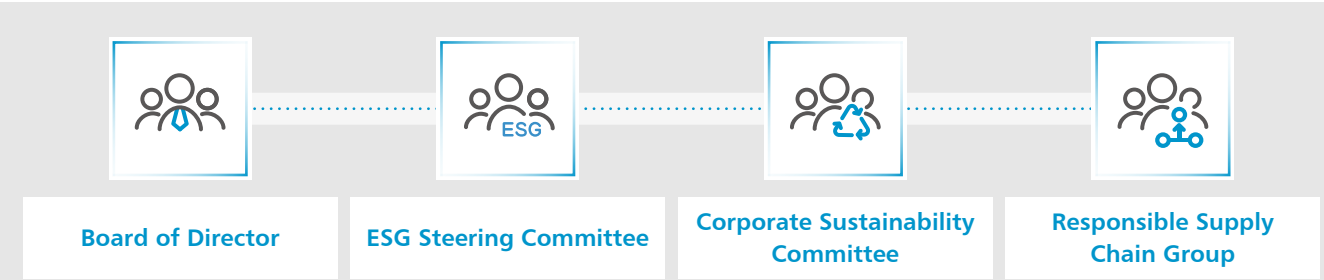
Ensure that products and supply chain are free from conflict mineral.

Please visit UMC website to review our commitment to suppliers



Responsible Supply Chain Organization Chart

The supplier policy of UMC is to view suppliers as partners and guide them towards long-term cooperation, planning a comprehensive management approach for suppliers. UMC places importance on sustainable supplier management and focuses on the requirements of suppliers in the areas of environment, society, and corporate governance. Therefore, the Responsible Supply Chain Committee has launched the Supply Chain Sustainability Blueprint and supply chain self-set goals and projects, with the results being reported quarterly to the Corporate Sustainability Committee and the ESG Steering Committee, and ultimately reported to the Board of Directors, which oversees the responsible supply chain projects.



Suppliers ESG Risk Assessment Criteria

UMC formulated its Supplier Code of Conduct based on the Responsible Business Alliance (RBA)'s Code of Conduct, the UNGP, and the CSDDD. Both existing and news suppliers are required to sign this code in order to become a supplier of UMC. In addition, subsequent risk assessments and onsite audits are conducted on a regular basis to facilitate continuous improvement.

At the same time, UMC encourages Tier 1 suppliers to require their previous-tier suppliers to adopt the UMC Supplier Code of Conduct in their management practices. In addition, UMC's supplier selection considerations also include compliance with the "Employee Code of Conduct" for UMC Suppliers, compliance with related regulations, and sustainable evaluation in four dimensions. UMC's implementation of supplier risk assessment criteria follow the "Supplier Sustainable Evaluation and Management Measures," using indicators such as "Supplier ESG Risk Screening Process" and "four major criteria of supplier sustainability assessment" to control and guide new and old suppliers, in order to establish a mutually beneficial and sustainable supply chain management mechanism.

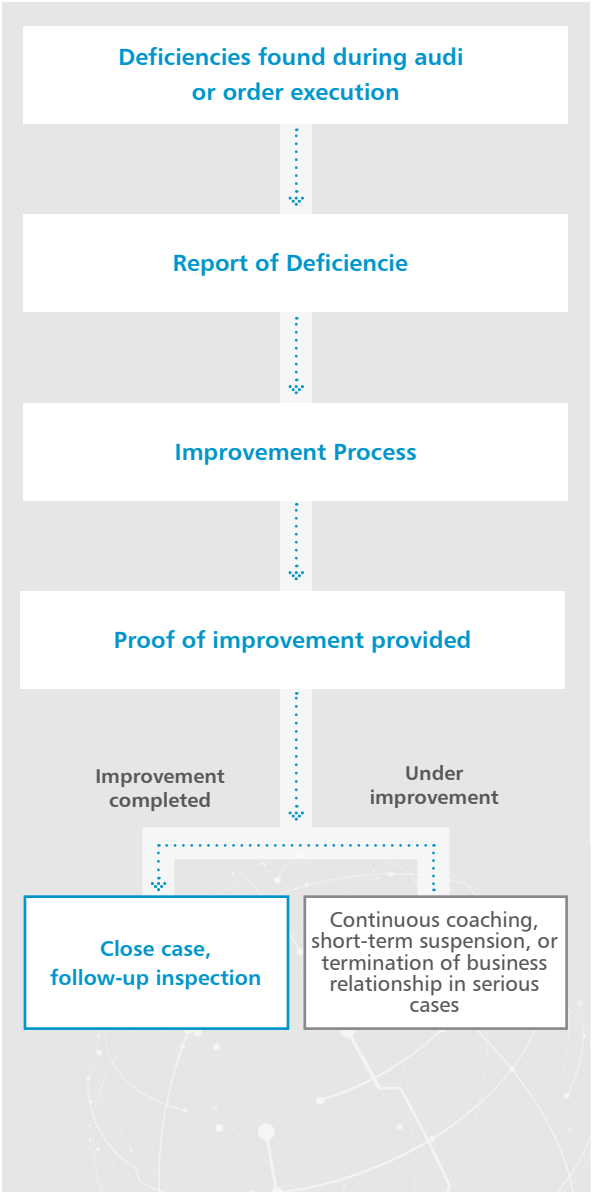
Four Major Criteria of Supplier Sustainability Assessment

Major Criteria 1	Environment	- Business continuity system - Health and safety management - Fire management	- Special Pest Management - GHG Management - Water resources and water risk management	- Energy management - Waste management - Biodiversity
Major Criteria 2	Society	- Human Rights and Labor Ethics - Human resources management	- Labor rights - Management system and communication channel	
Major Criteria 3	Governance	- Business ethics - Conflict Mineral Management	- Information Security Management - Business continuity plan	Sustainable Governance Policy
Major Criteria 4	Business	- Quality - Cost	- Delivery - Service	

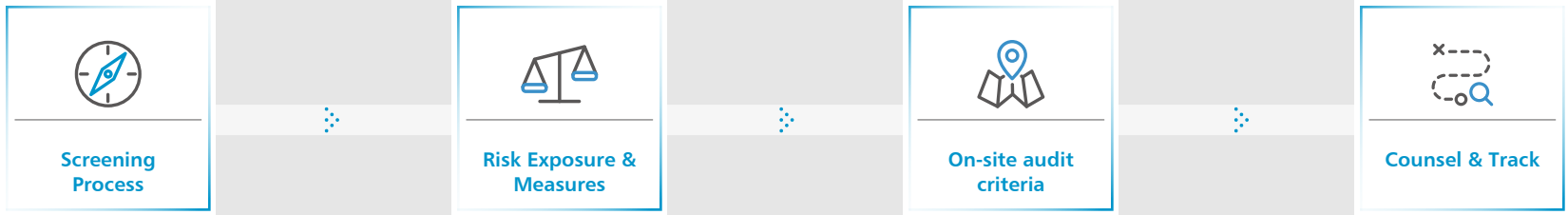
Supplier ESG Risk Screening Process

In order to regularly manage, assess, guide, and track supplier improvements, UMC has established and formulated the "Supplier Sustainable Evaluation and Management Measures," incorporating sustainable management requirements into the evaluation indicators. The Tier 1 suppliers supporting production are classified and scored according to risk levels, and regular management, assessment, guidance, and tracking of supplier improvements are carried out. By establishing the "Sustainable Supplier Rating and Response Measures," sustainable management requirements are included in the rating items according to their importance. Every year, UMC sends out an "ESG Risk Assessment Questionnaire" to Tier 1 suppliers supporting production for them to respond to in questionnaire form. The responses are then classified according to the supplier's characteristics and risk level to facilitate graded guidance and improvement during the improvement period. UMC has the "Supplier Sustainable Evaluation and Management Measures" according to the supplier's sustainable risk level. For significant high-risk suppliers, UMC identifies missing items and assists in guidance to achieve supplier improvement. If improvement cannot be achieved, UMC may continue to provide guidance or temporarily suspend or terminate the business relationship with the supplier. If a supplier cannot meet UMC's requirements even after guidance, UMC will provide continued guidance and notify the relevant department of the order, place the supplier on an observation list, and conduct a physical audit the following year. If the supplier still cannot meet the requirements for two consecutive years, their supplier qualification will be cancelled, and they will be removed from the supply chain. Suppliers who cannot comply with improvements within two years will be disqualified.

Process of Suppliers Failure Improvement Management



Supplier ESG risk assessment and Screening Process



Criteria

✦ The screening process includes environmental, social, governance, and business closeness ✦ Assessment Methodology : 1. Major incidents or governmental, environmental, or social violation record 2. Potential negative impact (Environmental: hazardous substance management; Social: child labor, forced labor; Governance: corruption, bribery or supply disruption risk) 3. Conduct preliminary assessment by reviewing purchase amount and category of supplier (including material, parts, facility, equipment supplier and service contractor) ✦ Sector-specific/Country specific/Commodity-specific risk^{note 1}	✦ Desktop Assessment Process 1. Send an ESG self-assessment questionnaire, and use the UMC's E-Platform to manage, control, and progress track and evaluate the survey 2. The questionnaire requires suppliers to attach supporting documents 3. A dedicated person will compare the supporting information to see if it meets UMC requirements. 4. If the requirements are met, the initial score of the questionnaire will be maintained; if the requirements are not met, the score of the questionnaire will be lowered. 5. Annual Supplier ESG Performance Score Summary ✦ Identify supplier risk levels: 1. Suppliers whose ESG self-assessment scores are lower than UMC's requirements are listed as potential high-risk vendors. 2. Suppliers with finding found during on-site audits will be included in significant high-risk suppliers 3. Suppliers whose ESG self-assessment scores meet UMC's requirements are considered low-to medium-risk suppliers.	✦ New suppliers will be subject to second-party audits by the purchasing division and related divisions. ✦ UMC entrusts a third-party organization for on-site audits. ✦ Provide suggestions for improvement during the audit	✦ Request suppliers to make improvements according to audit reports ✦ Suppliers whose audit results are lower than UMC's requirements will be audited in the following year ✦ For suppliers that have failed to meet the standards for two consecutive years, UMC will halt transactions according to company policy ✦ Suppliers with good performance will be recognized in the annual supplier awards or. UMC will award contracts to suppliers with good ESG performance.
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Scope

✦ All suppliers	✦ Tier1 Suppliers, Significant Tier1 Suppliers, Significant High Risk Suppliers^{note 2}	✦ Randomly inspect medium/low-risk suppliers during the year ✦ Significant high risk suppliers in the past year	✦ Potential high risk suppliers
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Note: 1. (1)Sector-specific risk suppliers screening: Enter the screening with the terms of the Responsible Business Alliance (Responsible Business Alliance, RBA). (2)Country specific risk suppliers screening: Conduct an annual BCP risk survey on key raw material suppliers, and classify the country with a supply chain risk map (such as frequent natural disasters, geopolitics, war...) (3) Commodity-specific risk suppliers screening: Conduct annual BCP risk surveys on key raw material suppliers, and classify key raw materials with supply chain risk maps (such as: high purchase volume, single source, non-substitutable,...)

2. Tier1 suppliers: Direct transactions with more than 2 orders per year and transaction amount greater than NT\$2 million Tier 1 suppliers are also classified by importance such as procurement volume, critical material, and critical/non-substitutable suppliers. In 2023 631pliers met the criteria. Significant Tier1 Suppliers: The annual purchase amount of the suppliers for more than 80%. Significant High Risk Suppliers: Suppliers who still do not meet UMC standards after providing corrective action plans in the previous year audit.

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Suppliers ESG Risk Evaluation and Screening Result

In 2024, UMC conducted a survey of all suppliers that support the supply of materials for UMC's production, excluding suppliers engaged in only a single transaction. The survey covered key suppliers for raw materials, equipment/machines, facilities, parts/components, etc. There were 718 Tier 1 suppliers included in the UMC Sustainability Assessment Questionnaire; of these, 116 suppliers were further inspected through onsite audits. 25 of the audited suppliers were identified as high-risk,including 24 high-risk suppliers for environmental and occupational safety and health, and 15 high-risk suppliers for human rights and ethics. UMC requires improvements within a deadline, and these 25 companies will also be listed as key observation targets in the next year. The audit ratio target of significant high-risk suppliers in 2024 is 100%.

Note: Suppliers who are excluded from 1 order include key raw materials suppliers, equipment, facility, and key components.

KPI of Suppliers Screening in the Past 4 Years

	2021	2022	2023	2024
Number of Tier 1 suppliers	673	579	631	718
Number of significant Tier 1 supplier	206	191	91	99
% of purchasing amount of significant Tier 1 supplier	80%	82%	80%	82%
Number of significant suppliers assessed via audited	206	191	153	116
% of significant high risk suppliers assessed via audited	100%	100%	100%	100%
Total number of significant suppliers in non Tier 1	80	146	64	17
Number of suppliers assessed with substantial actual negative impacts	17	18	20	25
Number of suppliers with substantial actual negative impacts that were terminated	0	0	0	0
Total number of suppliers supported in corrective action plan implementation	17	18	20	25
% of suppliers assessed with substantial actual negative impacts supported in corrective action plan implementation	100%	100%	100%	100%
Total number of suppliers in capacity building programs	206	191	153	116
% of significant suppliers in capacity building programs	100%	100%	100%	100%

Corrective Action Plan

Regarding the deficiencies found during the audit, a corrective and preventive action plan was issued by a third party on the day of the audit, UMC assists in coaching suppliers and completes improvement plans within one month after the audit date. In 2024, 100% of the audited suppliers completed the improvement of deficiencies within the specified time limit, and no supplier terminated their cooperation due to non-compliance with company regulations.

Via the 2024 sustainability assessment mechanism, and by introducing third-party certification body audits, the degree of supply chain compliance with the UMC Supplier Code of Conduct can be summarized. In view of the continual growth in the global semiconductor industry, UMC regards supply chain management as an important sustainability goal. UMC is committed to improving supply chain management performance and promoting continuous improvement, and requires suppliers to follow the UMC Supplier Code of Conduct to improve performance in five areas: Labor rights and ethics, the environment and climate change, occupational safety and health, supplier management, business continuity planning, and information security control. Based on requirements in these five areas, UMC also continues to implement several action plans, in hopes of working with suppliers to drive the gears of the sustainable supply chain.

Finding and Improvement Result of UMC suppliers in 2024

Category	Major Findings	Improvement Results
Labor rights and ethics	Lack of management for labor and ethical business practices	Establish relevant management procedures
	Lack of regular risk assessment, identification and control of labor and moral risks related to operations	Establish a supplier risk assessment mechanism
	Lack of documented standard requirements related to antitrust, fair trade, and business integrity	Requires supervision of upstream suppliers' compliance with the UMC Supplier Code of Conduct
	Lack of social performance indicators and implementation plans	Establish social performance indicators
Occupational safety and health	Lack of contractor safety management requirements	Establish construction safety controls
	Chemicals are not stored according to regulations	Effective control of flammable chemicals
	No mechanism for regular inspection of fire protection equipment has been established	Establish a regular inspection mechanism for factory/company fire protection
Business continuity planning and information security control	effectively implement a complete business continuity plan	Establish an operations continuity plan and update it regularly
	Missing information equipment management program	regular backups and ensure the security of storage locations

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Sustainable Supply Chain Projects

UMC believes that a responsible supply chain is an important part of implementing the Company's sustainable blueprint. We continue to leverage UMC's influence and promote various ESG projects to build a sustainable supply chain. This includes holding annual sustainability briefings to convey UMC's requirements for corporate sustainability and ESG performance, as well as various international standards, to suppliers and disseminate the concept of a sustainable supply chain. In addition, to raise awareness of sustainability among UMC's internal procurement personnel, we also provide employee education and training on international sustainability standards and issues.

Project ▶ UMC 2030 low-carbon supply chain

Purpose

In order to achieve the goal of net zero emissions by 2050, and to build up a low-carbon sustainable supply chain in response to the RE100 international renewable energy initiative, the goal has been set at 20% carbon reduction by 2030.

Action

The kick-off meeting was held in 2022 to conduct carbon emissions investigations on participating low-carbon suppliers, and to engage consultant companies to provide education and training to guide them in conducting greenhouse gas inventories. In addition, all suppliers are required to establish carbon reduction plans.

2024 Outcome

- Participating suppliers' carbon emission intensity reaches annual targets
- ✓ Achieved

Project ▶ Supply chain water reduction plan

Purpose

Enhance green sustainability awareness and reduce water resource consumption of semiconductor raw materials in the manufacturing process

Action

Tier 1 Suppliers who use more than 5,000 cubic meters of water resources annually will be identified as key water resource suppliers. By 2024, 22 such suppliers will be identified, and they will be guided/required to reduce water. UMC also requires non-key water resources suppliers to set water-saving goals. UMC investigate the establishment of water-saving goals among suppliers by annual sustainable ESG evaluations

2024 Outcome

- The water withdrawal by key water resource suppliers was reduced by 1%

Project ▶ Tier 2 significant supplier management

Purpose

To ensure that UMC's suppliers are diligent about their supply chain management, and are taking actions to manage their suppliers and ensure their suppliers are abiding by the Company's ethics agreement

Action

In SAQ and audit, UMC suppliers are required to sign the "Code of Ethics" for their significant suppliers.

2024 Outcome

- 49.7% of UMC's Tier 2 supplier are signed the Code of Ethics. 80% of UMC's Tier 2 suppliers are required to sign the Code of Ethics by 2025



2024 UMC Taiwan Supply Chain Sustainability Communication

Project ▶ Sustainable supply chain sharing session

Purpose

Hold sharing sessions with the theme of "Sustainability & Co-prosperity" to promote the concept of sustainable development to suppliers

Action

- Taiwan, China, Japan and other regions take turns to hold annual sharing sessions:
- Promote UMC's human-oriented sustainable conduct and co-prosperity in society
 - Continue to require ESG alignment in supply chain
 - New challenges for sustainable supply chains
 - Sharing of sustainable management experience from outstanding suppliers

2024 Outcome

- Held in Singapore in December, the main targets for promoting international sustainable development trends and norms are local suppliers in Singapore.

2024

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Project Sustainable education and training for purchasing agent

Purpose

Improve procurement agent's sustainability expertise

Action

Every year, courses are offered in accordance with international standards such as RBA, and questionnaires are sent to trainees as a basis for evaluating the results of the training.

2024 Outcome

Held 1 session, the purchasing agent training completion rate is 100%

Project Green procurement

Purpose

UMC views the development of green products as its responsibility and promotes green procurement starting from upstream manufacturing to conserve resources and protect the environment

Action

Green procurement indicators:

- Domestic: Category no. 1 to no. 3 green-labeled products, including Energy Label, Water Label, Green Building Material, Carbon Footprint Label, Carbon Label
- Overseas: green-labeled products originated from countries that have agreements with Taiwan, Energy Star, FSC, PEFC

2024 Outcome

- Green procurement amount in 2024 reached NT\$70 million
- Received the 2024 Green Procurement Award from the Environmental Protection Administration,
- Received the 2024 Green Procurement Award from the Hsinchu City government
- Recognized by Tainan City government as an Excellent Green Procurement Enterprise in 2024.

1.3.3 Sustainable Raw Materials Management

Raw Material Evaluation methods to Determine Management Priorities

UMC adopts the Scale approach, prioritizing the assessment of raw materials in the product. Through product component analysis, over 99% is silicon, while the rest consists of trace metals, including Al, Ti, Fe, Cu, Ni, and Co. UMC prioritizes the management of these materials to ensure that environmental and social impacts are avoided during the extraction or manufacturing stages.

The Amount Used for Each Metal Material by UMC

Metal/Mineral	Amount (Unit: tonne)	Share of material used that is recycled
Aluminum	15.356	48.4%
Cobalt	3.79	31%
Copper	26.846	49.6%
Iron/Steel	0.066	15%
Nickel	0.651	75%
Lithium	N/A ^註	N/A
Titanium	16.879	72%

Note: UMC does not use lithium metal in products

Sustainable Raw Materials Policy



Sustainable Raw Materials Management Process

UMC tracks environmental and social risks in the raw material supply chain through Supplier ESG risk assessment and Screening Process. The screening process includes environmental, social, governance, business closeness, Sector-specific/Country specific, and Commodity-specific risk related suppliers.

At the same time, UMC also requires raw material suppliers to be traced back to the source supplier during the annual ESG risk assessment process, and requires suppliers to be included in the relevant assessment process to reduce social and environmental impacts. In 2024, 100% of the audited suppliers completed the improvement of deficiencies within the specified time limit, and no supplier terminated their cooperation due to non-compliance with company regulations.

The Raw Materials Suppliers Assessment Process

- Desktop Assessment Process
 - Send ESG self-assessment questionnaire and manage and track the progress of the survey through an electronic questionnaire platform.
 - The questionnaire requires suppliers to attach supporting documents, which will be reviewed by dedicated personnel to ensure compliance with UMC's requirements.
 - If the requirements are met, the initial score of the questionnaire will be kept; if the requirements are not met, the score of the questionnaire will be lowered.
 - Consolidate the annual ESG performance scores of suppliers

- Identify Supplier Risk Levels
 - Suppliers whose ESG self-assessment scores are lower than UMC's requirements are listed as potential high-risk vendors.
 - Suppliers identified with findings during on-site audits last year will be classified as significant high-risk suppliers.
 - Suppliers whose ESG self-assessment scores meet UMC's requirements are considered low- to medium-risk suppliers.
 - Suppliers who meet UMC's ESG self-assessment requirements but fail to provide sufficient documentation on raw material traceability across the supply chain to the production site are listed as potential high-risk vendors.

High risk suppliers will be audited on-site by third party

- Tracking and Suppliers Improvement Mechanism
 - Request suppliers to make Corrective Action Plan (CAP) according to audit reports.
 - Suppliers whose audit results are lower than UMC's requirements will be audited in the following year.
 - For suppliers that have failed to meet the standards for two consecutive years, UMC will halt transactions according to company policy.

Action to Reduce Negative Environmental Impacts of Raw Material Production

- UMC 2030 low-carbon supply chain: The kick-off meeting was held in 2022 to conduct carbon emissions investigations on participating low-carbon suppliers, and to engage consultant companies to provide education and training to guide them in conducting greenhouse gas inventories. In addition, all suppliers are required to establish carbon reduction plans.
- UMC completed biodiversity survey for Tier 1 suppliers. No supplier's factories are located in biodiversity areas. UMC also requires suppliers to conduct biodiversity surveys on their upstream suppliers. The completion rate of the Tier 1 supplier biodiversity survey is 100%, totaling 718 suppliers in 2024.
- Conduct RBA, UNGP and UMC supplier code of conduct investigations on Tier 1 suppliers to reduce the impact of the upstream supply chain on human rights and occupational safety and health.

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Targets for the Share of
Sustainable Raw Materials

UMC set the following targets for the sustainable proportion of relevant metals with third-party sustainable certification in 2025 and 2030. UMC will continue to collaborate with suppliers and their upstream smelters to promote sustainable certification and will include this item in the evaluation for future new supplier selections.

Targets for UMC's Share of Sustainable
Raw Materials

Metal/Mineral	2025	2030
Aluminum	20%	24%
Tantalum	11%	13%
Copper	32%	37%
Titanium	20%	23%
Cobalt	68%	79%

Reporting Progress on Sustainable Raw Materials Sourcing
Targets

Regarding the procurement of third-party verified sustainable raw materials, the Responsible Supply Chain Group will continue to conduct regular tracking and reporting on the progress of procurement targets every six months at the Corporate Sustainability Committee quarterly meeting.

Training of Sustainable Raw Materials Awareness

To enhance the sustainability awareness of internal procurement personnel, UMC conducts employee training annually on international sustainability standards and other sustainability related topics. Two sustainable raw material awareness training sessions were conducted in 2024 for employees in relevant divisions.

Identification of Key Raw Material

The key raw materials refer to raw materials that have a single source or production area and are not easily replaceable. UMC completed its supply chain risk database and the management system of earthquake spare parts (quartz and fragile parts) for over 2,000 key raw materials supplied by approximately 100 vendors.

- Establish an automated system for risk warning notifications, report monitoring, and risk identification in the production are
- 95% of UMC's 8-inch fabs and 99% of 12-inch fabs have completed earthquake spare parts preparation.

1.3.4 Conflict Minerals Management

Conflict Minerals Management
Policy

In order to avoid inadvertently obtaining conflict minerals from the Democratic Republic of Congo and surrounding countries during the product manufacturing process, such as Tantalum, Tin, Tungsten, and Gold (referred to as 3T1G), UMC has developed the "UMC Responsible Sourcing Policy" and requires suppliers of products containing 3TG to adhere to the policy and sign the "Declaration of Minerals Conflict-Free." Every year, a supply chain investigation is conducted to review the signing of the "Declaration of Avoidance Use Conflict Minerals," and UMC has also proactively joined the Responsible Minerals Initiative (RMI) as a member. UMC continues to actively participate in international initiatives and implement relevant management mechanisms to ensure that the products provided by suppliers are not manufactured using conflict minerals.

Results of Recent Surveys

UMC follows the principles of the "Model Supply Chain Policy for a Responsible Global Supply Chain of Mineral from Conflict-Affected and High-Risk Areas," published by the Organization for Economic Co-operation and Development (OECD), to establish a "due diligence" process. Through this process, UMC can identify and evaluate the risks of its own suppliers and respond to and mitigate identified risks.

UMC mainly obtains information on suppliers that meet the requirements for conflict minerals through the "Responsible Minerals Assurance Process (RMAP)" and uses it as a basis for source management in the supply chain. UMC also requires suppliers to provide information on conflict minerals using the "Conflict Minerals Reporting Template (CMRT)" every year and compares it with RBA-recognized qualified smelters to ensure that no conflict minerals from conflict areas are used. If non-compliant smelters are found, suppliers are required to improve within a specified period.

UMC Conflict Minerals Management Process

- Conflict Minerals Specification
 - Comply with UMC's Responsible Sourcing Policy
- Reasonable Country of Origin Inquiry (RCOI) investigation of mineral sources
 - Conflict Minerals Reporting Template (CMRT)
 - Sign Conflict Minerals Representation Letters
 - Review suppliers' Declaration of Minerals Conflict-Free
 - On-site audit
- Due Diligence Investigation*(Please refer to UMC Conflict Minerals Due Diligence Investigation Process)
 - Comply with OECD Due Diligence Guidelines
- Independent Third-Party Audit (ISPA)
 - Conducting an independent third-party audit
- Public Disclosure
 - Disclosure of conflict minerals management results through the Company's sustainability reports and official website.

UMC Conflict Minerals Due Diligence
Investigation Process

- Establishment of Management System
 - A. Development of Responsible Sourcing Policy Referring to RMI requirements. (UMC RMI Due Diligence Investigation Procedure)
 - B. Establish a team responsible for allocation of responsibilities - Responsible Supply Chain Group
 - C. Risk Management System - If a supplier uses non-compliant smelters, they will be required to improve within a specified period, otherwise the transaction with the supplier will be suspended
 - D. Supplier Engagement - Supplier fills out CMRT form.
 - E. Supplier Grievance Mechanism -Responsible Supply Chain Group contact person
- Identification and Evaluation of Supply Chain Risk
- Issuance of Conflict Minerals Questionnaires to Suppliers and Subsidiaries
- Review Questionnaire Responses.

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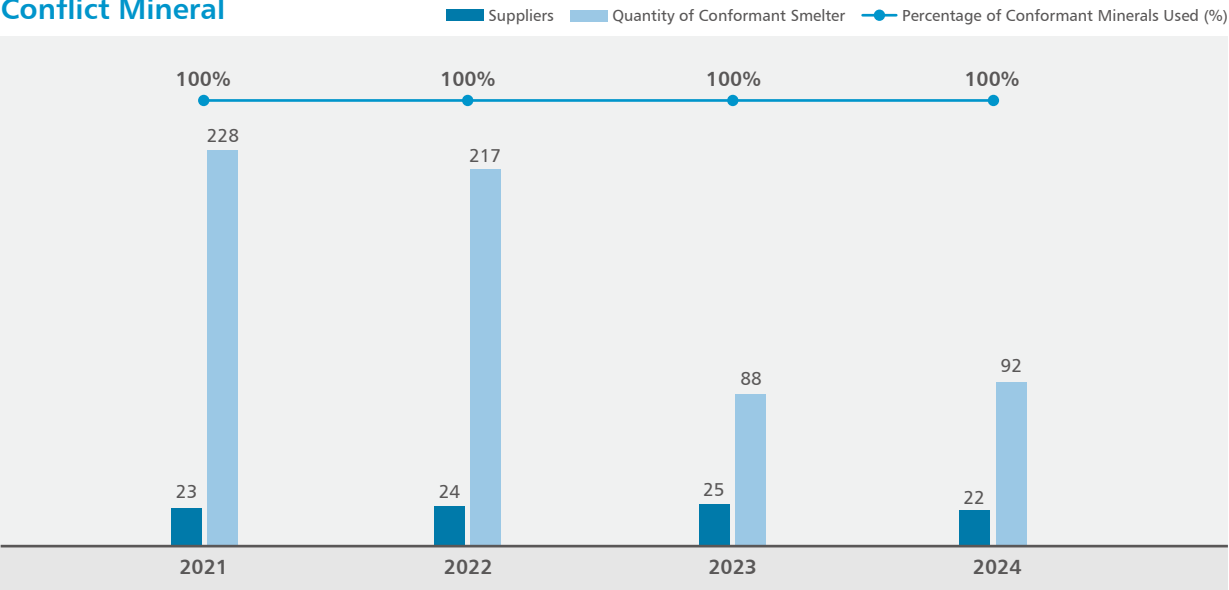
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UMC RMI Due Diligence Investigation Procedure

- Notify suppliers to sign the Declaration of Non-Use of Conflict Metals and provide the latest version of the CMRT to identify whether the products they provide contain 3T1G, Cobalt, Mica, etc.,
- Define the supplier risk level and develop related strategies based on the results of the investigation of the use of conflict metals.
- Conduct supplier audits to check the source of conflict metals and have specialized personnel compare the smelter information provided by suppliers to see if it is from non-conflict areas rated by RBA.
- Public Disclosure - Submit a supply chain conflict metal usage investigation report and provide the SEC SD form annually.

In 2024, UMC had a total of 718 suppliers, of which 22 (3%) provided products containing 3T1G, Cobalt, and Mica. Therefore, this year's "RMI Conflict Minerals Due Diligence Investigation" was conducted on the 22 suppliers providing such products. All 22 suppliers (100%) provided complete CMRT and related supporting documents. A total of 92 smelters were identified, and after comparison by specialized personnel, all 92 (100%) were conformant to RMI. This means that 100% of the smelters used by UMC have been verified by RMAP. Therefore, the 22 suppliers are qualified conflict minerals suppliers (100% of all suppliers providing conflict minerals) and provide products without conflict minerals.

Conflict Mineral



Note: Regulation-compliant minerals" means that smelters producing the 3T1G used in UMC's supply chain have been reviewed and certified in accordance with the Responsible Minerals Assurance Process (RMAP) standard and SORs are in conformant smelters list.

Independent Private Sector Audit (IPSA) and Public Disclosure

In addition, UMC designs a "Supplier Conflict Mineral Audit Form" based on the OECD "Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas", and through on-site audits, assists suppliers in establishing management mechanisms that comply with the OECD's due diligence guidance. In 2024, a total of 4 suppliers are audited, and all smelters used by these suppliers are "non-conflict (DRC Conflict-Free)".

According to Section 1502 of the Dodd-Frank Wall Street Reform and Consumer Protection Act, as announced by the U.S. Securities and Exchange Commission (SEC) on August 22, 2012, UMC has added the "conflict-free minerals" logo to its product packaging to declare that its products do not use conflict minerals. Since April 1, 2018, UMC's Taiwan factories have implemented this policy, and a formal statement has been issued on the official website. Additionally, UMC completes a Specialized Disclosure Report every May. For the latest disclosure documents, you can refer to UMC.com or the U.S. Securities and Exchange Commission (SEC) website.

For more information about UMC's conflict mineral procurement policy please refer to



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Innovative Products and Services



- 2.1 Innovation & Technology Development
- 2.2 Customer Satisfaction
- 2.3 Product Responsibility

Innovative Products and Services

Important Stakeholders | Shareholders | Suppliers | Customers |

The customer-oriented UMC will continue to provide professional wafer solutions that meet market trends and customer needs. Also, constant refinement and introduction of innovative technologies will be undergone to strengthen UMC's core competencies. UMC's long-term operations rely mainly on its own and its customers' competitiveness, and UMC will be able to achieve sustainable impact with its products and services.

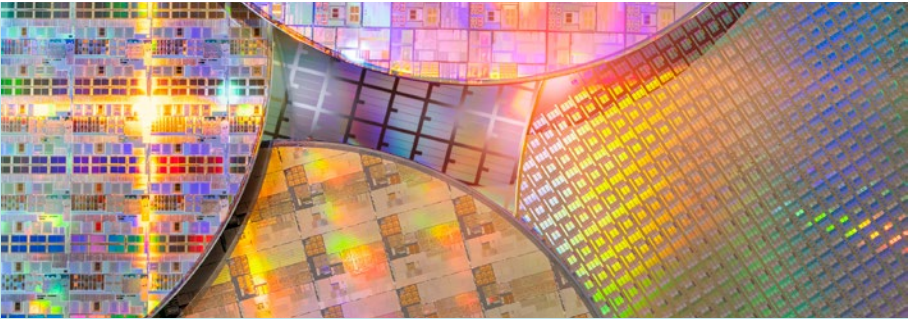
SDGs |



Performance Highlights 2024



364 patents granted this year





92% of satisfied customer



Fab 12A has been selected as "Smart Manufacturing Lighthouses" by the World Economic Forum



82.3% revenue from sustainable products as % of total revenue



53.0% of products produced by Industry 4.0 Smart Manufacturing Systems

2.1 Innovation & Technology Development

- 2.1.1 Robust Wafer Manufacturing Services
- 2.1.2 Key Technologies
- 2.1.3 Smart Manufacturing Execution



Material Topic Innovation Management and Intellectual Property Rights Protection

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Positive Impact Technology and Product Innovation		Negative Impact Intellectual Property Infringement
Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
Positive: Comprehensive innovation management, along with technology enhancement and intellectual property protection, ensures the delivery of products that align with market trends and meet customer demands.	- Follow annual corporate governance guidelines to initiate key technology R&D projects based on company-wide KPIs, and designate personnel with a rank of vice president or above to be in charge of the projects. - Develop sustainable technology/product R&D plans, and designate a Sustainable Innovation Taskforce to review implementation performance. - Set R&D schedule KPIs and progressive targets on a yearly basis, measure points every quarter, and regularly review R&D issues and solutions. - Customer Engineering Division regularly compiles customers' feedback and discusses solutions to meet customers' needs in PLM (product line management) meetings.	- Hold regular PLM meetings to discuss market demand trends, thereby developing response strategies and fixing outdated deficiencies in a timely manner to adhere to the goals of product innovation. - Hold project working group meetings on a weekly basis to examine problems identified during task implementation, thereby adjusting resource allocation and dealing with difficulties in a timely manner. - Review the achievement status of the goals set in company-wide KPIs on a quarterly basis, the result of which shall be signed and approved by the responsible vice president and submitted to the President.
Negative: Inadequate innovation management fails to provide products that align with market trends and meet customer needs.	- Orientate intellectual property development direction toward mutual protection of patents and trade secrets, and promote the steady growth of the number of patents granted and the number of key trade secrets registered. - In accordance with the annual corporate governance policy, Technology Development division and Intellectual Property division conduct patent map analysis and brainstorming meetings on issues related to important technology research and development projects, and identify opportunities for obtaining intellectual property rights and analyze possible infringement risks during this meeting. - In the case of intellectual property disputes (including cases where customers seek assistance and intellectual property risk assessments for suppliers), a dedicated person will be responsible for planning, evaluating and responding, and regularly reporting the progress of high-risk cases to the top management on a weekly basis, so as to reduce the impact of intellectual property dispute cases. - Improve employees' awareness of intellectual property protection, and regularly conduct intellectual property education and training courses on topics such as patents and trade secrets.	- Regularly follow up on a monthly basis to confirm the achievement status of intellectual property rights KPI in each division. - Introduced "Taiwan Intellectual Property Management System (TIPS)" from 2021, and in response to certification requirements, carried out internal and external audits every year to confirm the implementation of various management measures and education and training courses.



Sustainability Strategy and Blueprint	2024		2025	2030
	Subject	Target	Goals	
	Revenue from sustainable products ^{Noe 2} as % of total revenue	≥78%*	82.3%*	≥79% ≥81%
	Cumulative number of energy-saving specialty process platforms developed (base year: 2020)	≥4	4	≥5 ≥10
	Number of Granted Patents (for the year)	>250*	364*	>280
	Number of Cumulative total patents	>15,400*	15,560*	>16,330 >17,730

Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC); * mark refers to UMC (the parent company only).
2. UMC's sustainable products encompasses both environmentally and socially friendly attributes, including products that reduce energy consumption such as low power, low leakage, power management, and micro-controller units, products with high durability such as automotive chips, and products related to epidemic prevention and medical treatment to maintain life and health.

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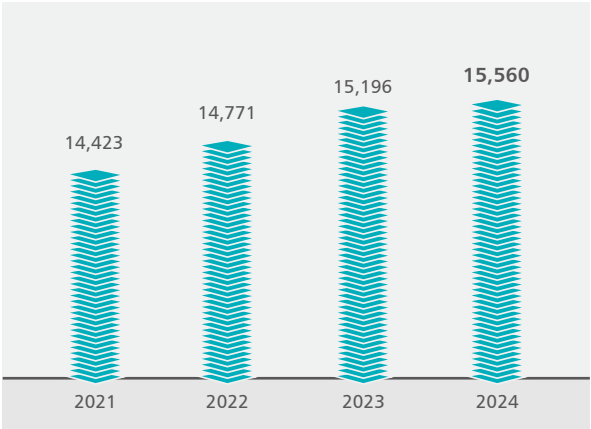
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2.1.1 Robust Wafer Manufacturing Services

UMC's R&D team has always been dedicated to promoting the development of logic process technology. With the adherence to the belief of "Customer's Demand Comes First," the Company has been providing pure-play wafer foundry solutions to meet market trends and customer needs, which include world-class technology, customer support services, and state-of-the-art manufacturing. As a leading worldwide semiconductor foundry, UMC provides comprehensive technologies across logic, RF, eHV, BCD and eNVM to allow customers to have more options to choose the right foundry process solutions and related devices for their mobile and wireless communication ICs, thus meeting the needs of future communication products. UMC provides process technologies that strengthen the devices energy-saving characteristics, enabling products to reduce carbon emissions in end-use applications in the value chain of relevant customers.

Total Number of Accumulated Patents



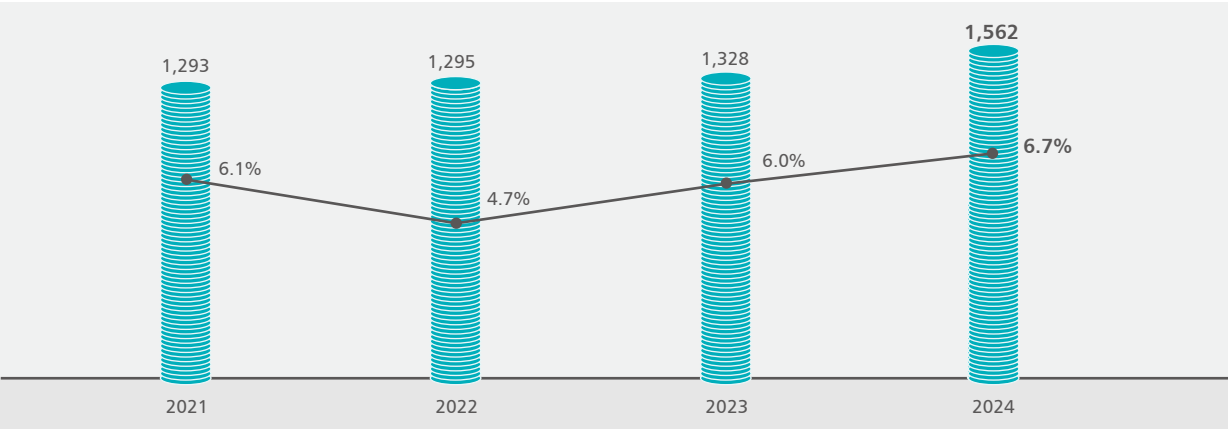
Scope: UMC

UMC's investment policy is based on the principle of long-term strategic investment that align with the Company's operations and development. They aim to mitigate risks and improve operational efficiency through a global layout. UMC continues to invest a large amount of R&D resources, with R&D expenses reaching NT\$15.6 billion in 2024. With global wafer demand outstripping supply, UMC secured mutually beneficial agreements with a number of world-leading customers on a NT\$100 billion expansion at its 12-inch Fab 12A in the Southern Taiwan Science Park between 2021 and 2025. Through the innovative win-win collaboration model, customers secure long-term chip supply at Fab 12A P6 by making a deposit at pre-determined pricing. For UMC, the expansion enables the Company's goal of achieving long-term profitability and increasing market share.

To expand its overseas footprint, UMC fully acquired Mie Fujitsu Semiconductor (MIFS) in 2019, and subsequently renamed it United Semiconductor Japan Co., Ltd. (USJC). The acquisition increased UMC's monthly production capacity by more than 30,000 12-inch wafers, expanded the Company's presence in the Japanese semiconductor market, and enhanced Taiwan's influence in the global semiconductor and foundry industry. Corresponding to China's vast IC market, UMC established its subsidiary United Semi in Xiamen China, as southern China's first 12-inch wafer fab, providing 22 to 90nm process technologies to meet the demand of local IC design companies.

In the face of intense technological competition, besides significantly increasing its key technology capabilities, UMC also focuses on patent distribution to protect its intellectual property rights, and has seen steady growth in its number of patents. In 2024, 364 domestic and foreign patents were granted. To date, UMC has a total of 15,560 granted patents, which offers comprehensive and robust protection of the Company's process technologies. To remain competitive, UMC has also significantly increased the patent quality of its key technologies, and continues to strengthen its customer service and competitive advantage.

R&D Expenses



Note: 1. Scope : UMC consolidated financial report
2. R&D expenses are compiled in accordance with the International Financial Reporting Standards approved by the Financial Supervisory Commission, Executive Yuan
3. UMC's R&D expenses in 2024 totaled NT\$13.15 billion, accounting for about 7.3% of entity revenue

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R&D Strategy

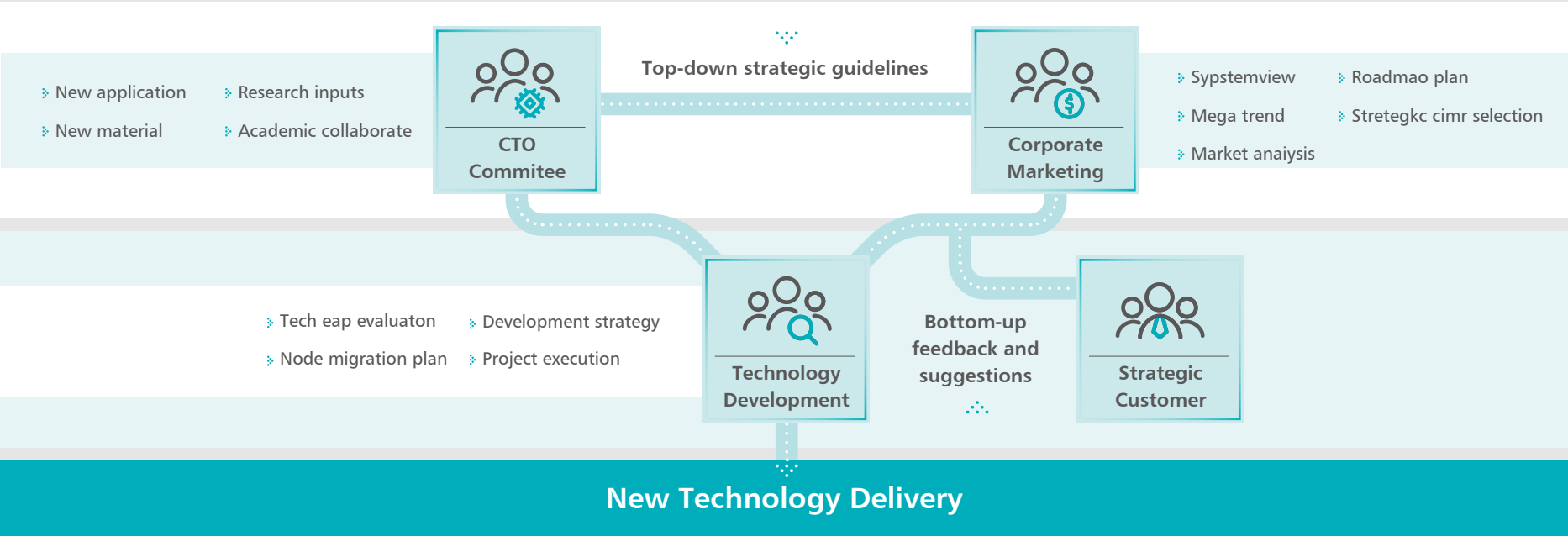
UMC's technology R&D innovation is primarily driven by the Corporate Marketing (CM) division, the Technology Development (TD) team and the CTO committee. The CM division collects and analyzes product market trends and customer needs to develop product portfolio strategies. The TD team analyzes product technology requirements to formulate preliminary technology innovation strategies and R&D goals, which are then reviewed in regular PLM meetings with high level cross-functional heads of the CTO committee to discuss market demand and industry trends. Based on in-depth research on industry development trends, the CTO committee issues the guidance and decisions on product portfolio strategies and technology R&D strategies. Moreover, UMC closely collaborates with strategic customers to obtain their feedback and guidance, ensuring that R&D projects remain consistent with market trends and meet customer requirements. These three solid pillars integrate top-down strategy guidelines tightly with bottom-up customer feedback through seamless communication. In addition to full support for R&D capabilities and proper allocation of resources, UMC continuously stimulates technological innovation, expands and strengthens intellectual property generation. These measures enable UMC's technology R&D innovation procedure to operate smoothly and sustainably.

In line with the United Nations Sustainable Development Goals (SDGs) and the Company's sustainability strategy blueprint, UMC is committed to increasing investment in energy-efficient product technologies, and providing semiconductor manufacturing services covering clean technology fields such as power electric vehicles, smart energy management systems, industrial automation, and optimization technologies and systems, so as to reduce carbon emissions in the industrial value chain and the carbon footprint of end products. In order to align technology development with global carbon reduction goals, UMC has set incremental goals through 2030 to expand the proportion of overall revenue derived from sustainable products. The annual goal is included in the KPI items of the operation strategy, and the implementation status is reviewed by the ESG Steering Committee quarterly and linked to the annual performance review. UMC has steadily promoted the development of various green process technologies from the strategy to the execution side, continuously launched high-performance, low-latency, and low-power solutions, and also continuously promoted the improvement of green process technologies to reduce the generation of harmful substances, and firmly moved towards the goal of net-zero emissions by 2050, contributing to the mitigation of global climate change.

UMC has actively developed advanced packaging solutions (2.5D and 3DIC) and achieved multiple milestones. In the field of RF component applications for mobile devices, UMC has leveraged wafer-to-wafer (W2W) hybrid bonding technology to achieve advantages in form factor reduction. The critical product verification for key customers has been successfully completed, with mass production expected to commence in 2025.

UMC has also partnered with Intel to collaborate on 12nm FinFET process manufacturing in the United States. This partnership represents a significant step in UMC's strategy to pursue cost-effective capacity expansion and process node upgrades, further underscoring UMC's ongoing commitment to its customers. The 12nm collaboration makes full use of the complementary strengths of UMC and Intel, expanding potential market opportunities while significantly accelerating the timeline for technology development. The process development is expected to be completed in 2026.

Intellectual Property



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Intellectual Property

UMC's IP team works closely with the TD team to select key projects as special IP projects in response to the annual product portfolio strategies and technology R&D strategy decision-making, and establish a competitive advantage through a unique IP mining system, including project meetings, brainstorming sessions and promotion meetings, to creatively mine core technology information such as potential patents and key trade secrets.

In 2021, UMC introduced the "Taiwan Intellectual Property Management System (TIPS)", which includes a review of the links between corporate intellectual property and operations and R&D planning. In 2023, with the perfect intellectual property management performance, UMC became the only company to receive the highest honor of TIPS AAA certification this year. It presents our unwavering commitment to IP and corporate governance best practices.

Green Process Development Plan

In response to the heightened global focus on corporate carbon reduction actions, UMC has gradually phased out old processes over the years, and instead focused on the development of low power (LP), ultra-low power (ULP), and ultra-low leakage (ULL) technologies, while also enhancing its power management (PMIC) technology for green and energy-saving products. In order to integrate the eco-friendlier technologies in new products and to increase the proportion of green manufacturing products, UMC has established the goal of developing new LL/LP, PMIC and microcontroller (MCU) platforms, continue to develop more green technology platforms, and provide more high-performance, low-latency, and low-power solutions, so as to produce more energy-saving and power-conservation products, and reduce carbon emissions from end product applications. The revenue contribution from energy-saving products in 2024 is estimated to be about 138 billion.

In the near future, UMC will continue to develop green and energy-saving process technologies with even lower power consumption to meet future demand driven by 5G, smart Internet of Things (AIoT), wearable devices, cloud applications, and automotive electronics applications. Through innovative green and energy-saving process platforms, the Company can bring substantial benefits to global greenhouse gas mitigation.

Please refer to UMC Web pages
for more information



2.1.2 Key Technologies

UMC not only engages in ongoing research and development to reduce chip power consumption, but also develops a range of power management chips, medical sensor chips, mobile communication chips, image sensing and display driver chips that decrease the environmental impact of end-products and advance communication, healthcare, and safety in society.

❖ The industry-leading 22nm embedded high-voltage (eHV) process has been successfully verified in cooperation with customers and has been introduced into mass production.

❖ Completed the development of the industry-leading 28nm embedded high-voltage and low-power 28eHV-LP process platform, which can reduce energy consumption by up to 15%.

❖ Launched the 40nm RFSOI process platform and its foundry design kit (FDK) to provide the millimeter wave (mmWave) RF front-end process solutions capable for mass production.

❖ The 22nm Image Signal Processor (ISP) technology has been mass production following the completion of the component model, the design specifications, and the related IPs (intellectual properties) construction.

❖ In the non-volatile memory process technology development, the 28nm eFlash memory (28ESF4) and the 22nm resistive memory (22RRAM) process platforms both have completed the verification procedures of Generic and IoT microprocessors.

❖ The next-generation customized PMIC for mobile phones development has completed product verification and entered trial production.

❖ The development of the new 0.11μm BCD 3.3V application platform has been completed, continuing in the FDK construction and the introduction of design finalization and verification.

❖ The RF GaN products and the 650V GaN power components both have entered the trial production stage.

❖ UMC's hybrid-bonding stacking technology has completed Cadence Integrity 3DIC design process verification, and the W2W 3DIC project has been established with supply chain partners to provide a comprehensive solution for customers' diverse needs.

2024		2024 UMC Sustainability Report Innovative Products and Services 78	
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	Key Technologies	Product Highlights	Applications
Message from the ESG Steering Committee Chairman About This Report About UMC Sustainable Governance Sustainability Strategy and Blueprint Enterprise Risk Management Sustainable Performance Highlights Determination of Material Topics Special Feature of the Year	12nn FinFET (12FFC) process technology	- Higher performance, smaller chip size, and lower power consumption - Wider range of applications connecting with various specialty processes, and more competitive	To drive the application of next-generation silicon chips in various fields such as networking, artificial intelligence and further high-end consumer electronic products Production of the 12 nm process is expected to begin in 2027
	14nm FinFET (14FFC) Process Technology	- High performance, low power loss, and low latency - Excellent energy efficiency in end-user application	The specialty process technologies expand the scope of product applications in electronic products including analog, mixed-signal, RF technology, Internet of Things, and mobile devices
	22nm Process Technology	The 22nm process technology, compared to the 28nm process, exhibits higher performance, lower power consumption, and reduced chip area, thus the cost competitiveness has been greatly improved	The specialty process technologies expand the scope of product applications in electronic products including analog, mixed-signal, RF technology, Internet of Things, and mobile devices The design specification for the automotive Grade-1 platform has been successfully established, and the reliability and durability verification have been completed
	28nm Process Technology	- UMC's 28nm gate-last High-K/Metal Gate process technology features superior low leakage and dynamic power performance 28HPC+ solution is ideal for high-speed mmWave devices for circuit applications up to 110GHz	Can be applied to products, such as processors, mobile phone basebands, WLAN, tablet PCs, FPGAs, Netcom ICs, image signal processors, mmWave system-on-chip (SoC), mobile devices, automotive electronics, Internet of Things (IoT), etc. 28HPC+ mmWave design reference flow has been certified through silicon tape-out
	Non-volatile memory (NVM)	- The embedded non-volatile memory (eNVM) process technology improves the computing and storage performance of the chip, and has the advantage of confidentiality - Providing a complete process and supporting IP solutions from 180nm to 22nm nodes	Can be widely applied in microprocessors and electronic products, covering the Internet of Things (IoT), industrial control, and various 3C products in the automotive field, and also extended to solutions for various bank cards and telecommunication cards
2.1 Innovation & Technology Development	22nm resistive RAM (RRAM)	- Resistive RAM (RRAM) is a novel NVM with simple structure, low operation voltage, low read current, fast program/erase speed and very good reliability - It is fully logic compatible with less add-on masks and low extra cost	RRAM can be applied to wearable components such as hearing aids, watches, etc., as well as edge computing applications
2.2 Customer Satisfaction	22nm Magnetoresistive RAM (MRAM)	- The new generation of Magnetoresistive Random Access Memory (MRAM) has the same size as the traditional charge-trapping memory, but has simpler process conditions - It has faster operation speed, and exhibits excellent thermal stability, read/write endurance, and data retention - MRAM products can be embedded in logic processes and operate with very low standby electrical current, reducing the power consumption of products in end applications, so as to reduce carbon emission in the entire value chain of UMC and its clients	Can be applied in artificial intelligence Internet of Things (AIoT), aerospace/LEO satellite communications The 22nm high-density 1Gb HDMRAM chip product used in aerospace/LEO satellite communications has entered the trial production stage
2.3 Product Responsibility			
3 Implementing Green Operations			
4 Providing a Friendly Workplace	High Voltage process technologies	- UMC has led the industry successfully in the development of 22nm embedded high-voltage (eHV) process technology - UMC's 28nm ultra-low power embedded high-voltage platform 28eHV-LP process technology, featuring the industry's smallest SRAM bit cell, offers up to 15% reduction in power consumption without compromising image quality or data rate, addressing the demand for longer battery life of devices	Can be applied to mid-to-high-end smartphones, VR/AR equipment mounted panels and automotive electronics Start a plan to integrate FinFET and High-Voltage processes to develop a new competitive display driver IC
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	Key Technologies	Product Highlights	Applications
	Power management IC (PMIC)	- Combined with the world's third-generation low-on-resistance high-voltage components in one of the process technologies, it also provides ultra-high voltage process to reduce energy loss during voltage conversion and achieve energy-saving effects for various power management needs - BCD (Bipolar-CMOS-DMOS) products provide a wide range of high-voltage operating capabilities (up to hundreds of volts) required by power management chips, as well as standard logic compatibility and a complete IP platform	The power management chip (PMIC) is in strong demand and is widely used in consumer electronics, communications, computing, industry, automobiles, such as 5G mobile phones, servers, electric vehicles, etc. The PMIC integrated with 0.11μm BCD platform and the embedded non-volatile memory platform has completed the automotive component model and design specifications, as well as automotive IP
	CMOS image sensor and image signal processor (ISP) process technologies	- The high-performance CMOS image sensor (CIS) solutions that have been adopted by global leaders in mainstream applications such as portable devices, medical devices, ambient light sensors, surveillance systems, and more - The 55nm back-illuminated image sensor (BSI) CIS process technology have entered mass production - The mass-produced 28HPC+ ISP technology provides the industry's highest-end 200-megapixel image sensor product with the smallest pixel size - The 22nm ISP process technology, which can provide lower core operating voltage and better energy performance, has entered mass production	- Can be applied to high dynamic range (HDR) related products, such as surveillance, driving record (Dash CAM) and Internet of Things (AIoT), etc., to meet the market demand of customers with high sensing resolution and fast focus function (PDAF), as well as product diversity - Continue to develop ISP technologies to be applied on the products related to automatic white balance, automatic focus, anti-shake, and high dynamic range (HDR) related products, such as surveillance, driving record (Dash CAM) and Internet of Things (AIoT), etc., to meet the market demand of customers with high sensing resolution and fast focus function (PDAF), as well as product diversity
	Micro-electro mechanical device technology (MEMS)	- UMC has established a series of fundamental MEMS process building blocks to help customers quickly construct MEMS structures in a CMOS foundry fab - UMC has developed advanced sensors with second-generation micro-electromechanical (CMOS-MEMS) technology	- Can be applied in medical-related sensor chips, flow sensor chips for medical respirators, vaccine temperature control chips produced by temperature and humidity sensors, and carbon dioxide sensing elements - Flow sensor chips can be used in medical purpose, such as life-sustaining respirators; temperature and humidity sensors can be used in the storage and transportation of medicine, food and chemicals; the carbon dioxide sensing element can be used to monitor the concentration and trend of carbon dioxide for air quality control
	Radio Frequency Silicon-on-Insulator process (RFSOI)	- For the requirements of RF communications such as 4G, 5G, WiFi, provide highly competitive component characteristics (low high frequency doubled harmonics and low power loss) for 4G/5G mobile phones with stringent requirements for RF switches and low noise amplifiers - UMC continues to develop the 40nm RFSOI technology platform, offering the foundry design kit (FDK) and the solutions for production of millimeter wave (mmWave) radio frequency (RF) front-end products	Can be widely used in RF Front End applications, such as antenna switch, antenna tuner and Low Noise Amplifier (LNA)
	Compound Semiconductors	- GaAs HBT power amplifier integrates SAW filter and RFCMOS switch, provides mobile phone RF front-end module (RF-FEM) and WiFi6/7 communication module integration solution - Developing GaAs pHEMT, RF GaN and power GaN products suitable for high power 5G communication, all are under trial production	Can be applied to mobile phone RF front-end module, WiFi6/7 communication module, 5G mobile communication mmWave band base station, and Ka/Ku-band low-orbit satellite receiver chips, and high-efficiency power supply component
	2.5D & 3DIC Heterogeneous Integration Packaging Technologies	- UMC newly launched wafer-level 3D W2W hybrid bonding solution offering advantages in both bandwidth and form factor reduction - The high-end RF communication product adopts 3D stacking to reduce the area by 40% and has completed customer product verification	It can meet various applications demands from customers in high-performance computing (HPC), cloud computing (Cloud AI) and edge computing (Edge AI) for artificial intelligence, high-end communications (4G/5G/6G), artificial intelligence of things (AIoT), virtual reality and augmented reality (AR/VR), crypto currency and other applications UMC has established a one-stop vertical integration project with our supply chain partners, providing customized low-power logic chip and memory module combination to meet the needs of customers in IoT, secure smart infrastructure, and other terminal devices

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Energy Efficient Products and End Applications

As part of its sustainability strategy, UMC aims to increase investment in technologies that deliver enhanced energy efficiency and expand the development of clean technology products, reducing carbon emissions along the value chain as well as the carbon footprints of end-products. To align its technology development roadmap with carbon reduction goals, the Company has set incremental goals through 2030 to expand the proportion of overall revenue derived from sustainable products. UMC offers ultra-low power (ULP) and ultra-low leakage (ULL) logic technology platforms, which can be integrated with the Company's leading specialty processes to enhance energy saving characteristics of end-products. UMC's technologies also contribute to global climate change mitigation by providing semiconductors to power electric vehicles, smart energy management systems, industrial automation, and optimization technology and systems.

Logic Process Technology

For its 28nm logic process, UMC provides two types of high-performance and ultra-low-power options to suit different customer demands, including automotive applications. The 22nm logic process platform provides lower core operating voltage, demonstrating ultra-low leakage and energy-saving features. The 14nm process is more suitable for computing and data processing due to its high performance, low power consumption and low latency.

Embedded Non-Volatile Memory Technology (eNVM)

UMC offers state-of-the-art eNVM solutions to meet the needs of a variety of embedded system applications. In addition to the standard eNVM solutions, UMC has also successfully developed resistive memory (RRAM) and magnetoresistive memory (MRAM) technologies, which can be embedded in logic processes and operate with very low standby electrical current, reducing the power consumption of products in end applications. Coupled with robust logic processes for peripheral devices, UMC produces energy efficient microcontrollers for a range of applications including automotive, smart factory, and consumer products.

Embedded High Voltage Technology (eHV)

UMC has the industry's broadest portfolio of eHV process technologies and corresponding intellectual property solutions for display panel driver chips of all sizes. UMC's ultra-low leakage solution enables customers to integrate control logic, gate drivers, source drivers, and memories on the same IC to form a system-on-chip (SoC), which can achieve multiplexing and improve product performance without increasing power consumption.

Bipolar-CMOS-DMOS Technology (BCD)

UMC's BCD platforms, backed by comprehensive eNVM, deliver complete power management functions in a single chip. Customers' power management ICs produced using BCD technology exhibit lower energy loss during voltage conversion and boast a smaller form factor, making it a preferred solution for developing end-products that is more energy efficient.

Radio-Frequency Silicon-on-Insulator Technology (RFSOI)

UMC's RFSOI technology, with its unique substrate characteristics capable of preventing low and high frequency doubled harmonics and reducing power loss, exhibits outstanding energy efficiency in end applications.

Clean Tech End Applications	14nm Logic	22nm Logic	28nm Logic	eNVM	eHV	BCD	RFSOI
Demand-Side Management Home energy management systems, DDI/LCD controller, HVAC, Intrusion/fire detection system, Security gate, etc.		●	●	●	●	●	
Industrial Automation Manufacturing systems, General-purpose automation modules, Environmental sensor, Compute electronics, Enterprise equipment, etc.		●	●	●			●
Optimization Technology & System Server with CPU sockets, Smart meter, Solid-State-Light, Storage system, Internet service provider, etc.	●	●	●	●			●
Hybrid/Electric Vehicles Driver information, Battery management, Power bank, Engine control, Powertrain, Remote keyless entry, Power steering, ADAS, etc.			●	●	●	●	●

2.1.3 Smart Manufacturing

Since 2016, the UMC Group has begun evaluating and taking stock of the key areas for enhancing Industry 4.0 and planning the future development direction of smart manufacturing. In 2017, the Company adjusted its internal organizational structure and established the "Smart Manufacturing Division " dedicated to promoting digital transformation. At the same time, the concept and technology of Industrial AI (IAI) created by Professor Li Jie of the University of Maryland in the United States was introduced to collaborate on the development of various applications for smart manufacturing and coordinate related OT (Operation Technology) and IT projects, integrating internal and external resources.

The Company's sustainable strategy blueprint also includes the strategy of "promoting the smartification of Industry 4.0," planning to continuously introduce various smart innovation solutions to improve delivery time and achieve energy-saving and environmentally friendly smart manufacturing factories. By incorporating relevant initiatives into the Company's overall goals and conducting regular tracking, we ensure their effective implementation.

Sustainability Strategy and Blueprint		2024		2025	2030
Goals	Goal	Achievement		Goal	
Ratio of products produced by Industry 4.0 Smart Manufacturing Systems	≥ 50%	✔ 53.0%		≥ 60%	≥ 80%

Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC)

The Core Concept of Smart Manufacturing Project Team

UMC integrates the DNA of Industry 4.0 into its corporate culture, applies core technology to develop and implement various smart manufacturing projects. These projects aim to achieve process optimization, flexible production, shortened delivery time, manpower quality and efficiency improvement, and address the increasingly challenging and complex manufacturing environment. This includes IAI projects, IT infrastructure construction, physical security of the smart supply chain, enhanced information protection, and integration with upstream and downstream industries related to semiconductors to jointly establish a smart manufacturing ecosystem.

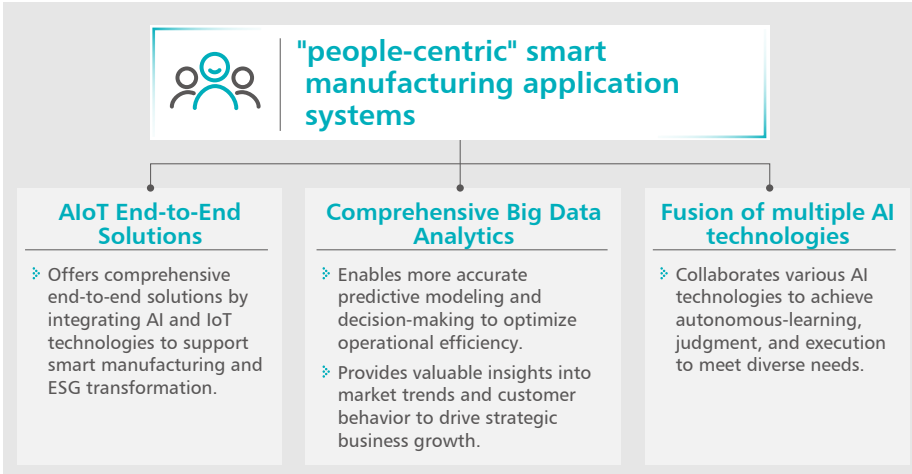
Smart Manufacturing Strategy



With reference to the methodology of Industry 4.0, UMC has formulated a development blueprint for smart manufacturing based on international and industrial benchmarks. Through the project team, we promote and implement relevant strategies not only to achieve digital optimization goals in work efficiency, product quality, and green manufacturing, but also to create innovative business models.

AI-driven Autonomous Green Manufacturing of the Future

UMC introduces IAI technology and concepts, and develops "people-centric" smart manufacturing application systems through strategies such as " AIoT End-to-End Solutions," "Comprehensive Big Data Analytics," and "fusion of multiple AI technologies." These systems focus on enhancing quality, efficiency, cost savings, and energy conservation while reducing waste. They optimize business processes through digital technology, strengthening operational efficiency and green manufacturing.



Build efficient intelligent decision-making/execution platforms by integrating industrial dual AI (analytical AI + generative AI), AIoT, big data, and the cloud technologies.

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Smart Manufacturing Innovation

From the perspective of business strategy, UMC has transformed into a specialty process solution provider. To meet the demands of long lifetime, low energy consumption, short cycle time, and small amount and high diversity, UMC actively promotes innovative approaches to digital transformation and digital optimization to achieve efficient workflows, rapid mass production, short production cycles, production flexibility, and green manufacturing requirements.

Transformational Thinking
People-Centric

UMC's digital transformation focuses on "people-centric" mindset. Under the premise of making full use of the operation technology (OT) process transformation, UMC keeps developing smart manufacturing applications to comprehensively improve the efficiency, process quality, and product yield, to effectively enhance the overall operating value.

Value Enhancement
Environmental Sustainability

1. Introduce digital technology into energy-saving applications and facility management, and promote all-round energy-saving measures in fab equipment, processes, systems and management.
2. Conduct big data analysis using intelligent energy management systems, review energy consumption data on a rolling basis and explore potential energy saving opportunities.
3. Optimize operation through control systems and process systems, effectively manage energy consumption, and promote smart energy conservation for the industry.

Partnerships
Diversified Cooperation

Introduction of IAI technology Join the National Science Foundation (NSF) Intelligent Maintenance System (IMS) Industry-Academia Collaboration Center to introduce related technologies required for industrial artificial intelligence and internalize the UMC Smart Manufacturing Systems.

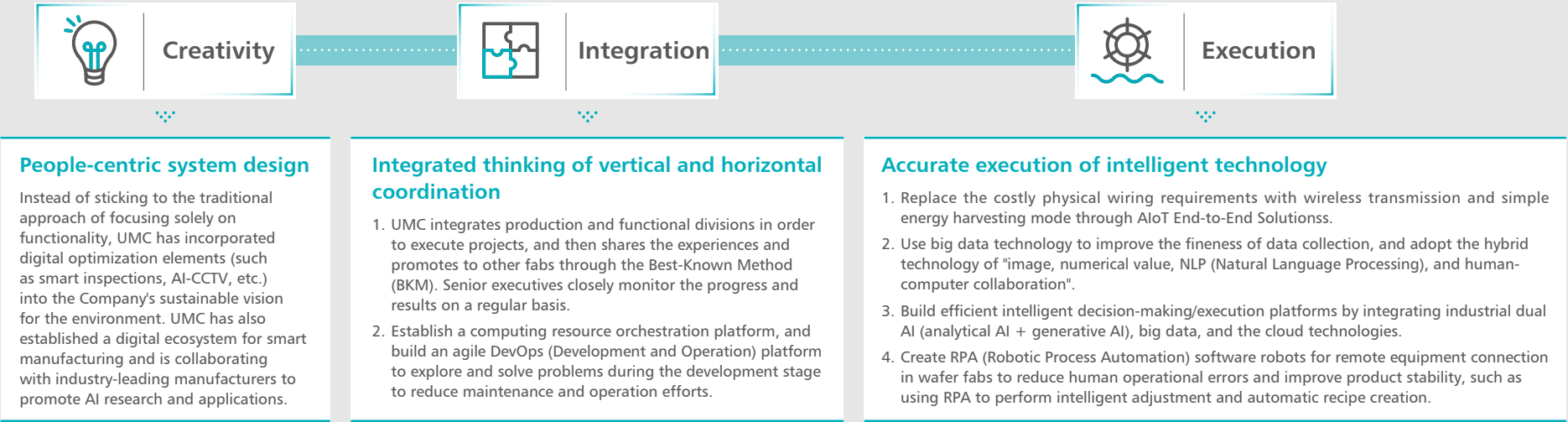
Establish AI research center In response to the future of new AI technologies, UMC collaborates with academia to conduct early-stage research and development in order to sustain the R&D momentum in AI.

Alliance Leader Select the most forward-looking and competitive manufacturers and open source software organizations in order to carry out cooperative development or product introduction.

Project Promotion

Location: Southern Taiwan Science Park Pilot Fab

To actively implement the smart manufacturing projects, UMC proposes the integration of three digital cores -- creativity, integration and execution. Since 2018, UMC Fab 12A, located in the Southern Taiwan Science Park, has been appointed as the pilot fab. It has been gradually implementing various intelligent innovation solutions to shorten product delivery schedules and promote energy-saving as well as eco-friendly smart manufacturing.



2.2 Customer Satisfaction

- 2.2.1 Strengthen Customer Communication
- 2.2.2 Quality Management System



Material Topic 2.2 Customer Relationship Management

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Positive Impact 2.2.1 Strengthen customer relationship management Negative Impact 2.2.2 Customer rights impacted

Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
Positive: Establish smooth communication mechanisms to understand customer opinions, enhance service quality, and improve customer satisfaction. Negative: Without smooth communication mechanisms, customer rights may be compromised, and their needs may not be met.	- Product quality is managed continuously by designated divisions according to quality management system regulations - Offer designated sales and customer service contact to communicate and serve customers in a timely manner - Voice of Customer (VOC) instant online feedback system managed by designated personnel to collect customer feedback and to strengthen two-way communication with customers	- Product quality and delivery schedule are the key performance indexes to be reviewed regularly - Understand customers' requirements and satisfaction through scorecards provided by customers in quarterly business reviews and other meetings - Collect customer requirements and respond through meetings with customers, while also improving internal procedures accordingly - Conduct annual customer satisfaction surveys with key customers through a third-party agency in order to obtain objective feedback and take action accordingly

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Ratio of satisfied customers	≥90%	✓ 92%	≥90%	

Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC)

2.2.1 Strengthen Customer Communication

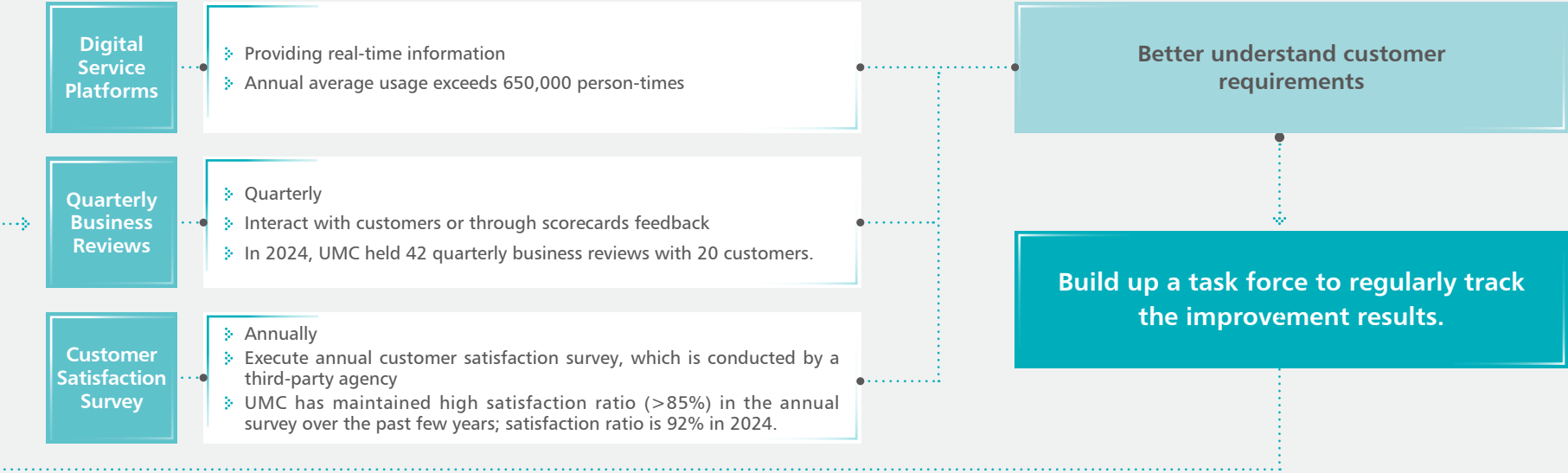
Multiple Communication Channels

UMC is committed to achieving customer satisfaction as its mission and prioritizes customer-oriented products and services. The Company's MyUMC, MyHJTC (reserved for customers of HJ), and MyUSC (reserved for customers of USCXM) online service platforms provide customers with complete and real-time supply chain information, including production status of orders, shipping dates, and product quality data. These service platforms are accessed and utilized over 650,000 times every year on average.

The Voice of Customer (VOC) instant online feedback system allows customers to make requests, comments, or suggestions for UMC products or services. Designated personnel re-direct, process and respond to requests made through the VOC system, and customers can enquire the status of their products online 24/7.

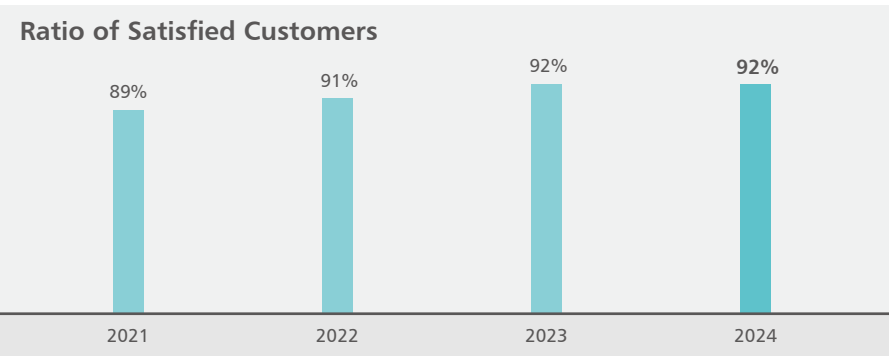
UMC also holds quarterly business review meetings with key customers, and asks customers to provide scorecards to enable the Company to better understand customer requirements and satisfaction. In 2024, UMC held 42 quarterly business reviews with 20 customers.

Digital Service Platforms Providing Real-Time Information



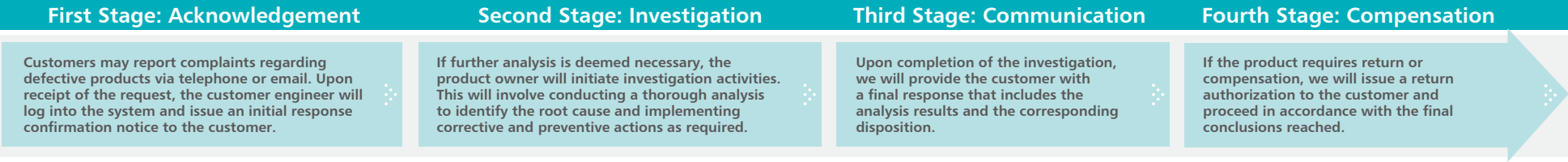
Customer Satisfaction Improvement

UMC executes annual customer satisfaction survey, which is conducted by a third-party agency and administered to customers contributing more than 75% of UMC's overall revenue. In the 2024 survey, we expand the customers who contributed 81.9% of revenue. The survey collects customer feedback on six key pillars, including strategy and partnership, technology, quality, manufacturing and yield, sales support and customer service, and design support. UMC has maintained high satisfaction ratio (>85%) in the annual survey over the past few years.



2.2.2 Quality Management System

UMC Group is a leader in the global semiconductor wafer industry. The Company has established a rigorous quality management system based on quality management system (ISO 9001) and the global automotive industry quality management system (IATF 16949). Before product delivery, we have supplier quality assurance, manufacturing quality assurance and outgoing quality assurance to proactively prevent the release of potentially defective products, thereby avoiding product recalls. In response to customer complaints about defective products, a customer quality assurance is established, with designated personnel responsible for subsequent distribution, handling, and responses, ensuring effective resolution. The defective products customer complaints handling procedure had 4 stages.



Additionally, the quality system integration encompasses the implementation of international quality management system standards and internal quality audits. It also provides product quality-related training courses for both non-supervisory and supervisory personnel, including IATF 16949 core tools^{Note} and automotive product management courses. For instance, we offer specialized training in R&D process quality control tools to our R&D quality management team, enabling them to effectively implement quality control measures within the R&D process. In response to changes in the market environment and the advancement of various technologies, UMC Group follows the PDCA (Plan-Do-Check-Act) method for continuous improvement, enabling the Company to provide high-quality wafer manufacturing services, and to develop a customer-oriented quality management system. UMC has passed the third-party ISO 9001:2015 and IATF 16949:2016 verifications every year, and will continue to strive for improvement in product quality.

Note: IATF1694 core tool: APQP (Advanced Product Quality Planning), CP (Control Plan), FMEA (Failure Mode and Effects Analysis), MSA (Measurement System Analysis), SPC (Statistical Process Control), PPAP (Production Part Approval Process)

Manufacturing Quality Assurance

- Manufacturing quality management
- Development and implementation new process and specialty process
- Statistical process management

Customer Quality Assurance

- Advancing quality related process to provide effective customer service and improve customer satisfaction



Supplier and Outgoing Quality Assurance

- Raw material management
- Mask management
- Testing and packing management
- Product outgoing quality management

Quality System Integration

- Promoting international standard quality management systems
- Internal quality audit
- Quality system integration
- Technical information center

Quality Policy

UMC Group has an effective communication and feedback system that facilitates quick response to customer demands and market changes. UMC encourages and requires all employees to engage and improve product and service processes to achieve the Company's quality requirements and goals according to the principle of continuous improvement.

Business Promotion

- Market-driven approach for total customer satisfaction

Research and Development

- Constant improvement to excel and create industry-leading technologies



Education and Training

- Full participation in strengthening quality awareness

Product Manufacturing

- Maintaining system performance at its peak to achieve the highest possible quality standards

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★ Material Topic 2.3 Product Responsibility and Lifecycle Assessment

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Hazardous Substances
Free Policy



Environmental
Protection Policy



Negative Impact 2.3 Product Responsibility Mismanagement

Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
Negative: It may violate relevant regulations or customer requirements, leading to product scrapping or recall.	- Establish an interdivisional Hazardous Substance Management Committee, and establish a hazardous substance management system, a hazardous substance-free management program, and implement a non-hazardous substance management from the source to the manufacturing process - Simplify procedures and conserve materials as much as possible when designing processes, and develop green products with high performance and low power consumption - Develop green chemicals and carry out hazardous substance substitution programs to proactively replace harmful chemicals at the source - Establish raw material green procurement procedures and new-material assessment systems; incorporate hazardous substance control into Material Receiving Inspection Procedures; no substance shall be released if there is a concern about hazards - In addition to following all applicable laws and standards, conduct self-evaluation according to even stricter standards and prioritize replacing hazardous substances - Conduct regular assessments of "Product Carbon/Water Footprints" and "Product Lifecycle Assessment" to understand their environmental impact and effectively improve the production process	- Conduct third-party verification annually: QC 080000, IATF 16949 - Regularly test the content of hazardous substances in products in independent laboratories to ensure that products meet international regulations and customer requirements for controlling hazardous substances - Establish hazardous material replacement plans and set up corresponding objectives. Review progress of implementation on a quarterly basis - Regularly conduct the inventory of "Product Carbon/Water Footprints" and "Product Lifecycle Assessment," verified by a third party

Future Goals	2024	2025	2030
Item	Achievement	Goals	
No customer complaints, product scrap and recalls due to violations of hazardous substance management.	✓ Achieved	On track	

Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC)

UMC is committed to providing products that meet customer demands and are competitive in the market. As a member of the global community, UMC values resources and strives to reduce the impact of its products on the environment and resource consumption. By obtaining green factory certification for clean production, developing replacements for hazardous substances, and reducing the environmental impact of products, UMC aims to lead responsible production efforts in the semiconductor industry.

Considerations for Green Products at Each Stage	Choice of raw materials or components	Product production and manufacturing	Distribution, storage and transportation	Use phase-operation and servicing/maintenance	End of life management
Implementation Direction	Promote the reuse of raw materials in the production process, and independently develop green chemicals to promote chemicals that are free of harmful substances and recyclable	Promote clean production, reduce the use of hazardous substances	- Reuse the package materials from raw materials and some products in order to reduce the demand for such materials and the generation of waste - Considering the location of customers and the amount of products that can be transported by available carriers, UMC adopts common-route delivery rather than frequent, small and un-routine delivery modes	Develop advanced process technologies that are eco-friendly and consume less power	To facilitate easy recycling and disposal, it is crucial to maintain uniformity in the composition of ICs and products, while avoiding the use of composite materials

2.3.1 Hazardous Substance Management

UMC implements the hazardous substance management system QC 080000 to ensure that products comply with the European Union's requirements for restricted substances in electronic products (such as EU RoHS) and global chemical control regulations. UMC also provides customers with green products to meet their needs and further expand the market for green products. UMC has also established a cross-division Hazardous Substances Process Management Committee (HSPM committee) to improve the efficiency of green product management.

Topic	Best Response
Monitor and management of controversial material	Hazardous Substances Process Management Committee established in 2006
Plan for elimination of controversial material	Review the usage status in processes and develop plan for elimination or replacement of controversial material

Exposure to Hazardous Substances in Products

The Company has evaluated the exposure to hazardous substances in its products and conducted risk assessments on its product portfolio. Upon the announcement of new hazardous substances by international regulations, the Company requires relevant divisions to confirm the scope of impact of the hazardous substances. From the raw material stage, the Company analyzes the raw materials of the products (such as photoresists, chemicals, etc.), confirms the composition, and lists all possible hazardous substances. At the product stage, by analyzing the hazardous substances in the product portfolio, the Company identifies the hazardous substances that may exist in the products, such as heavy metals, persistent organic pollutants (POPs), toxic chemicals, etc., and entrusts the Company's product portfolio to a third-party international certification and testing company to analyze the hazardous substance components of the products. The analysis of the product portfolio shows that no harmful substances were detected, and it complies with relevant legal standards, ensuring that the products meet specifications and reduce negative impacts on the environment. It not only demonstrates the company's commitment to public safety and environmental protection, but also reflects its responsibility for sustainable development.

The percentage of company revenue that tracks products containing persistent organic pollutants (POPs) or substances classified as hazardous by regulatory agencies is 0 in 2024. EU POP regulations define persistent organic pollutants as a class of chemical substances that are difficult to decompose and may accumulate in the environment, causing long-term harm to ecosystems and humans. Substances classified as hazardous by regulatory agencies are usually toxic or hazardous. The Company actively monitors the latest information regarding persistent organic pollutants (POPs) from the EU website and requires relevant divisions to assess the impact of these hazardous substances, demonstrating its commitment to prioritizing materials that may affect human health and the environment in its operational management practices. By tracking the revenue share of products related to these substances, the Company can effectively assess the impact of its product portfolio on health and the environment, and take appropriate improvement measures to reduce potential risks and ensure compliance with regulatory standards.

Risk Assessment of Potential Impacts of Hazardous Substances in Products on Human Health and the Environment

The Company conducts an annual risk assessment on 100% of its products to evaluate the potential impact of hazardous substances on human health and the environment. The following are specific practices and steps that the Company follows to ensure the comprehensiveness and effectiveness of risk assessment.

1 Create a hazardous materials list

Identify hazardous substances

Analyze the raw materials of the product (such as photoresist, chemicals, etc.), confirm the composition, and list all possible hazardous substances.

Reference to international standards

Determine the hazardous substances that require special attention according to relevant laws, regulations and standards, such as the EU REACH Regulation and the RoHS Directive.

Supply chain communication

Request suppliers to provide material composition information to fully understand the material composition of the product.

2 Determine the scope of the risk assessment

Regarding human health and environmental impacts

Through a comprehensive analysis of harmful substances within the product portfolio, we identify potential contaminants, including heavy metals, persistent organic pollutants (POPs), and toxic chemicals. To ensure accuracy, we engage a third-party international certification and testing company to evaluate the composition of harmful substances in our products. The results of this analysis indicate that no harmful substances were detected, confirming compliance with relevant legal standards and ensuring that the Company's products do not pose a risk to human health or the environment.

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Hazardous Substance Free Policy

UMC Group established the Hazardous Substances Free Policy to avoid product scrapping or customer complaints caused by the violation of hazardous substance-related laws and regulations, as well as the requirements of customers for controlling hazardous substances. On the other hand, by raising the awareness of all employees, UMC provides products that comply with regulations and customer demands through the control and improvement of design and production, fulfilling its corporate social responsibility of protecting the environment as well as public health and safety.

Management



Response to the International Hazardous Substance Management Requirements

EU Restriction of the Use of Certain Hazardous substances in Electrical and Electronic Equipment Directives (EU RoHS)	Halogen-free Requirement	EU Regulation on Registration, Evaluation, Authorization and Restriction of Chemicals. (EU REACH)	EU Waste of electrical and electronic equipment (WEEE)	Persistent Organic Pollutants, POPs	IEC 62474 Declarable Substance List
❖ Able to comply with the requirements	❖ Able to comply with the requirements	❖ Able to comply with the requirements	❖ These regulations do not directly apply to UMC as the Company manufactures semiconductor chips which are not end products. ❖ The end product dealer is the one responsible for recycling end products containing semiconductor components which are discarded as waste after use.	❖ UMC led the industry to achieve the overall replacement of PFOS, PFOA and PFOA-related chemicals. ❖ Since 2018, UMC began to pay attention to PFBS to keep up with international environmental protection regulations, and to collect data on PFBS-related raw materials and alternative chemicals.	❖ Based on the calculation of the total sales amount of products, the percentage of non-compliance with IEC62474 is 0%. UMC fully complies with the requirements of the IEC 62474 Declarable Substance List.

Hazardous Substances Replacement Program

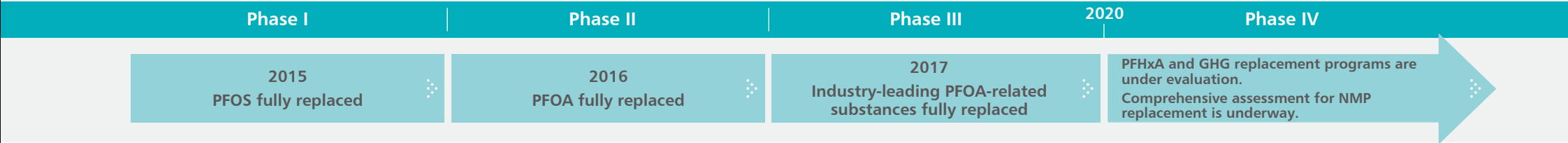
UMC Group has long been paying attention to the development trends of international environmental protection regulations. During the drafting stage of relevant regulations, UMC proactively made arrangements to launch the "assessment plan" for hazardous substances. After the relevant regulations are finalized and announced, the "replacement plan" will be implemented immediately in fabs. During the initial assessment phase, the UMC Group invested relevant R&D resources and budget, and committed to seek alternatives or gradually phasing out hazardous substances.

Assessment Plan

- 1. Set up a review/filter mechanism at the early stage of new material evaluation to prohibit the use of hazardous substances listed in the announcement.
- 2. Take inventory of usage in fabs.
- 3. Request suppliers to provide alternative products or work together to develop alternatives.

Replacement Plan

- 1. Ask fabs to arrange experiments to collect engineering data.
- 2. Complete the certification process on products.
- 3. Modify the inventory preparation plan and complete the ban as early as possible.



Perfluoroalkyl Substances (PFAS) Management

Per/Poly fluoroalkyl substances (PFAS) are a class of chemically stable synthetic substances that are waterproof, oil-proof, and have low friction. They are widely used in the manufacturing process of many products. However, because PFAS does not easily decompose in the environment and may cause harm to the human body, more and more countries and regions have begun to regulate it. UMC Group completed the replacement of PFOS and PFOA in 2015 and 2016 respectively, and led the industry in completing the replacement of PFOA-related substances in 2017.

With the expansion of the scope of international control of PFAS substances, UMC Group began to make advance arrangements during the proposal stage, and successively took stock of the materials used in the manufacturing process. For existing materials, we monitor the presence of PFAS in raw materials through supplier questionnaires. Additionally, we collaborate with suppliers to discuss the possibility of substituting related substances, enabling us to proactively address potential concerns. For materials that are difficult to replace (such as photoresist), suppliers are required to provide proof and seek exemptions through the ECHA platform. For substitutable materials, suppliers are required to provide substitutes and successive evaluations of substitutions.

Through the review mechanism of new material assessment, UMC Group has blocked the source of input materials and prohibited the new introduction of process raw materials containing PFAS above C5 (inclusive) to reduce the possibility of using PFAS banned substances. UMC Group is committed that all materials are prohibited from containing PFOS, PFOA, PFOA-related substances, PFHxS and PFHxA. Currently, only one photoresist used by the subsidiary, USJC, contains PFHxA. No chemicals containing PFOS, PFOA, PFOA-related substances, PFHxA, and PFHxS are used in other fab processes. In addition to the PFHxA-free alternative plan provided by the photoresist supplier, UMC Group is also looking for suitable alternative photoresists from other suppliers, and found a suitable target in 2024, arrange for preliminary testing in 2025, and target to achieve phase-out on PFHxA by the end of 2028. In addition, the UMC Group also participates in discussions with the Taiwan Semiconductor Industry Association and the Chemical Working Group of the World Semiconductor Council. The company is committed to collaborating with these organizations to research and identify alternatives or reduction technologies for PFAS.

In the future, UMC Group will continue to pay attention to the development trends of international regulations and participate in discussions at domestic and foreign organizations to respond to relevant regulatory impacts in advance, ensure the stability of operations and production, and implement the sustainable strategic direction of minimizing environmental footprints.

NMP Replacement Plan

In addition to extending the acid replacement cycle of N-methylpyrrolidone (NMP) and simplifying the process to reduce the usage of NMP, UMC Group has launched a complete replacement plan for NMP in 2020 and successfully developed a weakly acidic solvent at the end of 2022. Solvents can replace the cleaning function of NMP. Each 8-inch fab of UMC Group has taken the lead in completing the electrical performance, yield and reliability verification of various products, and has gradually started replacement operations starting in 2023.

UMC Group 8-inch fabs have collectively reduced the use of NMP by 80.5% in 2024 compared to 2022. And by the end of 2027, the goal is to complete the NMP replacement of all 8-inch fabs of UMC Group. UMC Fab 12A is currently evaluating its feasibility, and aims to complete the electrical, yield and reliability verification of various products by the end of 2025, and begin import replacement operations. Subsequently, Fab12i will also refer to the evaluation data from Fab12A and introduce an assessment of the feasibility of replacing NMP by the end of 2025.

GHG Replacement Action

UMC Group continues to invest in the research and development of green processes to reduce greenhouse gas use. Through process simplification, the company has successfully developed a streamlined post-Hard Mask process applicable to the 28/22nm process platform and expandable to mature processes. This innovation further reduces N₂O usage, with estimated benefits of lowering approximately 385.4 metric tons of CO₂ equivalent emissions annually, contributing to global climate action efforts. Additionally, UMC has collaborated with ITRI to enhance the conversion efficiency of N₂O exhaust by incorporating compounds.

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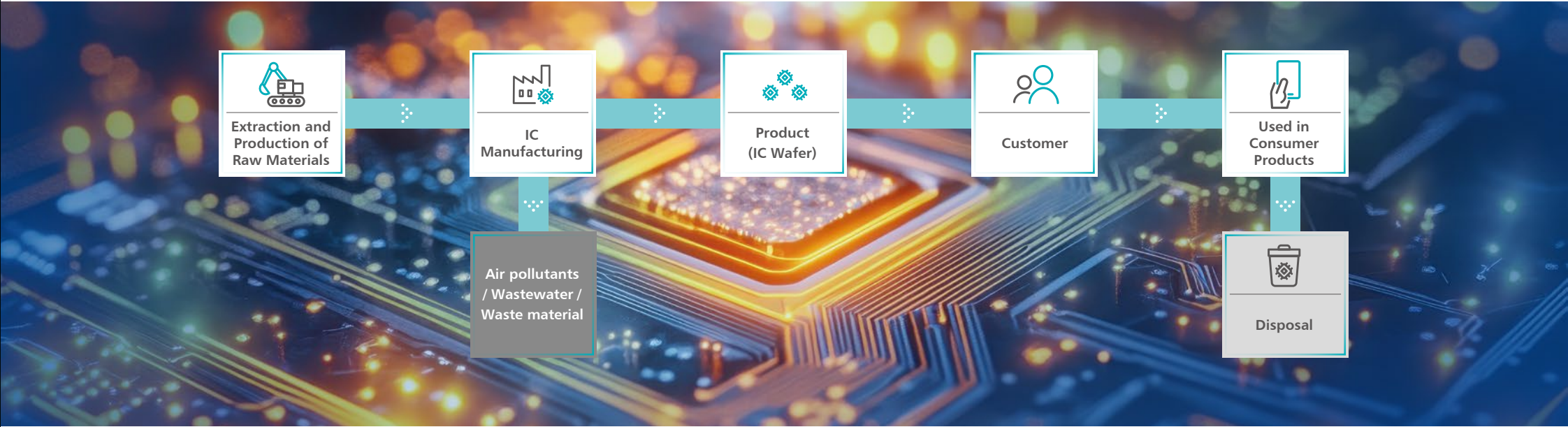
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Appendix

2.3.2 Product Environmental Impact Assessment

UMC is committed to maximizing production efficiency while also minimizing environmental impact. To achieve this goal, the Company employs various environmental assessment tools to assist in strategy formulation and implementing measures.

Product Lifecycle of Semiconductor

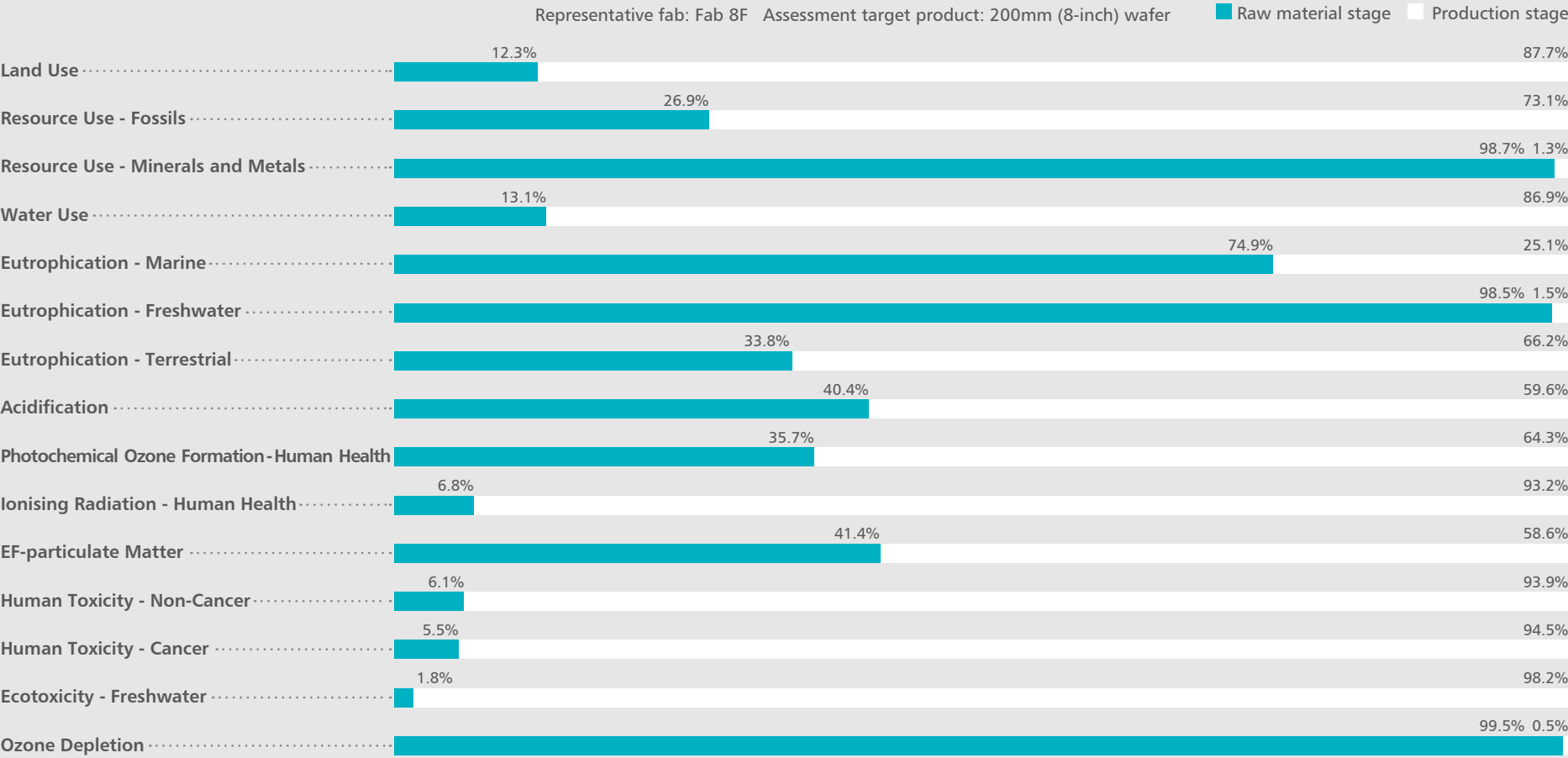


Product Lifecycle Assessment

Since 2005, UMC has implemented lifecycle assessment in all its fabs, which covers "from cradle to gate" (UMC shipment) and assesses items including energy consumption, raw materials usage, and emissions. The Company systematically and constantly monitors the environmental impact of its products through the inventory results of supply chain and manufacture process. To assist with this effort, UMC utilizes the DoITPro software with the assessment method EF3.1 to map out the environmental impact assessment by metho.

2005	2009	2010	2011	2015
Implemented Life Cycle Assessment (LCA) in all fabs	Delivered the world's first wafer with carbon footprint verification	Led the industry to complete water footprint verification for 200mm (8-inch) and 300mm (12-inch) wafer fabs	Commenced annual and comprehensive carbon footprint check throughout the Company and conducted verifications on selected fabs.	Led the industry in passing the ISO 14046:2014 Water Footprint Assessments verification of UMC's Taiwan fabs in cooperation with suppliers

Results of 2024 Environmental Impact Assessment



Product Carbon Footprint

As an upstream manufacturer in the semiconductor industry, providing customers with high-quality and eco-friendly products that comply with environmental regulations has always been one of UMC's most regarded business values. Therefore, the Company not only conducts inventory and verification of greenhouse gas (GHG) emissions in its operations, but also implements a comprehensive GHG management plan. In addition, UMC proactively promotes the product carbon footprint inventory program, following ISO 14067:2018 to complete the carbon footprint inventory of wafer manufactured in all UMC fabs, and obtaining third-party verification to identify key stages in the product lifecycle with potential for GHG reduction.

For more information on UMC's GHG achievements, please refer to Chapter 3.1 Climate Action.

Product Water Footprint

Affected by climate change and extreme weather, global water resources have become one of the most discussed topics. Water resources are critical and essential in semiconductor manufacturing processes. Following the Business Water Footprint Accounting standards developed by international non-profit organization Water Footprint Network, UMC led the industry in completing the water footprint verification of its 8-inch and 12-inch fabs in 2010. UMC carried out product water footprint inventory, targeting the 12-inch wafers produced by Fab 12A in 2024 for verification, and obtain the ISO 14046:2014 certificate.

For more water-saving measures and results, please refer to Chapter 3.3 Water Resources Management.

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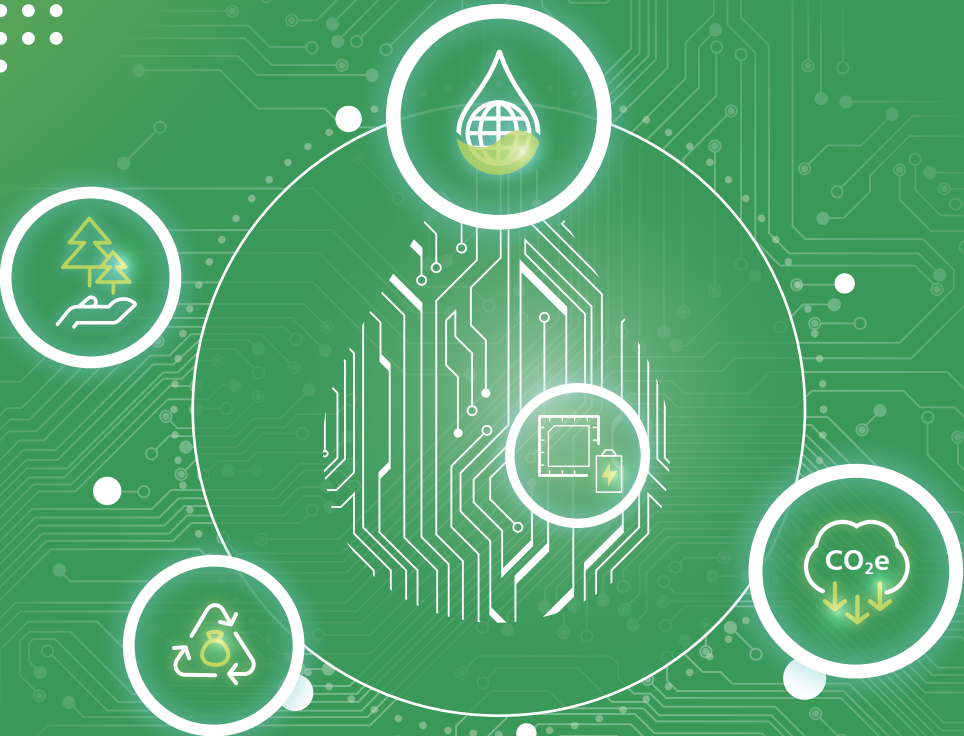
- 3.1 Climate Action
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Implementing Green Operations



- 3.1 Climate Action
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Implementing Green Operations

Important Stakeholders | Government | Suppliers | Community |

Actions are urgently needed to address climate change. UMC is committed to minimizing the environmental impact of its operations, and has set environmental goals/indicators for carbon emissions, water, and waste as part of the Company's climate change strategy. Over the years, UMC has optimized energy/resources utilization efficiency, as well as introduced eco-friendly designs to make the Company's operations more sustainable, prospering together with the environment.

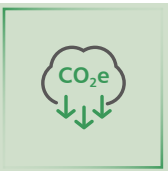
SDGs |



Performance Highlights 2024



The only semiconductor company in the world earned "Double A" in CDP's both Climate & Water rating for three consecutive years



31% absolute reduction in Scope 1 & 2 GHG emissions (base year: 2020)



Ratio of renewable energy in total electricity used 17.1%



Awarded the Gold Award, as an "Outstanding Company Performance in Resource Circulation" by the Ministry of Environment



Reclaimed water usage rate 17.2%



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UMC Group follows the environmental policy signed by the chairman and is committed to environmental protection in the long term. UMC expects to create a green operating model of "maximizing production efficiency and minimizing environmental impact" in order to achieve a harmonious win-win situation between the economy and the environment. The ESG Steering Committee, composed of the management team, supervises and manages the Company's environmental protection work. It regularly reports to the highest governance unit of the Company's major environmental strategy, the board of directors, including management and implementation results of climate change, circular economy, biodiversity, and other aspects. The Corporate Sustainability Committee under its jurisdiction is responsible for integrating and promoting the Company's environmental protection work on energy conservation, carbon reduction, water conservation, waste reduction, and other aspects. It continuously improves environmental performance and reduces environmental impact. For more information on the detailed organization and operation of the sustainability-related committees, please refer to the "Sustainable Governance" section.

In 2010, the Company launched the three-phase Green Action Plan, which has improved energy saving, water conversation, waste reduction, and greenhouse gas reduction indicators through source reduction and enhanced utilization. At the end of 2020, the Company's president revealed the fourth phase of the Plan, Green 2025, which aims to achieve 15% energy saving, 15% water saving, 25% waste reduction, and 65% reduction in fluorinated greenhouse gas (F-GHG) emissions intensity. In 2021, based on the latest sustainability trends and in combination with the Company's development direction, aforementioned goals are upgraded to 25% energy saving, 25% water saving, and 35% waste reduction. Meanwhile, these goals are also involved in UMC's Sustainability Strategy and Blueprint to track the progress and are quarterly monitored by the highest unit of the Company's ESG decision-making – the ESG Steering Committee to ensure the achievement of environmental objectives.



Vision

❖ Improve the utilization efficiency of energy and resources, while minimize the impact of using energy and resources on the environment

Project of Each Phase

❖ Phase 1 Project 333 Reduction (2010~2012) ... Goal Achieved

❖ Phase 2 369⁺ Reduction (2013~2015) Goal Achieved

❖ Phase 3 Green 2020 (2016~2020) Goal Achieved

❖ Phase 4 Green 2025 (2021~2025) Ongoing

Progress of Green 2025

	Base Year		Achievement in 2024
	2015	❖	-28.0%
	2015	❖	-30.3%
	2015	❖	-44.2%
	2010	❖	-80.3%

Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC); with * remark is UMC (the parent company only).
2. Calculation method: Please refer to the explanation in each chapter for details.

UMC Group Implementation and Performance in 2024

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Item		 Electricity Saving	 Water Saving	 Waste Reduction	 F-GHG Emissions Reduction
Main Reduction Measure		- Improvement on energy efficiency - Improvement on air conditioning system - Improvement on lighting system	- Water saving from production machine - Increase in wastewater recovery - Improvement in pure water production - Reduce Municipal water replenishment	- Reduction of waste sulfuric acid and waste solvent - Reducing the water content of waste IPA and extend the lifetime of solvent - Waste recycle and reuse	Installing high efficiency F-GHGs abatement systems
Reduction Performance	Environmental Benefi	255,530 MWh of additional electricity savings - Around 122,000 tonnes of CO₂e reduction	1,023,000 tonnes of new water savings	1,498 tonnes of additional waste reduction	UMC's annual reduction of F-GHG emissions exceeds 790,000 tonnes of CO₂e.
	Economic Benefi	Savings of NT\$1,010 million		Savings of NT\$7.52 million	Taking the example of Fab 12i located in Singapore where carbon tax is levied, the newly installed reduction equipment saved more than NT\$9 million in carbon tax costs.

Note: The statistics on water conservation are the savings achieved in the year of implementation of water-saving measures from 2015 to present. The calculation method adopts an estimated value



UMC is committed to sustainability and demonstrates determination to minimize environmental impact through alliances with green financing partners

Green Bond

UMC issued green bonds for the first time in 2021. The funds raised are designated for expenditures related to environmental protection, such as the development of renewable energy systems, the improvement of equipment efficiency, and emissions control of manufacturing processes. These projects will effectively reduce carbon emissions and waste generation, and lessening the environmental burden of the Company's operations. The interests from the green bonds will go to the "Special Fund for Environmental Protection" established by the Company in 2013. The fund supports initiatives by UMC related to environmental protection and social welfare, enabling the Company to give back to the society and the planet through meaningful actions.

Sustainability Linked Loans

UMC signed sustainability linked loan agreements with CTBC. The loan structure has been reviewed by financial institutions and followed the "Sustainability-Linked Loan Principles" formulated by the International Capital Market Association (ICMA), which is in line with several indicative key performance targets of UMC for sustainability. Through collaboration with financial institutions, the Company strives to minimize its environmental impact and advance its sustainability vision.

Item	2024 Goal	Performance
GHG Emissions Management	35% reduction for GHG emission per unit product compared to the base year 2010	-57%

Note: 1. Includes direct emission (Scope 1) and indirect emission from purchased energy (Scope 2)
2. Scope: UMC Group (UMC and its subsidiaries)

3.1 Climate Action

- 3.1.1 Net Zero Emissions Commitment and Targets
- 3.1.2 GHG Emissions Management
- 3.1.3 GHG Emissions Reduction

★ Material Topic 3.1 Climate Strategy and Action

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

UMC Climate Change Policy



Positive Impact 3.1 Climate-change-related opportunities Negative Impact 3.1 Climate actions failure

Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
Positive: In response to climate change-related risks and opportunities, set short, medium, and long-term goals and action plans to mitigate greenhouse gas emissions or enhance climate resilience through adaptation strategies.	- Promote the "3 Resolutions to Achieve Net Zero": Voluntary carbon emissions reductions in manufacturing processes; increase use of renewable energy; investment and participate in carbon offset projects. - Join organizations such as the RE100 initiative, Association of Taiwan Net Zero Emissions, and Climate Partnership to stay on top of global climate issue development trends. - Promote a sustainable supply chain, and reinforce supply chain resilience through joint carbon-reduction action. For details on the sustainable supply chain, please refer to Chapter 1.3 Building Sustainable Supply Chain.	- Report the progress of performance by the "The Environmental Management and Planning Group" to the Corporate Sustainability Committee for review quarterly. - Regularly participate in association meetings; engage in exchange with industrial/governmental/academia/research organizations.

Negative: If greenhouse gas emissions are not effectively managed, it will contribute to accelerating global warming.	- Conduct annual ISO 14064-1 GHG inventories and promote ISO 14067 product carbon footprint inventories in all fabs in order to identify emissions hotspots and areas with reduction potential. - Sign and support the Task Force on Climate-Related Financial Disclosures (TCFD) initiative; conduct annual TCFD project on risks and opportunities arising from climate change, and from this adopt further management and response measures. - Set science-based targets, and practically plan and implement carbon reduction pathways via official approval. In addition, set GHG reduction and renewable energy utilization indicators, and conduct quarterly reviews. - Through annual Sustainability Reports, externally disclose response measures and progress toward objectives to stakeholders.	- Annual third-party certifications: ISO 14064, ISO 14067, and TCFD. - Response to stakeholders: Respond to customers'/investors' questionnaire surveys on climate issues. - Participate in international CDP Climate Project for transparency.
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Sustainability Strategy and Blueprint		2024		2025	2030
Item	Baseline	Goal	Achievement	Goal	
Reduction of GHG emissions per unit product (base year: 2010) ^{Note1}	17.4	48%	✓ 56.7%	55%	60%
Absolute reduction of GHG emissions (base year: 2020) ^{Note2}	Scope 1 & Scope 2	288.3 ^{Note3}	✓ 31.0%	42%	
	Scope 3	224.5	✓ 23.7%	25%	

Note: 1. Unit: tCO₂e/wafer-m²
2. Unit: 10,000 tCO₂e
3. Considering changes in the disclosure scope, simultaneously adjust the baseline emissions for Scope 1 and Scope 2.
4. Scope: UMC and all subsidiaries.

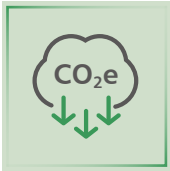
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3.1.1 Net Zero Emissions Commitment and Targets

★ Performance Highlights 2024



Introducing internal carbon pricing, with a price of USD\$ 100 per tCO₂e.

Obtained 261,000 tonnes of carbon reduction credit through the offset project approved by Taiwan Ministry of Environment

Awarded the highest rating, "A List", on CDP Climate Change category.

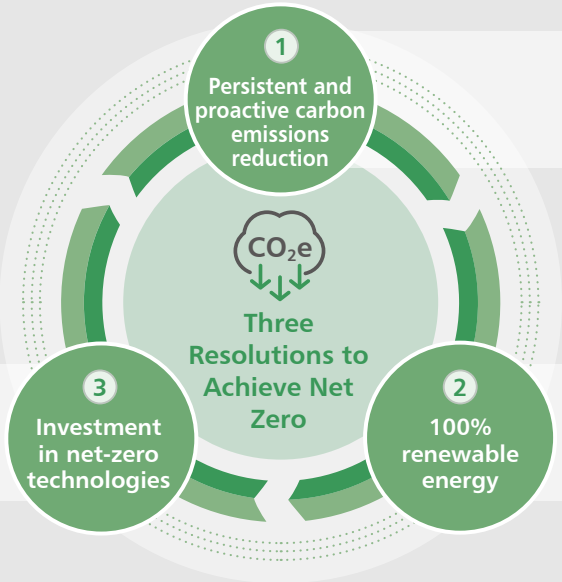
UMC is a pioneer in carbon reduction among companies in Taiwan, initiating a program back in 1999 to decrease the use of fluorinated greenhouse gases (F-GHGs). In 2021, UMC took its climate commitment to the next level by pledging to achieve net zero emissions by 2050.

For details on the risks and opportunities of climate change, please refer to Chapter 1.2.5 Managing Climate Risks and Opportunities.
For details on the supply chain carbon reduction, please refer to Chapter 1.3 Building Sustainable Supply Chain.

Three Resolutions to Achieve Net Zero

Over the past two decades, UMC Group has carried out several programs for carbon reduction. For the next 30 years, UMC will continue to strive for its Science-Based Targets (SBT) by enhancing its process technologies, increasing the percentage of renewable energy usage, and investing in emerging carbon reduction measures.

To offset carbon emissions that cannot be avoided, UMC Group will invest in net zero emissions technologies and participate in carbon offset projects.



Through continually developing advanced foundry process technologies, enhancing productivity and improving energy efficiency, UMC Group can minimize carbon emissions in both the manufacturing stage of wafers and usage of IC end products.

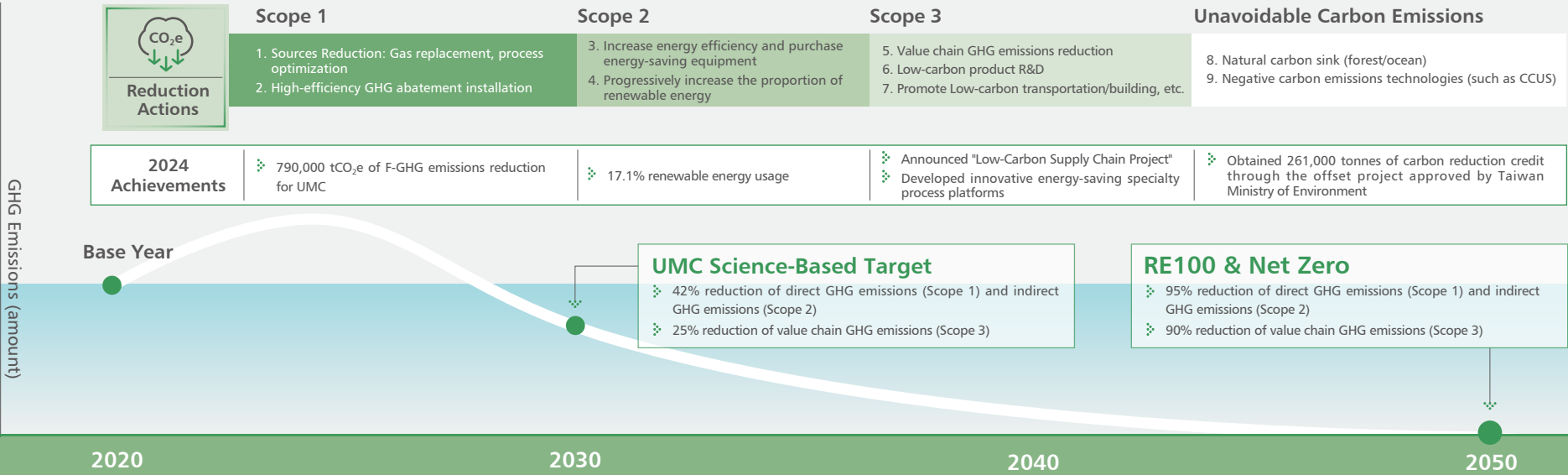
Support the low-carbon energy transition with actual actions through joining international renewable energy initiative RE100 as a member and setting mid-/long-term goals to boost the usage of renewable energy.

In addition, UMC Group also participates in organizations such as Taiwan Climate Partnership, Association of Taiwan Net Zero Emissions, Taiwan Carbon Capture Storage, and Utilization Association, in hope that through the collaboration and exchange with academia-industry-government agencies, solutions can be found to reinforce the resources and actions required to cope with climate change and facilitate the implementation of net zero transition.

SBTi Net Zero Target Commitment

Following the approval of UMC Group's 2030 absolute GHG reduction target by the Science-Based Targets Initiative (SBTi) in 2022, UMC Group further accelerated its carbon reduction efforts by signing the SBT net-zero target commitment in 2023. In accordance with the SBT net-zero standards, UMC Group adjusted its 2030 GHG reduction targets to reduce direct (Scope 1) and indirect (Scope 2) carbon emissions by 42% with 2020 as the base year by 2030. Additionally, a 25% reduction in carbon emissions within the value chain (Scope 3) is targeted. By 2050, UMC Group will reduce at least 90% of emissions and neutralize or offset the remaining unavoidable carbon emissions to achieve the net-zero target.

UMC Group Net Zero Blueprint



SBT 2030 Reduction Target

To achieve the SBT 2030 reduction target (based on 2020, a 42% reduction in Scope 1 & Scope 2 by 2030, and a 25% reduction in Scope 3), UMC Group has formulated corresponding reduction measures and estimated reduction proportions for each scope. Due to the significant differences in the emission proportions of various categories within Scope 3, we only focus on planning reductions for categories with larger emission proportions to ensure effective resource allocation and maximize carbon reduction effects. These measures not only align with global climate action trends but also demonstrate our commitment to the SBT 2030 reduction target and lay a solid foundation for future net-zero goals.

Scope	Reduction Measures	Estimated Contribution (%)
Scope 1 & Scope2	Install High-efficiency GHG abatement equipment	5~10%
	Increase the proportion of renewable energy usage	25~35%
	Install solar panels and implement energy-saving projects	2~5%
Scope 3	Conduct carbon footprint inventory and reduction for raw materials	5~10%
	1. Reduce unnecessary capital expenditures and increase the proportion of low-carbon, energy-saving capital expenditures	10~15%
	2. Reduce investments in companies or products with high carbon intensity	
	Increase the proportion of renewable energy usage, install solar panels, and implement energy-saving projects	2~5%

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3.1.2 GHG Emissions Management

All UMC Group fabs have carried out GHG inventories for many years. We will continue to expand the scope of GHG management by initiating carbon emissions inventories in UMC Group's non-wafer foundry subsidiaries. Through this, carbon emissions hotspots can be controlled, and analysis can be carried out to develop carbon reduction countermeasures. In addition, through the introduction of internal carbon pricing mechanisms and establishing a carbon management platform, links between daily operations and carbon reduction actions can be strengthened, and GHG emissions can be continuously reduced.

For details on the related information of energy efficiency improvement and renewable energy utilization, please refer to Chapter 3.2 Energy Management

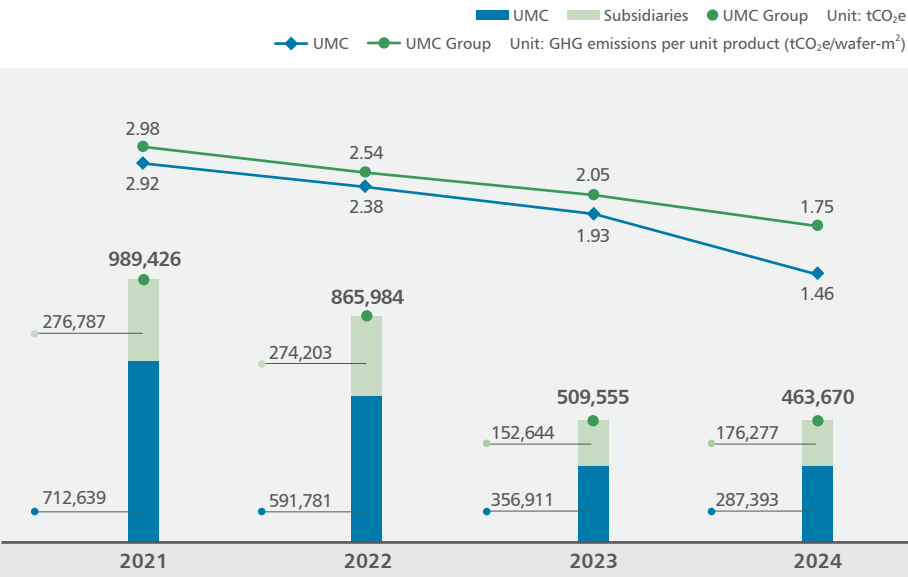
Greenhouse Gas Inventory

UMC Group established its GHG inventory standard mechanism in accordance with inventory guidelines defined by domestic and international organizations such as the ISO14064-1:2018, World Resources Institute's Greenhouse Gas Protocol and GHG Inventory Guidance in Taiwan. Since 2006, the Company has inventoried the GHG emissions of all fabs annually to verify the effectiveness of reduction measures. Furthermore, UMC Group's consolidated 27 subsidiaries all introduced the GHG inventory mechanism and conduct third-party verification to fully determine the status of GHG emissions.

In 2024, UMC Group's Scope 1 direct GHG emissions totaled 463,670 tCO₂e, and Scope 2 indirect GHG emissions amounted to 1,523,855 tCO₂e. The statistics above is calculated in accordance with ISO 14064-1: 2018 and the results had been verified by a third party. The emissions per unit product defined by the UMC Green 2025 indicator was 1.75 tCO₂e/wafer-m² and 5.76 tCO₂e/wafer-m², respectively for Scope 1 and Scope 2. Due to the effectiveness of GHG reduction, per unit product emissions of Scope 1 and Scope 2 decreased compared with the previous year.

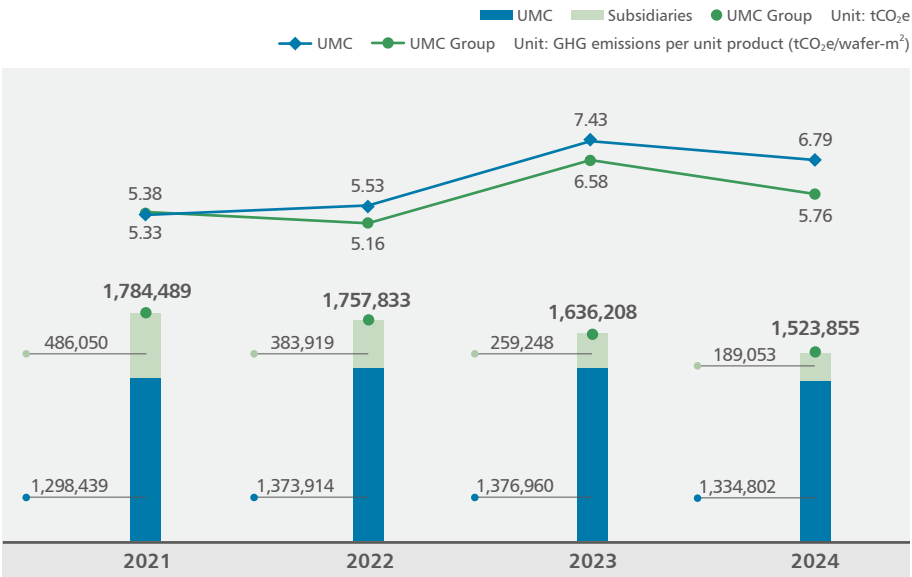
Note: After conducting an inventory, excluding the four foundry subsidiaries (HJ, USCXM, Wavetek, and USJC), other non-wafer foundry subsidiaries accounted for less than 5% of the total emissions reported in the consolidated financial statements, so based on materiality criteria, they are not included in the scope of disclosure.

Direct GHG Emissions (Scope 1)



Note: The approach for consolidating our GHG inventory is operational control. Global warming potentials (GWPs) used in the calculation is according to the definitions in the IPCC Fifth Assessment (IPCC AR5) (2014), and the GHG emissions is based on the Ministry of Environment Greenhouse Gas Emissions Factor Management Table.

Indirect GHG Emissions from Purchased Energy (Scope 2)



Note: The approach for consolidating our GHG inventory is operational control. CO₂e emissions are estimated based on the latest local power factor available in the year.

1. Taiwan: 0.494 kgCO₂e/kWh;
2. Singapore: 0.412 kgCO₂e/kWh;
3. China: 0.5366 kgCO₂e/kWh;
4. Japan: Mie 0.421 kgCO₂e/kWh, Yokohama 0.431 kgCO₂e/kWh

To account for other indirect GHG emissions (Scope 3), UMC Group referred to "Corporate Value Chain (Scope 3) Accounting and Reporting Standard" published by GHG Protocol and ISO14064-1:2018 to estimate the amount of Scope 3 GHG emissions, which has passed third-party verification. Due to the decrease in emissions from purchased goods and services, capital goods, and fuel-and-energy-related activities in 2024, Scope 3 emissions was reduced compared to the previous year.

Other Indirect GHG Emissions (Scope 3)

Category	GHG Emissions (tCO ₂ e)
 Purchased goods and services	656,013
 Capital goods	424,251
 Fuel-and-energy-related activities	370,697
 Upstream transportation and distribution	52,229
 Waste generated in operations	31,025
 Business travel	3,255
 Employee commuting	11,652
 Downstream transportation and distribution	3,267
 Downstream leased assets	9,418
 Investments	151,700
Total Amount	1,713,507

Note: 1. Scope: UMC and all subsidiaries
2. Global warming potentials (GWPs) used in the calculation is according to the definitions in the IPCC Fifth Assessment (IPCC AR5) (2014), and the GHG emission factors referred to the Ministry of Environment Greenhouse Gas Emissions Factor Management Table, Taiwan Ministry of Environment Carbon Footprint Calculation Platform, supplier-specific factors, and SimaPro database, etc.
3. The categories that are not applicable or have no emissions are not included in the table above.

Carbon Management Mechanism

UMC Group has integrated carbon management into our operational strategies. This includes introducing internal carbon pricing mechanisms, a carbon management platform, mechanisms to link ESG performance with employee remuneration, and more. Through these measures, we seek breakthroughs in carbon reduction.

In response to global carbon pricing trends, UMC Group has set our internal carbon prices, which is a shadow price, at 100 USD per tonnes by considering prices prevailing in global carbon markets, carbon prices stipulated in GHG regulations, and internal carbon reduction costs of Scope 1, 2 and 3. UMC group applies the shadow price to decision-making processes related to Scope 1, 2, and 3. These also serve as a reference for our carbon reduction management and planning. Through the carbon pricing mechanism, UMC Group takes advance response measures for internal/external carbon reduction pressures and risks arising from factors such as Singapore's carbon tax (UMC's Fab 12i is located in Singapore); Taiwan's future carbon fee levied on emission sources in accordance with the Regulations Governing the Collection of Carbon Fees; carbon emission supervision mechanisms being adopted by countries around the world; and requirements to increase the share of renewable energy utilization, as regulated by the Renewable Energy Development Act.

Furthermore, internal carbon pricing introduces the concept of carbon valuation incorporating carbon pricing into the return on investment (ROI) considerations for capital expenditures on Scope 3 capital goods. This influences the financial decisions related to capital expenditure budgets, reducing the purchase of high carbon-intensive products, promoting the replacement of high-energy-consuming systems and fostering more carbon reduction projects. By aligning capital expenditure control through carbon pricing with the Paris Agreement's goal of limiting global warming to 1.5°C, it also reflects the consideration of climate change issues in decision-making. Through the carbon pricing mechanism, UMC identifies opportunities for low-carbon practices, driving improvements in energy efficiency, reducing fluorinated greenhouse gas (F-GHGs) in processes, and investing in low-carbon initiatives such as carbon capture and storage.

Additionally, through the carbon management platform, we systematically integrate the emission information of the entire group, linking data from over 1,000 machines. The data update frequency has been increased from annually to monthly. This allows us to obtain the carbon management information in real-time and accelerate our response to production conditions. This platform comprehensively monitors various greenhouse gas reduction projects by ranking machine carbon emissions and comparing carbon cost expenses. UMC already planned to improve the number of high-efficiency greenhouse gas abatement devices (Local Scrubber) by 2030, proactively addressing carbon fee policies and reducing future carbon fee charges.

In order to encourage management, executive teams, and employees as a whole to achieve GHG reduction goals and improve energy and resource efficiency, major program objectives have been included within UMC's operating performance indicators. These serve as a basis for employee performance appraisals and bonuses, helping realize the spirit of accountability management. Since 2021, we have also established a new mechanism that links ESG performance with employee remuneration. Based on this, employees receive a certain proportion of additional bonuses if overall company target performance is achieved.

With stimuli from the aforementioned internal carbon pricing and other mechanisms, as well as planting sustainability deeply within UMC's culture, employees and subsidiaries are encouraged to work together to step by step realize UMC Group's net zero emission pathways and targets.

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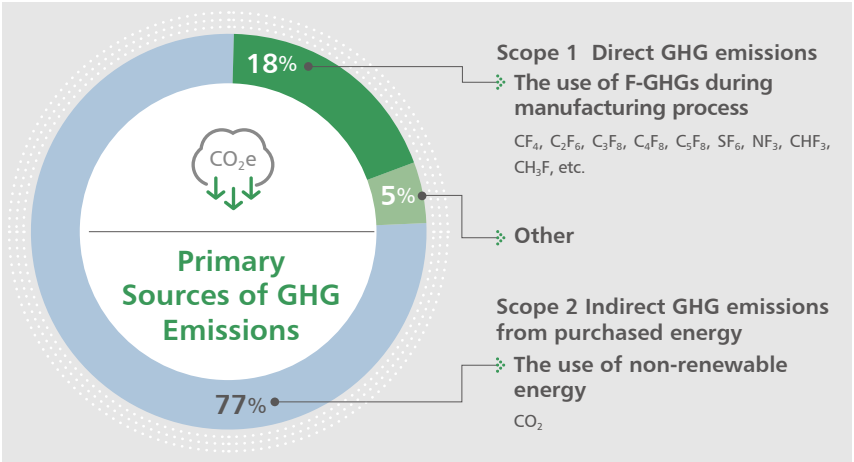
3.1.3 GHG Emissions Reduction

Based on the result of the GHG inventories, the Company learned that the main GHG sources are the fluorinated greenhouse gases (F-GHGs) used in the manufacturing process in Scope 1, and from purchased electricity in Scope 2. These emissions account for about 95% of UMC Group Scope 1 and Scope 2 GHG emissions. Therefore, GHG reduction of both F-GHGs and purchased electricity have become the priority for UMC.

Through low-carbon production processes, UMC produces products with relatively low carbon footprints. All UMC facilities have implemented reduction plans, and all products use low-carbon processes. Additionally, UMC continues to develop low-power consumption products through the research and development of energy-saving innovative process platforms, which helps avoid carbon emissions during the product end-use phase.

In 2024, based on actual production capacity, UMC Group's Scope 1 and Scope 2 greenhouse gas emissions presented an absolute reduction of 31% as compared to the base year (2020). Even under full capacity scenarios, a reduction of 17% is still achievable, demonstrating the Group's commitment to decarbonization regardless of capacity fluctuations, and making a tangible contribution to global greenhouse gas reduction.

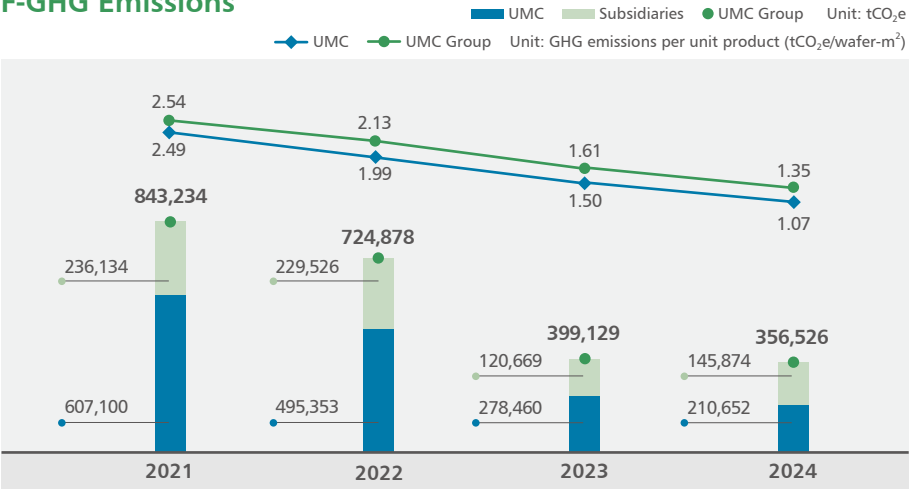
Note: For details on the related information of indirect GHG emissions from purchased electricity (Scope 2) or emissions reduction during the use phase of end products, please refer to related chapters: 3.2 Energy Management, 2.1 Innovation and Technology Development.



F-GHGs Reduction

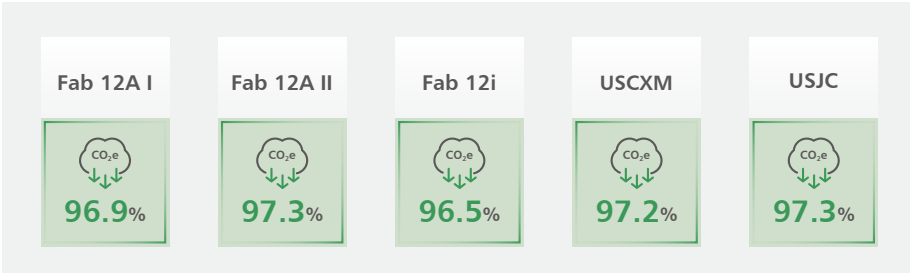
In 2024, UMC Group's F-GHG emissions amounted to 356,526 tCO₂e, which was lower than the emissions in 2023. UMC Group has continually conducted reduction measures such as installing high performance local scrubbers for F-GHGs in all new equipment, and utilizing NF₃ gases in new CVD equipment to keep emission intensity in a downward trend. In addition, based on the Electronic Product Environmental Assessment Tool (EPEAT) 1680.1:2018 standards, jointly created by the US Environmental Protection Agency (EPA) and Institute of Electrical and Electronics Engineers (IEEE), the overall F-GHGs reduction rates achieved at all 12-inch wafer fabs of UMC and its subsidiaries were all higher than the standard threshold requirement of 75%.

F-GHG Emissions



Note: F-GHG emissions was calculated by the methodology of IPCC tier2c and was verified by a third party.

F-GHGs Reduction Rates Achieved at 12-inch Wafer Fabs



Note: 1. The reduction rates were calculated using the formulae in the Electronic Product Environmental Assessment Tool (EPEAT) 1680.1:2018 standards, jointly created by the US EPA and the Institute of Electrical and Electronics Engineers (IEEE). These calculations apply only to processes at 12-inch wafer fabs.

2. The F-GHG emissions were calculated based on the IPCC 2019 tier2c approach, and the GWPs used in the calculation were obtained by adopting the definitions in the IPCC's Fifth Assessment Report (IPCC AR5) (2014). All data has been verified by third-party assurance agencies.

3. The plasma etching and chamber cleaning processes were included in this calculation. The approach used in reduction calculations was Default and Measured DREs, excluding calculation of fluorinated heat transfer fluids.

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As a pioneer in carbon reduction among companies in Taiwan, UMC established the Fluorinated Greenhouse Gas Reduction Taskforce in 1999, and continued to implement several phased green initiatives – including the 333 Reduction Plan (2010–2012), the 369+ Reduction Plan (2013–2020), the Green 2025 Reduction Plan (2021–2025), etc. – to reduce fluorinated greenhouse gases (F-GHGs) generated in semiconductor manufacturing processes.

F-GHGs Reduction Current Status

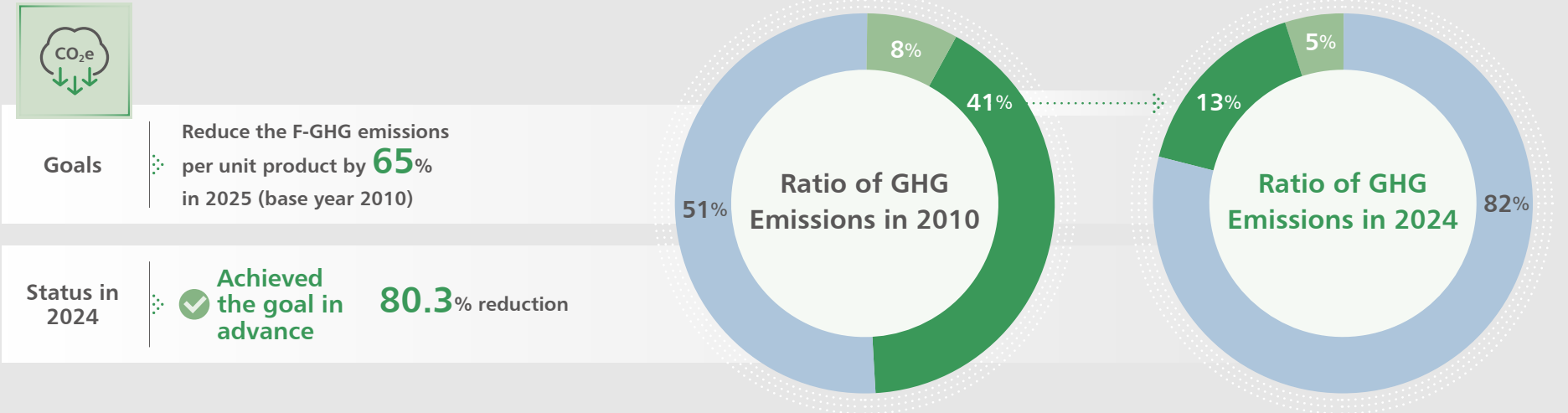
Phase 1 Goal Achieved	Phase 2 Goal Achieved	Phase 3 Goal Achieved	Phase 4 Achieve
1998	2010~2012	2013~2020	2021~2025
Comply with TSIA initiatives Decrease total annual F-GHG emissions in Taiwan fabs by 0.216 MMTCE	333 Voluntary Reduction Plan Compared to 2009, reduce F-GHG emissions per unit product by 33% in 2012	369+ Voluntary Reduction Plan Compared to 2010, reduce F-GHG emissions per unit product by 36% in 2020	Green 2025 Action Plan Compared to 2010, reduce F-GHG emissions per unit product by 65% in 2025

Note: The timeline for voluntary F-GHGs reductions in the 369+ Reduction Plan was 2013–2020, so the Phase 4 reduction actions and goals were specified in the Green 2025 Plan.

Through these measures, emission of F-GHGs has been de-linked from production capacity and changed the proportion of each greenhouse gas emission source. Moreover, UMC's F-GHGs substitution strategies are widely emulated by peer companies in the industry; together with the industry, we are making breakthroughs in carbon reductions.

In 2024, due to the installation of high-performance local scrubbers and the promotion of optimal treatment efficiency, the local scrubbers achieved a reduction of 790,047 tCO₂e. UMC's F-GHG emissions per unit product have decreased by 80.3% since 2010, achieving the 2025 target ahead of schedule.

F-GHGs Reduction Current Status



Note: The scope is UMC

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Carbon Assets and Carbon Trading

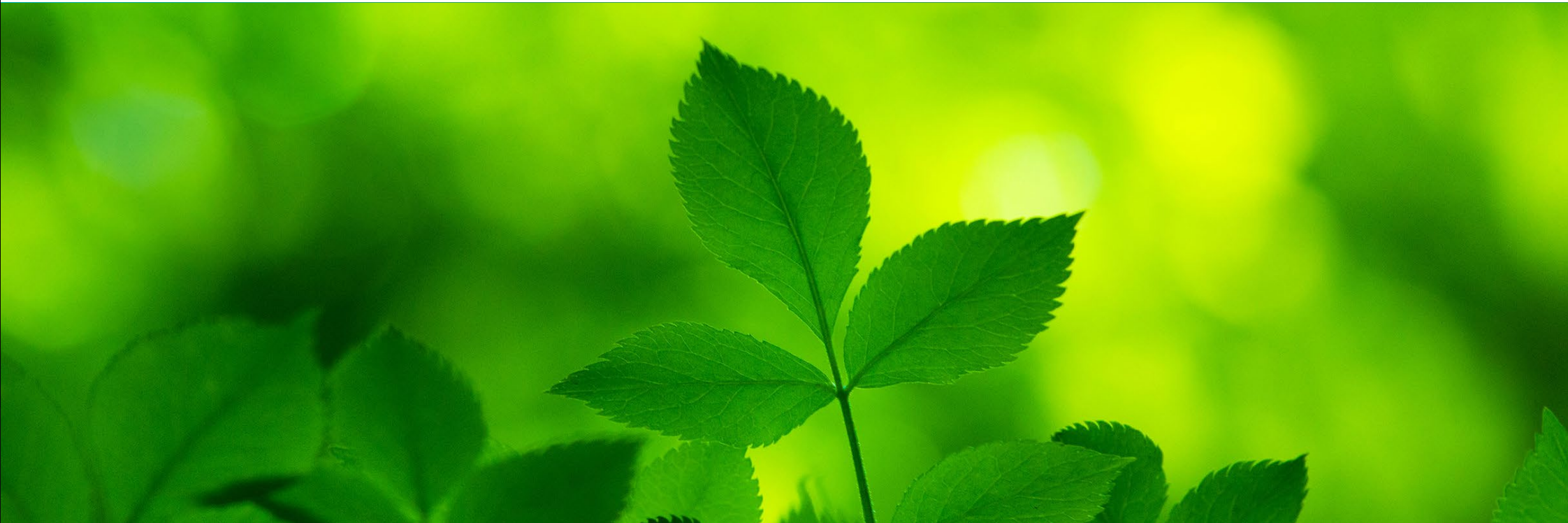
UMC has actively diversified its carbon assets portfolio. Following the first carbon trading transaction recognized by the Ministry of Environment in 2014, UMC participated in the first batch of carbon credit transactions at the Taiwan Carbon Solution Exchange (TCX) in 2023. There, UMC purchased international carbon credits with multiple sustainability benefits, marking a significant milestone in the domestic carbon emission trading market in Taiwan.

UMC's purchase of international carbon credits by UMC will finance Clean Water programs, which focus on assisting local residents to obtain clean drinking water sources by restoring wells and related facilities in African countries. It contributes to carbon reduction by eliminating the need to burn wood for water boiling and sterilization. Additionally, it ameliorates the plight of women in rural areas, who must otherwise devote considerable time and energy searching for clean water sources. Meanwhile, this purchase achieves multiple United Nations sustainable development goals (SDGs), including Good Health and Well-being, Gender Equality, Clean Water and Sanitation, and Climate Action.

As the carbon emission trading market matures, UMC has established a carbon credit trading and offset procedures, standardizing internal carbon credit trading and offset processes. UMC have screened criteria for carbon credit purchasing, incorporating verification standards, issuance years, additionality, stability, and other benefits into the assessment criteria. We hope to exert a sustainable impact by supporting carbon reduction projects, acquiring high-quality carbon credits, further promoting low-carbon production technologies and the development of innovative industries, and driving the green economic cycle.

UMC actively responds to the government's carbon reduction policies. As early as 2013, UMC participated in the Taiwan Ministry of Environment's "Early Reduction Project" and acquired a carbon reduction credit of 3.02 million tonnes. In 2020, UMC participated in Ministry of Environment's "greenhouse gas offset project" based on the approach of "Emission Reduction Methodology for Semiconductor Industry Fluorinated Greenhouse Gases and N₂O Abatement Technology." Through this project, the GHG reduction in UMC's fabs in Taiwan can be used to establish a 10-year carbon asset. UMC successfully obtained approximately 120,000 tonnes of carbon reduction credits for the first year in March 2024, and approximately 141,000 tonnes for the second year in September 2024. Currently, UMC holds a total of 261,000 tonnes of carbon reduction credits from the project. These carbon reduction credit can be widely used for various purposes. UMC will continue to apply for and assess the usage of the obtained carbon emission quotas.

Moreover, UMC is also paying constant attention to different countries' carbon credit markets and trading platforms, participate in carbon trading or develop carbon reduction projects to stay on top of market trends and make early deployments. By doing this, we proactively prepare for future carbon fees, tax offsets, and even carbon credit investments, to address the risks and opportunities arising from climate change.



3.2 Energy Management

3.2.1 Energy Mix 3.2.2 Energy Efficiency Improvement

★ Material Topic 3.2 Energy Management

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Positive Impact ❸ Renewable energy deployment		Negative Impact ❸ Energy/resource consumptionfailure
Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
Positive: Improve energy efficiency, increase the use of renewable energy, and set goals and measures to reduce the use of fossil fuels	❖ Install solar energy systems ❖ Purchase renewable energy and attend to needs of customers	❖ Monthly update related KPIs and quarterly report the progress of performance by the "Environmental Management and Planning Group" the Corporate Sustainability Committee for review.
Negative: If energy efficiency is not effectively controlled, it will increase energy consumption	❖ Replace old and inefficient equipment ❖ Implement energy saving measures through internal assessment ❖ Introduce ISO 50001 Energy Management System	❖ Monthly update related KPIs and quarterly report the progress of performance by the "Environmental Management and Planning Group" the Corporate Sustainability Committee for review. ❖ Promote energy saving projects and experience sharing by Energy Saving Committee. ❖ ISO 50001 verified by a third party

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Reduction of electricity consumption per unit product (base year: 2015) ^{Note2}	14%	✓ 28%	25%	30%
Cumulative energy savings (base year: 2015)	770 GWh	✓ 1,007 GWh	818 GWh	1,000 GWh
Ratio of renewable energy in total electricity used ^{Note3}	13%	✓ 17.1%	25%	50%

Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC)
2. The performance is a comparison of the performance under the BAU scenario with the baseline value; the baseline value is calculated from 2015 as the base year and includes the annual data of the foundry subsidiaries.
3. Ratio of renewable energy in total electricity used = renewable energy used ÷ total electricity used

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3.2.1 Energy Mix

In order to lessen environmental impact of greenhouse effect and minimize energy consumption, UMC has established company-wide carbon reduction goals and development plans through the Corporate Sustainability Committee, which coordinates and integrates energy saving and carbon reduction strategies and initiatives of various divisions. Regular committee meetings are held to review progress. The Company continually introduces energy saving technologies and implements energy efficiency improvement programs in all facilities. In addition to its strategy of energy savings, increasing energy efficiency, and expanding installation of renewables, UMC also promotes the implementation of energy management in its offices and common areas through activities, education, and training to cultivate a mindset and habit of energy conservation and GHG emission reduction among employees

Bulk purchases of electricity dominate UMC Group's energy usage, accounting for 77.9% of the total consumption, followed by natural gas (including LPG) which account for 5.6% of the total. Therefore, energy saving is mainly aimed at reducing the use of bulk electricity and natural gas. UMC Group's total energy consumption in 2024 was 4,129,707 MWh. Of the total, electricity consumption was 3,218,294 MWh, while natural gas consumption was 230,382 MWh. The electricity consumption per unit product was 12.2 MWh/wafer-m², while the natural gas consumption per unit product was 0.9 MWh/wafer-m².

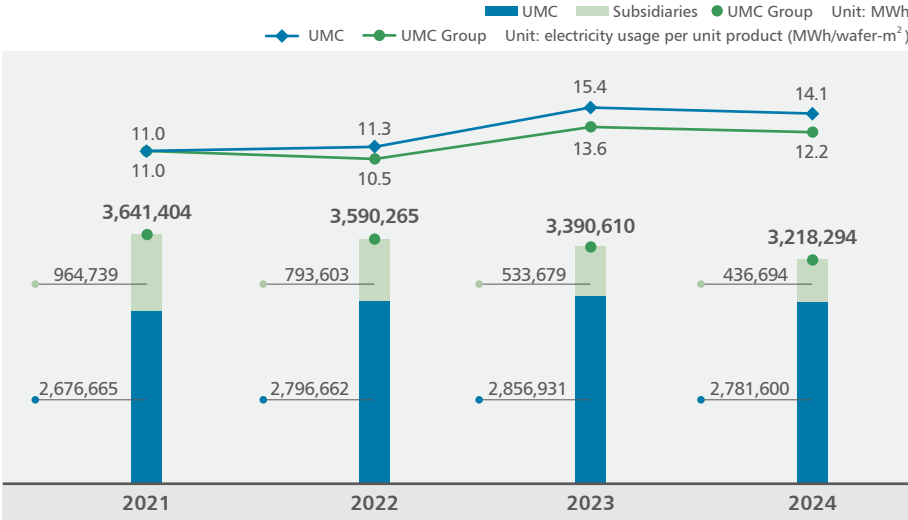
Total Energy Consumption in 2024

Unit: MWh

		UMC	UMC Group
Non-renewable Energy	Purchased Electricity	2,781,600	3,218,294
	Nature Gas	166,043	230,382
	Diesel, LPG and Other Purchased Energy	15,364	15,777
Total Non-renewable Energy Consumption (a)		2,963,007	3,464,453
Total Renewable Energy (b)		122,407	665,254
Total Energy Consumption (a+b)		3,085,414	4,129,707

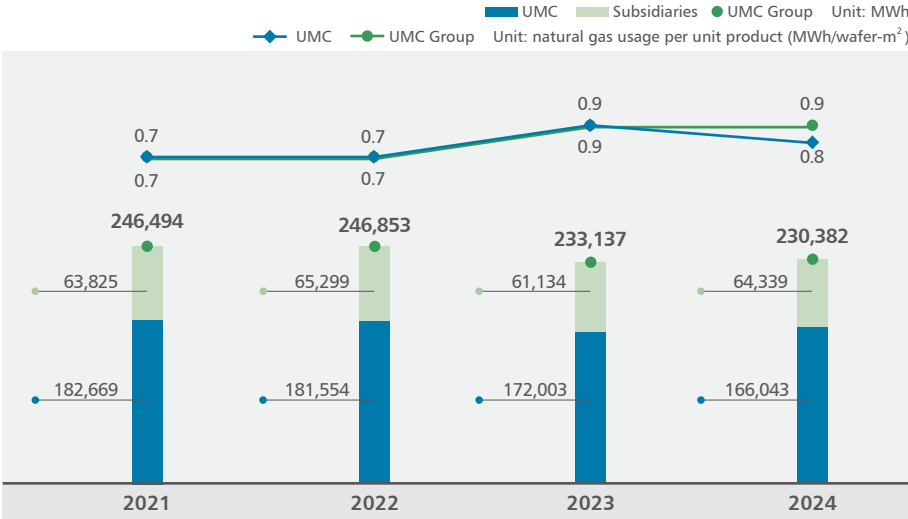
- Note: 1. UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.
2. The sources of renewable energy are self-generated solar energy, renewable energy purchased (such as PPA) and RECs.
3. Data for natural gas and diesel is calculated based on the heating value(s) provided by the Energy Administration, MOEA.
4. The total energy consumption in 2024 was about 14,866,944 GJ; the percentage of grid electricity is >90%; the percentage of renewable energy in total energy consumption is 16.1%.

Purchased Electricity Consumption



- Note: 1. UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.
2. Renewable energy usage was excluded.

Natural Gas Consumption



- Note: 1. UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.
2. Data for natural gas is calculated based on the heating value(s) provided by the Energy Administration, MOEA.

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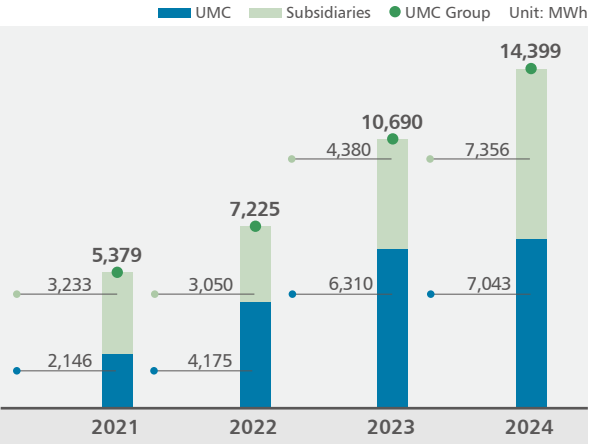
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Promotion of Renewable Energy

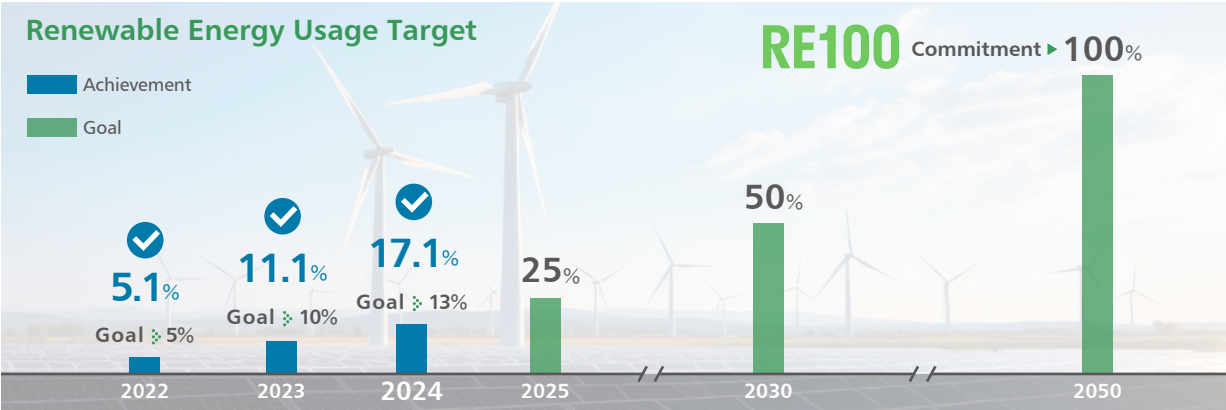
UMC Group is enhancing energy efficiency and diversifying energy use by actively installing renewable energy facilities in its fabs. Solar power systems are now a standard feature in new fabs, with power generation increasing for five consecutive years. The original plan to install 500kW of solar power in 2024 has been adjusted and consolidated into a total of approximately 900kW to be installed in 2025. By the end of 2024, UMC Group had installed over 14,000 kWp of solar power systems, with an estimated annual generation of 15,000 MWh. However, solar power generation is affected by climate factors, such as the multiple typhoons in Taiwan in 2024, making this an estimated, not fixed, value. In 2024, UMC Group's solar power systems generated 14,399 MWh for self-use. From the end of 2024, all solar power will be used for self-consumption, and the Group will continue to increase self-generated green electricity to support clean energy.

Self-use Solar Power Generation



UMC Group joined RE100 as a member in 2021, becoming the second semiconductor wafer foundry in this international renewable energy initiative. The Company is committed to using 100% renewable energy by 2050. In 2024, UMC Group achieved a renewable energy usage ratio of 17.1%. In addition to maximizing the installation of photovoltaic systems at its Taiwan facilities, UMC Group also procured over 650,000 MWh of renewable energy. The Company continues to actively support and promote energy transition policies, having recently signed a power purchase agreement (PPA) for offshore wind energy. This agreement will secure over 30 billion kWh of renewable energy over the next 30 years, steadily advancing towards UMC's goal of 50% renewable energy usage by 2030.

In the future, UMC will continue to increase the proportion of renewable energy usage through various actions such as expanding power purchase agreements, participating in the construction of green power plants, and evaluating the adoption of emerging renewable energy technologies. By diversifying its supply sources and collaborating with value chain partners, UMC aims to drive the robust development of the green energy industry, gradually achieving the RE100 and 2050 net-zero emission targets.



UMC Signs 30-Year, 30 Billion kWh Offshore Wind Power Agreement with CIP's Fengmiao I Offshore Wind Farm

UMC signed a Corporate Power Purchase Agreement (CPPA) with Fengmiao I Offshore Wind Farm ("Fengmiao I"), developed by Copenhagen Infrastructure Partners' ("CIP") flagship fund, CI V. UMC will purchase more than 30 billion kilowatt-hours of power from Fengmiao I over least 30 years, marking the largest renewable energy transaction in UMC's history. This agreement will help UMC Group achieve its goal of 50% renewable energy use by 2030 as part of the Company's roadmap to net zero emissions and 100% renewable energy by 2050.



3.2.2 Energy Efficiency Improvement

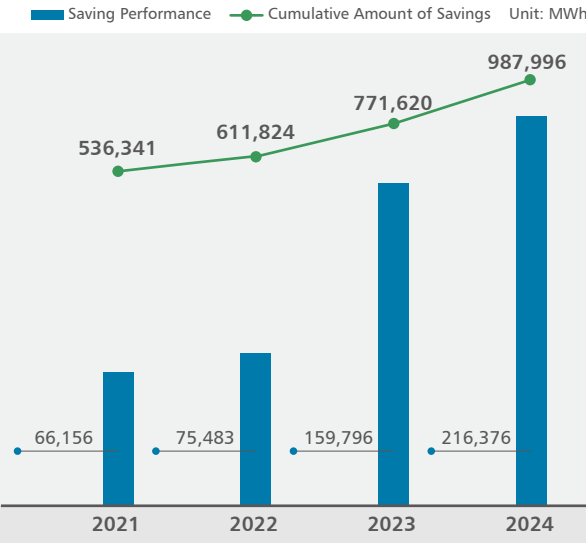
UMC continues to improve the energy utilization efficiency of all fabs through various energy-saving measures. In 2024, the total electricity and natural gas saved was 222,798 MWh, which is equivalent to a reduction of 103,409 tonnes of CO₂e emissions.

Major Energy Conservation Measures in 2024

Performance in 2024	
Electricity - Energy saving measures for production machine and water treatment systems - Energy saving measures for cold water and AC systems - Compressed gas reduction to conserve energy	- AC units clean-up for energy conservation - Energy saving measures for production process cooling water - Energy efficiency in water and wastewater treatment systems - Energy saving measures for illumination 8.6% reduction of electricity usage 216,376MWh of electricity savings 102,112 tonnes of CO₂e reduction
Natural Gas (NG) - HDI temperature optimization - VOC upgrading - Boiler upgrading	2.53% reduction of NG usage 6,422 MWh of NG usage saving 1,297 tonnes of CO₂e reduction

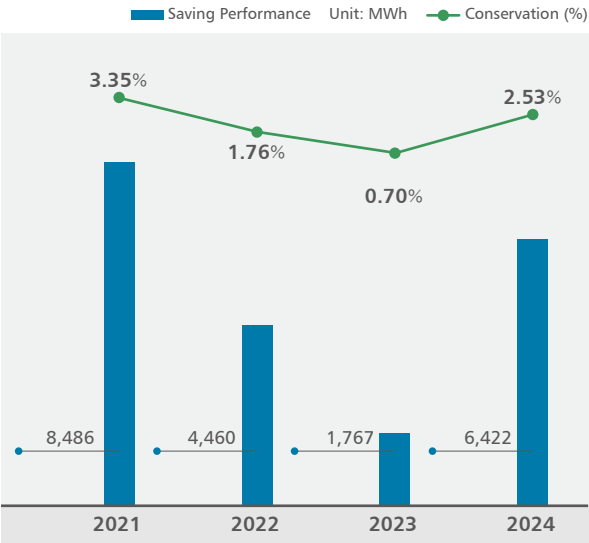
Note: 1. The statistics of calculation is based on electricity consumption per unit product by 2015 and the scope covers UMC.
2. CO₂ emissions are calculated using the latest local electricity coefficient. Taiwan: 0.494 kg CO₂e/kWh; Singapore: 0.412 kgCO₂e/kWh
3. The gas-saving amount estimated is a theoretical amount. The CO₂, CH₄, N₂O emission coefficient of the 2006 IPCC fixed and mobile resource (gas) and gas caloric value from the various local fabs are also incorporated into the carbon emission calculation.
4. The newly added improvement item was calculated based only on the 12-month performance. For the cross-year project, only the performance generated in 2024 was counted.

Electricity Savings



Note: The scope covers UMC.

Natural Gas Savings



Note: The scope covers UMC.

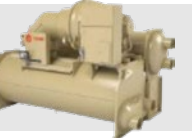
Energy Saving Plans

UMC's energy saving plan covers approximately 400 energy saving measures, including the complete introduction of equipment energy saving features, provision of inverters, replacing online uninterruptible power supplies (UPS) with offline UPS systems, and ice water systems. The Company's reduction goal for 2025 is 42,000 MWh, equivalent to CO₂ emissions reduction of about 20,200 tonnes. UMC will continue its program to reduce the HDI temperature and VOC rotor upgrading and other saving measures which is expected to reduce natural gas consumption by 6,422 MWh, equivalent to CO₂ emissions reduction of about 1,297 tonnes.

❖ Inverter air compressor control



❖ Intelligence chiller



❖ Cooling tower replacement



❖ Energy saving motor



ISO 50001 Certificatio

UMC plans to fully introduce ISO 50001 Energy Management System Certification to all fabs in Taiwan. Fab 12A P5/6, Fab 8E, Fab 8CD has obtained the certificate in September 2024. It is expected that all UMC's fabs in Taiwan will be verified by third-party for ISO 50001 by 2026.



★ Material Topic 3.3 Water and Wastewater Management

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Water Management Policy and Commitment



Positive Impact 3.3 Water resource management and reuse

Negative Impact 3.3 Water resource consumption and pollution

Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
Positive: Effective utilization of water resources and wastewater recycling increase industrial resilience and reduce dependence on natural water sources.	- Develop and adopt diverse sources, such as reclaimed water and desalinated water. - Adopt multipronged approaches such as increasing the recycling of in-fab used water and investing in recycling technologies. - Participate in public affairs, and cooperate with government and industry to facilitate water-resource-related communication, interaction and guiding assistance.	- A quarterly report on the performance progress of "The Environmental Management and Planning Group" is submitted to the Corporate Sustainability Committee for review. - Compile monthly statistics on water consumption, water-saving achievements, and recovery rates. - Regularly participate in meetings held by the industry's professional associations and the Water Resources Agency.
Negative: Excessive consumption of water resources and insufficient regulation lead to pollution discharge issues	Water Consumption - Continue the reinforcement of in-fab water saving measures, and increase appropriate water-saving measures. - Install U water generation equipment in response to sudden droughts. Wastewater Pollution - Invest in the R&D of wastewater-related treatment technologies. - Monitor and optimize wastewater discharge. - Monitor the environmental development trends in the world, and assess relevant opportunities and risks.	Water Resource Management - A quarterly report on the performance progress of "The Environmental Management and Planning Group" is submitted to the Corporate Sustainability Committee for review. - Check the establishment progress of U water equipment. - Report water risk status monitoring results on a monthly basis. - Participate in international CDP Water Security Project for transparency. Wastewater Treatment - Conduct wastewater/discharge testing and acquire relevant data on a monthly basis. - Participate in external regulations-related seminars, and environmental protection meetings held by the Allied Association for Science Park Industries. - The Environmental Team reports and shares the results of regulatory compliance checks in monthly meetings. - Implement ISO 14001:2015 Environmental Management System.

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Reduction of water withdrawal per unit product	14%	30.3%	25%	30%
Cumulative new water savings (base year 2015)	5.6 million tonnes	6.5 million tonnes	6.1 million tonnes	8.6 million tonnes
Reclaimed water and desalinated water usage rate	16%	17.2%	18%	32%
Key water pollution indicators better than regulated standards	56%**	74.3%	60%	70%

Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC); With ** remark is UMC fabs in Taiwan.
2. The actual performance of water consumption per unit product is the performance under BAU scenario compared with the base value; the base value is calculated from 2015 as the base year and is included in the annual data of the foundry subsidiaries.
3. The water-saving statistics refer to the current year's saving performance achieved through the water-saving measures implemented since 2015. The statistics were calculated based on estimated values.

4. The usage ratio of reclaimed water and desalinated seawater is calculated as (reclaimed water + desalinated seawater) ÷ (third-party water withdrawal) x 100%
5. The main indicators of water pollution are superior to the regulatory standards, which refers to the reduction rate of pollutants in UMC's wastewater discharge compared to the main indicators of local regulatory emission standards. The calculations are based on the data collected by the local government's testing agency in December of the current year. The reduction rate of key water pollution indicators is calculated as [1 - (total emissions ÷ regulatory total emissions)] x 100%.

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3.3.1 Water Structure

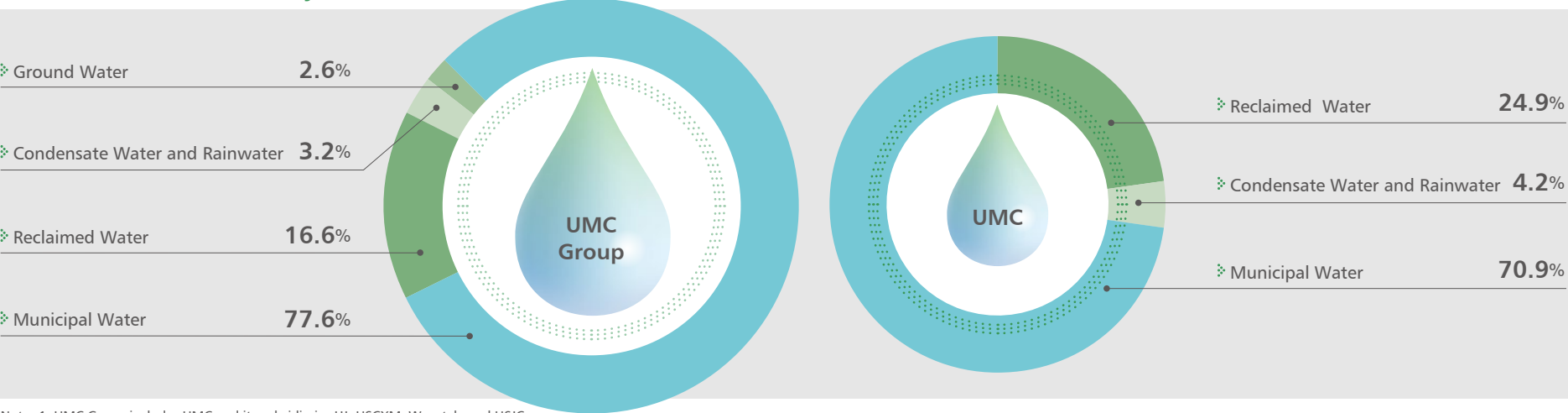
Due to the increasingly frequent occurrence of floods and droughts caused by climate change in recent years, stakeholders have become more concerned about water resource issues. Water resources are also an indispensable resource for semiconductor manufacturing processes. To proactively address complex water resource issues, UMC established the "Water Resources Management Policy and Commitment" and signed off by the President in 2015, which serves as the Company's highest guiding principle for water resource management. The Company's water resource management action plan, which includes all operations of UMC Group (100%), aligns with this policy, aiming to maximize water efficiency, strengthen the resilience of the upstream and downstream value chain against water risks, and promote water resource saving.

The Environmental Planning and Management Group, which consist of Group Risk Management & Environment Safety Health Division and Facility Operation Division that spans across different divisions and fabs, is account for the day to day water management, including responsible for promoting and setting management goals, (such as reducing water consumption per unit of product, and increasing the proportion of reclaimed water use, etc.). They regularly report their progress to the Chairman of the Corporate Sustainability Committee, who also serve as the Company's President and is responsible for the relevant performance management. Over the years, Environmental Planning and Management Group has successfully implemented the "369+ Reduction" and "Green 2020" programs, achieved the water resource goals, while also launching the "Green 2025" program. In 2023, to develop and apply diversified water sources, it has been evaluated that desalinated water will be one of the water sources, and the goal is set to achieve a ratio of 32% for the use of reclaimed water and desalinated water by 2030. In response to the potential risk of drought and water shortage in Taiwan, UMC co-presidents approved additional investment in the "UMC U Water" wastewater recycling system in 2023. This system can recycle wastewater into reclaimed water for internal use, effectively reducing the risk of water supply restrictions during water shortages.

The UMC Group's water sources primarily come from municipal water and reclaimed water, which are freshwater sources, accounting for 77.6% and 16.6% respectively; UMC's main water sources are the same as the Group's, with municipal water about 70.9% and reclaimed water about 24.9%. In 2024, the UMC Group's total water withdrawal was 27.56 million tonnes, while UMC's was 18.41 million tonnes. Due to the expansion of Fab 12A II and Fab 12i, an additional 1.29 million tonnes were added compared to the previous year.

Organization	Responsibilities of UMC Group Water Resources Management
Corporate Sustainability Committee	❖ Responsible for water management strategy and performance
Environmental Planning and Management Group	❖ Set water resources management goals
Facility Operation Division of Headquarters	❖ Dedicated unit for water management at a non-executive and non-committee level. ❖ Consolidate the water resource management performance of the UMC Group and monitor the achievement of objectives. ❖ Expand the implementation of exemplary water-saving measures to all fabs and subsidiary companies. ❖ Assessment, adoption, and promotion of water resource-related technologies.
Facility Operation Divisions of All Fabs and Subsidiaries	❖ Implement monitoring of water resources in fabs: compile statistics on water usage, recycled water, and water saving to grasp the actual water consumption situation. ❖ Execute water resource enhancement projects in fabs: water-saving programs and water pollution prevention plans.

2024 Water Withdrawal by Source



Note: 1. UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.
2. Reclaimed water amount is used by UMC's fabs in Taiwan and Singapore. The groundwater withdrawal amount is used by subsidiary USJC.

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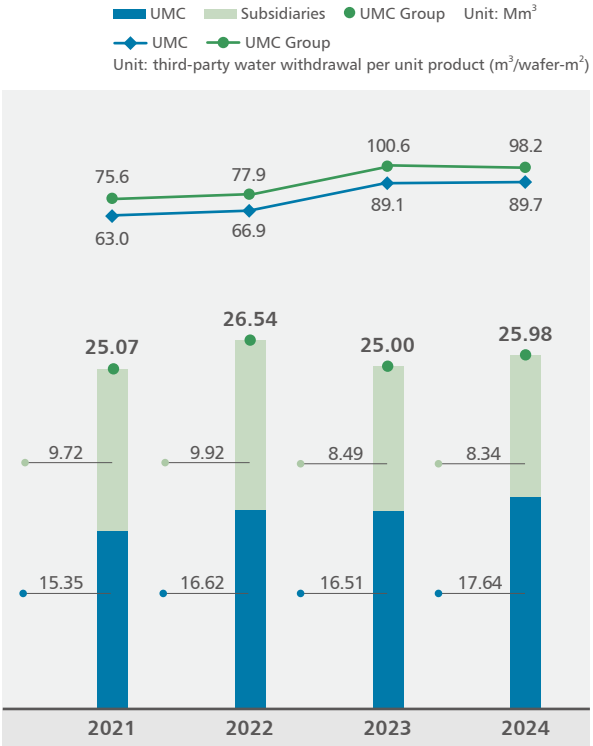
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Third-Party Water Sources

Municipal and reclaimed water are both sourced from third-party suppliers. In 2024, UMC Group's water withdrawal from third-party sources amounted to 25.98 million tonnes, with a water withdrawal per unit product is 98.2 m³/wafer-m². UMC's own water withdrawal was 17.64 million tonnes, with a water withdrawal per unit product is 89.7 m³/wafer-m², marking an increase from 2023. The primary reasons for this increase were adjustments in production capacity and the rise in water usage due to new equipment installations at Fab 12A P6 and Fab 12i P3.

Third-party Water Withdrawal



Note: 1. Sources of third-party withdrawal include municipal water and reclaimed water.
2. UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.
3. Water meter readings are recorded daily. Annual water amount is calculated based on average monthly water usage.

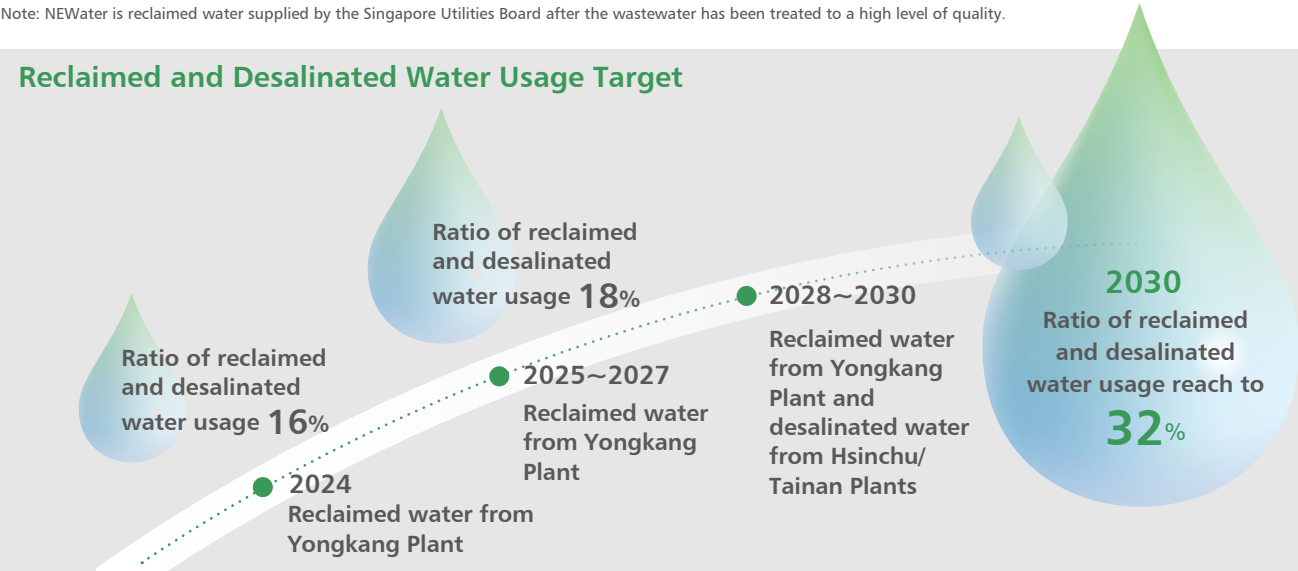
Reclaimed Water and Desalinated Seawater

To reduce dependency on municipal water, the Company has diversified its water sources, include reclaimed water as alternative water sources. UMC Group's reclaimed water consumption is mainly at Fab 12i in Singapore. When Fab 12i was built, reclaimed water (NEWater^{Note}) was introduced in the production of 300mm wafers. In 2024, Fab 12i's reclaimed water reached 4.00 million tonnes, accounting for 97.6% of the fab's total water withdrawal. UMC's fabs in Taiwan are also actively assessing reclaimed water utilization. Fab 12A in Taiwan introduced the use of reclaimed water in December 2022. Fab12A's reclaimed water usage reached 0.58 million tonnes. The reclaimed water used in Fab 12A and Fab 12i totaled 4.59 million tonnes, an increase of 16.9% compared to 2023.

In addition to reclaimed water, UMC Group is also actively planning to introduce desalinated seawater to implement diversified water resource utilization. UMC's Taiwan facilities will participate in the government's desalination plant construction projects in Hsinchu and Tainan in 2024. The total purchase of desalinated seawater is expected to provide approximately 15,000 tons of water per day. Once the desalination plants are completed and operational, they will further reduce dependence on natural water resources and mitigate regional water resource impacts.

Note: NEWater is reclaimed water supplied by the Singapore Utilities Board after the wastewater has been treated to a high level of quality.

Reclaimed and Desalinated Water Usage Target



Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC)
2. Reclaimed water from Yongkang Plant has officially in operation. The Hsinchu desalination plant is expected to start supplying water in 2028, and the Tainan desalination plant is projected to begin water supply in 2029.

Water Quality Risk Control

Effective monitoring and control of water quality can decrease the amount of resources consumed during the manufacturing process, optimize product yield, and increase the efficiency of water usage. Before entering fabs, water is first tested with pH devices and continuously monitored with a conductivity meter to ensure stable quality. During the pure water production process, each unit of the water production facility is equipped with inspection instruments, which are connected to the statistical process control (SPC) system to ensure the water quality and reliability. In Singapore, Fab 12i increased its hydrofluoric acid (HF) treatment systems to improve the quality of the discharged water (fluoride ion< 15ppm), and strengthened source classification management to control sulfates to <1,000ppm in order to increase the stability of reclaimed water quality.

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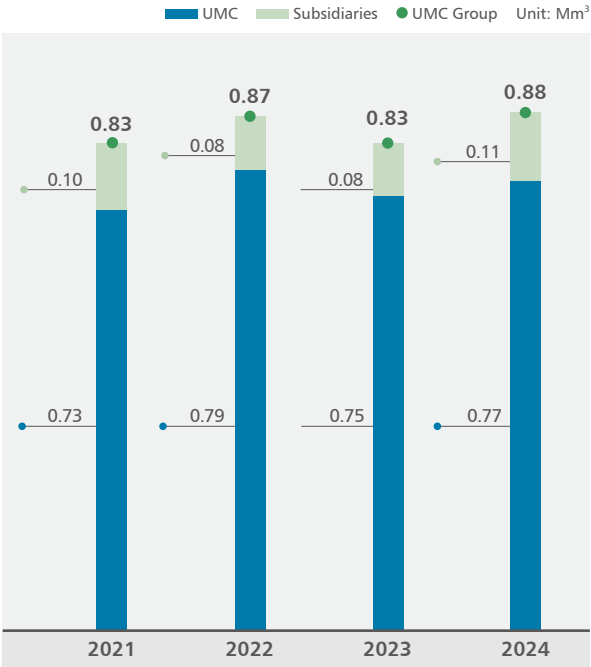
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Condensate Water and Rainwater

Condensate water and rainwater are precious natural water sources and are easily subjected to seasonal climate changes. While accounting for a small proportion of UMC's total water withdrawal, making good use of condensate water and rainwater can reduce the impact on water resources and the environment. Condensate water and rainwater are mainly used in secondary manufacturing processes and watering of green spaces at UMC Group, and the total usage in 2024 was 0.88 million tonnes.

Condensate Water and Rainwater



Note: 1. UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.
2. Annual water amount is calculated using flow meters and estimates.

Water Stress Area Assessment

To further understand the water stress and water use risks of each fab, UMC used the water risk assessment tool developed by the World Resource Institute (WRI) and combined it with indicators of local water resources, including water source stability and regional supply-demand ratio, to identify the percentage of UMC Group fabs that are located in water-deficient areas, serving as reference for formulating water risk management strategies.

According to the WRI Aqueduct Tool, only the subsidiary HJ's location in Suzhou, China is identified as an area with extremely high level of water deficiency. HJ accounts for about 7.8% (2.16 Mm³) and 3.8% (1.84 Mm³) respectively of the Group's water intake and water consumption. Further evaluation based on indicators including water source stability and regional supply-demand ratio showed that HJ's water risks are low. In addition to the ample water resources provided by the nearby Taihu Lake, which has a storage capacity of 4.5 billion tonnes, and a dual water source and dual water plant safety supply model, the dual water sources (Taihu Lake and Jiuyang Lake) have a combined supply capacity of 650,000 tonnes/day. HJ also implements water saving measures to improve water use efficiency and reduce the impact of water risks on operations. In 2024, there were no incidents affecting the operation of the UMC Group due to water-related issues.

In addition, UMC and its subsidiaries have no significant impact on the use of water sources in the region, and the percentage of water consumption in the region (the impact of regional water use) is less than 5%.

UMC Indicators of Water Related Risks Assessment

Water Related Risks Assessment



- ❖ The WRI (Water Resource Institute) Aqueduct Tool is used to assess water risk.
- ❖ **Water Source Stability:** Considering the stability of water supply, cross-regional dispatching flexibility, rainfall variability, etc.; high level means relatively stable water resources.
- ❖ **Regional Supply-demand Ratio:** Calculated based on the water supply of the region and the water demand of the fab. The bigger the ratio the lower competence between region's peers.

For the identification of water risk factors, please refer to Chapter 1.2.5 Managing Climate Risks and Opportunities.

Impact of Regional Water Withdrawal

Site	Location	Region Water Withdrawal ^{Note 1} (10,000 tonnes/day)	Water Withdrawal ^{Note 2} (10,000 tonnes/day)	Impact ^{Note 3}	
UMC	Hsinchu Science Park	61.0	1.46	2.40%	
	Southern Taiwan Science Park	97.2	2.08	2.14%	
	Singapore	23.0	1.12	4.87%	
Subsidiary	HJ	China Suzhou	65.0	0.58	0.90%
	USCXM	China Xiamen	27.0	0.62	2.31%
	Wavetek	Hsinchu Science Park	61.0	0.14	0.23%
	USJC	Japan Mie-Ken	88.9	1.12	1.26%

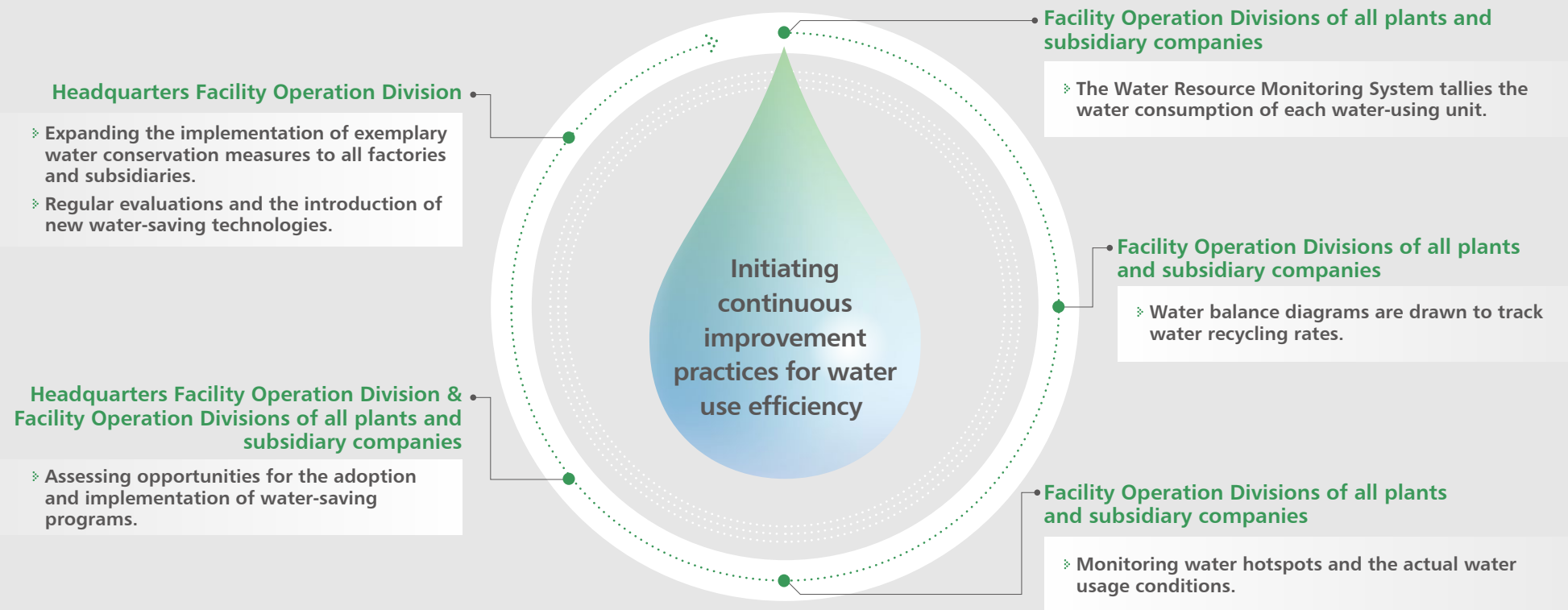
Note: 1. Source of regional data: official data announced by the local government.
2. Source of UMC's data: measured by each fab's on-site flowmeter.
3. Impact (%) = water withdrawal ÷ regional water withdrawal x 100%

3.3.2 Water Saving

Water Efficiency Management Program

UMC Group identifies opportunities to improve water resource efficiency through its water resource management program.

UMC Group has established a water resource monitoring system, supported by on-site facility management personnel, to control the consumption of major water-using units within the fab. Monthly water usage statistics for each unit are compiled into water balance diagrams and calculations of fab-wide and process water recycling rates. Using this information, water hotspots are identified and monitored, leading to the implementation of water-saving opportunities. Ultimately, the dedicated water management team at headquarters expands successful water-saving measures to all sites. Regular evaluations and the introduction of new water-saving technologies ensure continuous improvement in water use efficiency across all sites.



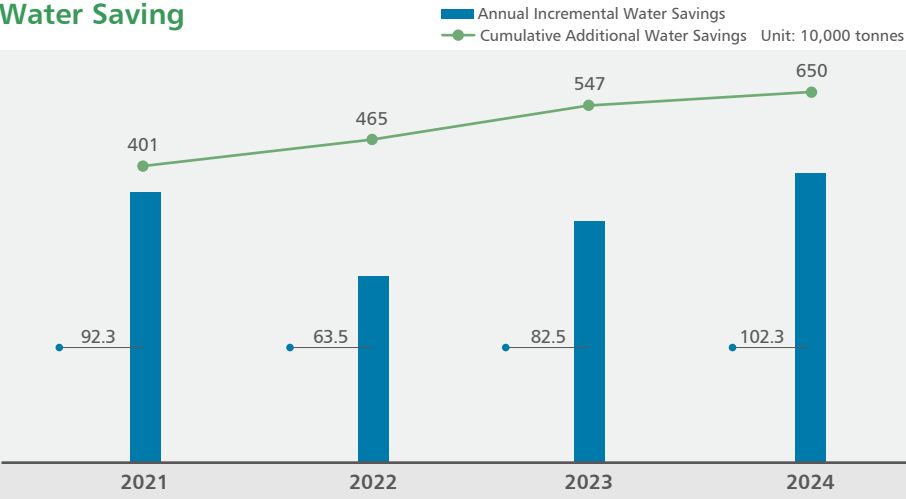
The Company's principle of water utilization is first to incorporate water-saving design in manufacturing processes, then focus on boosting water recycling and reuse. Combined with efficient water management, consumption can be effectively reduced. UMC continues to implement improvement measures for water recycling and reuse, and actively sets progressive water saving goals to guide policies and performance. UMC also regularly conducts environmental education courses related to water resource efficiency and invites external professional instructors to provide specialized training on water systems. These courses aim to enhance system efficiency, including water conservation, improving wastewater quality, optimizing operations, and introducing new technologies for improvement. Through professional education and training, employees' knowledge of water resource efficiency and innovative solutions will be enhanced, thereby aiding in the identification of water-saving opportunities and the ability to propose improvement plans.

Water Recycling and Reuse

With the water-saving, consumption-reduction and recovery measures continuously promoted over the years, UMC Group achieved an additional water saving of approximately 1,023,000 tonnes in 2024, with a cumulative water saving reaching 6.50 million tonnes. Moreover, the recycling rates of manufacturing processes in all fabs exceed local regulatory standards, with UMC's process water recycling rate reaching 83.3%^{Note}. Through strategies such as water saving, recycling and reuse, and reduction of ultra-pure water in processes, UMC Group successfully met the environmental phased targets for reducing water usage per unit product. In addition, in response to global climate change and the impact of extreme weather, UMC has been actively promoting measures for coping with water-related disasters such as floods and droughts, to minimize potential operational impacts.

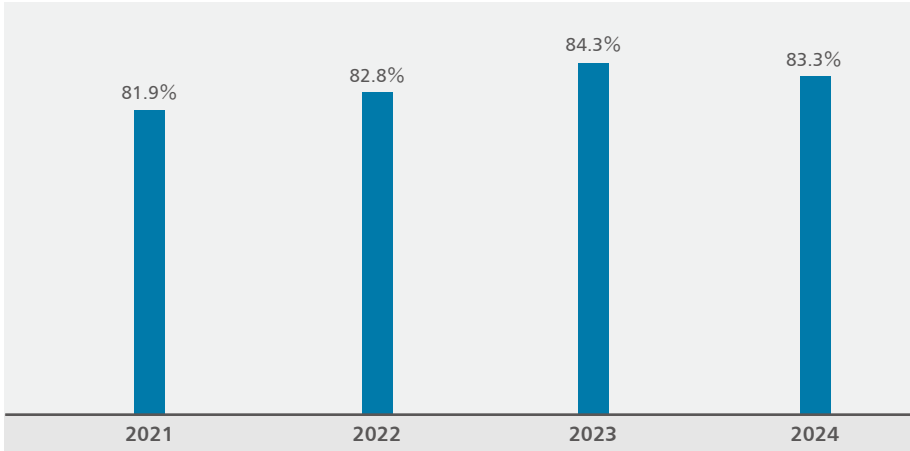
Note: Manufacturing-process water recycling rate = (All fabs' manufacturing-process water withdrawal ÷ UMC's total manufacturing process water withdrawal) x Sum of all fabs' manufacturing-process water recycling rate.

Water Saving



Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC)
2. The water-saving statistics refer to the current year's saving performance achieved through the water-saving measures implemented since 2015. The statistics were calculated based on estimated values.

Water Recycling Rate for UMC Manufacturing Process



Note: 1. Manufacturing-process water recycling rate for UMC (company-wide) = (All fabs' manufacturing-process water withdrawal ÷ UMC's total manufacturing process water withdrawal) x Sum of all fabs' manufacturing-process water recycling rate.
2. The calculation method for process water recycling rate in 2024 will be adjusted in accordance with the revised water balance diagram regulated by the Science Park. Starting from March 2024, the version will be updated from R8 to R9.

Manufacturing-process Water Recycling Rate in 2024

Site		Standard	Actual
Taiwan	Fab 8A	≥ 70%	84.6%
	Fab 8D	≥ 70%	90.5%
	Fab 8E	≥ 85%	86.5%
	Fab 8F	≥ 85%	86.1%
	Fab 8S	≥ 85%	87.6%
	Fab 12A	≥ 85%	87.3%
	Fab 12A II	≥ 85%	86.0%
Singapore	Fab 12i	NA	67.8%

Note: NA means there is no local standard available. All fabs are installed with basic recycling equipment.

Decrease Intake

The 2024 water conservation plan, applied throughout the Company, includes all operations of UMC Group. Through the implementation of various water-saving measures and diverse water sources, the plan involves adjustments and improvements in the cleaning time and flow rate of production equipment, the addition of RO recovery systems for wastewater, the re-purification and recovery of discharge water from pure water systems, and the increase of backup pipelines for recycled water to reduce the supplementation of tap water. As a result, overall reduction of UMC Group's water withdrawal reached 1,023,000 tonnes in 2024. It is expected that through relevant measures, the withdrawal from natural water resources will continue to decline in 2025.

UMC Group New Water Saving Performance in 2024

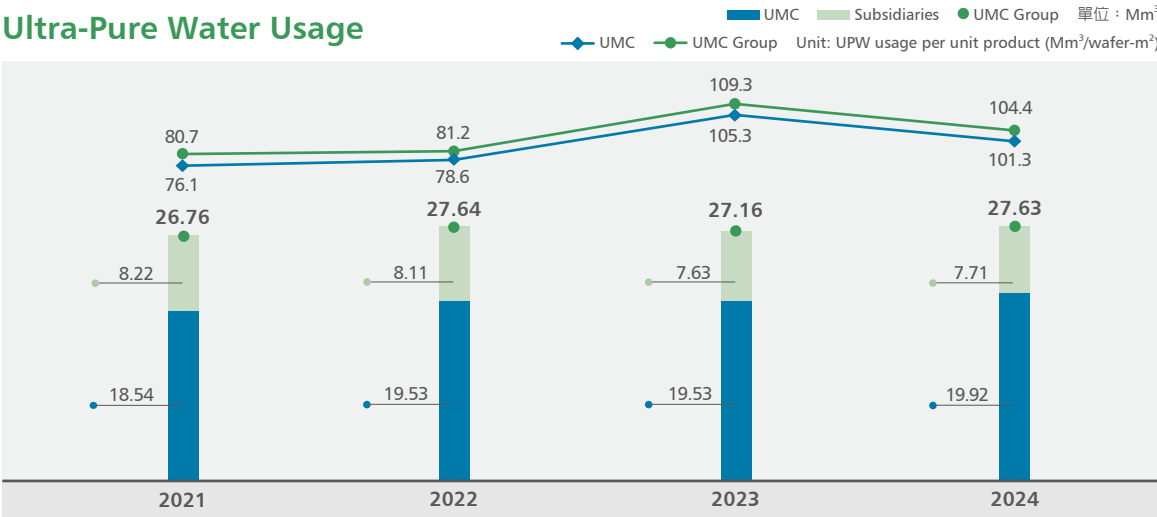
Site		Site				Site Coverage (%)
		Water saving from production machines	Wastewater recovery increasing	Pure water production improving	Reduce supplement from municipal water	
UMC	Fab 8A, Fab 8C, Fab 8D, Fab 8E, Fab 8F, Fab 8S, 12A I&II, Fab 12i	●	●	●	●	100%
Subsidiaries	HJ	●	●	●	●	100%
	USCXM			●	●	100%
	Wavetek		●			100%
	USJC			●	●	100%
Water Savings (10,000 tonnes)		2.0	15.4	19.4	65.6	Total Amount of the newly added water-saving: 1.02 million tonnes

Note: The newly added improvement items were calculated based only on 12-month performance. For multiyear projects, only the performance generated in 2024 was counted.

Ultra-Pure Water Reduction

UMC Group has gradually decreased its per unit product's ultra-pure water (UPW) consumption over the years by pursuing continuous improvement, increasing internal awareness, as well as enhancing production controls and technology. Water usage for the manufacturing process is mainly used by Wet Bench equipment, including Single-Wafer and Batch-Type Wet Bench. The commonly used reduction methods are to optimize the manufacturing process water flow, process water clean time and standby water flow, process simplification or to increase the time interval between each flush during standby, etc. The Green Operation Group under the Corporate Sustainability Committee has consolidated the reduction plans of fabs, and applied them to other sites where UPW reduction measures have not yet been implemented. UMC Group's UPW usage in 2024 was 27.63 million tones and the UPW usage per unit product in 2024 was 104.4 m³/wafer-m². Compared to the ultra-pure water usage of 109.3 m³/wafer-m² in 2023, there is a benefit.

Ultra-Pure Water Usage



Note: UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.

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3.3.3 Water Pollution Control

UMC's water pollution control includes process source reduction and waste liquid diversion, followed by treatment classification. New fab areas have up to 27 categories of wastewater diversion, which are divided into solvent-based (reused or incinerated depending on flash points), inorganic acids (reused), and wastewater (multiple reuse according to classification). The process simplifies the composition of wastewater before it is treated in the fab's wastewater treatment facilities to the control standards of the Science Park Administration and discharged into the Science Park sewage systems. For real-time monitoring and response, equipment for continuous monitoring of water quality (pH, fluoride ion concentration) and water quantity are installed, and SPC management is adopted for self and early prevention to ensure that the quality of water discharged into the sewage complies with control regulations. In addition, the Science Park Administration conducts monthly unscheduled and random quality inspection of water discharged by different companies to reaffirm the quality of discharged water.

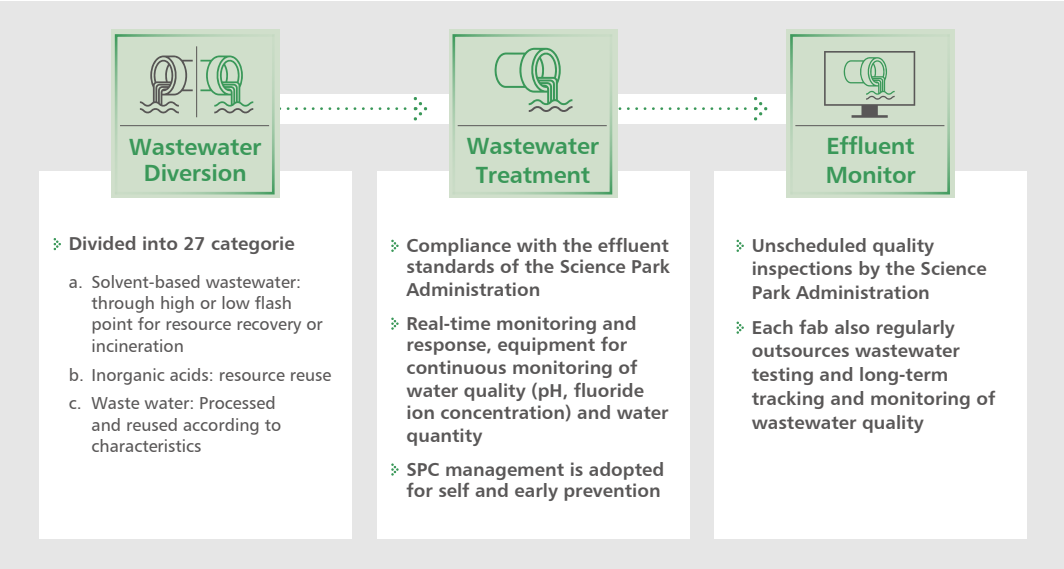
In addition, UMC has planned wastewater quality improvement targets, with the aim that key water pollution indicators will exceed regulatory standards by achieving a reduction rate of 70% by year 2030.

Wastewater Treatment

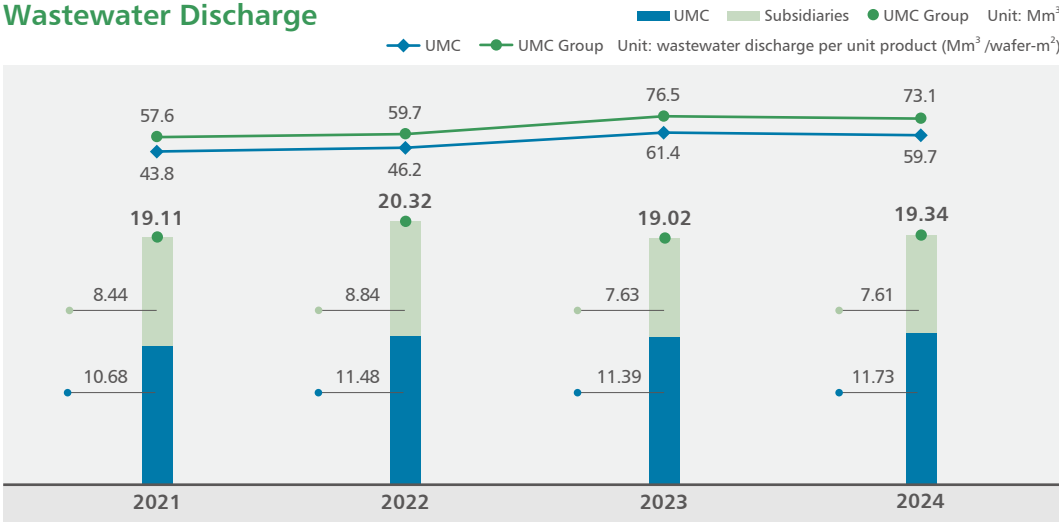
In 2024, the wastewater discharge of UMC Group was 19.34 million tonnes, which was higher than the previous year. Wastewater discharge per unit product in 2024 was 73.1 m³ /wafer-m².

To mitigate environmental impact, UMC Group continuously enhances wastewater diversion rates and persistently invests in the research, development, and implementation of new wastewater treatment technologies. In recent years, advanced ammonia-nitrogen wastewater treatment technologies have been introduced to reduce the load on water bodies. Additionally, Fab 12A has officially implemented copper sulfate waste liquid recovery and treatment technology. This technology extracts pure copper from copper sulfate waste liquid through electrolytic reduction and reprocesses it into copper pipes, reducing 1,680 tonnes of copper sulfate waste liquid and significantly improving water resource utilization.

For more details, please visit »



Wastewater Discharge



Note: UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC

UMC Group Wastewater Discharge

Site					Combined Waste Water Treatment			
Site	Location	Discharge Amount (10,000 tonnes/day) Note 1	Impact (%)		Treatment Department	Current Treatment Amount (10,000 tonnes/day) Note 2	Re-use	Discharged into Drainage Area
UMC	Hsinchu Science Park	Fab 8A, Fab 8D, Fab 8E, Fab 8F, Fab 8S	1.05	8.8%	Hsinchu Science Park Administration sewage treatment plant	12.0	-	Ke-Ya River
	Southern Taiwan Science Park	Fab 12A	1.43	8.2%	Southern Taiwan Science Park Administration sewage treatment plant	17.5	-	Yanshuei River
	Singapore	Fab 12i	0.73	0.9%	Public Utilities Board (PUB)	84.8	Industrial reuse	South China Sea
Subsidiaries	China Suzhou	HJ	0.11	0.9%	Suzhou Industrial Park, Hua Yan Water Ltd.	12.0	-	Wusong River-
	China Xiamen	USCXM	0.50	1.0%	Xiamen City, Xiang-an (Ma Xin) sewage treatment plant	50	-	Dongkeng Bay
	Hsinchu Science Park	Wavetek	0.47	4.7%	Hsinchu Science Park Administration sewage treatment plant	10	-	Ke-Ya River
	Japan Mie-ken	USJC	1.00	-	USJC's wastewater treatment facilities can convert wastewater to a quality in compliance with local regulations and standards (Water Pollution Prevention Act). A real-time monitoring system is installed in the facilities, through which the discharge shall be immediately stopped if any anomalies are detected.	1.26	-	Osugidani River

Note: 1. Sources of discharge data: Water meter readings are recorded daily, and annual water consumption is calculated based on monthly amount.
2. Sources of data:
(1) Fabs in Hsinchu and Southern Taiwan Science Parks, subsidiary Wavetek: Ministry of Science and Technology Statistics and Data Bank.
(2) Fab in Singapore: Based on PUB website information.
(3) HJ: Suzhou Industrial Park, Hua Yan Water Ltd. sewage treatment plant; USCXM: Xiamen City, Xiang-an (Ma Xin) sewage treatment plant.
(4) USJC: Refer to the approved treatment capacity of USJC's emission control equipment

UMC and its subsidiaries, including Wavetek, HJ, and USCXM, treat their wastewater at third-party sewage treatment facilities to ensure compliance with local discharge standards before releasing it. UMC subsidiary in Japan, USJC, conducts secondary wastewater treatment on-site and discharges the treated effluent into the original water body only after meeting local regulatory requirements. UMC will continue to monitor wastewater discharge related data and develop wastewater treatment technologies to reduce the possible impact of wastewater discharge on regional watersheds.

Volume of UMC group Water Discharge by Source in 2024

Unit: megaliters

Level of Treatment		Site	Volume of Discharge
No treatment		NA	0
Secondary treatment in accordance with Hsinchu Science Park administration standards		Fab8A, Fab8D, Fab8E, Fab8F, Fab8S, and the subsidiary Wavetek	4,242
Secondary treatment in accordance with Southern Taiwan Science Park administration standards		Fab 12A, Fab 12A II	5,227
Secondary treatment in accordance with Singapore government administration standards		Fab12i	2,658
Secondary treatment in accordance with Suzhou Industrial Park administration standards		Subsidiary HJ	1,844
Secondary treatment in accordance with Hsinchu Science Park administration standards		Subsidiary USCXM	1,705
Secondary treatment in accordance with Local regulations administration standards		Subsidiary USJC	3,668

Note: 1. 1 Mm³ = 1,000 megaliters
2. With the exception of our subsidiary in Japan, USJC, all fabs pre-treat their wastewater before discharging it to local treatment facilities, with no direct discharge into receiving water bodies. The effluent from USJC is discharged in compliance with local regulatory standards after undergoing our proprietary treatment processes, such as the removal of heavy metals, nitrogen, and phosphorus.

UMC Group's Water Withdrawal, Discharge, and Consumption in 2024

Water consumption is calculated as the total water withdrawal minus the total water discharge. The primary sources of water consumption are the evaporation from cooling towers (due to process cooling and cleanroom environmental control), and other factors such as natural evaporation from water storage systems, landscape ponds and watering, cafeteria cooking, environmental cleaning, and employee drinking water. To reduce the impact of water consumption on the ecological environment, UMC sets annual water consumption targets ^{Note1} (the target for 2024 is 6,690 megaliters) and gradually reduces water consumption through measures such as improving the efficiency of chiller systems and reducing evaporation and splashing from cooling tower operations. In 2024, UMC's actual water consumption was 6,678 megaliters, achieving the annual target.

Unit: megaliters

			Subsidiary			
			UMC	HJ	USCXM	Wavetek
				USJC		
Unit: megaliters			18,412	2,159	2,329	523
Source	Municipal Water		13,058	2,141	2,288	503
	Reclaimed Water		4,586	0	0	0
	Condensate Water and Rainwater		768	18	41	20
	Others (such as groundwater)		0	0	0	707
Total Water Discharge (b)			11,734	1,844	1,705	393
Destination	Surface water, Groundwater, Seawater		0	0	0	3,668
	Third-party Water		9,076	1,844	1,705	393
	Third-party water sent for use to other organizations		2,658	0	0	0
Water Consumption (a-b)			6,678	315	624	130

Note: 1. The target of water consumption is set based on climate conditions, the capacity of each fab, changes in water usage due to production equipment and facility infrastructure, and the implementation of water-saving plans.
2. 1 Mm³ = 1,000 megaliters
3. The wastewater discharged is not fresh water (≤1,000 mg/L total dissolved solids).
4. Singapore's NEWater uses wastewater from UMC Fab 12i as source water and then provides the treated water to other organization for industrial water usage.
5. The figures in the table are presented as integers, so the actual total may differ from the sum shown in the table.

Communication and Public Participation

In addition to internal water resource management and active promotion of water saving, UMC also serves as a representative in the industry's professional associations. The Company works closely with the government and in collaboration with other companies in communicating and advising on water management. UMC also contribute to initiatives by the Taiwan Water Corporation, including emergency responses to water shortage and improvement in reservoir water quality, to jointly stabilize the supply and quality of regional water bodies and ensure the continuity of municipal water supply.

Participation with Public Policy

- 1.Attended water resources communication meetings convened by the Water Resources Agency and manufacturers' associations.
- 2.Supported coordination of water resources by the Water Resources Agency (WRA) during dry season.
- 3.Assessed the utilization of reclaimed water to align with the Water Resources Agency's reclaimed water facilities.
- 4.Participated in the construction of seawater desalination plants in Hsinchu and Tainan, promoted by the Water Resources Agency to increase water supply stability. UMC has committed to purchasing a total of 15,000 tonnes per day of desalinated water.
- 5.Established a communication platform with the WRA to meet future water supply and demand.

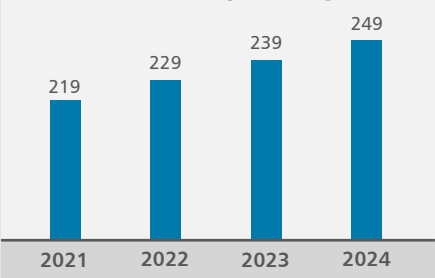
Exchange and Counseling with Industry Peers

UMC has participated annually in water saving coaching sessions in the Science Park since 2002. As of 2024, the program has conducted a total of 249 sessions and coached more than 100 companies. water saving by companies coached by UMC has reached an accumulated amount of 1.67 million tonnes over 4 years (2021~2024). Companies counseled by UMC in 2024 are estimated to have potential water saving of 382,000 tonnes per year. Tracking the 10 companies which were coached by UMC in 2024, the actual water-saving volume during year 2023 was 341,000 tonnes of water.

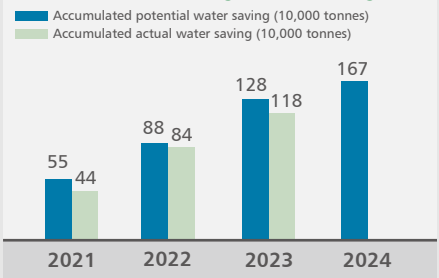
By sharing the Company's valuable experience with peers, it can not only reduce the overall industry's dependence on natural water resources, but also promote exchanges and cooperation among the regional networks.



Accumulated Water-saving Counseling Cases



Accumulated Water-saving from Counseling Cases



Note: The water saving achieved in 2024 will be tracked in 2025.

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3.4 Waste Management

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- 3.4.4 Contractor Management

★ Material Topic 3.4 Waste Management and Circular Economy

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Environmental Protection Policy



Positive Impact ▸ Circular economy		Negative Impact ▸ Waste not being used properly
Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
Positive: Waste management and reduction implementation status, increase the resource recovery rate of waste, and establish a circular economy business model.	▸ Collaborate with global operation partners to achieve waste recycling and reuse. ▸ Pay attention to the development of international trends, evaluate opportunities and risks, handle waste properly, and make good use of energy resources to promote a circular economy. ▸ Establish a low-carbon circular business model.	▸ A quarterly report on the performance progress of "The Environmental Management and Planning Group" is submitted to the Corporate Sustainability Committee for review. ▸ Plan the establishment of the "Circular Economy & Recycling Innovation Center" to increase the percentage of recycled products and convert waste into usable resources. ▸ Participate in external regulatory seminars and conferences.
Negative: Improper waste management can lead to environmental pollution, affecting ecological conservation and human health.	▸ Promote process technology improvement and source management measures to reduce raw material consumption and waste generation. ▸ Reduce raw material consumption through process design and technology improvement. ▸ Review environmental protection and international environmental trends/government regulations. ▸ Select priority targets for waste reduction based on the raw materials that contribute the most to major waste generation, adopt cost-reduction policies, learn from benchmarking cases in the industry to evaluate technically feasible plan, and conduct small-scale experiment to confirm its feasibility. If the plan is deemed feasible, it will be implemented in fabs.	▸ A quarterly report on the performance progress of "The Environmental Management and Planning Group" is submitted to the Corporate Sustainability Committee for review. ▸ Regularly audit waste handling vendors. ▸ Perform monthly regulatory compliance checks. ▸ To prevent any violation of regulations, environmental protection cases are reviewed every quarter and employees are informed at the Safety Committee meeting.

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Reduction of waste generated per unit product ^{Note 2}	24%	44.2%	35%	50%
Cumulative waste reduction (base year 2015) ^{Note 3}	Cumulative additional waste reduction: 16,400 tonnes	17,684 tonnes	118,000 tonnes	273,000 tonnes
Ratio of in-fab resource conversion ^{Note 4}	13.5%	19.7%	22%	35%
Develop circular economy solutions/products (in total)	2 items	2 items	4 items	6 items

Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC); with * remark is UMC (the parent company only).
2. Considering the proportion of 12-inch production capacity in the current year and the base year of 2015, regression correction is conducted.
3. The cumulative reduction benefits for each year from 2015 to the target year.
4. Ratio of in-fab resources conversion = (volume of in-fab resource conversion ÷ volume of total waste generated)

★ Performance Highlights 2024

At the 7th Green Innovation Awards, PackAge+ won with their proposal for "E-commerce Circular Packaging." With the support of the award's funding and expert guidance, they expanded their existing business model from e-commerce packaging to include supplier packaging in circular applications. This Green Award initiative established a collaboration between UMC and PackAge+.

In 2023, UMC implemented a "Packaging Material Circular Model Pilot Run" at three of its facilities, replacing cardboard boxes with reusable containers for transporting cleanroom consumables such as masks and gloves. Starting in 2024, the award-winning company became a partner of UMC, further expanding the use of reusable containers to seven UMC facilities. Currently, 1,000 containers are in circulation each month, initially for gloves and masks, with plans to gradually extend to cleanroom hairnets. This collaboration not only reduces waste generation but also contributes to environmental sustainability.

It is estimated that this initiative can save 12,000 cardboard boxes annually. For example, using reusable containers for gloves can reduce carbon emissions by approximately 50% compared to cardboard boxes. On the social welfare front, PackAge+ collaborates with social service units, such as the "Hehao Workshop" in Hsinchu, which assists with cleaning the reusable containers. The unit can help approximately 20 disadvantaged individuals each month by providing job opportunities and stable income.

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3.4.1 Waste Output Structure

Through process design and technology improvement, UMC Group reduces the consumption of raw materials, which not only decreases the emission of pollutants from the source, but also lowers operating costs and minimize resource consumption and the corresponding environmental impact. Considering chemicals that are concerns for environmental protection and government regulations, the Company prioritizes the reduction of raw materials that are significant waste generators. Referencing benchmark players in the industry, the Company then designs technically feasible plans and conducts small-scale tests. The plans will be extended to fabs for implementation after their feasibility is verified.

UMC Group total waste output in 2024 was 62,566 tonnes^{Note}. Of which, 58,080 tonnes of waste was recycled or reused, accounting for 92.8%. A total of 4,486 tonnes of waste were treated during the year.

Note: excluding routine office waste

Waste Generation Status in 2024

Unit: tonnes

Category	Composition	Total Amount	Recycling	Treatment
Hazardous Waste	Waste acid	31,251	31,251	0
	Waste solvent	6,290	4,145	2,145
	Waste copper sulfate	2,164	2,164	0
	Container	669	648	21
	Others	494	104	390
	Subtotal	40,868	38,312	2,556
General Waste	Waste solvent	1,747	866	881
	Waste ammonium sulfate	6,243	6,243	0
	Sludge	7,049	7,009	40
	Container	617	617	0
	Waste mixed hardware	305	305	0
	Others	5,737	4,728	1,009
	Subtotal	21,698	19,768	1,930
Total Amount		62,566	58,080	4,486

Note: 1. Scope: UMC Group
2. The categories are in accordance with Taiwan "Waste Disposal Act," which also used by all sites.
3. Waste acid includes waste sulfuric acid, waste phosphoric acid, BOE, Buffered hydrofluoric acid, and mixed acid waste liquid.
4. Direct disposal methods include incineration, landfill, solidification and chemical treatment.
5. Scrap metal is classified as general industrial waste during stages of storage and disposal according to "Standards for Defining Hazardous Industrial Waste", Ministry of Environment.

Waste Recycling Status

Unit: tonnes

Category	Method of Recovery	On-site	Off-site	Total Amount
Hazardous Waste	Reuse for original purpose	0	0	0
	Regeneration	11,226	27,086	38,312
	Others	0	0	0
	Subtotal	11,226	27,086	38,312
General Waste	Reuse for original purpose	0	0	0
	Regeneration	2	19,766	19,768
	Others	0	0	0
	Subtotal	2	19,766	19,768
Total Amount		11,228	46,852	58,080

Note: 1. Scope: UMC Group
2. Categories are in accordance with Taiwan "Waste Disposal Act," which also used by all sites.
3. The percentage of hazardous waste recycled or reuse is 93.7%.

Waste Treatment Status

Unit: tonnes

Category	Method of Treatment	On-site	Off-site	Total Amount
Hazardous Waste	Incineration, serving as fuel	0	1,942	1,942
	Incineration only	0	586	586
	Landfil	0	0	0
	Others	0	28	28
	Subtotal	0	2,556	2,556
General Waste	Incineration, serving as fuel	0	419	419
	Incineration only	0	1,226	1,226
	Landfil	0	285	285
	Others	0	0	0
	Subtotal	0	1,930	1,930
Total Amount		0	4,486	4,486

Note: 1. Scope: UMC Group (UMC and its subsidiaries HJ, USCXM, Wavetek and USJC)
2. For treatment performed outside the fabs by outsourced operators, the calculation was based on the actual weight measured. For treatment performed inside the fabs, the calculation was based on the weight estimated by flow meter.
3. Part of the waste that has been incinerated can be used as auxiliary fuel or regenerated as derived fuel, which is regarded as recovered energy.
4. Other methods of disposal include solidification, chemical, and overseas treatment.

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3.4.2 Source Reduction

"Zero waste" is UMC's ultimate waste management goal. By improving process technologies, raw material source reduction, and other source management measures, the Company can effectively reduce waste generation from its operations.

In 2024, UMC Group total waste outsourced for treatment was 51,336 tonnes^{Note}, and waste output per unit product was 194.0 kg/wafer-m2, which represents a decrease of 14.7% compared to 2023. In 2024, hazardous waste outsourced for treatment amounted to 29,641 tonnes, and the per unit product of hazardous waste output was 112.0 kg/wafer-m2, which represents a decrease of 8% compared to 2023. Through various reduction plans and measures, the total waste reduction in 2024 reached 1,498 tonnes.

Note: Excluding routine office waste

2024 New Waste Reduction Measures and Performances

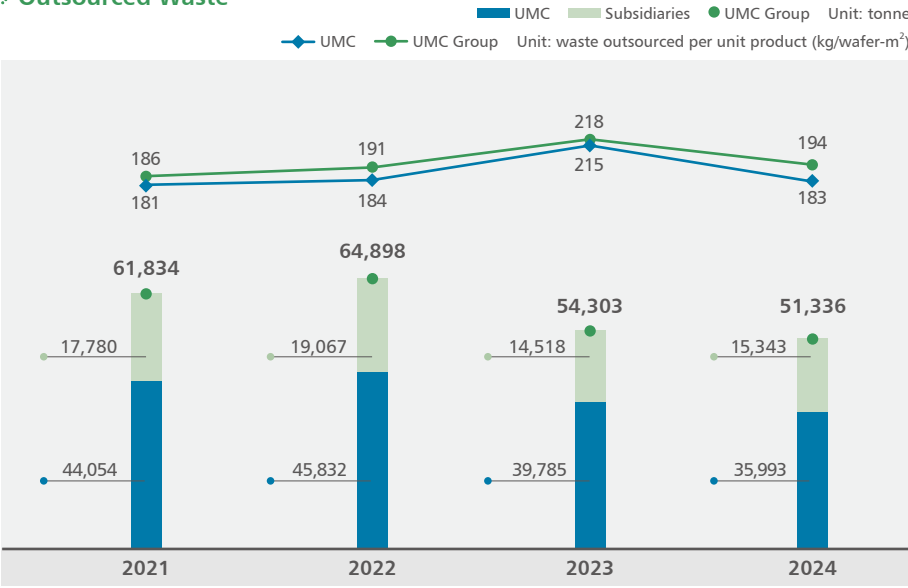
Plan	Measures	Reduction (tonne)
Reduction of waste sulfuric acid	Optimize usage for manufacturing process	1307
Reduction of waste solvent	Lower backside rinse flow, extending the lifetime of chemicals	100
Reduction of waste Ammonium sulfate	Ammonia nitrogen wastewater temperature adjustment	47
Reduction of sludge	Optimal dosage and pH adjustment	38
Others	Extending the lifetime of chemicals, empty barrels, spare parts, filter, etc.	6
Total Amount		1,498

Note: The scope of the above data statistics is UMC Group.

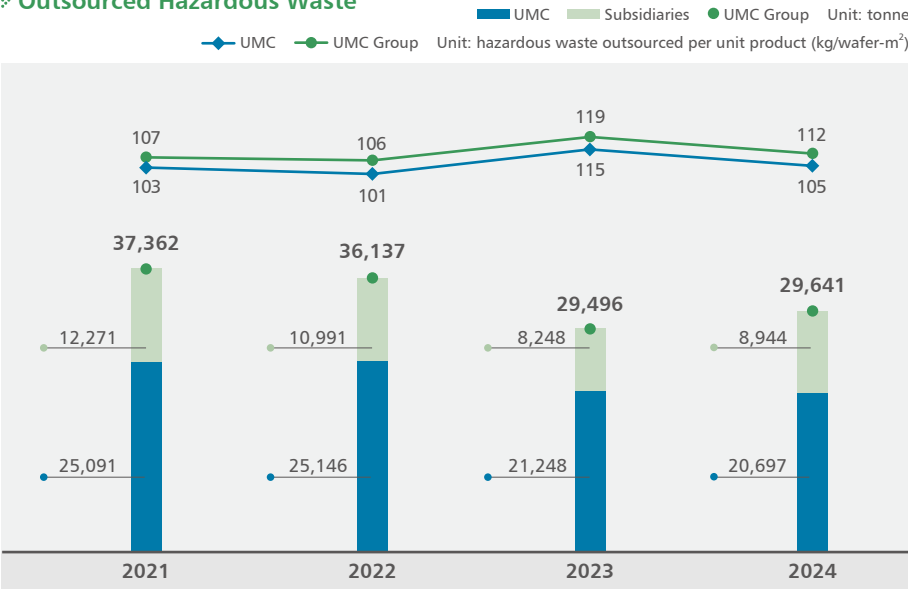
Planned Reduction Measures in 2025

- Continue to extend the lifetime of solvents to reduce the output of waste solvents
- Replace the acid-base neutralizing agent in wastewater treatment plants with waste sulfuric acid to reduce the amount of waste sulfuric acid generated
- Extend the usage of waste sulfuric acid to reduce the amount of waste sulfuric acid generated
- Extending the service life of equipment parts to reduce waste from scrapped items
- Improving the wastewater pressure filtration system to reduce the sludge moisture content and the amount of outsourced sludge treatment

Outsourced Waste



Outsourced Hazardous Waste



3.4.3 Circular Economy Promotion

In addition to reducing waste at the source, UMC Group integrates various recycling and reuse programs, investments in innovation or R&D to minimize high-impact landfill treatments. In 2024, UMC Group waste diversion rate reached 95.8%. Through the development of a circular economy model, UMC transforms waste into useful resources and can achieve three major benefits: waste reduction, reducing energy and cost consumption in waste treatment, and recycling waste into regenerated resources. Besides promoting source reduction in the production process, UMC also requires suppliers to set "waste reduction and recycling" goals in the sustainability supplier management aspect and environmental aspect to minimize waste production in the entire value chain.

Conversion of In-fab Resources

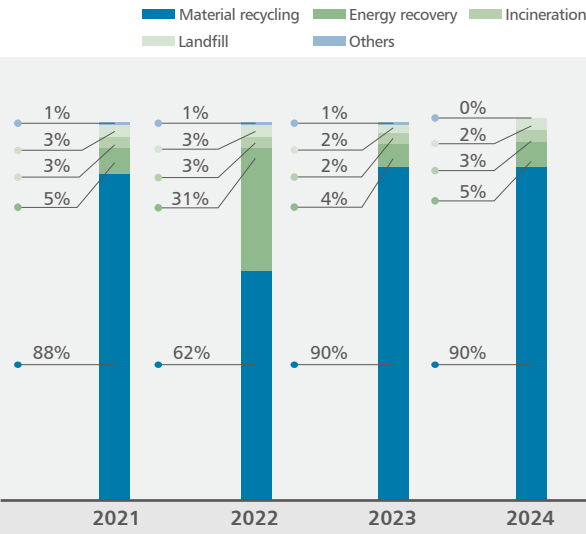
UMC Group also promotes waste recycling to its subsidiaries. In 2024, UMC Group removed hydrogen peroxide from waste dilute sulfuric acid in fabs, which is then used for ammonia nitrogen wastewater treatment or as a chemical for air pollution control equipment. Together, waste sulfuric acid was reduced by 11,700 tonnes, and the purchase of industrial grade sulfuric acid was reduced by 10,761 tonnes, resulting in total economic benefit of NT\$60.87 million for the Company.

In 2024, UMC launched a waste photomask recycling project. By removing patterns from used photomasks with cleaning equipment, they are converted into quartz substrates that can be resold. This project has reduced waste photomask generation by 2 tonnes, and the economic benefit from selling the recycled products reached about NT\$5.56 million. In the same year, UMC also launched a circular economy project for converting waste copper sulfate liquid through electrolysis into copper rods with resale value. This project reduced copper sulfate output by a total of 924 metric tons, with 27.3 metric tons of recycled copper rods produced. The economic benefit from sales amounted to approximately NT\$7.43 million.

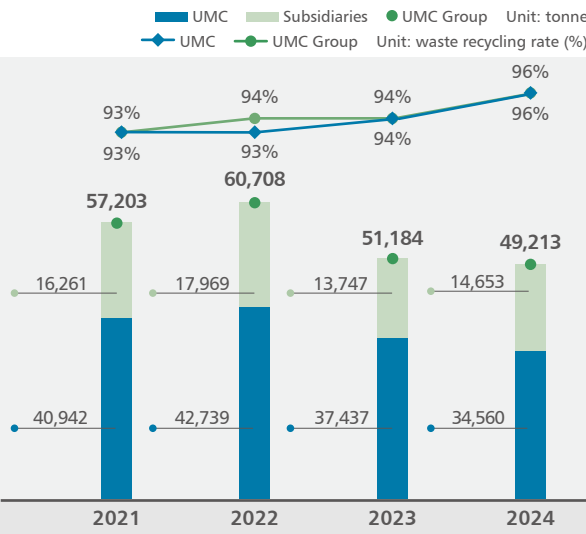
Conversion of Off-site Resources

In 2024, UMC Group amount of waste recycled outside the fabs was 48,371 tonnes, achieving a waste recycling rate of 94.2%, which includes not only material reuse but also reuse through incineration as a fuel source. The amount of hazardous waste reused outside the fabs was 29,028 tonnes, achieving a hazardous waste recycling rate of 97.9%, which also includes reuse through incineration as a resource. In the same year, approximately NT\$24 million of financial benefit was earned by fabs in Taiwan with the outsourced recycling treatment.

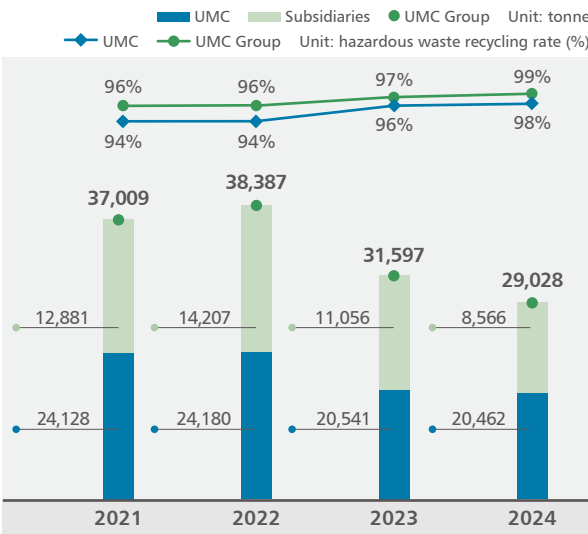
Waste Recycling Outsourcing Trend



Waste Recycling Outsourcing Status



Hazardous Waste Recycling Outsourcing Status



Note: 1. Other waste recycling methods include solidification, overseas treatment and chemical treatment.
2. Energy Recovery means that waste solvent or waste wood were recycled as an auxiliary fuel.
3. Waste diversion rate=Waste Recycling ÷ all waste; the waste diversion rate is 95.8% in 2024.

Circular Economy & Recycling Innovation Center

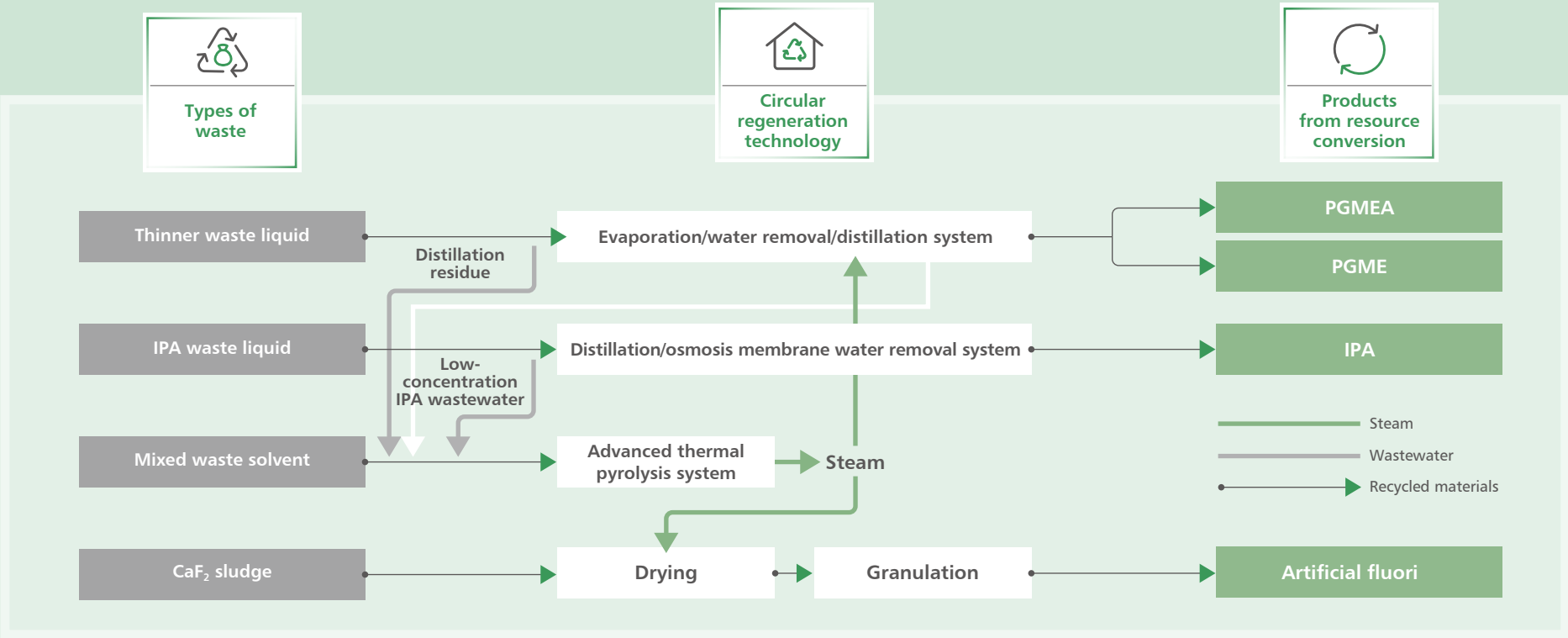
In March, 2023, UMC announced an investment of NT\$1.8 billion to establish the Circular Economy & Recycling Innovation Center at its Fab 12A. The Center will not only be the first R&D center for waste recycling in the Southern Taiwan Science Park, it will also serve as an important benchmark for circular economy and sustainable operations for Taiwan's industries. It is expected to reduce 15,000 tonnes of semiconductor manufacturing waste annually after its official launch in 2025. This investment demonstrates UMC's determination to move towards a circular economy and a zero-waste future.

The processing facilities were installed in March 2024, and by the end of 2024, the installation of processing equipment was nearly complete. Various permits will be applied for in succession, moving towards the goal of officially commencing operations in 2025.

The Circular Economy & Recycling Innovation Center is designed according to green building standards and serves as both an eco-friendly facility and a center for sustainable education.



Schematic Diagram of Circular-economy Technology



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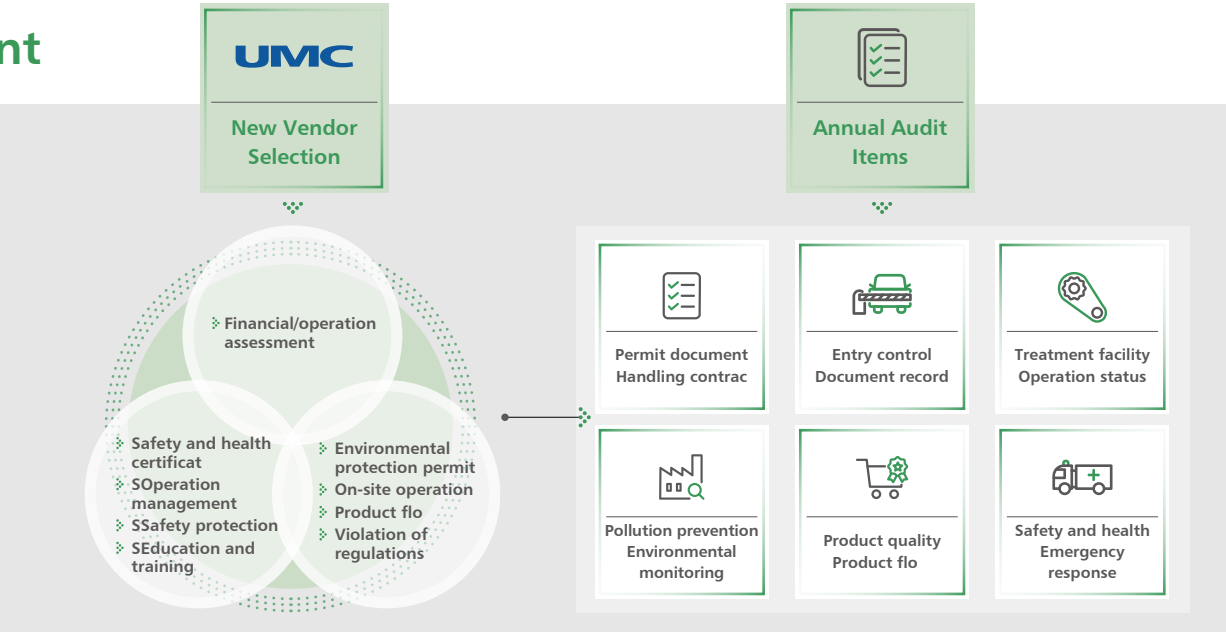
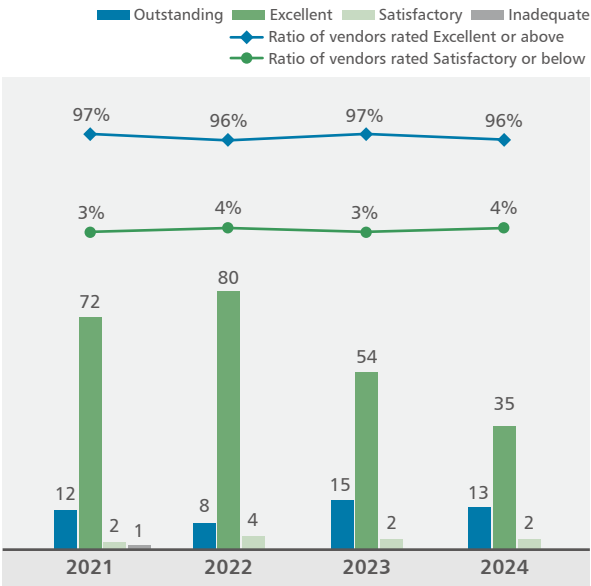
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3.4.4 Contractor Management

UMC conducts on-site audits of its waste clearance/disposal/reuse vendors to inspect their management, storage areas, treatment facilities management and pollution control, site safety management, and operating conditions (including sales flow of recycled products). Based on the audit results, the Company determines whether to maintain cooperation or increase the frequency of audits.

For the selection of new waste treatment vendors, the assessment must include the vendor's financial stability, environmental protection permits, safety and health certificates, and previous records of regulation violations. In addition, operation management, safety protection, education and training of the new vendors must be included in the assessment criteria.

In 2024, UMC completed audits of 86 waste management vendors, including 36 transportation vendors and 50 processing and recycling vendors. Audit results show that 96% vendors were rated excellent or higher, and only two vendors received satisfactory rating and zero vendor received inadequate rating. The Company will continue to conduct related audits to ensure all vendors follow UMC's requirements and regulations.



Rating and Follow-up of Waste Treatment Vendors Audit

Level	Score	Follow-up
Outstanding	≥ 90	Maintain cooperation ✦ Audit once every 2 years
Excellent	80~89	Maintain cooperation ✦ Audit once every year
Satisfactory	70~79	Maintain cooperation (list as a substitute vendor) ✦ Audit once every year
Inadequate	60~69	Put on a watch list. Increase audit frequency. Require improvement before the deadline. If no improvement made before the deadline, contract will be terminated. ✦ Audit once every 6 months
Unsatisfactory	≤ 60	Replace immediately

The disposal of UMC's waste by contractors is carried out in accordance with the regulations of the Ministry of Environment. Before disposal, the Company is required to visit the "Industrial Waste Report and Management System (IWR&MS)" on the Ministry of Environment website to make declaration in advance, confirm and revise the content on the declaration form within two days of waste transportation, and track the progress of the disposal operation within 30 days after the waste leaves the fabs. UMC has reinforced the management and control of commissioned waste disposal procedures to ensure that waste is properly delivered to the designated disposal or reuse organizations. UMC cooperated with a GPS system manufacturer to develop the "UMC GPS Virtual Vehicle Fleet System. " Since 2017, transportation vehicles for waste solvents, waste sulfuric acid, waste effluent sludge, waste phosphoric acid, copper sulfate, ammonium sulfate, waste containers, and other hazardous waste are monitored through the system. No major abnormal incidents were observed in 2024. Subsidiaries HJ, USCXM, Wavetek and USJC also track the flow of waste disposal companies to ensure proper waste management.

3.5 Air Pollution Control

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
VOC reduction	>94%*	✓ 97.3%	>95%	>97% (new fabs) >95% (existing fabs)

Note: the scope cover UMC Group (UMC, subsidiaries HJ, USCXM, Wavetek and USJC); *UMC

UMC's air pollution control strategies involve using high-performance equipment to treat air pollutants that have already been reasonably reduced to a level that complies with, or is less than, the government's environmental stipulations. Test results over the years showed that UMC air pollutant emissions is less than the emission standard set by the Ministry of Environment.

UMC categorizes waste gas from manufacturing processes into acidic exhaust, alkaline exhaust, volatile organic compound (VOC), and general exhaust. The primary management policy is to adopt and develop environment-friendly technologies, and to strengthen source reduction and pollutant prevention.

In 2024, the new VOC control equipment at Fab 8S became operational, resulting in UMC's average VOC treatment efficiency maintaining at 97.3%, which is superior to the regulatory standard of 90% and exceeds the self-imposed control target (average treatment efficiency above 94%). Consequently, UMC's total VOC emissions in 2024 were controlled at 23.95 tonnes, with a reduction of 852.02 tonnes. The VOC emissions per unit product were 0.122 kg/wafer-m2, showing a decrease compared to 2023 and indicating a year-on-year downward trend in the total amount of emissions.

Acidic and Alkaline Exhaust

❖ Stage 1: Install abatement equipment on tools to treat toxic, flammable, and fluorinated greenhouse gases.

❖ Stage 2: After undergoing treatment in a central exhaust treatment system, the gas is released into the atmosphere through a stack.



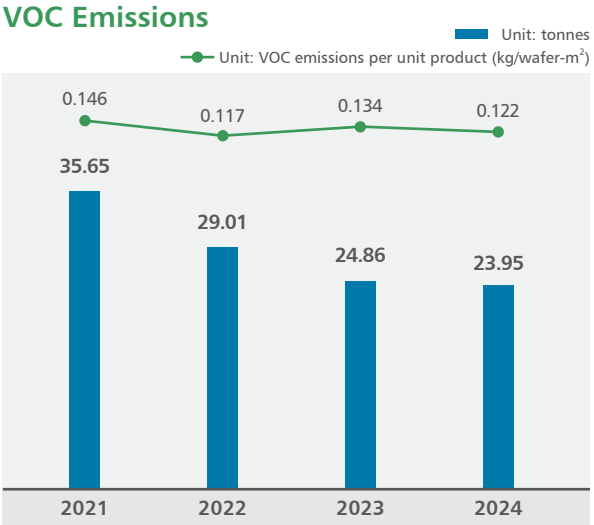
Acid and alkaline waste gas treatment facilities

Volatile Organic Compounds

❖ Volatile organic compounds exhaust (solvent exhaust) is treated using VOC Zeolite carousel wheels, which treat and release gases using low temperature adsorption and high temperature desorption.



Volatile organic compounds waste gas treatment facilities



Note: 1. Scope: UMC
2. Measurement of VOC emission: For fabs in Taiwan, emissions is the daily volume monitored by continuous emission monitoring system multiplied by the number of days. For 12i in Singapore, is the emissions monitored in December 2024 multiplied with the number of days in year 2024.

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Sustainability Strategy and Blueprint

2024

2025

2030

Item

Goal

Achievement

Goal

Protect nature and promote biodiversity

Completed biodiversity surveys at Taiwan sites

Achieved

Conduct TNFD application assessment

Cumulative execution of 5 native species conservation or restoration projects (base year: 2024)

All UMC Group fabs are located in industrial zones, and none of its operating bases are located in globally or nationally agreed "Key Biodiversity Areas (KBA)". With the aim of contributing to the United Nations' Sustainable Development Goals (SDG) "Life below water" (conserving marine ecosystems) and "Life on Land" (protecting terrestrial ecosystems), UMC's Chairman approved the "Biodiversity and No Deforestation Commitment" to demonstrate the Company's commitment to ecosystem conservation under the supervision of its sustainability governance body. In addition, biodiversity and zero deforestation requirements have also been incorporated into UMC's Supplier Code of Conduct, based on which suppliers are required to comply with relevant rules and standards, thereby promoting the value chain's joint effort to support biodiversity.

UMC Biodiversity and No Deforestation Commitment

We have established UMC Biodiversity and No Deforestation Commitment, pledging to regularly conducts biodiversity risk assessments and sets priorities to implement mitigation measures such as avoidance, minimization, restoration, and offsetting to reduce ecological impacts. The goal is to achieve a Net Positive Impact on the environment and no gross deforestation by 2050. UMC actively collaborates with suppliers and external stakeholders to respond to the United Nations Convention on Biological Diversity (CBD) initiative.



Our Actions

Biodiversity Risk Assessment

UMC began using the WWF Biodiversity Risk Filter tool in Q4 2022 to assess natural dependency and impact risks at its Taiwan facilities and other operational sites. The Company prioritizes avoidance and minimization measures for areas with medium to high risk, reducing ecological impact. The assessment showed that UMC's Taiwan facilities are not located within 2 km of any globally or nationally designated areas of biodiversity importance. UMC's environmental policy aims for zero pollution, complying with or exceeding environmental laws and international conventions. The Company continuously introduces high-performance pollution prevention technology and equipment and maintains a zero-penalty record for environmental violations, minimizing the impact of pollution on biodiversity. UMC issued the "UMC Water Resource Management Policy and Commitment" in 2015, aiming to maximize water efficiency and achieve a process water recycling rate of over 80%, exceeding regulatory standards. To reduce reliance on tap water sources, UMC has set water conservation targets in its sustainability strategy blueprint and expanded multiple sources of water supply. By 2030, the Company aims to reduce unit product water usage by 30% and achieve a 32% usage ratio of recycled water and desalinated seawater, reducing its operational reliance on natural water resources. UMC joined the TNFD (Taskforce on Nature-related Financial Disclosures) Adopters in November 2024 and plans to conduct a new biodiversity risk assessment, with the first TNFD report to be published in 2025.

Introducing Green Fab

UMC Group introduced the green fab concept in 2007. By engaging external experts in green architecture and ecology, and collaborating with academia through joint research programs, UMC constructed green buildings in accordance with the U.S. LEED standards and Taiwan's EEWH guidelines. The design features of UMC green buildings that aim to promote biodiversity are as follows:

Ecological (Green) Network: Retain continuous green belts around fabs to increase opportunities for creatures to migrate, forage, or live.

Habitat for little creatures: Create green floodwater detention ponds, design highly porous revetments, maintain a mixed forest, and preserve a meadow/shrub area to meet the needs of different creatures.

Plant diversity: In principle, choose native and diverse plants which can attract butterflies or birds to create diversified habitat conditions for creatures.

Not only UMC Group's new fabs are designed based on the principles of green buildings, green factories and smart buildings, existing fabs are also assessed based on the principles of green buildings and green factories, for which relevant green design and construction are gradually introduced.

Status of Each Fab in 2024

U.S. LEED	Gold Level: Fab 12A P3&4; Fab 12A P5&6; Subsidiary-USCXM
Industrial Development Bureau, MOEA - Intelligent Building	Diamond Level: Fab 12A P5&6; Fab 12A P5&P6 offices
Industrial Development Bureau, MOEA - EEWH Green Building	Certified Level: Fab 8A; Fab 8F; Fab 8S
	Gold Level: Fab 12A P3&4
	Diamond Level: Fab 12A P5&6
Industrial Development Bureau, MOEA - Green Factory	Fab8A; Fab8F; Fab8S; Fab12A; Fab12A P2
Industrial Development Bureau, MOEA - Cleaner Production Assessment System Certificatio	All 8-inch fabs in Taiwan, Fab 12A and Fab 12A P2

Note: Currently the Industrial Development Agency

Engagement with stakeholders on biodiversity

UMC has long supported Taiwan's environment and ecosystem conservation through efforts to promote ecological biodiversity and raise awareness of environmental protection among the general public. Since 1999, UMC has implemented voluntary greenhouse gas (GHG) emissions reduction plans and measures. The reduction achievements had passed the review of the Ministry of Environmental, and enabled UMC to obtain carbon credits. In 2014, UMC signed Taiwan's first competent authority approved carbon credits transaction with Dragon Steel Corporation, and used all the proceeds from the carbon credits transaction to launch the Eco-Echo Ecological Conservation Hope Project. Under the project, the Sauter's Frog Habitat Conservation Project and the Eco Echo Award were established, demonstrating that UMC can contribute positively to our society and ecological environment.

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4.1 Enforce Human Rights Management

4.2 Diversity, Equity and Inclusion in the Workplace

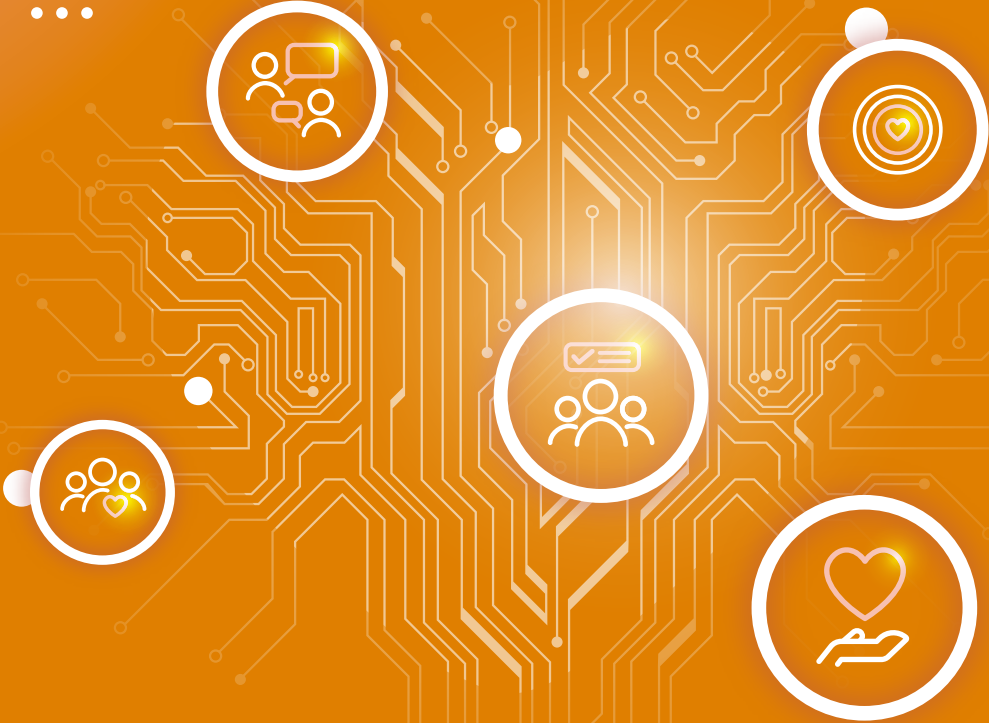
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Providing a Friendly Workplace



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- 4.2 Diversity, Equity and Inclusion in the Workplace
- 4.3 Protecting the Employees and the Work Environment

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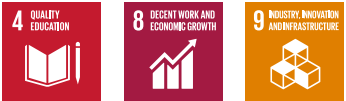
Providing a Friendly Workplace

Important Stakeholders | Employee | Supplier | Government |

Talent is the most important asset at UMC, thus UMC is committed to creating a workplace environment where employees can work in roles that suit their abilities and feel secure. UMC offers a competitive compensation and benefits system, cares about the physical and mental health of each employee, and creates a comprehensive talent development system, thereby enabling employees to give full play to their potentials in the workplace. Other than that, maintaining industrial safety is also considered a basic requirement at UMC. The UMC's corporate culture that takes industrial safety as the first priority is built with the aim to identify potential workplace safety risks in advance and create a safe workplace environment, along with a transparent and smooth worker management communication mechanism to protect employees' rights and workplace safety.

Performance Highlights 2024

SDGs



All employees completed "Human Rights" training courses

100%

Proportion of females in all UMC management positions

22.3%

Awarded the "World's Best Employer" by Forbes Magazine

UMC implements an employee stock ownership plan



The turnover rate for UMC Group is the lowest in the past 4 years

4.1 Enforce Human Rights Management

- 4.1.1 Enforce Human Rights Management
- 4.1.2 Employer-employee Communication



Material Topic : Human Rights

Related Policy

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Human Rights
Policy Declaration



Anti-Discrimination and
Anti-Harassment Policy



Negative Impact : Human rights violation

Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
If human rights issues are not properly managed, including comprehensive management policies, systems, and training, it may fail to promote and implement human rights awareness among employees and suppliers.	- Formulate anti-discrimination and anti-harassment policies. - Establish channels for submitting wrongful harm appeals. - Conduct regular human rights risk assessments, and complete the RBA Self-Assessment Questionnaire (SAQ) on a regular basis. - Participate in the RBA "Validated Assessment Program" (VAP) implemented by a third-party agency	- Designate June of each year as the month for wrongful harm prevention; track and confirm whether employees understand relevant rights and interests. - Conduct educational training sessions on human rights. - Complete the RBA SAQ on a yearly basis, and assess the risk levels. - Participate in the RBA VAP implemented by a third-party agency

Achievements in 2024

Goals	Achievements
Complete the RBA SAQ on a regular basis	All UMC fabs' scores were above 80, indicating a low risk level
Pass the RBA VAP certification	Fab12i, Fab 12A, Fab 8E, Fab 8F and Fab 8S have all completed the VAP evaluation; the audit reports have been provided to customers through the RBA-Online platform.
All employees complete the training session of human rights*	100% of completion rate for training session
All employees complete the training session of illegal violation/workplace violence**	100% of completion rate for training session ^{Note2}

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Promoting human rights equity and employer-employee communication: Continue the Promotion of Work Engagement Questionnaire Surveys***				
Coverage rate	98.7%	98.9%	98.9%*	99%*
Score	Score: 80	Score: 82.3	Score: 80.2*	Score: 80.5*

Note: 1. Scope: UMC (the parent company only); With * remark is UMC Group (UMC, its subsidiaries, HJ, USCXM, Wavetek, and USJC); **UMC fabs in Taiwan; ***UMC, its subsidiaries, HJ, and USCXM

2. Excluding those who are unable to complete the test during the testing period due to reasons such as long holidays, returning home, or resignation, etc.

4.1.1 Enforce Human Rights Management

★ Performance Highlights 2024

100% employees
completed "Human Rights"
training courses

All UMC fabs' scores of RBA SAQ were
above 80, indicating a low risk level

Human Rights Policy

UMC supports and values governing international labor and human rights stipulations, and formulated the "UMC Code of Conduct" as well as the "UMC Human Rights Policy Declaration" in accordance with the spirit of the "UN Guiding Principles on Business and Human Rights" and the "Responsible Business Alliance Code of Conduct", hoping that all employees can comply with applicable regulations while engaging in their daily work and performing the required businesses, so as to ensure the sustainable growth and development of the Company.

To ensure that the human rights policy is well implemented, UMC complies with four major principles:

- Protection**

Abide by the local laws and regulations of the operating base, UMC promises to safeguard and protect the human rights of workers, and provide a safe and healthy workplace environment.
- Respect**

UMC prohibits forced labor and child labor, provides equal working opportunities, allows freedom of association, respects privacy, and opposes all kinds of discrimination, bullying, and harassment.
- Improvement**

UMC provides clear grievance procedure, formulates and continuously improves mitigation and remedial measures to reduce the impact of human rights risk.
- Management**

UMC conducts regular human rights risk assessments, reviews potential risks in business activities, and discloses the governance of human rights issues to ensure information transparency.

More details for UMC Human Rights Policy



Human Rights Promotion Practices

UMC's human rights policy applies to UMC and its reinvested businesses, subsidiaries, joint ventures, suppliers, customers, and other business partners to ensure that daily operations and all business activities meet the requirements. UMC incorporates human rights considerations in all aspects of operations, including employees, customers, suppliers, business partners, and the social environment where UMC's operating bases are located.

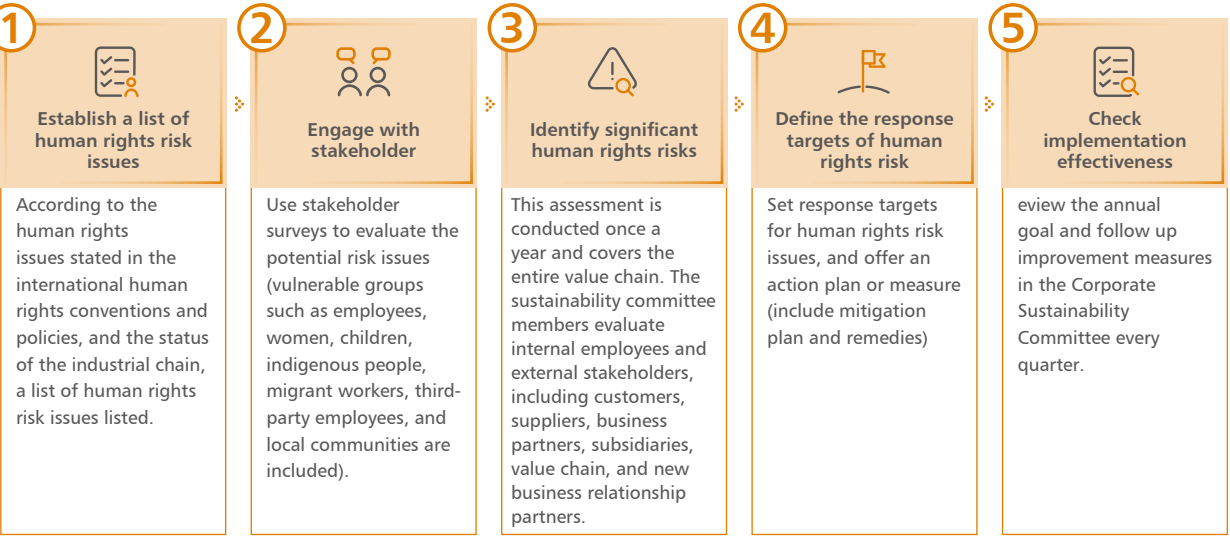
UMC and its subsidiaries, HJ and USCXM, and Wavetek, attach great importance to the promotion of labor and business ethics policies in the "Responsible Business Alliance Code of Conduct" (hereinafter referred to as the RBA Code of Conduct), and regularly review the disciplinary code of employees, highlighting the importance of labor, ethics, child labor avoidance, labor relation, freely chosen employment, working hours and anti-discrimination.

In 2024, UMC and its subsidiaries, HJ, USCXM, Wavetek and USJC provided a total of 16,254.4 hours of human-rights-related training sessions to all employees, with a training completion rate reaching 100%. In addition, to protect the basic rights and interests of workers, when major changes in operations occur, UMC will follow the relevant provisions of the Labor Standards Act in Taiwan, the Employment Act in Singapore, and the Labor Contract Law in China.

Human Rights Risk Identification Process

Every year, risk identification is conducted in accordance with the "Human Rights Risk Identification Process", and surveys on corporate social responsibility related issues are sent to stakeholders including employees, communities, and government agencies through online and paper questionnaires. Through systematic data collection and analysis, we gain deeper insights into the issues of concerns to stakeholder, which form the foundation for subsequent human rights risk management. Additionally, we develop and implement internal human rights management initiatives informed by stakeholder recommendations to ensure the comprehensiveness and effectiveness of our management strategies.

Due diligence identification objects include their own operations, value chain, suppliers, subsidiaries, business partners and new business relationships (such as mergers, joint ventures, and acquisitions). According to the risk identification process, a comprehensive due diligence assessment is conducted before the establishment of new business relationships to ensure that there are no potential risks, and is included in the Human Rights Risk Assessment disclosure scope in the year following the formal establishment of the new business relationship.



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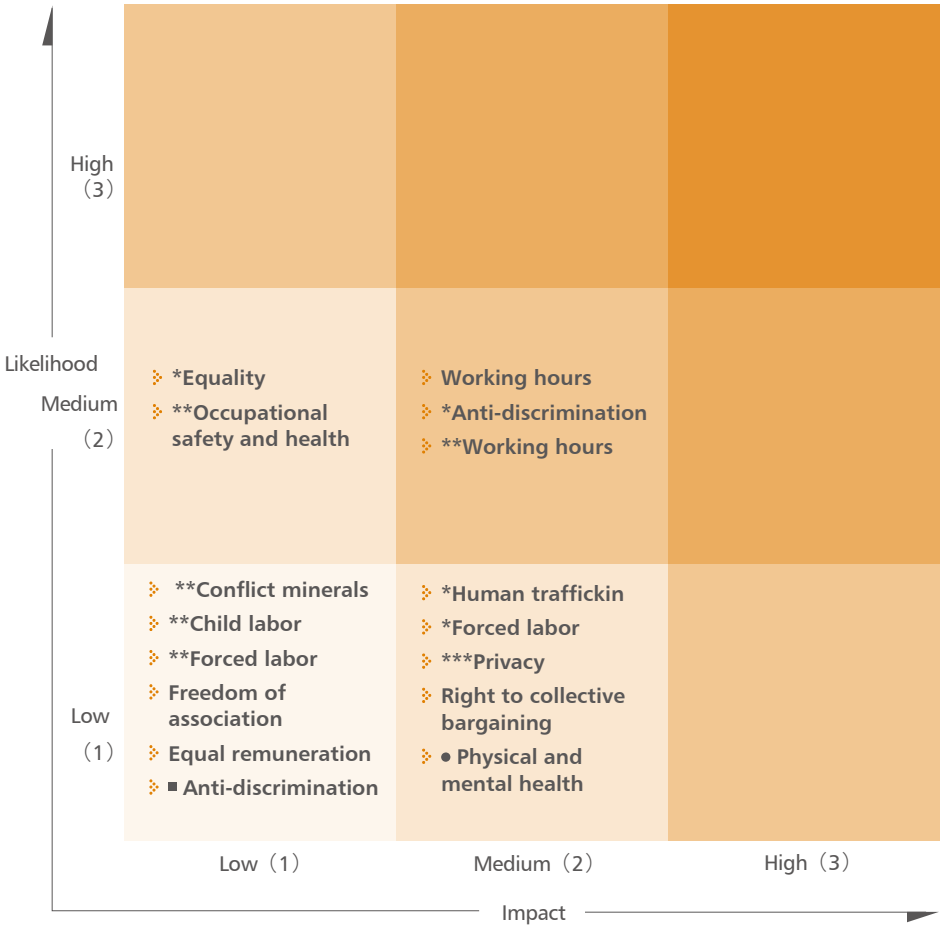
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Human Rights Risk Matrix

Human rights risk assessments are conducted by the execution members and officers of the Corporate Sustainability Committee according to the probability and severity of the risks. The results of the assessments are then used to prepare the "Human Rights Risk Matrix," and the "Major Human Rights Risk Issues" are identified according to the ranking.



Note: 1. Risk = Degree of Impact (Impact) × Probability of Risk (Likelihood). 3 points or less is low risk; between 3 points (including) to 5 points is moderate risk; and 6 points (including) or higher is high risk.

2. The relevant risk issues are indicated by symbols corresponding to specific groups: unmarked issues are common to all employees; * foreign employees and female employees; ** suppliers; *** customers; ● community; ■ indigenous peoples and persons with disabilities.

Human Rights Issue Management

UMC's board of directors serves as the highest governance body for managing human rights issues, playing a supervisory and guiding role. They regularly review the progress of human rights policy management guidelines and urge adjustments when necessary. The ESG Steering Committee is responsible for decision-making and management of human rights issues, reporting to the board of directors annually. The Corporate Sustainability Committee supervises the implementation of human rights issue management by the human rights and employee development/care group under its supervision, tracking performance on a quarterly basis.

Human Rights Governance



UMC established the "RBA (Responsible Business Alliance) Committee" in 2013 to manage human rights related issues, including labor, health and safety, environment, ethics, etc., and formulate relevant policies as well as performance targets, and regularly track and review the results of the implementation plan. The RBA Committee annually reviews whether the internal system of UMC complies with the latest RBA Code of Conduct. The risks of each fab are assessed through the RBA-Online platform to prevent potential risks from occurring. In 2024, the RBA SAQ (Self-Assessment Questionnaire) scores of all UMC fabs are higher than 80 points, indicating that the risk is low.

In addition to the internal self-inspection mechanism, UMC also commissions third-party agencies to carry out the RBA "Validated Assessment Process" in response to customers' requests. Through the checking of external audit, the overall management system can be further strengthened. By the end of 2024, all fabs in Taiwan and Singapore have obtained the VAP certification and the corresponding audit reports were provided to the customers through the RBA-Online platform.

RBA Risk Assessment Performance

Fab		SAQ	VAP	
		2024	2023	2024
12-inch wafer fab	Fab 12A	96.0	—	200
	Fab 12i	80.0	187.0	200
8-inch wafer fab	Fab 8AB	97.9	200	—
	Fab 8CD	97.3	200	—
	Fab 8E	95.9	—	181.5
	Fab 8F	87.3	—	181.4
	Fab 8S	87.9	—	193.8
Subsidiaries	HJ	75.5	—	—
	USCXM	74.8	—	—
	Wavetek	—	—	—
	USJC	77.3	—	—

Note: 1. SAQ: RBA modified the SAQ scoring structure in 2024. Full score is 100 points, ≥80 points is low risk, ≥60 points and <80 points is moderate risk, and <60 points is high risk. The subsidiary Wavetek has not implemented SAQ.

2. VAP: The certification is valid for two years and the perfect score is 200 points. The subsidiaries HJ, USCXM, Wavetek and USJC have not implemented VAP.

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Management of Human Rights Issues

According to the results of human rights risk assessment, UMC establishes mitigation and remedial measures, identifies the human rights issues related to each division, notes management and improvement measures, and tracks the results of implementation. Human rights mitigation measures and remediation measures are implemented in all 12 fabs in Taiwan, China, Japan and Singapore. There were no high-risk risks nor human rights violations identified in 2024; thus, no remedial measures were implemented. The results of such assessments enable UMC to maintain frequent and regular dialogues with stakeholders, and their feedback can serve as the basis for review and adjustment of the Human Rights commitments and policies to ensure that they meet the stakeholders' expectations and requirements.



There were no high-risk risks nor human rights violations identified in 2024; thus, no remedial measures were implemented.

2024 Human Right Risk Assessment Result

Human rights mitigation measures and remediation measures are implemented in 12 fabs in UMC, HJ, USCXM, Wavetek and USJC .

	Party of Concern	Core Issues	Risk Identification	Impact Assessment and Mitigation Measures	Remediation Action	2024 Management Goals	Achievements in 2024	Targets for 2025
 Migrant employees and female employees		Equality	Low	- The RBA Committee was established in the Human Rights Committee to regularly review human rights-related issues. - The inspection and evaluation of the "Prohibition of Forced Labor" management procedures and various grievance procedures are carried out through internal audit.	- Issue handled in accordance with wrongful harm handling procedures. Corresponding punitive measures taken for perpetrators; compensatory measures organized after negotiation based on complaint-filing employees' preference - Actively construct a comprehensive working hours management mechanism and system. If working hours exceed the legal limit, the system sends a warning to alert supervisors. In case of violations, employees' rights will be restored	- The annual online completion rate of new recruits for the "Labor/Ethics" education and training in the RBA Code of Conduct to reach 100% - All UMC fabs to complete the RBA SAQ assessment and score higher than 80 points	 Achieved	- The annual online completion rate of new recruits for the "Labor/Ethics" education and training in the RBA Code of Conduct to reach 100% - Maintain that all UMC fabs complete the RBA SAQ with a score that meets low risk
		Anti-discrimination	Medium					
		Human Trafficking	Low					
		Forced labor	Low					
 Suppliers		Conflict minerals	Low	Conduct questionnaire surveys and on-site audits on suppliers' compliance with the RBA Code of Conduct	Ensure that the suppliers have a medium-to-high risk compensation policy in place and promise to provide immediate and reciprocal compensation in the event of such a situation	- Ensure all new suppliers to sign the UMC Supplier Code of Ethics and Supplier Code of Conduct - Third-party agencies conduct questionnaire surveys and on-site audits on key suppliers' compliance with sustainability (ESG) and RBA requirements	 Achieved	- Ensure all new suppliers to sign the UMC Supplier Code of Ethics and Supplier Code of Conduct - Third-party agencies conduct questionnaire surveys and on-site audits on key suppliers' compliance with sustainability (ESG) and RBA requirements (may be conducted through video conferencing due to pandemic). Audit coverage rate reached 100%.
		Child labor	Low					
		Forced labor	Low					
		Working hours	Medium					
		Safety and health	Low					
 Customers		Privacy	Low	Regularly communicate with customers for feedback and periodically obtain customer satisfaction ratings provided by them	In case of leakage of private information, UMC will make improvement and compensation according to the terms of the confidentiality agreement	- All employees, business partners, and customers must sign confidentiality contracts, requiring mutual protection of confidential information - Internally formulate the "Confidential Information/Data Management Regulations", and all customer information is handled by dedicated units - No incidents related to violations of customer privacy or leakage of customer information	 Achieved	- All employees, business partners, and customers must sign confidentiality contracts, requiring mutual protection of confidential information - Internally formulate the "Confidential Information/Data Management Regulations", and all customer information is handled by dedicated units - No incidents related to violations of customer privacy or leakage of customer information have occurred

	Party of Concern	Core Issues	Risk Identificatio	Impact Assessment and Mitigation Measures	Remediation Action	2024 Management Goals	Achievements in 2024	Targets for 2025
 All employees	Labor relation	Freedom of association	Low	❖ The inspection and evaluation of labor relation are carried out through the labor-management meetings, quarterly company wide town hall meetings, and various grievance procedures	❖ Hold a review committee meeting to provide compensation to employees for relevant losses based on the resolutions from labor disputes or complaint records	❖ "Labor-management meetings" and "company-wide town hall meetings" are held on a quarterly basis ❖ 100% of appeal cases are handled and settled ❖ Continue to implement the "company-wide engagement survey" and strengthen organizational identification	❖ A total of 36 labor-management meetings and 4 company-wide town hall meetings were held. ❖ Achieved ❖ Implementing as planned	❖ Hold regular "Labor-management meetings" and quarterly "company-wide town hall meetings" ❖ Settlement of appeal cases: 100% ❖ Continue to implement the "company-wide engagement survey" and strengthen organizational identification
							❖ No employees proposed to organize a labor union in 2024 ❖ Formed a total of 31 employee social clubs with a total participation of 2,441 people	❖ Continue to encourage employees to balance work and life, participate in leisure and welfare activities, and continue to handle evaluations in accordance with the management measures of the social clubs to help them operate smoothly
	Equal Remuneration /Equal Pay for Equal Work		Low	❖ The establishment and implementation of various salary systems shall not differ based on gender, race, religion, political stance, marital status, etc. This ensures that employees receive fair and reasonable compensation. Regular reviews of salary and welfare systems are conducted in accordance with legal regulations and basic salary provisions to protect the rights and interests of employees	❖ Adjustments to compensation are made based on the principle of equal pay for equal work, and the rationality of the system is reviewed and preventive measures are formulated to prevent future issues	❖ We regularly monitor market salary information and provide competitive compensation packages that include differentiated pay and long-term incentive rewards. We also implement diverse tools to ensure that our compensation remains competitive. If necessary, we will adjust our salary strategy to remain competitive in the market	❖ The salary level aligns with the organization's compensation strategy and regulatory requirements	❖ The salary level aligns with the organization's compensation strategy and regulatory requirements
	Working hours		Medium	❖ Take the initiative to construct a sound working hour management mechanism and system	❖ Actively construct a comprehensive working hours management mechanism and system. If working hours exceed the legal limit, the system sends a warning to alert supervisors. In case of violations, employees' rights will be restored	❖ Continue to strengthen the working hour management system to bring the benefits of managing the working hours and caring for employees	❖ Overtime management was promoted in corporate secretary forum and labor-management meetings ❖ The overtime management system is strengthened to promote the communication and planning of overtime needs ❖ Send monthly reminders to supervisors to show care for employees in vacation planning	❖ Continue to strengthen the working hour management system to bring the benefits of managing the working hours and caring for employees
 Community residents	Physical and mental health		Low	❖ Carry out regular surrounding noise measurement and air pollution monitoring to properly manage these intrusions and reduce local residents' health risks	❖ If any equipment is found to be abnormal, it will be instantly repaired and replaced with spare equipment	❖ Noise: Contract special agency to perform surrounding noise measurement every year, and the results shall meet the governing standards ❖ Air pollution: Monitor the operating parameters of the air pollution control equipment, and the operating range meets the requirements of the licensing regulations	✔ Achieved ❖ All operating parameters of the air pollution control equipment have met the requirements ❖ The VOC removal efficiency is higher than the governing standard (90%)	❖ Noise: Contract special agency to perform surrounding noise measurement every year, and the results shall meet the governing standards ❖ Air pollution: Monitor the operating parameters of the air pollution control equipment, and the operating range shall meet the requirements of the licensing regulations

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	Party of Concern	Core Issues	Risk Identification	Impact Assessment and Mitigation Measures	Remediation Action	2024 Management Goals	Achievements in 2024	Targets for 2025
 Indigenous people and the physically challenged	Anti-discrimination	Low	Pay attention to equal employment rights during recruitment and provide fair employment opportunities	Hold open recruitment activities, and review monthly on the People with Disabilities Rights Protection Act to ensure the adequate quota of employment	- We ensure compliance with regulations regarding the employment of individuals with disabilities - Our job vacancies are publicly advertised with a complete explanation of salary and benefits information	 - Achieved - In 2024, a total of 146 employees with disabilities were recruited, complying with and exceeding the governing regulations - Providing 8 hours of ritual leave to indigenous people each year to show respects to the unique culture of different ethnic groups. In 2024, 108 indigenous people were recruited	- Meet the recruitment requirements specified in the People with Disabilities Rights Protection Act - Job are offered through open recruitment. Recruitment channels fully explain the compensation and benefits - Providing 8 hours of ritual leave to indigenous people each year to show respects to the unique culture of different ethnic groups	
Subsidiary company-USCXM All employee	Working hours	Low	- We conduct an annual review of working hours through RBA regulations and self-assessment questionnaires (SAQs) - We continuously build and improve our working hour management mechanisms and systems	- Pay corresponding overtime pay in accordance with mainland laws and regulations - If the working hours exceed the legal limit, the system will provide a warning to remind the supervisor to pay attention	Continue to strengthen the working hour management system to bring the benefits of managing the working hours and caring for employees	- Overtime management was promoted in meetings every month - Strengthen system management: If work hours exceed statutory limits, division supervisors are reminded to review anomalous situations. - Proactively remind supervisors on a monthly/quarterly basis to remind and encourage employees to plan holidays	Continue to strengthen the working hour management system to bring the benefits of managing the working hours and caring for employees	
Subsidiary company-HJ All employee	Working hours	Medium	- Each month, the number of employees exceeding the overtime limit is calculated and reported by each division. Managers are reminded to reasonably schedule and manage overtime - According to labor regulations, employees should have at least one day of rest every seven days. The attendance system will remind employees based on their card swipe records, and HR will review this information on a monthly basis	We provide overtime pay in accordance with legal regulations and review any abnormal overtime situations on a monthly basis. Managers are reminded to manage these situations and must explain the reasons for the overtime and propose measures to prevent it from happening again	Continue to strengthen the working hour management system to bring the benefits of managing the working hours and caring for employees	- Overtime management was promoted in meetings every month - Strengthen system management: If work hours exceed statutory limits, division supervisors are reminded to review anomalous situations - Proactively remind supervisors on a monthly/quarterly basis to remind and encourage employees to plan holidays	Continue to strengthen the working hour management system to bring the benefits of managing the working hours and caring for employees	
Subsidiary company-Wavetek All employee	Anti-discrimination	Low	Take notice of equal employment rights in recruitment, and provide equal employment opportunities	Maintain the practice of open recruitment, and check recruitment requirements stipulated in disability-related regulations	Conduct open recruitment for job vacancies; hiring decisions shall not consider ethnicity, nor shall employees be discriminated against in terms of salary, benefits, promotion, salary adjustment or other working conditions and employment rights	Open recruitment is conducted; salary and benefit information is clearly disclosed through recruitment channels	Conduct open recruitment for job vacancies; hiring decisions shall not consider ethnicity, nor shall employees be discriminated against in terms of salary, benefits, promotion, salary adjustment or other working conditions and employment rights	
	Right to collective bargaining	Low	Review and evaluate through quarterly labor-management meetings and different complaint channels	Hold a review committee meeting based on labor-management disputes or complaint resolution records; provide compensation for employees' losses	- Hold labor-management meetings on a quarterly basis 100% of complaint cases have been resolved and closed	 - Achieved - No complaint filed in 2024	- Hold labor-management meetings on a quarterly basis 100% of complaint cases have been resolved and closed	
Subsidiary company-USJC Young employee Female employee	Prevention of harassment	Low	Evaluate through the harassment prevention training program and the contents of the consultation desk	Appropriate penalties will be imposed on the perpetrators, and compensation measures will be coordinated based on the wishes of the complainants	- Implementation of harassment prevention web training revised in 2024 - Continuation of harassment prevention training for Leader class	- Completion rate 99% - Completed	"Harassment prevention online training completion rate 100%" "Target group: Supervisor harassment prevention completion rate 100%"	

Note: 1. UMC human rights risk identification covers its own operations, value chain, and all new business relationships (such as mergers, acquisitions, or joint ventures).
2. When conducting risk assessments, vulnerable groups such as employees, women, children, indigenous people, migrant workers, third-party employees, child labor, underage workers and local communities.

Channels of Communication

Important UMC Communication and Complaint Channels:



Based on the standpoint of safeguarding the human rights of employees, the Company also has a confidentiality mechanism for employees in complaints or incidents to ensure that employees can file complaints freely and confidentially. Among UMC's communication channels, the "e-Suggestion" platform has the highest utilization rate. In 2024, it received 191 opinions and suggestions from colleagues, and the settle and closing rate was 100%.



- ✧ "UMCWe United Microelectronics Website" is an external website that provides a platform that colleagues can use anytime, anywhere. It integrates benefits, societies, and special stores to strengthen the interaction between the organization and colleagues.

Freedom of Association

According to the rights granted to workers by the law, the employees of UMC shall have the right to form associations freely, and UMC shall not interfere or intervene with the employees' freedom of association. UMC provides a diversified and sound communication mechanism to effectively understand the employees' needs and deal with the problems they are facing. In Taiwan, labor-management meetings are held according to Regulations for Implementing Labor-Management Meeting of Ministry of Labor. Labor representatives are elected through free elections, and the "Labor-Management Meeting Labor Representative Online Voting" is conducted by the company's voting system based on the legal gender ratio, with the voting results publicly shared with the colleagues. Elected employee representatives communicate and express employees' concerns and opinions for discussion in the quarterly employer-employee meeting. Over the years, internal communication has been smooth, achieving a 100% employee coverage rate. In addition, the China subsidiary HJ and the Japan subsidiary USJC also established labor unions, and sign collective contracts to coordinate labor relations and safeguard the labor rights and interests of enterprise employees. The Company continues to communicate with employees through forums, divisional meetings and suggestion boxes.

Area	Item	2021	2022	2023	2024	Details	Frequency
UMC Group	Town Hall Meeting	4	4	4	4	Interactive video conference for 7 sites spanning multiple countries and regions hosted by the president himself, allowing UMC colleagues and external parties to review the latest company policies, directives, and performance.	Held quarterly after the investor conference.
UMC (Taiwan)	Fab and Divisional Communication Meetings	96	88	87	90	Share overall operational performance as well as developmental highlights of the fabs and divisions with all employees.	Held semi-annually by each fab and division.
	Secretary Conference	8	8	8	8	The Human Resource (HR) division shall collect key topics and discuss them with all secretaries. And secretaries shall convey key issues of the meeting to colleagues to achieve two-way communication.	Held quarterly at fabs in Hsinchu Science Park / Southern Taiwan Science Park
	Employer-employee Meeting	36	36	36	36	The HR division shall delegate colleagues and employees familiar with relevant regulations to organize and assemble cross-unit and cross-functional consulting team. The team shall follow up on specific meeting topics and facilitate subsequent improvement plans to effectively promote harmonious employer-employee relationships.	Held quarterly at 9 fabs.
	Employee Benefit Committee Meeting	4	4	4	4	The employee well-being activities as well as the usage of employee benefit funds shall be reported to the employee benefit committee member delegated by each fab in the quarterly meeting.	Quarterly
HJ	Employer-employee Meeting	12	19	4	4	Any questions that employees have, such as difficulties and problems at work, can be raised at the meeting, and the supervisor and the person in charge will answer them.	Aperiodic
USCXM		4	4	3	2	Manufacturing division direct labor conferences: Two conference sessions attended by HR and manufacturing division direct labor to discuss issues mainly related to compensation and benefits, food/housing/transportation, appraisal/attendance, promotion, etc. Two meeting sessions related to benefits matters	Aperiodic
Wavetek		4	4	5	4	The HR division appoints employees and supervisors who are familiar with relevant laws and regulations to form a cross-unit and cross-functional consulting team to be responsible for the follow-up and improvement of meeting topics, effectively promoting harmonious labor relations.	Aperiodic
USJC		2	4	3	4	Hold aperiodic meetings and communicate with the trade union	Aperiodic

Note: UMC Group includes UMC and its subsidiaries HJ, USCXM, Wavetek, and USJC.

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Employee Work Engagement Survey

UMC has always adhered to a people-oriented spirit by regarding UMC employees as the Company's most important asset. Therefore, caring about employees, listening to their feedback and making corresponding improvements are important to the Company. In the past, UMC used the "employee satisfaction survey" for analysis, but at the end of 2018, the "employee engagement survey" was introduced in line with international trends. In 2022, UMC has already expanded the targets of the survey from UMC Taiwan and Singapore fabs to include all employees of its subsidiaries in China, and added all employees in Japan in 2024. The overall survey result in 2024 was 80.9 points, which is higher than the semiconductor industry norm of 76 points. It is hoped that the organization's key issues can be identified through the survey report, based on which improvement countermeasures can be taken to retain key talents.

Achievements of the Year

In 2024, efforts to strengthen work-rest balance and policy promotion are implemented. Systematic reports and work hour management are established to achieve workload balance. Additionally, physical courses are conducted, including upward and cross-division communication, organizational problem diagnosis, and other communication and agility courses, to enhance communication and problem-solving skills. The results indicate that considerable improvement had been achieved.

Post-survey Improvement Plans and Practices

As for the collected survey results, in addition to confirming the effectiveness through cross-comparison of annual variations, corresponding communication was also carried out based on various division functions for the implementation of quarterly progress tracking. In regards to the less-than-satisfactory items indicated in specific units' 2024 survey results, the extent of corresponding improvements was greater than the Company's average improvement extent, demonstrating the effectiveness of the countermeasures. Moreover, in terms of the projects proposed to improve work efficiency and labor balance for the Company's overall operating environment, work patterns have been adjusted through further review of work hours and diagnosis of reasons for employee turnover in the hope of enhancing retention rates at fabs using advanced process technologies.

2024 Employee Work Engagement Survey Outcome - by age

Age	Scope	Average	<30	30-39	40-49	50-59	≥60
Engagement Survey	UMC	81.6	79.9	80.0	82.0	84.6	86.2
	UMC, Subsidiary HJ and USCXM 、USJC	80.9	80.4	81.0	81.6	79.4	86.8

Note: The distribution population is the number of full-time employees within the scope of the survey conducted in 2024

Engagement Questionnaire Topics



» **Work Relationships (12 questions)**
Employee Cooperation 5
Leadership Relationships 7



» **Work Content (11 questions)**
The Work Itself 7
Working Environment 4



» **DEI (2 questions)**
DEI 2



» **Working Conditions (14 questions)**
Remuneration and Rewards 6
Appraisal and Promotion 4
Learning and Growth 4



» **Employee Engagement (15 questions)**
Employee Involvement 15



» **Organizational Factors (13 questions)**
Corporate Culture 6
Business Operation 7

The key points we inquire through the questions of the Work Engagement Questionnaire encompass the following concepts:

Satisfaction at work

As far as my job responsibilities are concerned, my salary and benefits are fairly good

Clarity in the workplace

The authority and responsibility of each role in our division are clearly defined, where situations of "having responsibility but no authority" or "having authority but no responsibility" do not exist.

Sense of happiness at work

I like this job, and am always happy at work.

Stress at work

Moderate work stress is acceptable to me.

Survey Target and Outcome

Year	Coverage Rate	Work Engagement Index			
		Target	Overall	Male	Female
2021	97.6%	77.1	79.5	80.6	77.9
2022	97.6%	79	79.4	80.2	78.0
2023	98.6%	79.6	81.7	82.3	80.8
2024	98.9%	80	80.9	80.7	81.2

Note: 1. In response to the rating standards stated in the official definition of Dow Jones Sustainability Index (DJSI), the calculation method of survey coverage rate has been adjusted to: Total number of collected questionnaires / Supposed number of respondents.
2. Statistics from 2020 to 2021 included all UMC employees in Taiwan and Singapore. Employees of the Company's subsidiaries HJ and USCXM have been included in the survey since 2022.
3. In 2024, the employee coverage rate of UMC's Taiwan and Singapore fabs is 98.9%; the engagement score is 81.6 points

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Comprehensive Appeal and Employee Support Systems and Channels

Based on whistleblower program and the "zero tolerance" principle for all kinds of illegal infringements in the workplace in accordance with relevant reporting measures, UMC encourages open communication between employees and stakeholders. When experiencing injustice or doubts, a whistleblower can report in a named or anonymous way. The Company provides access to file complaints 24/7 in both Chinese and English, allowing stakeholders to report issues without any language or time barriers.

The whole investigation process for grievances is carried out with full confidentiality. Each complaint channel has a dedicated window to ensure that the information is properly preserved and only relevant personnel are informed. After receiving the complaint, an investigation team will be set up to handle it within thirty days after receiving the relevant complaint or notification. During the process, the privacy protection of the complainant will be paid attention to, and the preliminary investigation results will be notified to the parties in writing. To protect the parties involved from improper retaliation and treatment, the company has established internal regulations such as "Workplace Sexual Harassment Prevention Measures, Complaints and Disciplinary Regulations," "Prevention for Workplace Wrongful Harm When Performing Job Duties," "Regulations on Whistleblowing," etc to explicitly safeguard the rights of complainants and prohibit retaliation against them and those who assist in the investigation. Employees who engage in improper conduct will be subjected to disciplinary actions such as warnings or dismissal, depending on the severity of the situation.

The grievance process is investigated in a confidential manne



All case records will be properly preserved in accordance with laws and company regulations, managed by the Human Resources Department. All case records will be kept for at least 3 years, which can only be accessed by personnel related to the handling of the cases to ensure that complainants' privacy is protected. With the aim of comprehensively communicating with employees and effectively solving problems, in addition to the aforementioned communication platforms, UMC has also set up the following grievance mechanisms and channels:

Employee Relationship (ER Service) Hotline
extension 12885
Human Resources division
Employee Relations Dept. Deputy Division Director

Report of sexual harassment and unfair treatment
extension 31995; mailbox: 31995@umc.com
Human Resources Division
Senior Division Director

Fraud and Ethics Violation Report system
External reporting hotline: 0800-024-399
Whistleblower Hot Lines: 03-5782258, ext. 31425
Legal And Compliance Senior Division Director

CHO's e-mail box
UMC_CHO_Mailbox@umc.com
Human Resources Division
Vice President

CSR mailbox
csr@umc.com
Corporate Sustainability Committee

Information Security and Confidentiality Protection Complaint
Infosec@umc.com
Corporate Security Deputy Division Director

In addition to establishing a comprehensive set of communication channels and platforms, UMC shall continue to improve upon the effectiveness of communication channels and carry out projects to enhance communication of key topics and information throughout the Company, ensure the comprehensiveness and depth of communications, and strengthen global communication capacities for every employee. A total of 156 formal and large-scale conferences were held in 2024 to effectively communicate key topics on UMC businesses.

UMC will investigate and implement follow-up improvements based on the issues related to the complaint cases, and use educational and training tools (such as e-mail promotion and online testing) to require employees to comply with the code of conducts. In 2024, UMC received no labor affairs related complaint.

In addition, to clearly express our "zero tolerance" stance against various forms of workplace misconduct and to commit to establishing a safe, dignified, non-discriminatory, mutually respectful, and inclusive workplace culture, UMC has designated June of each year as the "Workplace Misconduct Prevention Month," during which comprehensive publicity and training will be conducted.

Regarding employee training on workplace misconduct, we provide an in-depth introduction to the basic concepts and types of workplace misconduct, as well as the Company's complaint channels and processes for handling such issues. Additionally, we offer knowledge related to gender, cultural diversity, and anti-discrimination to enhance employees' sensitivity to these issues, ensuring that all colleagues share the responsibility of maintaining a workplace free from misconduct.

For supervisors, we strengthen their ability to prevent and address workplace misconduct incidents at least once a year, particularly in identifying changes in employee behavior, responding to potential violence risks, effectively handling aggression, and eliminating hostile situations. Through the "Workplace Misconduct Self-Assessment Checklist," we require supervisors to self-reflect on inappropriate behaviors and to model appropriate conduct, fulfilling their supervisory role.

In 2024, a total of 8,526 indirect employees and 4,182 direct employees participated in the training, achieving a completion rate of 100%. Additionally, 1,466 supervisors at the section level and above used eCheck to self-assess whether they had engaged in inappropriate treatment of employees, also achieving a completion rate of 100%. These efforts not only demonstrate our commitment to workplace safety but also lay a solid foundation for creating a more inclusive and respectful work environment.

Numbers of Grievances

Category / Year	Human Rights Issues				Labor Affairs			
	2021	2022	2023	2024	2021	2022	2023	2024
Filed through the mechanism	9	6	8	13	11	10	20	10
Addressed	9	6	8	13	11	10	20	10
Reviewed	8	4	8	11	11	8	11	6
Officially undergoing judiciary proceeding	1	0	0	1	0	0	0	0
Resolved	9	6	8	13	11	10	20	10

Note: 1. The human rights category includes complaints such as sexual harassment, illegal infringement, and forced labor. There was 1 sexual harassment case filed in 2024.
2. The labor affairs category includes work hours and salary, occupational safety, education and training, and promotion/benefits. Human Rights Issues include: workplace violence, sexual harassment and discrimination.
3. Please refer to the UMC Code of Conduct https://www.umc.com/en/Html/UMC_code_of_conduct

4.2 Diversity, Equity and Inclusion in the Workplace

- 4.2.1 Human Resource Development
- 4.2.2 Compensation and Benefit
- 4.2.3 Strengthening Talent Cultivation

Material Topic Employee Diversity and Inclusiveness

In accordance with GRI Standards (2021) for identifying impacts of sustainability issue s, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Diversity and Inclusion Statement



Positive Impact workplace diversity and equity

Impact	Management Actions	Processes Used to Track the Effectiveness of Action
Issue a diversity and inclusion declaration, ensure equal opportunity in talent selection, promote employment for various groups, and strive to facilitate employee communication, enhancing recognition and cohesion.	- Formulate the Diversity and Inclusion Statement, and regularly review the effectiveness of acts for achieving diversity and inclusion. - Set a target for the percentage of women in managerial positions, and organize lectures to promote gender equity. - Prioritize the career development of women and improve the underrepresentation of women: Setting a target for the proportion of newly hired female engineers to reach 15% by 2025 and 20% by 2030 as a challenge. - Develop "Workplace of Happiness" measures: Formulate maternity leave/paternity leave standards that are better than mandatory requirements.	- Management KPIs are established and progress towards achieving them is summarized by the "The Human Rights and Employee Development/Welfare Team" each quarter and reported to the Corporate Sustainability Committee. - Participate in international ESG ratings such as DJSI, MSCI, etc.

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Proportion of women among managerial personnel	22%*	22.3%	23%	24%

Note: the scope covers UMC Group (UMC, subsidiaries HJ, USCXM, Wavetek and USJC); *UMC

Material Topic Talent Attraction and Retention

In accordance with GRI Standards (2021) for identifying impacts of sustainability issue s, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Positive Impact Enhancing corporate competitiveness and providing high-quality job opportunities

Negative Impact Talent drain

Impact	Management Actions	Processes Used to Track the Effectiveness of Action
Positive: in response to business operation needs, enhance talent recruitment and retention through improved welfare measures and competitive salaries. Negative: creating a manpower gap, which in turn increases recruitment and training costs, leading to a decline in business competitiveness.	- Regularly participate in campus recruiting activities and interact with target groups for recruitment. - Carry out strategic manpower planning and develop a complete recruitment blueprint based on the Company's business focuses. - Keep track of market salary information, provide differentiated compensation and introduce diversified and long-term incentive/rewarding tools to ensure that an overall remuneration system with market competitiveness is being offered, and adjust compensation strategies when necessary. - Work with divisions with high turnover rates, and establish employee care mechanisms, through which the division can be alerted in advance about employees' intention to resign. - Offer diversified career development paths, e.g. internal reassignment. - Promote key talent development plans.	- Regularly examine the progress and results of talent development programs. - Conduct at least one market competitiveness analysis per year. - Participate in monthly meetings held by divisions with high turnover rates, and conduct turnover rate analysis reports. - Continue the promotion of Work Engagement Questionnaire survey - Explore key weaknesses in the organizational climate and take countermeasures. Keep focusing on efforts to improve the issues to enhance employee retention. - Management KPIs are established and progress towards achieving them is summarized by the "The Human Rights and Employee Development/Welfare Team" each quarter and reported to the Corporate Sustainability Committee.

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Mid-high-performance employee retention rate	≥92.8%	96.1%	≥93.5%	≥94%

Note: Scope covers UMC; * means UMC fabs in Taiwan only



Connect Accept Respect Embrace all

Diversity, Equity and Inclusion Commitment and Practices

Fostering an inclusive workplace that nurtures a sense of belonging and personal achievement, together we build a brighter future.

We attract diverse talents through multiple channels, foster innovative thinking, and respect individual differences.

We offer equal opportunities and resources for all employees, ensuring fair treatment and the realization of their full potential.

We prioritize workplace inclusion, value employee opinions, and provide respect and support, to ensure that employees experience a sense of self-worth and belonging.

UMC encourages a diverse and open workplace culture and respects the uniqueness of each employee. When recruiting employees, prioritize selection based on the professional skills required for each position, regardless of race, color, age, gender, sexual orientation, gender identity or expression, ethnicity or national origin, disability, pregnancy, religion, political affiliation, union membership, covered veteran status, protected genetic information or marital status and other factors. In 2024, the Company hired a total of 1,021 new employees, including young generation, aboriginals and people with disabilities. UMC believes that diversified employee composition will bring different perspectives and opportunities for progress. By establishing a diverse and inclusive workplace culture, it will help employees exert their strengths at work and put forward different thinking, which will become the key to corporate growth.

Starting from 2022, efforts to enhance a diverse, equal, and inclusive workplace have been launched, including setting goals for increasing the proportion of female executives and continuously promoting various strategies. In 2023, under the banner of "U Care, We Care," UMC pledges to actively create an inclusive workplace environment that fosters a sense of belonging and self-fulfillment, and collaborates with employees to build a better future together.

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Appendix

UMC Diversity and Inclusion Statement

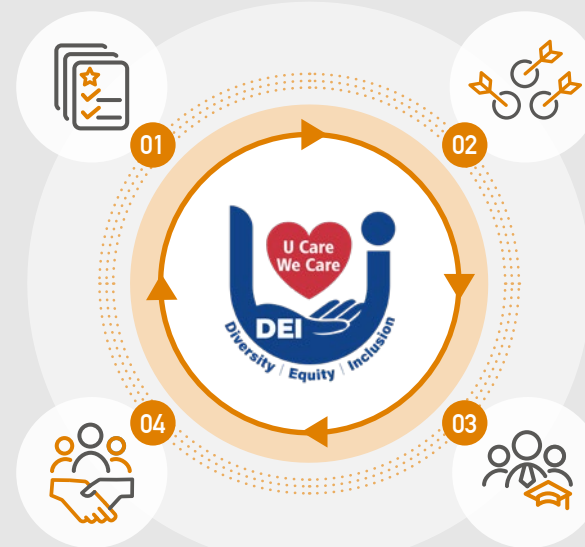
The Diversity and Inclusion Declaration was signed by the co-president and chairman of the Corporate Sustainability Committee in 2022, and announced on the Company's official website. UMC respects the uniqueness of every employee, and emphasizes equal treatment and the provision of a friendly work environment. Work rights equity is valued when recruiting employees. Recruitment prioritizes the required professional skills for the position, and does not discriminate based on race, color, nationality, ethnicity or ethnic origin, gender, marital status, age, disability status, family status, sexual orientation, political beliefs, or other factors.

In terms of creating a diversified, equal and inclusive workplace environment, UMC has systematically built up a sense of equality and inclusion through the establishment of a high-level supervision mechanism, the setting of diversified goals, the promotion of education and training, and the enhancement of friendly workplace welfare practices. Meanwhile, considering the characteristics of the semiconductor industry, where the ratio of women in management and engineering is under-represented, UMC has set a target for the employment of female engineers to ensure diversified employment, and organized activities such as outstanding women's lectures to assist female employees' career development.

01

Senior Management Oversight

- ✦ The Human Rights and Employee Development/Welfare Team promoted the implementation of tasks for diversity and inclusion, and the Chief HR Officer supervised the promotion progress. Presentations on progress and objective achievement status were given at the quarterly meetings of the Corporate Sustainability Committee and ESG Steering Committee



04

Reinforcing Acts to Enhance Friendly Workplace Welfare

- ✦ On March 8, 2024, International Women's Day, a women's employee resource group (ERG) called "United for Women" was established, with 476 employees in attendance to show their support. Through 3 tea gatherings, 2 career seminars, and 9 women's empowerment article sharing sessions, we successfully raised awareness among employees. Over 2,846 participants took part in these events.
- ✦ Promoting "Workplace of Happiness" program: The Company planned and promoted self-determined working hours in 2023, allowing employees to arrange their working hours in a flexible way. The Company formulated new maternity leave and paternity leave standards that exceed the mandatory requirements (maternity leave: increased from the mandatory 448 hours to 480 hours; paternity leave: increased from the mandatory 56 hours to 80 hours).
- ✦ To take care of newly recruited female employees, UMC provides better-than-required benefits by law, offering full salary maternity leave to those who have worked for less than 6 months.
- ✦ Encouraging employees to have children: Measures ranging from wedding subsidy, care and nursing consultation during pregnancy, maternity leave, maternity inspection leave, exclusive parking space, priority meal pickup, maternity subsidy, unpaid parental leave, certified breastfeeding room, night shift transfer and other measures have been adopted.
- ✦ In order to meet employees' needs for children's education, the Company has signed collaborative agreements with neighboring kindergartens and provided employees with assistance in registering at experimental elementary and high schools in HSP and STSP. In 2022, the Company collaborated with the STSP Administration to set up a nonprofit kindergarten at Lian-Yuan Dormitory Complex to provide more comprehensive assistance for employees in taking care of their families.

02

Setting up Strategic Objectives

- To improve the diversity of manpower composition in the workplace and to increase the proportion of under-represented groups through target setting:
 - ✦ Making efforts to achieve gender equity: Focus on issues related to female employees' career development and set a target for the percentage of women in senior managerial positions.
 - ✦ Ongoing planning for long-term goals: Set targets for the overall ratio of women in all managerial positions.
 - ✦ Compliance with regulations for the employment of persons with disabilities.
 - ✦ Set a target for the proportion of new female engineers to reach 15% by 2025 and 20% by 2030 as a challenge. Implement various proactive measures, including additional referral bonuses, internship and pre-employment mechanisms, internal promotion target setting, and encouragement, to strengthen the recruitment of female talent in technology.

03

Promoting Educational Training

- Based on the diversity and inclusion declaration, commitments, and practices outlined on the official website, implement education and training at all levels to ensure that all employees establish a consensus on diversity and equality
 - ✦ Starting with awareness and understanding, we launched diverse and inclusive training programs at UMC, HJ, and USCXM. These programs aim to help participants recognize and overcome unconscious biases, fostering better collaboration and communication among colleagues in the workplace. A total of 15,641 people completed the training in 2024.
 - ✦ Emphasizing female employees' career development: Two experience-sharing lectures related to exemplary female leadership and management were held.
 - ✦ Strengthening gender equality awareness: Designate June of each year as the month for wrongful harm prevention, and provide all employees with relevant educational training. The training completion rate in 2024 was 100%.
 - ✦ Compliance with the RBA (Responsible Business Alliance) Code of Conduct: Provide tests for employees on a yearly basis to ensure employees' awareness of core content such as laborers, ethics and integrity, child labor, labor-management relations, forced labor, working hours, and anti-discrimination. In 2024, UMC provided a total of 16,254.4 hours of human-rights-related training sessions to all employees, attaining a training completion rate of 100%.

4.2.1 Human Resource Development

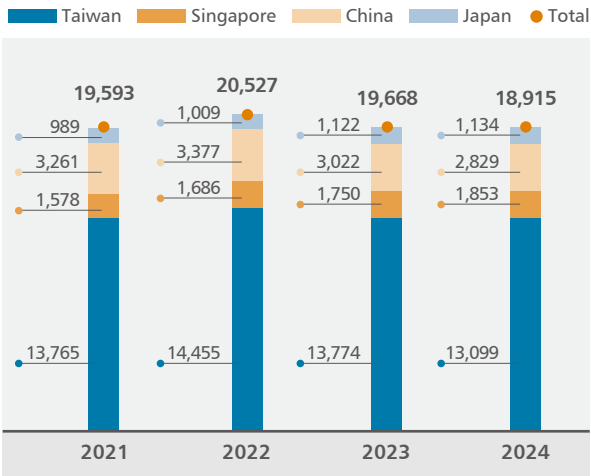
Human Resource Distribution

By the end of 2024, the total number of formal employees at UMC Group was 18,915. This figure included 2,140 managers, 10,079 engineers, 643 administrators, 6,052 technicians, and 1 clerk.

The working population within the Company can be divided into 2 categories by type of employment, namely formal employees (98.7%) and non-formal employees (1.3%). Formal employees can be further subdivided according to the type of their contracts, namely non-regular contracts (97.9%) and periodic contracts (2.1%). Non-formal employees include temporary staff (46.2%) and dispatched staff delegated by external vendors to provide services in UMC (53.8%). Temporary staff positions were offered to temporary replacement for employees taking maternity / paternity leaves. These positions will be kept open for the said employees when they return to UMC.

For work area distribution, almost 69.2% of employees work in the primary business locations in Taiwan. For age distribution, 66.0% of the total employee population was between 30 and 50 years of age. The overall average age was 40.0 years old.

UMC Group Manpower Trend



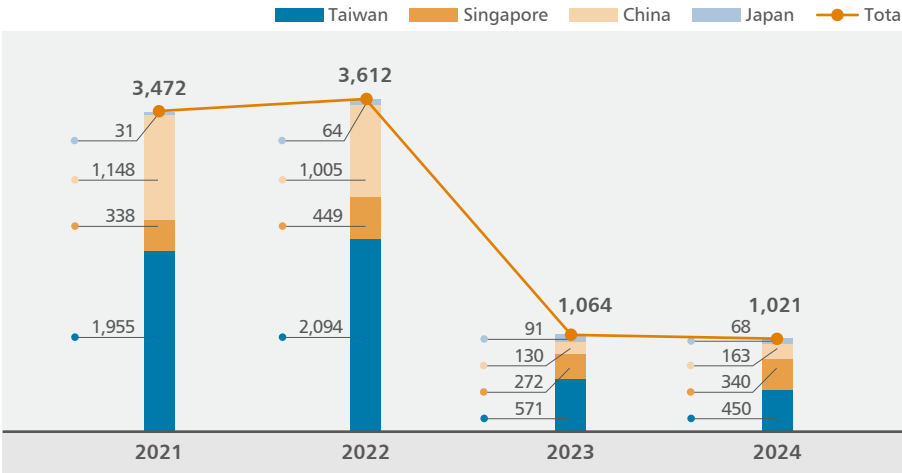
Category	Type		Male		Female		Total	
			No. of Persons	Percentage	No. of Persons	Percentage	No. of Persons	Percentage
Total Employees Type of Employment	Formal employees	Non-regular contracts	11,259	60.8%	7,264	39.2%	18,523	96.7%
		Periodic contract	3	0.8%	389	99.2%	392	2.0%
	Temporary staff		81	69.8%	35	30.2%	116	0.6%
	Dispatched staff		97	71.9%	38	28.1%	135	0.7%
Formal Employees Job Category	Managers		1,862	87.0%	278	13.0%	2,140	11.3%
	Engineers		8,007	79.4%	2,072	20.6%	10,079	53.3%
	Administrators		138	21.5%	505	78.5%	643	3.4%
	Technicians		1,254	20.7%	4,798	79.3%	6,052	32.0%
	Clerks		1	100.0%	0	0.0%	1	0.0%
Formal Employees Job Site	Taiwan		7,486	57.1%	5,613	42.9%	13,099	69.2%
	Singapore		1,214	65.5%	639	34.5%	1,853	9.8%
	China		1,556	55.0%	1,273	45.0%	2,829	15.0%
	Japan		1,006	88.7%	128	11.3%	1,134	6.0%
Formal Employees Age	Under 30		2,101	67.9%	994	32.1%	3,095	16.4%
	30-50		6,931	55.5%	5,552	44.5%	12,483	66.0%
	Above 50		2,230	66.8%	1,107	33.2%	3,337	17.6%

Note: 1. UMC Group includes subsidiaries HJ, USCXM, Wavetek and USJC.
2. The gender ratio of employees is calculated on the same work category, while the total ratio is calculated based on work type.
3. As for the temporary staff, based on the regional distribution, there are 72 people in Taiwan, 7 people in Singapore, 32 people in China, and 5 people in Japan.
4. As for the dispatched staff, based on the regional distribution, there are 19 people in Taiwan, 1 person in Singapore, and 115 people in Japan. Their main job function is production line support.
5. There is no part-time or non-guaranteed-hours employees in UMC Group.
6. In 2024, 14.2% of UMC's formal employees were under 30 years old, 69.1% were between 30 and 50 years old, and 16.7% were above 50 years old.

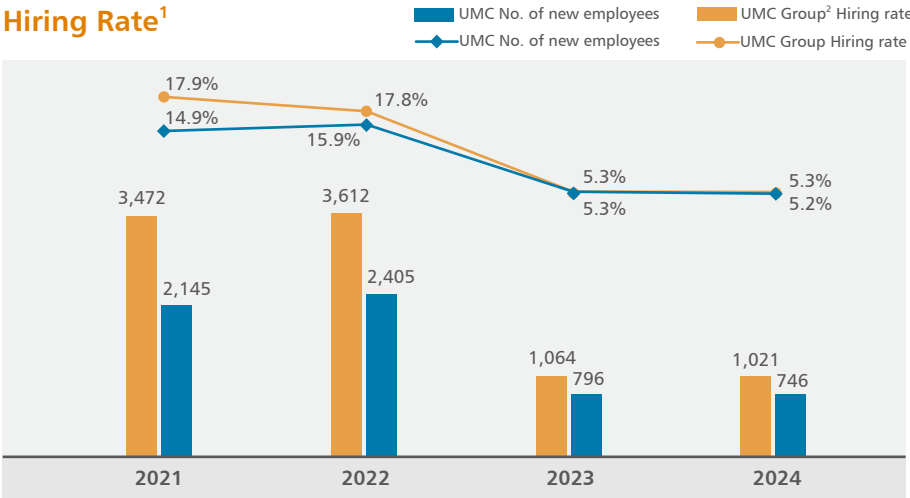
New Employees

In 2024, UMC Group hired a total of 1,021 new employees, with an average cost of around NT\$87,204 per hire. Out of the total number of new employees, 746 were hired by UMC, with an average recruitment cost of around NT\$73,052 per hire.

Number of New Employees



Hiring Rate¹

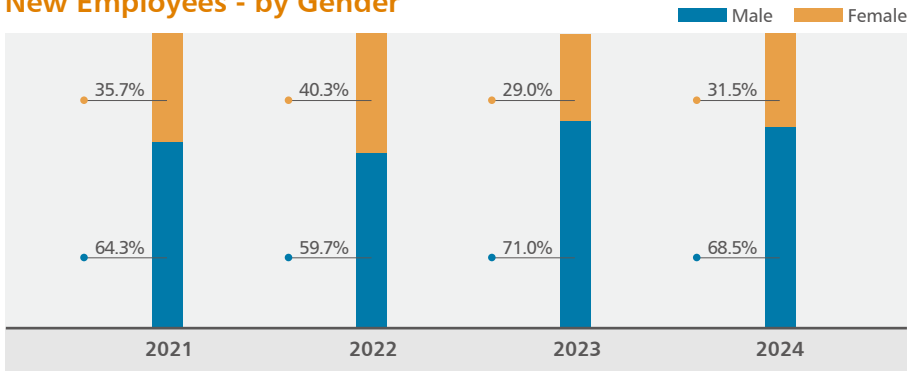


Note: 1. Annual employee hiring rate refers to the total number of newly hired employees/average number of existing employees. Annual average number of employees = (number of employees at the end of December of the previous year + total number of employees at the end of each month of the current year) ÷ 13

2. Including UMC and its subsidiaries HJ, USCXM, Wavetek and USJC.

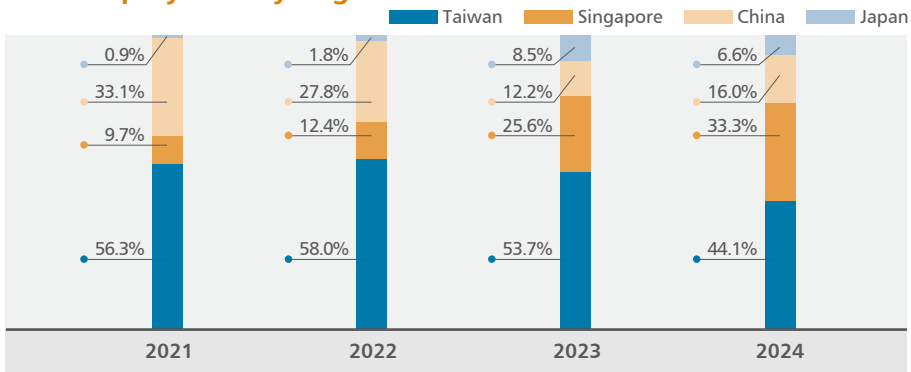
3. The number of new employees has been adjusted to exclude employees returning from unpaid leave and corporate transfer employees. The reported information of 2021~2023 has also been adjusted.

New Employees - by Gender

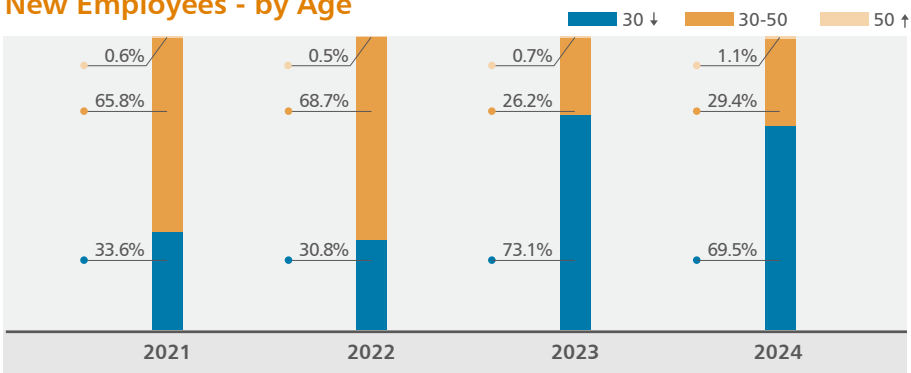


Note: In 2023, 70.9% of UMC's new employees were males and 29.1% were females. In 2024, 66.0% of UMC's new employees were male and 34.0% were female.

New Employees - by Region



New Employees - by Age

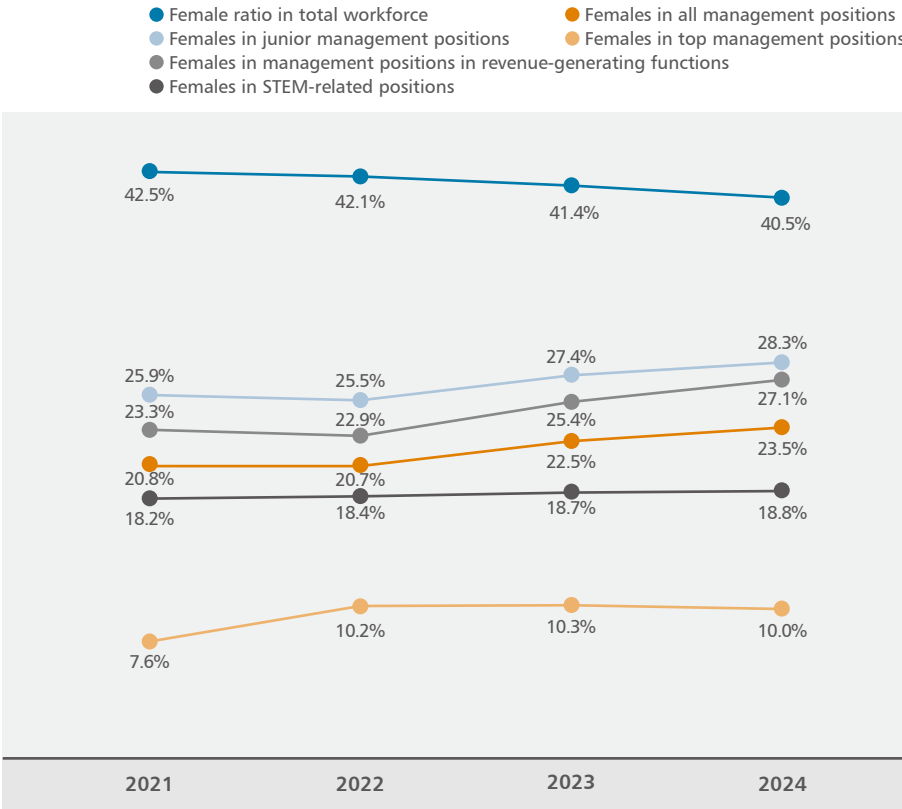


Note: In 2023, 72.1% of UMC new employees were under 30 years old; 27.1% were between 30 and 50 years old; and 0.8% were above 50 years old. In 2024, 67.8% of UMC new employees were under 30 years old; 31.4% were between 30 and 50 years old; and 0.8% were above 50 years old.

Female Workers

UMC Group ensures gender equality in employment rights and values the promotion and development of female employees. In view of the characteristics of semiconductor industry, most of the UMC managers and engineers are male. Meanwhile, administrators and technicians are mostly females. UMC keeps holding sharing sessions on exemplary female leadership and management to enhance the willingness of female employees to assume management positions.

Proportion of Female Workers by Position

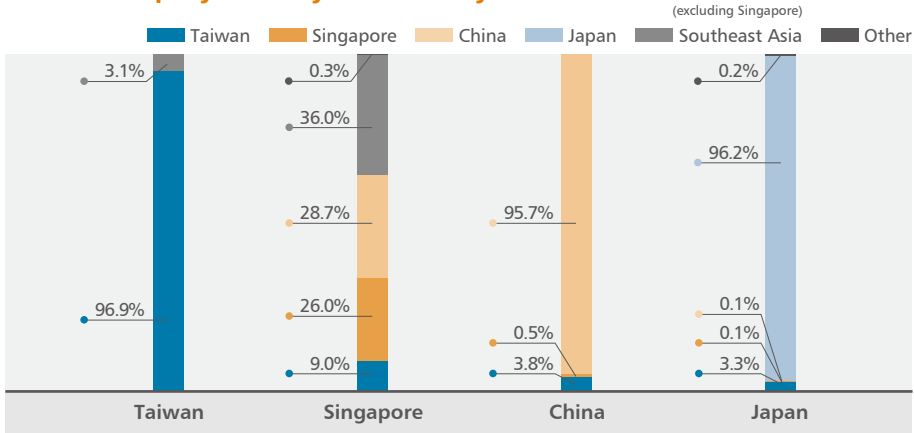


Note: 1. The female proportion statistics includes UMC and its subsidiaries HJ, US CXM, Wavetek and USJC.
2. The management positions include junior, middle and top management positions. Junior management positions include first line manager and those managers with job title as supervisor or leader. Top management positions include two levels away from the President. (Senior FAB/Division Director and above).
3. Management positions in revenue-generating functions: Fab and sales & marketing managers include first line manager and those managers with job title as supervisor or leader.
4. STEM-related positions: Engineering related jobs.
5. In 2024, UMC's female made up 41.4% of the total workforce, with 22.3% in management positions, 26.6% in junior management positions, 11.5% in top management positions, 26.3% in revenue-generating functions, and 19.0% in STEM-related positions.

Local Employment

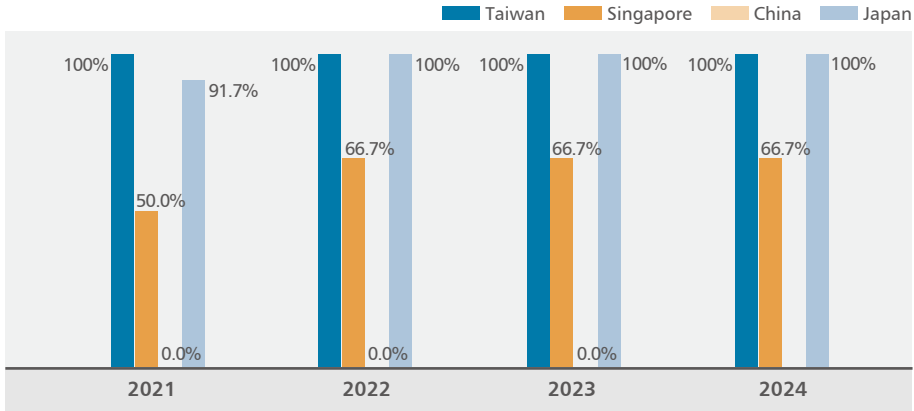
To fulfill social responsibility and create sufficient employment opportunities to local residents, the staff at the UMC headquarters and Wavetek in Taiwan comprised of 96.9% local employees as of 2024. Since Singapore is ethnically diversified, 26.0% of the employees are locally hired. In UMC China fabs, namely HJ and USCXM, 95.7% of the staff are locally hired. The staff at USJC in Japan, 96.2% of the staff are locally hired. Among the employees employed by UMC, Asians accounted for 92.9% of the total labor force, migrants accounted for 6.7%, and indigenous people accounted for 0.4%.

Formal Employees – by Nationality



Note: The percentages of Chinese, Singaporean, and Japanese employees in Taiwan and Southeast Asia (excluding Singapore) employees in China are all less than 0.1%, and are not shown in the chart.

Locally Hired Top Management Positions



Note: 1. Local hire refers to employees who are nationals of the main operating location/region.
2. Top management positions include two levels below the President. (Senior Fab/Division Director and above).

Foreign Talent Recruitment

To internationalize the talent pool in the semiconductor industry, UMC has established a comprehensive international talent mechanism, from recruiting and hiring overseas talent to assisting international hires in acclimating to the local environment, while proactively following the latest policies and regulations to avoid risks associated with hiring foreign workers. This effort demonstrates UMC's commitment to creating a diverse and friendly work environment.

International Talent Recruitment

To expand the recruitment of talents with global perspectives, UMC has dedicated personnel for overseas recruitment, and actively participated in international talent acquisition and campus job fairs in 2024 to broaden the talent pool. Through the "Talent Recruitment in New Southbound Countries Program" organized by the Industrial Development Administration Ministry of Economic Affairs, UMC has engaged in communication with engineering and technical talent from Southeast Asian countries such as Malaysia, Singapore, and the Philippines, promoting Taiwan's semiconductor employment environment. By increasing awareness of UMC, the program aims to increase the chances of attracting talent to Taiwan and provide incentives. Additionally, UMC established an overseas recruitment booth at campus job fairs, inviting managers and HR representatives from overseas sites to provide multilingual consultation services in English and Japanese to students. This effort not only strengthens campus relationships but also connects foreign students with overseas job opportunities, expanding the Company's diverse workforce.

Compliant Employment Application

Under the policy of globalization and cross-border operations, UMC considers the risks that may arise from the protection of trade secrets, export control restrictions and sanctioned lists in the process of hiring foreign personnel. To ensure that hiring practices comply with international regulations, UMC has implemented three measures. First, UMC has established rules for the employment of foreign technical workers. Secondly, UMC prioritizes the regulations for hiring foreign personnel and ensures that all procedures, such as applying for visas and work permits, are managed and controlled by dedicated personnel to comply with government regulations. Lastly, UMC provides education and training for local and foreign colleagues on various norms such as the protection of trade secrets, intellectual property rights, information security, and export control on an irregular basis to ensure compliance with these regulations.

Create a Suitable Environment

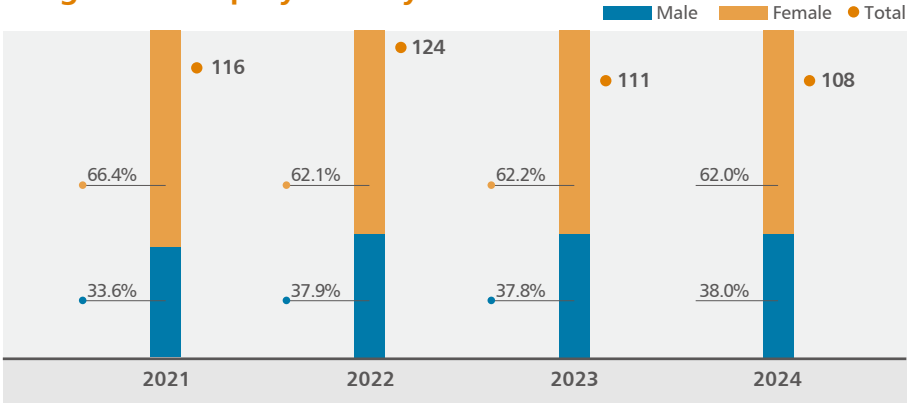
To help foreign colleagues adapt to the environment quickly, UMC provides comprehensive dormitory services and new employee education and training programs. The Company also has translation teachers in the dormitory to assist foreign colleagues in their work and work environment, avoiding poor adaptation caused by language barriers. Every year, UMC holds festival activities exclusively for foreign colleagues, allowing supervisors and foreign colleagues to celebrate festivals such as Dragon Boat Festival, Mid-Autumn Festival, and Christmas together, promoting cultural exchange.

Indigenous Employment and Ethnic Minority

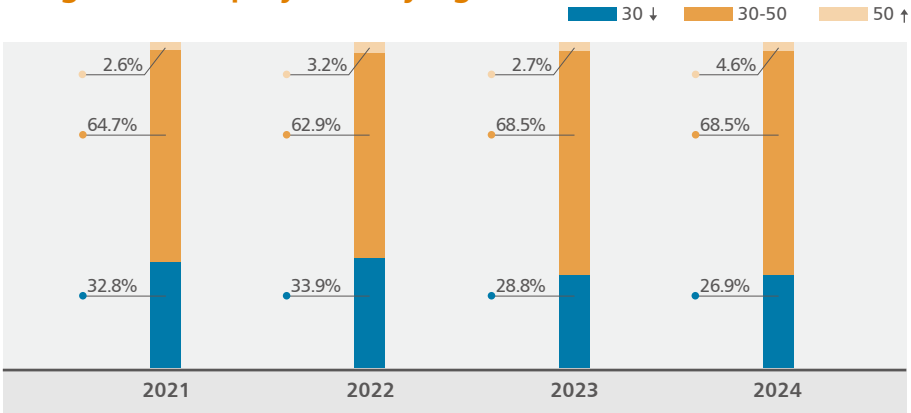
UMC Group respects the unique cultures of ethnic minorities. By the end of 2024, UMC Group has employed a total of 108 indigenous employees, except for those in Singapore and Japan where there are no indigenous people. Indigenous employees in Taiwan (including subsidiary Wavetek) receive 8 hours of ritual leave every year, which can be freely and flexibly used as a holiday.

Note: 1. The indigenous population statistics includes UMC and its subsidiaries HJ, USCXM and Wavetek.
2. UMC has a total of 51 formal employees with indigenous identity, and including the temporary staff and the person who is on parental leave, the total number is 53.

Indigenous Employees – by Gender



Indigenous Employees – by Age



Employment of People with Disabilities

UMC Group supports the employment of people with disabilities, and has established channels for hiring people with disabilities. Through the Student Ambassador Project, people with disabilities such as physical handicap, visual impairment, hearing impairment, functional loss in vital organs and chromosomal abnormalities are hired. By the end of 2024, UMC Group hired a total of 146 employees with disabilities. The headquarters in Taiwan has already hired 124 employees with disabilities with the ratio 1.0%, which exceeds the target ratio 1.0% specified by laws and regulations. This demonstrates UMC's commitment to caring for the disadvantaged groups. In the future, UMC will continue to actively carry out job evaluations internally to create more job opportunities for people with disabilities.

Note: The calculation method for the number of hires with disabilities follows the Taiwan regulation. If a person with severe or above disabilities is hired, s/he is counted as two hires.

Employee Turnover Rate and Internal Transfer Rate

From 2021 to 2024, the annual average turnover rate of is 7.7%. (As for 2024 turnover rate by gender, 5.5% was female and 5.3% was male. And for age group, 13.0% was under 30 years old, 4.6% was 30-50 years old, and 1.3% was above 50 years old.) The average turnover rate of the UMC Group is 9.6%. In recent years, UMC Group has been actively promoting transformation plan. Through the revision of technology and market strategies, as well as a reshaping of UMC culture, UMC has achieved outstanding operational profitability and gained recognition of employees. In 2024, the UMC Group had a turnover rate of 6.1%, while UMC had a rate of 5.4%, both of which were the lowest rates in the past four years. In particular, UMC's turnover rate in 2024 decreased by 1.8% compared to the previous year, which is under the target turnover rate of 7.0% continuously.

By closely observing and comprehending the organizational atmosphere, Human Resource division can promptly detect when employees are displaying indications of wanting to resign. Subsequently, HR collaborates with supervisors to promptly initiate employee care and provide necessary aid at early stages to improve employee retention.

When employees apply for resignation, they are interviewed separately by their direct supervisor and the Human Resources division to understand the reason(s) for resignation. After understanding the reasons for employee's resignation, HR not only offers the flexibility of adjustment on work content and workplace location based on the personal situation, but also provides internal position transfer opportunities based on the employee's job experience and expertise to achieve the goal of retaining employees.

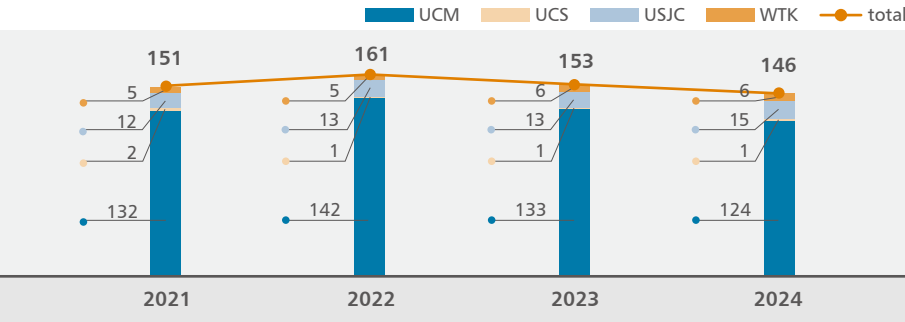
The Human Resources division also regularly follows up and cares for employees who have resigned to keep long-term relationships and provide an open mechanism for them to return to the Company.

UMC Group has strengthened the retention of talent by promoting activities such as UMC employee well-being supporting programs, UMC culture communication activities and internal transfer system to provide employee job skill enhancement, career development, and opportunities to take on challenging tasks to value up themselves, thus enhancing the retention of mid- and high-performance talents.

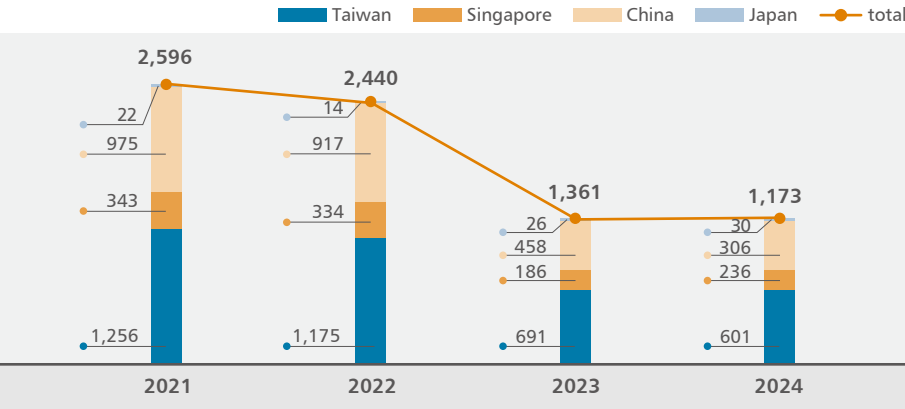
In 2024, the internal transfer rate in UMC was 77.8%, which is the percentage of open positions filled by internal candidates, exceeding the target value of 55.5%, and in UMC Group, it was 65.7%. The retention rate of mid- and high-performance employees of UMC in 2024 was 96.1%, surpassing the set target value of 92.8%.

UMC will continue to provide multiple channels and incentives of internal transfer, as well as implementing a comprehensive exit interview process and follow-up care for departing employees to maintain a healthy turnover rate.

Disability Employment at UMC Taiwan Headquarters



Resignations in UMC Group



Years		2021	2022	2023	2024
UMC Group ²	Turnover rate	13.5%	12.1%	6.8%	6.1%
	Voluntary turnover rate	13.4%	12.1%	6.8%	6.1%
	Internal transfer rate ³	23.8%	32.4%	63.4%	65.7%
UMC	Turnover rate	10.5%	9.4%	5.5%	5.4%
	Voluntary turnover rate	10.4%	9.3%	5.4%	5.4%
	Internal transfer rate	57.1%	58.6%	65.1%	77.8%

Note: 1. Annual employee turnover rate refers to the total number of employees who resigned/ average number of existing employees. Annual average number of employees = (number of employees at the end of December of the previous year + total number of employees at the end of each month of the current year) ÷ 13
2. Including UMC and its subsidiaries HJ, USCXM, Wavetek and USJC.
3. Internal transfer rate = number of internal transfers ÷ number of job vacancies.

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4.1 Enforce Human Rights Management

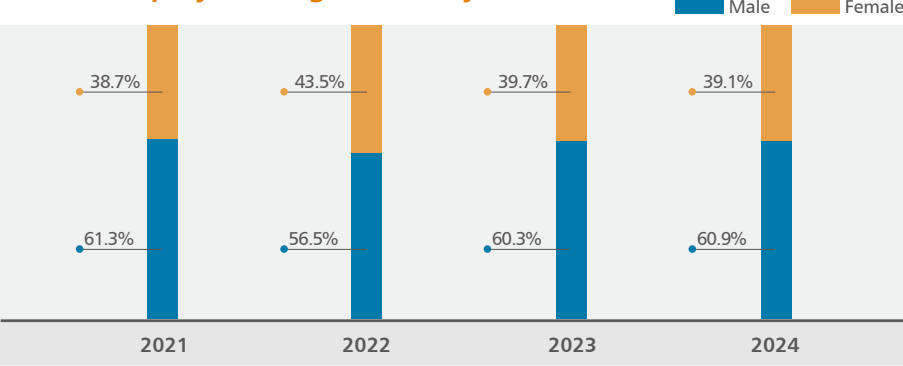
4.2 Diversity, Equity and Inclusion in the Workplace

4.3 Protecting the Employees and the Work Environment

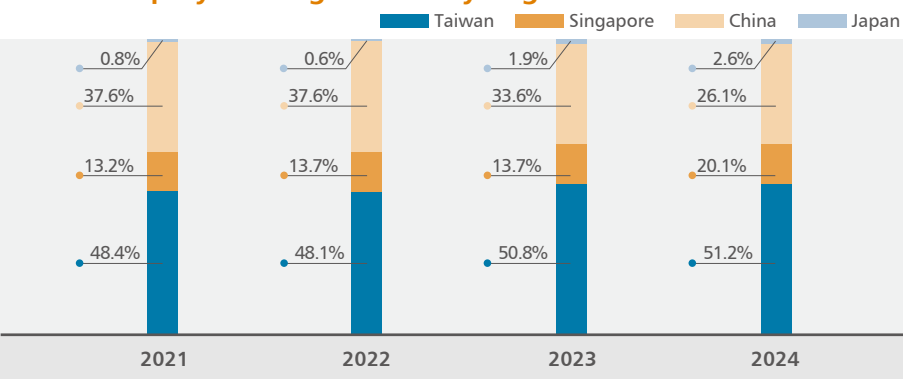
5 Enhancing Society Mutual-Prosperity

Appendix

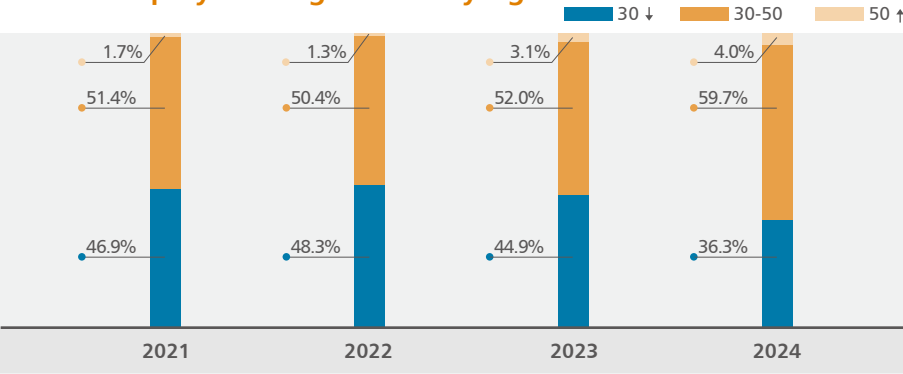
Formal Employee Resignation—by Gender



Formal Employee Resignation—by Region



Formal Employee Resignation—by Age



Parental Leave

According to the Act of Gender Equality in Employment in Taiwan for UMC headquarters and Wavetek and Japanese law in USJC, employees may apply for parenting leave without pay. In 2024, a total of 70 female employees returned to their positions when their parental leave contract expired, indicating a return rate of 92.1%, and 19 male employees returned to their positions, indicating a return rate of 79.2% after parental leave. The work situation of those who returned after parental leave was observed. In 2023, 89.7% of female employees and 68.4% of male employees who took parental leave returned to work and continued to work over 1 year. The employees who have not returned after parental leave all voluntarily resigned because their parental leave have expired but the needs of the family remained.

2024 Parental Leave	Female	Male	Total
Total eligible number for parental leave (A)	307	740	1,047
Actual number of parental leave (B)	135	37	172
Application rate for parental leave (B ÷ A)	44.0%	5.0%	16.4%
Total number of end of contract for parental leave (C)	76	24	100
Total number of returning from parental leave (D)	70	19	89
Return rate after parental leave (D ÷ C)	92.1%	79.2%	89.0%
2023 Total number of returning from parental leave (E)	58	19	77
2023 One year retention rate after return from parental leave (F)	52	13	65
2023 Number of returning from parental leave and retention rate. (F ÷ E)	89.7%	68.4%	84.4%

Note: Parental leave is provided to UMC Taiwan and Japan employees according to local laws and regulations. Employees in UMC fabs in Singapore and China are not included in this table.

4.2.2 Compensation and Benefit

UMC has always regarded its employees as the most important asset, and provides a competitive overall compensation and benefits package to attract excellent talent to join the UMC team. UMC's employee compensation is based on educational level, performance and market value, regardless of gender, race, religion, political position or marital status. To ensure that employees receive fair and reasonable compensation, UMC annually reviews the compensation obtained during the year and monitor the gender pay gap between male and female employees in the same positions, job duties and performances, thereby ensuring equal remuneration for men and women.

UMC employees worldwide enjoy the compensation and benefits that comply with or exceed all applicable local laws and regulations in terms of minimum wage, work hours (including overtime), social insurance, retirement pension, and other mandatory benefits (such as paid annual leave). The working hours that exceed the legal requirements are considered overtime. Employees who report overtime based on actual overtime work will receive corresponding overtime pay. In terms of leave benefits, employees are granted statutory paid leave each year. UMC annually reviews and makes sure that entry-level employees' wage is higher than legal minimum wage. In Singapore and Japan, there are no prescribed minimum wage laws, and salary levels are determined based on local government or union recommendations according to employee skills and market conditions. In 2024, based on the same positions, job duties, and performances, the average fixed pay for female entry-level employees in Taiwan is 1.7 times the minimum wage, while for male employees, it is 1.6 times. The salary gap is due to employees' seniority, and is not correlated with gender.



To provide more diversified remuneration and improve employee well-being, UMC implements an employee stock ownership plan in 2024. UMC provides subsidies for our permanent employees in Taiwan to purchase company stocks for assisting employees in long-term investment and reduce investment risks, so as to attract and retain talent.

Performance-Oriented Compensation System

In order to achieve personal, divisional and company performance goals as well as understand the job performance of employees for management by objectives, UMC conducts annual performance appraisals for all employees regardless of gender and job category twice a year. The annual performance appraisal focuses on past work achievement and the future goal setting. The types of appraisals include management by objectives, multidimensional performance appraisal, agile conversations and feedback. In 2024, the completion rate of multidimensional performance appraisal is 47.0%, and 42.1% including subsidiaries.^{Note}

In performance appraisals, supervisors and employees work together to agree on the priority of focuses, and devise developmental plans based on current working style, capability, career goals and project evaluation. Employees with poor performance will be supplemented with key improvement plans to improve work efficiency.

Note: Including HJ, USCXM and Wavetek

Performance Appraisals System

Type of Appraisals	Applicable Target	Frequency	Execution Method
Management by objectives	All employees	Once a year	- At the beginning of the period, employees propose work plans and quantifiable performance indicators, and set annual goals after discussing with their supervisors. At the end of the period, employees and supervisors jointly review the status of goal achievement and receive feedback from the supervisor - For team-based performance appraisal, employees who participate in projects have to include personal objectives and team objectives in the goal setting.
Multidimensional performance appraisal	IDL (indirect labor)	Once a year	- Employees will receive multi-dimensional feedback from cross-divisional supervisors or employees who have shared work experience. - If employees encounter organizational changes during the performance appraisals period, feedback will be provided by their original unit supervisor.
Agile conversations and feedback	All Employees	Once every quarter/ immediately	A quarterly Town hall Meeting is held by the co-president to keep employees informed about the Company's goals and work priorities. Supervisors can initiate performance communication with employees at any time, providing necessary support and guidance immediately through a one-on-one communication system.

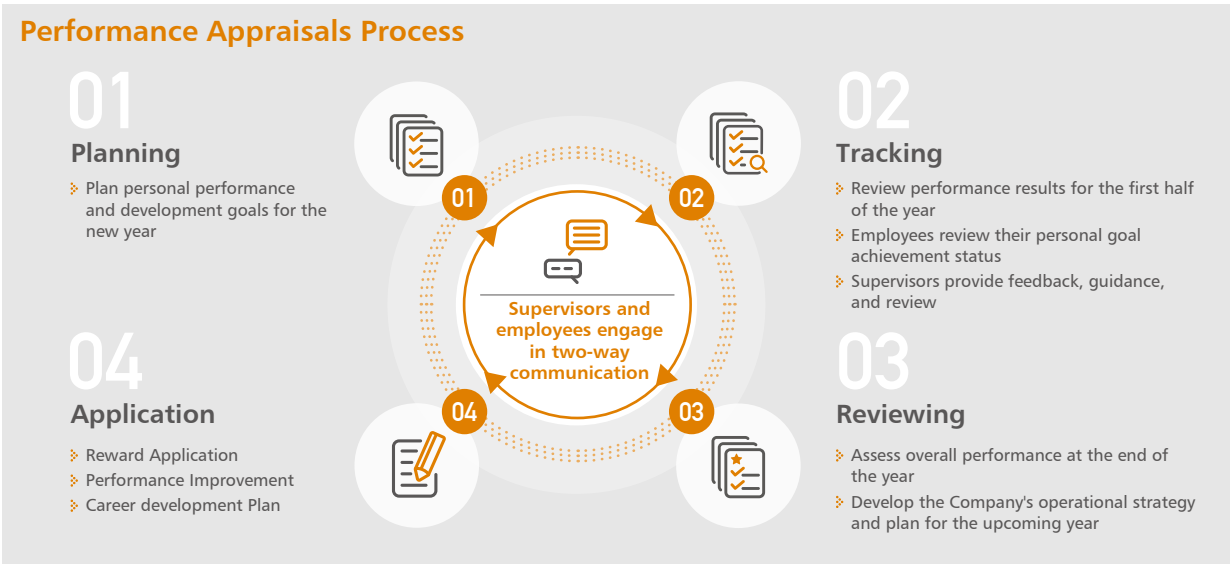
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Appendix



UMC offers salary adjustment, differentiated bonus, profit distribution as employee's compensation and stock remuneration (employee stock option certificates, treasury shares and restricted stock award) based on individual performance, job responsibilities and future development potential to attract, retain and motivate outstanding employees. UMC also actively joins the remuneration surveys of well-known domestic and international enterprises, annually reviews and analyzes our compensation policy based on compensation survey results and adjusts the policy if necessary to ensure that the overall remuneration offered by UMC is competitive in the market.

Moreover, in order to internalize the concept of ESG into the core of operation and enhance employees' sense of responsibilities for corporate sustainability and risk management, ESG targets are set and linked to employees' variable compensation. By aligning the interests of employees and the Company, corporate sustainability can become a common goal. In 2024, UMC has achieved the ESG targets and the allocation of employee profit sharing percentage would be increased accordingly, so that all employees can enjoy the benefit of ESG achievement.

Basic Salary and Annual Total Compensation by Gender in 2024

Male																				Female																			
UMC								Subsidiaries																															
	TW				Singapore			HJ			USCXM			Wavetek			USJC																						
	Executive level	Management level	Non-management level	Technicians	Management level	Non-management level	Technicians	Management level	Non-management level	Technicians	Management level	Non-management level	Technicians	Management level	Non-management level	Technicians	Management level	Non-management level	Technicians																				
Base Salary	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%																				
	98.5%	98.1%	100.8%	100.8%	98.4%	98.6%	104.5%	104.2%	101.9%	104.9%	103.9%	98.7%	105.7%	98.5%	98.9%	100.3%	103.7%	96.9%	98.7%																				
Total compensation	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%																				
	99.7%	99.2%	99.3%	96.7%	97.9%	98.9%	103.0%	103.9%	100.1%	99.5%	102.9%	99.7%	104.1%	97.3%	98.7%	90.2%	106.2%	95.3%	91.80%																				

Note: 1. Management level: From first-line/junior managers up to top/senior managers (Executives excluded); Non-management level: Professional engineers; Technicians: Local technicians.
2. The calculation is based on the actual cash rewards received in the current year. Female and male salaries are compared based on equivalent positions, job duties and performance rating.

3. The executives' compensation is determined by job duties and relative importance of the position, individual performance, and market value, regardless of gender.
4. There are no female executive in Singapore and subsidiaries HJ, USCXM, Wavetek and USJC.

The number of full-time, non-executive employees in Taiwan, their average and median salary, and the difference over the years

No. of Employee			
2024	12,122	Down	386 people
2023	12,508		
Average Salary (ten thousand NT\$)			
2024	180.0	Down	5.1
2023	185.1		
Median Salary (ten thousand NT\$)			
2024	161.1	Up	3.4
2023	157.7		

Note: The above information is based on the regulations of Taiwan Stock Exchange, deducting managers (executives) and employees eligible for exemption. Those who have been employed for 6 months but less than 1 year, their compensation is calculated based on their weighted days of employment. Employee compensation is accrued based on an accrual basis, and the unpaid employee compensation is an estimated amount. These numbers have been audited by CPAs and will be disclosed on Taiwan's market observation post system.

Comprehensive Benefits Superior to Statutory Regulatio

UMC is committed to creating a work-life balance workplace and offers all employees various welfare benefits that are superior to statutory regulation including holidays, insurance, financial assistance for emergencies, discounts in designated shops and employee assistance service. Temporary staff and dispatched staff enjoy the same benefits. To show the care for employees, the Company actively reminds employees to take vacations and holds diverse activities to relieve stress. Moreover, UMC provides parenting measures and breastfeeding rooms to take care of the day-care needs of employees' children.



Paid Leave that is Superior to Local Regulations

- Consider the reduction of national holidays after the revision of the Labor Standards Law, 7 days of supplemental holidays are given and can be used for personal and family care needs. UMC Singapore fab and other overseas subsidiaries also provide paid annual leave that is superior to local regulations of paid annual leave of employees.
- Provide extended maternity leave of 480 hours, which is an increase from the statutory requirement of 448 hours, and extend paternity leave of 80 hours, which is an increase from the statutory requirement of 56 hours.
- UMC provides fully-paid maternity leave to newly hired female employees, regardless of their length of service, as part of our commitment to better care for our employees.
- As a way of providing support, if the grandparents of an employee's spouse pass away, the employee is entitled to a funeral leave of 48 hours (up from the statutory requirement of 24 hours). If the great-grandparents of an employee's spouse pass away, the employee is entitled to a funeral leave of 24 hours at full pay, which exceeds the requirements of the Labor Standards Act.



Parenting Measures and Breastfeeding-Friendly Environment

- To encourage employees to have more children, UMC provides measures ranging from care and nursing consultation during pregnancy, maternity leave, maternity inspection leave, exclusive parking space, priority meal pickup, maternity subsidy, unpaid parental leave, certified breastfeeding room, night shift transfer and other measures.
- To support employees with young children under the age of 3, UMC offers the option to apply for parental leave based on their needs. During this period, working hours may be reduced by 1 hour per day. To provide comprehensive assistance, UMC signed contracts with neighboring kindergartens to take care of the day-care needs of employees' children, and assists in the registration work of the experimental primary and secondary schools in the Science Parks. In 2022, in cooperation with the Southern Taiwan Science Park Bureau, a non-profit kindergarten is set up to offer employees more comprehensive support for family care.



Health Care and Emergency Assistance

- Regularly conduct employee health checks, plan health promotion activities, set up a health care system, and actively track and care for employee health.
- Equipped with employee assistance programs (EAP), injury and illness treatment and care, taking care of employees' health comprehensively.
- Provide company group insurance, cancer insurance, medical insurance, accident insurance, hospitalization condolences, emergency relief funds to assist employees in emergency situations.



Diverse Activities and Discounts in Designated Shops

- Employee Benefit Committee holds diverse activities such as year-end party, Secretary's Day, the Company's anniversary day, Engineers' Day, one-day trip, family day, and art festival, etc.
- Employee social club evaluations and subsidies for club activities.
- Corporate agreements with stores all over Taiwan, provide APP for employees so they can enjoy discounts for food, clothing, housing and travel.



Friendly Working Environment

- Employee cafeteria, free shuttle bus, activity center, telecommunications/banking/ insurance on-site services, convenience stores/coffee shops/bakeries, and other stores are available in offices and fabs.
- Provide flexible working shifts and hours.

Comprehensive Insurance and Retirement Policy

UMC provides insurance coverage that is consistent with local laws and regulations to ensure the basic rights and interests of employees. In accordance with the law, the Company's headquarters in Taiwan also provides labor insurance (including Employment Insurance) and national health insurance. In addition, UMC provides employees with additional group insurance, including life insurance, major illness insurance, health insurance, accident insurance, cancer insurance, and travel insurance for overseas business trips to ensure work and life security for its employees. The Company also provides a selection of group insurance for employee families so that employees can work with peace of mind.



Over 50% of employees choose to include their spouse and family members into UMC's group insurance



100% of the regular employees in Taiwan participate the relevant pension plans



On-site insurance service

Over 50% of UMC employees choose to include their spouse and family members into UMC's group insurance program to provide their family an additional layer of protection. UMC also provides an insurance company service office inside the Company buildings, making it very convenient for employees to make inquiries on insurance services and apply for claims. Furthermore, the Company regularly posts e-newsletters on insurance benefits, and introduces information on insurance and compensation rights so that employees are clearly aware of their actual insurance content and benefits.

Retirement Benefits Pla

UMC complies with local statutory regulations and systems related to retirement to safeguard the retirement rights of our employees. In Taiwan, the Labor Standards Act was used as the basis to stipulate regulations for the calculations and payment rules of retirement pensions. 100% of the full-time employees in Taiwan participate in the relevant pension plans. Since July 1, 2005, the Labor Pension Act became effective as a defined contribution plan. According to the regulations, the Company has to contribute a certain percentage of salary to an individual labor pension account. Employees may select to keep applying to pension regulations prescribed by the Labor Standards Act or switching to the new pension systems applicable to the Labor Pension Act. They also can retain their seniority on the previous pension system. The pension benefits mandated by the Labor Standards Act are disbursed based on the units of service years and average monthly salary prior to retirement. The Company contributes an amount equivalent to 2% of the employees' total salaries and wages on a monthly basis to the pension fund deposited with the Bank of Taiwan under the name of a pension fund supervisory committee. In 2024, total pension expenses of NT\$34.896 million were recognized by the Company. In addition, for the pension benefits under the Labor Pension Act, the Company makes monthly contributions of 6% based on each individual employee's salary or wage to employees' pension accounts. Pension benefits for employees of the Singapore fab are provided in accordance with the local regulations. Total pension expenses of NT\$1,166.866 million were contributed by the Company in 2024.

When employees apply for retirement, the Company not only provides pension application service, but also awards "UMC Recreation Center Lifetime Membership," which offers a selection of healthful and interesting activities for retirement life to show appreciation for their long-term effort and contribution. In order to ensure those who retire from or terminate employment with UMC to obtain necessary assistance, UMC also provides severance pay for terminated employees in accordance with relevant local laws and regulations, as well as information on related employment service channel.

Note: 1. Employees also have the paid leave hours for public welfare activities.
2. Working from home (WFH) is adopted under applicable condition according to the Company's regulation.

4.2.3 Strengthening Talent Cultivation

UMC Talent Development Path

Based on the training and development model, UMC defines the competency base of each level in accordance with the Company's business strategy to plan the training and development plan, provides a comprehensive and diversified learning environment according to the type of work, cultivates multiple talents, continuously improves the Company's growth, achieves the goal of retaining and nurturing talents, and implements the energy of career sustainable development. In the design of internal training curriculum, UMC conducts a company-wide training needs survey in the fourth quarter of each year, and plans corresponding development courses based on the professional needs of supervisors and employees. In addition to required training to help employees achieve job performance, employees can also prepare for their career planning and development by participating in other training courses based on their personal needs and future development plans.

In terms of talent development planning, in addition to arranging internal instructors for knowledge transfer, UMC also collaborates with renowned training consulting firms, government agencies, benchmark enterprises, and academic experts both domestically and internationally. The aim is to enhance UMC's learning comprehensiveness through rich external resources. All courses must undergo rigorous evaluation and execution procedures to maintain learning quality, including aspects such as the instructor's teaching experience, certification, the consulting firm's reputation and evaluations, and contract signing to ensure the rights and obligations of both parties. In 2024, UMC collaborated with external organizations to offer company-wide courses covering semiconductor expertise, the introduction of DEI (Diversity, Equity, and Inclusion) concepts, compliance with safety regulations, enhancement of management and language skills, and the latest trends in AI. After the completion of the courses, continuous optimization is ensured through course satisfaction surveys, observations by course managers, and evaluations of learning outcomes, guaranteeing that employees receive the best learning experience.

UMC Talent Development Path



Training Effectiveness

In order to confirm the effectiveness of training, UMC refers to the Kirkpatrick model of learning effectiveness evaluation proposed by Donald. L. Kirkpatrick, a professor at the University of Wisconsin, and added the per capita output value as an indicator of the highest level of training effectiveness. It is used as a measure of the effectiveness of all training activities, so that training keeps improving and achieves talent development and retention.

Feedback Level 1	Learning Level 2	Behavior Level 3	Result Level 4	ROI Assessment Level 5
❖ Satisfaction of learning activities - the indicators are course satisfaction, instructor satisfaction, and course completion rate.	❖ What knowledge, skills and attitudes are learned from learning activities? The indicators are course test scores and training program completion rate.	❖ Use the learned knowledge, skills and attitudes to transform into actions and show them - the indicator is the SPC implementation success rate.	❖ The impact or result of the learning activity - the indicators are the retention rate of key talent and the turnover rate of new employees.	❖ Measure training investment returns with per capita output value.

In 2024, UMC Group offered a total of 12,489 classes with a total of 768,145 person-times participant. The overall satisfaction rate is over 93.8%.

Course Satisfaction

Type/Year	2021	2022	2023	2024
Number of Courses	8,248	8,570	8,861	12,489
Total number of Participants	161,231	189,617	677,298	768,145
Overall Satisfaction	95.2	94.9	94.3	94.3
Satisfaction with Instructor	95.2	94.9	94.4	94.4
Satisfaction with Teaching Materials	95.1	94.8	94.2	94.2
Beneficial to Wor	95.1	94.9	93.8	93.8

Note: The scope of statistics includes UMC's fabs in Taiwan and Singapore, UMC mainland China subsidiaries HJ and USCXM, and UMC Japan subsidiary USJC and the Taiwan subsidiary Wavetek. And it covers all regular, temporary and dispatched staff.

UMC Group provides complete education and training for employees of different job categories and levels. In 2024, the total number of training (persons) hours reached 1,155,880 hours. The average training expense was NT\$4,053 per person and the average training hours was 60.3 hours. The average training hours for all employees of UMC in 2024 has reached 77.5 hours, mainly due to the expansion of training hours collection for various lecture activities, external training, and certification programs and so on.

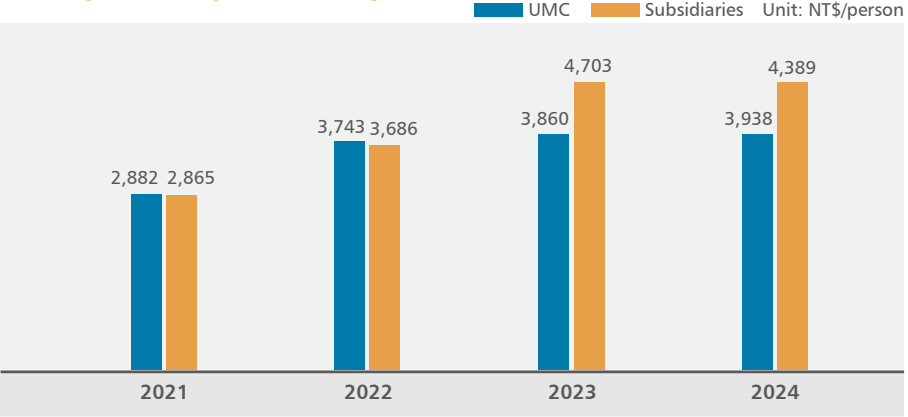
Note: Training Hours calculated to one decimal place, but are rounded to whole numbers in the table. Therefore, the actual total may differ from the sum displayed in the table.

UMC Group Average Employee Training Cost

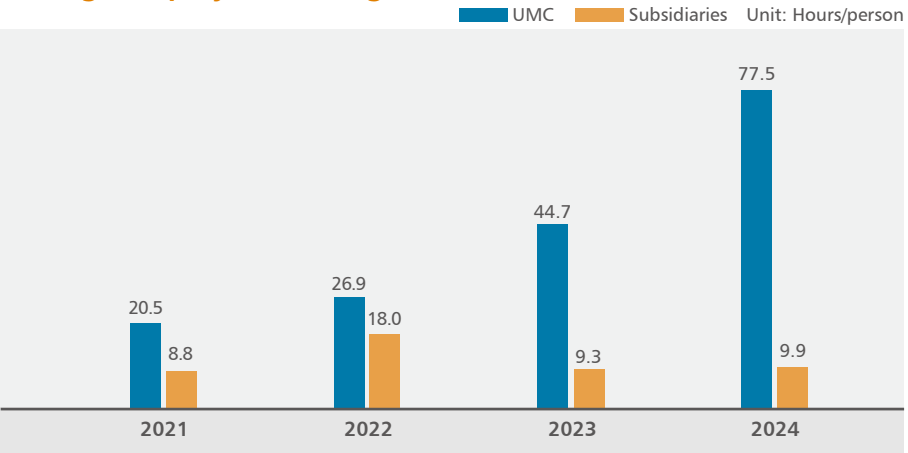
Type/Year	2021	2022	2023	2024
Total Training Cost (NT\$)	42,120,553	57,450,399	81,200,340	77,653,106
Number of Employees (Persons)	14,613	15,347	19,929	19,161
Training Cost per Employee (NT\$)	2,882	3,743	4,074	4,053

Note: The scope of statistic included UMC fabs in Taiwan and Singapore from 2020-2022, From 2023, the scope of statistics for UMC's subsidiaries includes mainland China subsidiaries HJ and USCXM, and UMC Japan subsidiary USJC and the Taiwan subsidiary Wavetek. And it covers all regular, temporary and dispatched staff.

Average Employee Training Cost



Average Employee Training Hours



Average Training Hours for Various Job Levels in 2024

Type	Training Time (Hours)	Number of Participants (Persons)	Average Training (Hours)
Director Level	126,472	2,140	59.1
Indirect Labor (non-director level)	740,584	10,979	67.5
Direct Labor	288,823	6,042	47.8

Note: The scope of statistics includes UMC's fabs in Taiwan and Singapore, UMC mainland China subsidiaries HJ and USCXM, and UMC Japan subsidiary USJC and the Taiwan subsidiary Wavetek. And it covers all regular, temporary and dispatched staff.

Average Training Hours for Various Type of Employment in 2024

Type	Training Time (Hours)	Number of Participants (Persons)	Average Training (Hours)
Formal Employees	1,142,016	18,915	60.4
Temporary Staff	9,034	111	73.7
Dispatched Staff	4,830	135	42.1

Note: The scope of statistics includes UMC's fabs in Taiwan and Singapore, UMC mainland China subsidiaries HJ and USCXM, and UMC Japan subsidiary USJC and the Taiwan subsidiary Wavetek. And it covers all regular, temporary and dispatched staff.

Average Training Hours for Gender in 2024

Type	Training Time (Hours)	Number of Participants (Persons)	Average Training (Hours)
Male	731,280	11,439	63.9
Female	424,600	7,722	55

Note: The scope of statistics includes UMC's fabs in Taiwan and Singapore, UMC mainland China subsidiaries HJ and USCXM, and UMC Japan subsidiary USJC and the Taiwan subsidiary Wavetek. And it covers all regular, temporary and dispatched staff.

Return on Human Resources Capital Investment

Unit: NT\$ Thousand

Item	2024
Operating Revenue(A)	179,681,047
Operating Expenses(B)	139,441,144
Employee Salaries and Benefits Expenses(C)	32,907,721
Return on Human Capital Investments (A -(B - C))/C	2.22
Total Number of Employees	14,297 人

Note: The scope of statistics includes UMC's fabs in Taiwan and Singapore. And it covers all regular, temporary and dispatched staff. The source of the cost calculation is the publicly available 2024 financial report.

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Employee Development Programs

Program Name	Leadership Development Program: Middle and Senior level Succession project	Continuous Improvement Activities
Program Goals Description & Business Benefit	- Establish an internal talent pool and management pipeline to ensure talent sustainability and strengthen talent retention. - Reduce the investment and timeline costs associated with external recruitment, while increasing the internal personnel replacement rate. - Enhance leaders' management skills and competencies, improve business operations and management thinking, lead teams to demonstrate effectiveness, and enhance the company's competitiveness. In response to organizational development needs and strategic directions, UMC has formulated a talent cultivation and succession development plan including, - Select high-potential talents, and use talent assessment tools to identify and analyzed the potential and opportunities of talents and teams. - According to the analysis of environmental trends, formulated exclusive development plans, including management knowledge training, team development, mentorship, job rotation, overseas assignment and job experience opportunities and other aspects of development	- Encourage all employees to participate in continuous improvement activities, including individual improvement and team improvement. - Enhance the efficiency of the Company's total quality system and make the Company more competitive. These teams included Quality Improvement Team (QIT) and Project Management (PM). By training employees to think systematically and use scientific methods to solve problems, employees can actively participate in teamwork. Project benefits include: - Develop new products and improve product quality - Increase production capacity and reduce operating costs
Quantitative Impact of Business Benefit	- There were a total of 97 managers participating in the succession project. - Accelerate the development of leadership in management with a complete training course, such as group development courses, online management knowledge courses, and personal development plans. A total of 341 participants attended the training courses with a total of 643.6 training hours - Among all participants in the project, there have been 6 managers have been promoted to key management positions of the Company, with internal employee promotion rate of 6.2%. According to the analysis report by the SHRM, the total cost of hiring new employees is approximately three to four times their salary. Based on the average annual salary of NT\$1.851 million for full-time employees at UMC in 2023, promoting internal talent can save at least NT\$33 million in costs. For the organization, this also helps retain critical knowledge and experience also increases retention willingness. Reference: SHRM The Real Costs of Recruitment	It is expected to generate more than NT\$3.075 billion in financial benefits for UMC. The calculation of benefits is based on the expense reduction, capital saving, and profit improvement achieved through each improvement activity. The benefits must be certified by the IE member (production planning division) to determine the final benefit amount.
% of FTEs Participating in the Program	Target: High-potential talents in all units Percentage: 0.68%	Target: Employees who participate in continuous improvement activities Percentage: 30.36%

Multiple Talent Training Program

UMC's talent development policy plans the training courses required by each employee (including temporary and dispatched staff) in their career development according to job category and job level, systematically provides employees with a comprehensive education and training, and implements talent development programs with a 70-20-10 Learning Framework to implement talent cultivation and provide a working environment for all employees to continuously learn and grow.

70% Experientia Learning

Training Committee

Promote experience inheritance, helping the organization grow the key momentum

- "Education and Training Committee" composed of education and training officers from various divisions to improve the quality of training for all employees. In line with corporate policy, the Company conducts a company-wide training needs survey every year and creates an annual training plan for the competency analysis of supervisors and associates. The Education and Training Committee holds quarterly meetings to review training operations.
- Sharing and exchange activities between fabs and function divisions were organized to promote the sharing of learning experiences between organizations, exchange training promotion methods, and help improve training effectiveness
- The Committee conducts selection activities of outstanding company-wide instructors to encourage employees to serve as lecturers.
- In terms of divisional and inter-divisional On-the-Job Training (OJT), the Education and Training Committee's downward education and training orientation allows division directors and their staff to fully participate in the planning, implementation and learning assessment.
- The diversity of self-learning and development channels, such as e-Learning, creates an atmosphere of mutual peer learning, development and team cooperation, thereby forming a comprehensive environment for learning, sharing and innovation.



70% Experientia Learning

Internal Promotion and Rotation

Multi-faceted job development and accumulation of practical work experience

- UMC plans corresponding training blueprints for different job categories and ranks, encourages employees to continue to learn and develop personal careers in line with industry development trends, continues to provide location transfer incentives, and implements internal transfer systems and promotion channels, support the work skills of in-service employees to improve their work skills, career development and challenge opportunities, and increase their work engagement.
- For more information on the internal transfer mechanism, please refer to Section 4.2.1 Human Resources Development

20% Experience Inheritance & Guidance Feedback 10% Classroom Training

Training for Newcomers

[UMCer R U Ready]

- To help new employees adapt quickly to the workplace and establish a common language and basic understanding of semiconductor processes, a complete learning plan is provided through UMC's new employee training program, supplemented by a mentoring system.

New-employee Experience Camp [We are the Batchmate, Friends!]

- By creating a good onboarding learning experience, we can enhance the sense of belonging of new UMC employees.
- In addition to the new-employee orientation that are required for all new employees, the new-employee experience camp is also organized. The program includes the
- Company's vision, strategy, and advantages that combine with the organizational characteristics, taught by senior executives.
- In addition, UMC's unique workplace anti-tress and positive attitude courses are available to accelerate the newcomers in blending into the corporate culture.



A total of 5 new-employee orientation camps were held in 2024, and a total of 210 new employees completed the camps.



Mentor System [A step-by-step study guide for newcomers]

- To help new employees adapt to their work and environment, appropriate mentors are selected to receive guidance training to assist newcomers in quickly integrating into the UMC team. Mentors assist new employees in drafting and implementing training plans after they join. During this period, direct supervisors and mentors conduct regular interviews and assessments with newcomers, using questionnaires to understand their learning and adaptation status. Supervisors and mentors provide appropriate feedback and comments to help solve any problems, reducing the uneasiness of newcomers entering the new environment and helping them feel the warm working atmosphere of the UMC team.

20% Experience Inheritance & Guidance Feedback 10% Classroom Training

Professional Courses

Special Engineer Skill Development Training Program

According to the needs of professional talents of each engineering unit and the inventory of professional functions, professional engineer training courses are planned to improve the overall professional level of engineers and the quality of engineering manufacturing, and hence get the manpower prepared for organizational development.

To keep up with modern new technologies, professional courses are presented in a more lively and vivid form using 2D/3D animations to simplify the previously rigid foundational knowledge. This fully enhances the understanding of the learners and greatly improves the learning experience and effectiveness.



From "Semiconductor Manufacturing Management"

Quality Management Training

Improve quality training and implement the organization spirit of continuous optimization

To implement the quality policy of UMC and carry out relevant training courses for quality requirements, such as: knowledge management, proposal improvement, project management, problem analysis methods, statistical process control, experiment design, etc.

Assessment System

UMC uses the internal and external evaluation channels including TOEIC English verification, professional skill evaluation system, and statistical process control (SPC) to effectively evaluate the core competencies each employee should possess in order to improve production quality and meet the needs of customers all over the world. Taking the SPC test as an example, the rule of Three-Six-Nine is established. In case of failure in the test, the individual will be forbidden from operating the equipment and will receive a negative evaluation in his/her employee performance review.

For the SPC course in 2024, the actual completion rate in Taiwan was 99.5%, and the actual passing rate in Singapore was 92%.

Knowledge Management, KM

To promote knowledge management, UMC has established a knowledge sharing platform. To take into account the protection of knowledge confidentiality, accessing of the knowledge is controlled according to the level of confidentiality, and knowledge documents are divided into 4 levels according to their completeness and contribution. The program further promotes the re-application of knowledge documents to enhance the flexibility of document application, thereby improving work efficiency and performance.

Performance	2021	2022	2023	2024
Read Rate (%)	93.4%	93.1%	93.2%	93%
Writing Penetration Rate (%)	74.9%	72.1%	74.1%	75%
Expected Re-application Benefits (NT\$10,000)	80,479	25,447	87,390	39,489

Note: 1. Read rate (%) is the rate of employees reading KM files.
2. Writing Penetration Rate (%) is the rate of employees writing KM files.
3. Expected Re-application Benefit (NT\$10,000) is to encourage employees to apply the KM knowledge they read to their work to show benefits.

Continued Improvement Activities

UMC encourages all employees to participate in continued improvement activities.

For related information, please refer to "Employee Development Programs" in this chapter and Section 2.2.2 Continuous Quality Improvement and Innovation.

Assessment System

SPC Assessment	In line with fab demand for manufacturing quality, engineers' understanding of SPC is emphasized
English Assessment	In line with the Company's international orientation, the strengthening of employee English competency is emphasized
Professional Skills Assessment	Professional skills assessment system is used for determining the professional competency of engineers

Skills Assessment

1 Category	Categorize according to facility, manufacturing and product
2 Content	Basic/Operations, Advanced/Learning from case examples, Connection with various tasks/Crisis management ability
3 Standard	Quarterly skills assessment 0-2 Continuing learning required 3-4 Independent operation 5-6 Completion of task through telephone instruction 7-10 Instructor
4 Content	Oral and written tests

SPC: 3-6-9 SPC Principle

Within 3 months	Within 6 months	Within 9 Application
Courses (Knowledge) New employees must complete SPC training courses within 3 months	Test (Knowledge) New employees must complete the SPC written test within 6 months	Practice (Practice) New employees must complete SPC practice within 9 months

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20% Experience Inheritance & Guidance Feedback 10% Classroom Training

Leadership Development

Developing leadership talents to ensure their sustainable development ability, in order to establish a management echelon

❖ The leadership development framework defines the required managerial competencies in four categories, including "Results," "Interaction," "Leadership" and "Potential" according to different levels of positions and roles. Corresponding development activities are planned and related learning resources are provided to encourage self-learning. To accelerate talent development, UMC employs diversified training and cultivating methods (70-20-10 learning rules) to strengthen leaders' management skills and competencies, lead teams to demonstrate effectiveness, and enhance UMC's competitiveness. The development of management skills is an important task for managers at all levels. Every year, according to the senior executives' expectations (top down), the results of the previous year's executive function demand survey (bottom up), and external trends, the annual management function training is planned, and a variety of learning resources are provided through expert lectures and forums, internal supervisor experience inheritance and sharing, customized management courses, succession projects at all levels and multiple online learning channels, etc., so as to assist managers in maximizing their effectiveness in their job and management roles.

❖ Executives

1. Talent Cultivation and Succession Planning for Senior Management: In line with the Company's future development strategy and operational needs, UMC has formulated a talent cultivation and succession planning for senior management that includes complete talent development policies and directions for executives and senior management trained under a succession plan framework, blueprint and development roadmap. This plan has also been clearly defined and included in the Nomination Committee Organizational Rules

In addition to reviewing implementation results within the Company, relevant plans are reviewed in management mechanisms such as policies and KPIs (Key Performance Index), and also included in implementation results. Progress and plans are reported to the responsible unit, senior executives, and the Nomination Committee/board of directors.

2. Succession projects at all levels: In response to organizational development needs and strategic directions, UMC has formulated a talent cultivation and succession development plan, used talent assessment tools to identify and select high-potential talents, and analyzed the potential and opportunities of talents and teams, and according to the analysis of environmental trends, formulated exclusive development plans, including management knowledge training, team development, mentorship, job rotation, overseas assignment and job experience opportunities and other aspects of development, and established an internal talent pool and leadership pipeline to meet employees' career development needs and strengthen talent retention.

❖ Middle Manager / First-line Manager

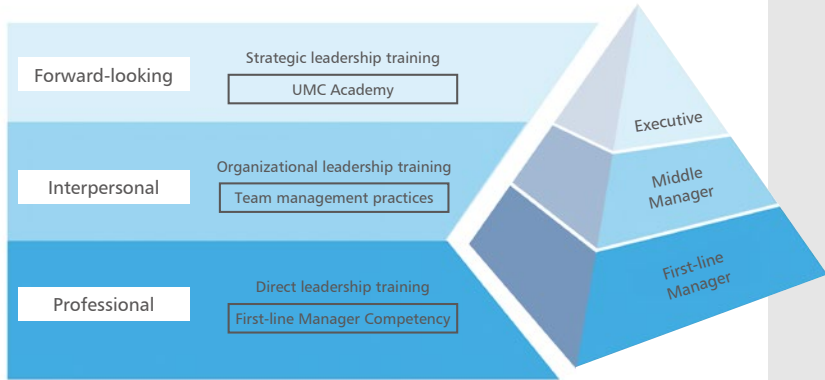
1. Succession projects at all levels: In response to organizational development needs and strategic directions, UMC has formulated a talent cultivation and succession development plan, used talent assessment tools to identify and select high-potential talents, and analyzed the potential and opportunities of talents and teams, and according to the analysis of environmental trends, formulated exclusive development plans, including management knowledge training, team development, mentorship, job rotation, overseas assignment and job experience opportunities and other aspects of development, and established an internal talent pool and leadership pipeline to meet employees' career development needs and strengthen talent retention.

2. Management Competency training at all levels: Through professional internal and external lecturers, from new first-line manager to middle-level manager, the learning map of management functions is planned according to the needs of management functions at all levels. The Company provides relevant learning resources for the management behaviors, skills, knowledge, and company culture integration that managers should possess, so as to assist managers in maximizing their effectiveness in their job and management roles.

Leadership Development Framework



Leadership Development model



20% Experience Inheritance & Guidance Feedback 10% Classroom Training

Leadership Development

Leadership Development Resources

- Supervisor experience inheritance and sharing:** Through the sharing and exchange of professional technology, management experience and career growth, UMC helps managers understand the management thinking at all levels and the practice of daily work, and promotes the inheritance of experience and the cultivation of management thinking.
- Excellent Leadership e-Newsletter:** UMC launches the Excellent Leadership e-newsletter every quarter. Through a themed, easy-to-read and regularly updated information platform, management related knowledge and tools are provided to managers. The e-newsletters, consisting of columns presenting the ideas of senior executives, aim to strengthen internal connections and build consensus. A total of 4 issues of e-newsletters were issued in 2024, reaching a total of 30,425 views.

The Video version of the Excellent Leadership e-Newsletter was launched in 2024, and can be learned in 5 minutes!

- Presentation** To enhance reading interest and comprehension, video and podcast versions of e-newsletters have been created. These versions feature two-person dialogue, Q&A and video infographic, allowing listeners to quickly grasp the key points of each article.
- Content** In 2024, following the principles and expectations of senior executives and focusing on organizational leadership and management, the Excellent Leadership e-Newsletter covered topics such as agility, constructive debate, team motivation, and strategic planning. Through practical case studies, this e-newsletter aimed to help executives at all levels understand the impact of these trends and promote continuous learning and application.



- "The UMC Business Management Thesis Award" Thematic lectures:** Awardees are regularly invited to share their latest research and ideas in management, improve their management skills, and bring new perspectives to business management. In 2024, Professor Su-Hua Ou from Soochow University, was invited to give a lecture on "Management Strategies from the Workplace to Life." in which she explained how enterprises should through the concepts of mutual learning and reverse mentorship in the workplace, emphasizing the importance of believing in and actively investing in the next generation to create a more progressive environment in both the workplace and society.



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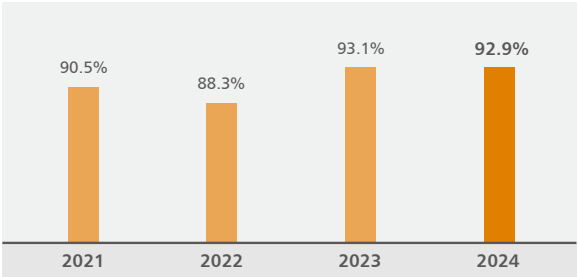
20% Experience Inheritance & Guidance Feedback 10% Classroom Training

The 7 Habits of Highly Effective People

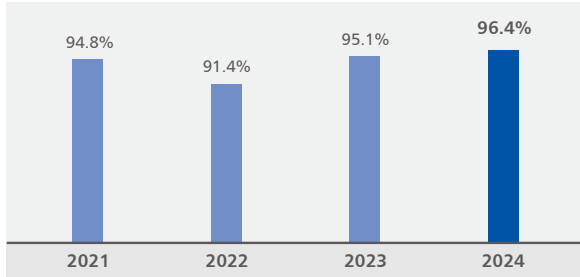
Establish core competency and strengthen a common language across the Company

❖ To implement the core values of the organization and practice the spirit of "accountability," UMC has continued to promote and deepen its implementation after introducing the "7 Habits of Highly Effective People" course in 2011. The program has since expanded to include the "7 Habits of Highly Effective Employees" course, which aims to cultivate the essential skills needed to lead a successful team, starting from personal success. This program helps create a common language across the Company. From 2021 to 2024, the overall training completion rate continued to increase and meet the required standards.

The 7 Habits of Highly Effective Employees



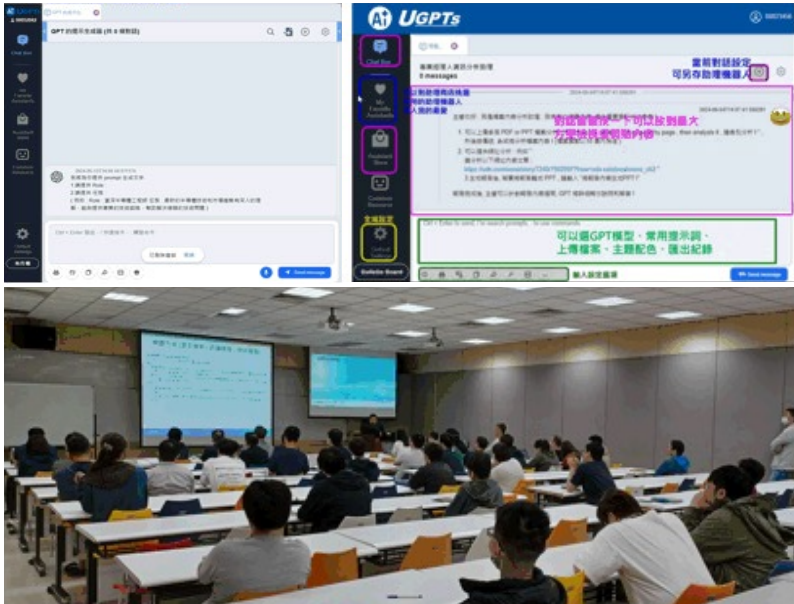
The 7 Habits of Highly Effective Managers



Digital Transformation Training Program

❖ In response to the changes in workflows and tool applications brought about by digital transformation, UMC's Smart Manufacturing Division has established a big data platform. Utilizing AI technology, the platform assists various divisions in production and cost reduction, helping employees adapt to the transformation and move towards the goal of an autonomous factory. To popularize the mindset and technology, a promotional website has been set up, incorporating various practical Gen AI tools, including translation, report or letter generation, quality inspection, data analysis, and knowledge databases. The aim is to empower employees, enabling them to focus on more creative and strategic tasks, thereby effectively enhancing work efficiency.

❖ To accelerate the implementation of AI culture transformation within the enterprise, starting from 2024, a "Generative AI Video Classroom" is issued weekly to all employees to cultivate foundational knowledge. Additionally, various training activities are arranged, such as generative AI courses, chatbot prompt teaching, and tool operation manuals, to deepen skills. In 2024, a total of 722 trainees completed training courses on the relevant topics. Moreover, to strengthen collaboration between organizations, a "Digital Transformation Seed System" has been established, appointing seed instructors from each division to serve as mentors for AI tool usage, ensuring employees receive timely support and assistance, making resource utilization more proficient. In 2024, 48 divisions participated in the seed system, cultivating 148 seed instructors. Participating divisions continue to grow through mutual exchange, grasping the pulse and trends of digital transformation.



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Diversified Learning Paths and Environmen

In addition to the training courses required for career development, UMC also integrates the Company's resources to provide a wealth of internal and external courses, On-the-job degree programs and other learning channels. By providing a complete training structure, UMC aims to create a comprehensive and diversified learning environment for its employees.

E-learning Platform

The new e-learning platform is now available and serves as a great learning companion

U-Learn e-Learning Platform: A new version of the digital learning platform has been created to enable the entire group companies to share learning resources and synchronize learning information with UMC employees around the world. In addition to working in the Company, UMC employees can learn anytime and anywhere according to their personal learning habits using mobile apps or home computers, breaking the limitations and boundaries of time and space.

Strengthening learning and establishing a self-learning atmosphere through the integration of internal and external resources

Diversified self-learning resources: Incorporate a variety of self-learning resources, such as Commonwealth Magazine Leader Campus and English learning e-newsletters, to establish diversified learning avenues.

Language Learning

Language cultivation is the best assist for enterprise internationalization

UMC offers free TOEIC test, TOEIC test preparation class and other courses, such as English letter writing, business conversation and other courses. Employees can sign up these courses according to their needs to continue to improve their English proficiency. In addition, online self-learning resources, such as TOEIC mock tests, speaking courses, and English e-weekly reports are provided for employees to study freely during their commute and enabling employees to learn freely during their commute and enhance their practical English speaking skills!

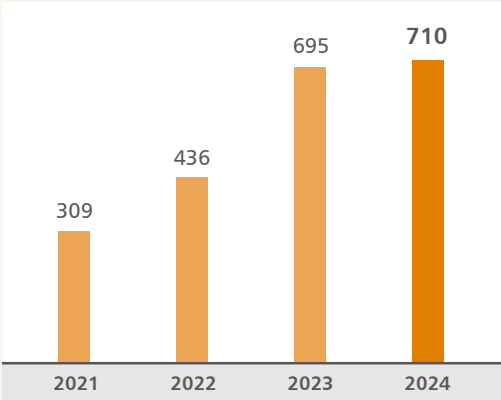
Moreover, tailored learning programs, such as one-on-one sessions and small-sized oral practice classes, are designed for functional divisions with specific language needs to accurately identify organizational needs and empower talent. Beginner and advanced Japanese courses are also available to those looking to enhance their Japanese language skills to foster multilingual abilities among internal talents.

External Training and Support for Degree Programs and Certifica es

To broaden professional horizons and increase awareness of industry trends, employees are provided with subsidies for utilizing external training resources.

- External Training: UMC provides external training opportunities, including domestic and international programs, to all employees (including dispatch and temporary staff). Relevant course information is also provided based on their job needs to improve their skills, so employees can enhance their self-abilities while staying informed about the latest industry trends. For example, employees are sent annually to attend the SEMICON Taiwan International Semiconductor Exhibition, helping them keep abreast of the latest developments in semiconductor technology and improve their technical competitiveness. In 2024, a total of 153 employees registered as a group, accumulating 1,071 training hours. After the training, employees bring back the results and new knowledge to the company for sharing and application in their work. Through this external training model, employees can further deepen and broaden their expertise in their professional fields. In 2024, UMC participated in 710 external training sessions and received a total of 7,705 training hours.
- promotes educational resources and learning channels, and also offers a "Measures of Subsidy for Degree Programs and Certificates," which all employees (including dispatched and temporary staff) are eligible for. This program encourages and supports employees to continuously learning and develop their professional abilities or management skills required for their careers. In 2024, employees participated in various training programs, including engineering courses, industrial safety and disaster prevention courses and sustainability courses, etc., and a total of NT\$598,000 of degree training subsidies was provided, and a total of 45 people have received subsidies since 2021.

Number of Participants in External Training Courses



4.3 Protecting the Employees and the Work Environment

- 4.3.1 Maintain Workplace Safety
- 4.3.2 Creating a Reassuring Workplace

★ Material Topic : Occupational Health and Safety

In accordance with GRI Standards (2021) for identifying impacts of sustainability issues, management actions in response to these impacts and processes used to track the effectiveness of the actions are listed in the table below:

Occupational Safety and Health Policy



Negative Impact : Work-related incidents

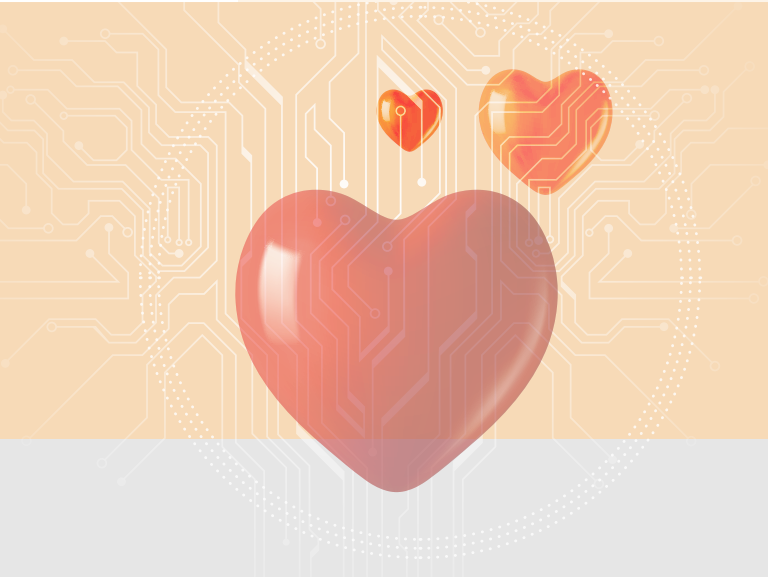
Impact	Management Actions	Processes Used to Track the Effectiveness of Actions
If the work environment is not properly managed, or if there are no relevant management policies/systems to prevent accidents, disasters, and occupational diseases, occupational injuries and illnesses may occur.	- Implement occupational safety and health management system, and obtain ISO 45001 certificate - Comply with the health and safety requirements of the Responsible Business Alliance Code of Conduct (RBA) - Review occupational safety and health management performance and legal compliance	- Conduct ISO 45001 third-party verification every year - Comply with external audit requirements, such as RBA audits - At the Occupational Safety and Health Committee, review the effectiveness of occupational safety and health management implementation

Sustainability Strategy and Blueprint	2024		2025	2030
Item	Goal	Achievement	Goal	
Disabling Injury Frequency Rate (FR) and Disabling Injury Severity Rate (SR)	Better than the 3-year average value for the semiconductor manufacturing industry (FR: 0.62, SR: 36) ^{Note 2}	✓ FR: 0.18 、 SR:2	FR: 0.56 、 SR: 31	FR: 0.27 、 SR: 6

Note: 1. Scope: UMC Group (UMC, its subsidiaries, HJ, USCXM, Wavetek, and USJC)
2. Based on the statistical model calculated by Occupational Safety and Health Administration of the Ministry of Labor and the 2022 announced data.



In line with the company's commitment to a "people-oriented" approach to occupational health management, UMC promotes an occupational disease prevention program aimed at achieving zero occupational illness



4.3.1 Maintain Workplace Safety



With the goal of zero accidents, UMC is committed to maintaining a stable growth of the Company under the priority of safety, preventing safety and health risks, avoiding major losses due to safety and health related issues, so the Company's competitiveness can be ensured.

The Safety-First Corporate Culture

UMC Group continues to implement a "Safety First" culture, requiring employees not to overlook safety in their work for the sake of speed. In addition, UMC has established the UMC "Safety Climate Initiative Index" to identify and improve unsafe behaviors and environments. Furthermore, each fab's performance index is compiled quarterly and the overall safety climate performance is presented in the form of light signals to prevent the occurrence of accidents, as well as to continuously reduce the Disabling Injury Frequency Rate (FR) and the Disabling Injury Severity Rate (SR).

Safety and Health Organization

All foundries of UMC Group have set up occupational safety and health committees in accordance with local regulations, and hold regular meetings to review the performance of occupational safety and health management. The status of the Occupational Safety and Health Committees in each site is shown in the table below:

Site		The frequency of meetings of the Occupational Safety and Health Committee	The proportion of labor representatives in the Occupational Safety and Health Committee
UMC	Taiwan	The committee meets every three months	33.3%
	Singapore	The committee meets every three months	51.4%
Subsidiary	Wavetek	The committee meets every three months	36.8%
	HJ	The committee meets every three months	7.7%
	USCXM	The committee meets every three months	44.4%
	USJC	The committee meets every month	50.0%

Safety and Health Management System

The safety and health management framework of UMC Group follows the management principles of ISO 45001 and is built on the "Plan-Do-Check-Action" framework, applicable to all workers in UMC Group. All UMC fabs have obtained ISO 45001 certifications, and the coverage rate of internal and external audits is 100%.

Hazard Identification and Risk Assessment

UMC Group conducts hazard identification and risk assessment for routine and non-routine operations. In addition to following the risk assessment methods of ISO 45001, it also introduces other risk assessment mechanisms or tools based on local regulatory requirements or fab engineering control needs. Currently commonly used hazard identification and risk assessment mechanisms and tools include:

1 Process and activity safety and health risk assessment

The safety and health risks of processes and tool transfer in fabs are evaluated by personnel who are trained and qualified in safety and health identification. The personnel assess the severity and frequency of risks, calculate safety and health risk indicators, and compile a list of major safety and health risks for improvement.

2 Hazardous workplace assessment

Taiwan and WTK fabs are inspected according to Hazardous Workplace Review and Inspection Rules as category A workplaces. The hazardous workplaces are evaluated by process safety assessment personnel who are trained and qualified in safety and health, and all applications have been completed. The Singapore, China HJ, and China USCXM have identified major hazard areas within their plants and completed corresponding risk assessments and implementation measures in accordance with local regulatory requirements, including the "MHI (Major Hazard Installation)," the "Jiangsu Province Industrial Enterprises Major Safety Production Risk Directory," and the "Interim Provisions on the Supervision and Administration of Major Hazard Sources of Hazardous Chemicals."

3 Identifying engineering risk using FMEA

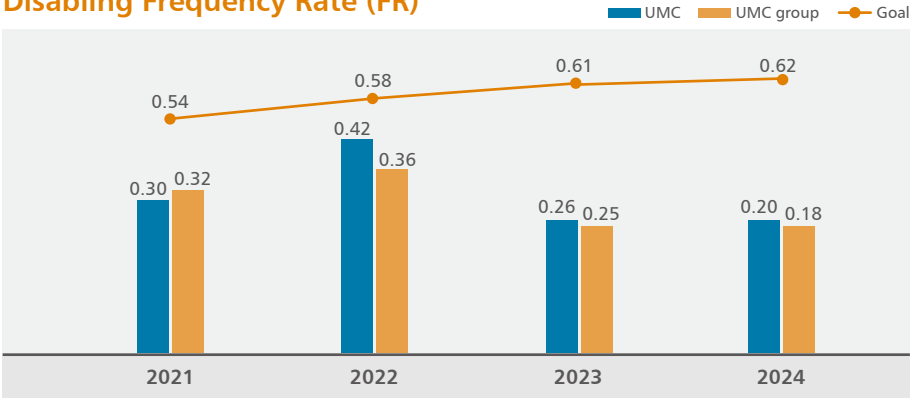
According to the engineering control requirements of fabs, failure mode & effect analysis (FMEA) is used to identify engineering risks, understand the engineering hazards before construction, and communicate with relevant personnel.

Occupational Disaster Management

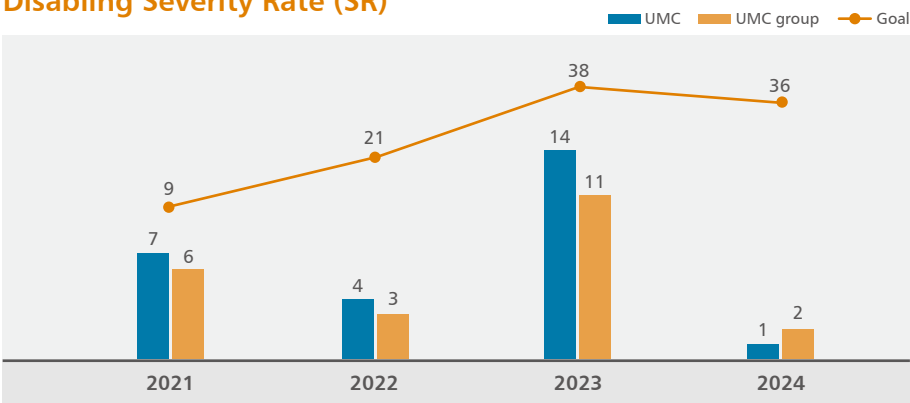
In 2024, UMC Group Disabling Injury Frequency Rate (FR) was 0.18 and Disabling Injury Severity Rate (SR) was 2, which are far below the average of the semiconductor industry^{Note}. UMC will continue to promote disaster reduction programs and move towards the goal of zero disasters.

Note: Based on the statistical model calculated by Occupational Safety and Health Administration, Ministry by Ministry of Labor (MOL). According to the 3-year disabling injury frequency (FR) and 3-year disabling injury severity rate (SR) average announced by the Occupational Safety and Health Administration of the Ministry of Labor in 2023; when the 2024 goals are set at the end of 2023, the latest announced industry data from the Occupational Safety and Health Administration is for 2022.

Disabling Frequency Rate (FR)



Disabling Severity Rate (SR)



Note: 1. FR = Number of person-times with disabling injuries x1,000,000 / total number of working hours [per million working hours].
2. SR = Number of lost day due to disability x1,000,000 / total number of working hours [per million working hours].
3. Goal: The three-year average of the national semiconductor manufacturing industry announced by the Occupational Safety and Health Administration two years ago in the same year; for example, in 2024, the three-year average of the national semiconductor manufacturing industry (2020-2022) in 2022 is used as the goal.

Safety and Health Education and Training

All UMC Group companies have clearly defined requirements for occupational safety and health (OSH) education and training. In Taiwan, UMC and Wavetek follow the Occupational Safety and Health Education and Training Rules. In Singapore, training is conducted in accordance with the Workplace Safety and Health Act. For HJTC and USCXM in China, the applicable regulations are the Provisions on Safety Training for Production and Operation Entities and State Administration of Work Safety Order No. 3. In Japan, USJC follows the Industrial Safety and Health Act. These regulations form the basis for their safety and health training systems, which include training for different types of operations and personnel. Examples include general OSH education (such as hazard communication) and specialized hazard operation training (e.g., forklift operation, handling of special chemicals/organics, and first-aid response). Employees responsible for conducting internal ISO 45001/14001 audits must receive proper training before performing such duties. To effectively control accidents and reduce harm, each UMC fab has established emergency response plans and regularly conducts scheduled and unannounced drills covering scenarios such as earthquakes, fires, gas leaks, and chemical spills. All fabs provide necessary safety and health education and training related to job duties and disaster prevention, aiming to cultivate a strong safety culture and enhance each employee's awareness and competency in workplace safety. This ensures a safe work environment while improving employees' safety awareness and skills, ultimately reducing accidents caused by unsafe behavior. In 2024, the UMC Group conducted a total of 2,643 training sessions, reaching 186,712 participants. Additionally, the Company offers online courses to help employees flexibly schedule their OSH training without being restricted by physical class times, which further enhances training convenience and coverage.

Consultation and Communication with Workers

UMC Group values the consultation and participation of employees. In 2024, there was no dispute related to occupational safety and health reported. UMC Group also conducted consultation and communication with workers in accordance with local regulations. In accordance with the election guidelines of safety and health labor representative regulated in the "Occupational Safety and Health Act," Taiwan fabs authorize employees to jointly elect labor representatives to participate in the quarterly Safety Committee with the consent of labor representatives through the labor management meeting. Additionally, they revise the Safety and Health Work Rules, investigate accidents, implement workplace environmental monitoring, and participate in decision-making on various safety and health related issues. The labor-management meetings also discuss the penalties for employee violations of labor safety discipline and reach consensus on safety and health related issues.

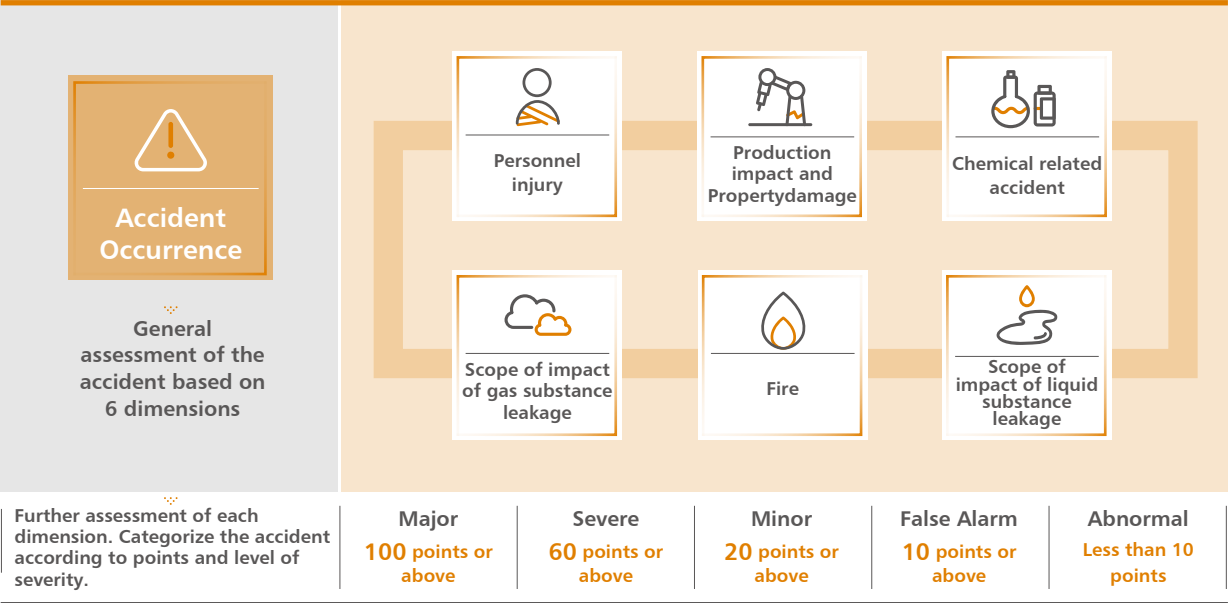
Apart from that, UMC Group also actively communicates with other non-employee workers. Through contractor alignment meetings and regular contractor briefing sessions, safety requirements are communicated, and a two-way communication channel is established. In addition to requiring all construction workers entering the factory to be informed of workplace hazards, UMC Group has also established a robust construction application system to effectively control construction applications and manage pre-, during- and post-construction stages.

Accident Analysis and Management

Accident Investigation and Analysis

UMC Group underscores the importance of immediate notification, rigorous investigation and recurrence prevention of each accident. Therefore, the Company has formulated the Accident Notification and Investigation Regulations to govern the procedures for accident notification, investigation and improvement. In addition, for systematic records and efficient control of accident notification, investigation, and improvement, an online system for accident notification and investigation has been installed to achieve instant transmission and storage of accident information.

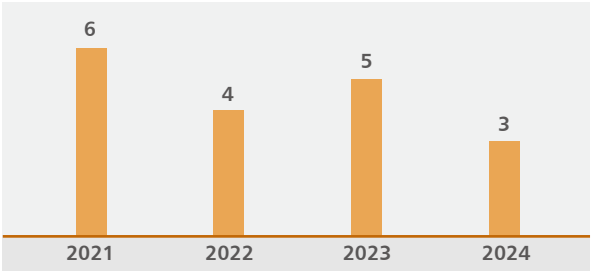
The Company also conducts a comprehensive assessment of the accident according to six dimensions, and quantifies the accident into major, severe, minor, false alarm or abnormal categories for more objective evaluation of the accident management performance.



Incident Management

UMC Group is committed to reducing the number of industrial safety accidents. In 2024, based on the six dimensions mentioned above, there were three minor accidents reported. As a result, the target was successfully achieved, and no regulatory violation related to safety and health incidents occurred. There have also been no fine for workers' health and safety, or financial losses due to judicial proceedings. In 2024, there were 24 fewer minor or above accidents than in 2011. The average financial loss caused by each minor accident from 2011 to 2024 is about NT\$2.47 million. Compared to 2011, UMC in 2024 has reduced its accidentrelated expenses by NT\$59 million. The three accidents in 2024 can be categorized: Arm injury due to pinching: 1, Finger injury due to pinching: 1, fire alarm: 1. One of the accidents required hospitalization following a physician's diagnosis. The accident type was an arm pinching injury, which was found to be due to insufficient guarding. In response, subsequent meetings were held to conduct comprehensive inspections and improvements on the mechanical guarding integrity across all fabs. In addition, we also promote accident prevention plans with "Zero FR & SR" as the main axis.

Number of Minor or Above Accidents



Note: UMC scores and classifies accidents according to injuries caused by people, impact on production, financial loss, involvement of chemical substances, range of impact, fire, or problematic employee behavior. Not all accidents result in human injury.

Key projects for accident prevention in 2024

- 1 Fabs included unsafe walking behaviors, unsafe walking environments, and temporary storage area/warehouse management as key items for internal operation observations and inspections by supervisors of various levels.
- 2 A comprehensive review was conducted on the suitability of the location and attributes of items kept in fabs' temporary storage areas/warehouses, with a focus on enhancing the management of items stored in these areas.
- 3 Three-step enhancement for walking safety
 - Step 1 List the fab's occupational safety competition evaluation items
 - Step 2 Case study and promotion
 - Step 3 Audit deficiencies result in point deduction for "repeated deficiencies" in the safety climate index
- 4 Implemented the Inter-fab Safety Performance Incentives and raised the FR & SR rewarding criteria, striving toward the goal of "zero accidents/zero injuries".

Occupational Disease and Illness Prevention and Management

The UMC Group reported zero cases of occupational disease. The basis for determination are as follows: UMC facilities in Taiwan and Wavetek follow the Regulations for Implementing Labor Occupational Accident Insurance Occupational Disease Appraisal; Singapore complies with the WSH (Medical Examination) Regulations; HJTC and USCXM in China follow the Administrative Measures for Diagnosis and Identification of Occupational Diseases; and USJC in Japan adheres to the Regulations for Enforcement of the Industrial Accident Compensation Insurance Act.

For occupational disease prevention, UMC facilities in Taiwan and Wavetek follow the Labor Health Protection Regulations; Singapore complies with the Workplace Safety and Health Act; HJTC and USCXM in China follow the Technical Specifications for Occupational Health Surveillance; and USJC in Japan follows the Industrial Safety and Health Act to conduct special health examinations. These examinations cover operations involving noise, ionizing radiation, dust, organic solvents, and specific chemical substances. In addition, HJTC and USCXM in China also include checks such as electrical work and pressure vessel operations. In 2024, the results of these special health examinations were graded or managed in accordance with local regulations. Only one individual at USCXM in China was found to be medically unfit for radiation-related work, due to personal health conditions, and was reassigned accordingly. Through high-risk job identification and occupational disease prevention management, UMC continues to identify high-risk individuals early and adopts a dual approach of engineering controls at the source and comprehensive health care. These efforts aim to create a safer, healthier, and more comfortable workplace.

The prevention management of high-risk jobs and occupational diseases and illnesses at UMC Group is in accordance with local regulations. In order to monitor the actual state of the labor working environment and assess workers' hazard exposure, Taiwan fabs have established a complete sampling strategy to monitor the operating environment, of which physical and chemical substances are monitored. The monitoring results of all areas (including clean rooms) were far below the concentration values allowed by law, and the results were announced to colleagues through fab's Safety Committee and posters on public bulletin boards. Taiwan fabs hold a Special Health Hazards Meeting every year to examine high-risk groups. In order to address the issues related to chronic injuries, diseases, illnesses or discomforts caused by long-term exposure to work hazards, UMC has established a notification and care mechanism for potential occupational chronic injuries, diseases and illnesses reporting, and set up an occupational disease and illness investigation team composed of labor health service physicians, the Health Development Division of the Human Resources Division, and the Risk Management and Safety and Environmental Protection Division. Furthermore, all aforementioned staff members conduct joint job observation and make recommendations and improvements as necessary. Hence engineering control and ventilation were improved, which prioritize the elimination of hazardous exposure factors from the source and make personal protective equipment as the last line of defense. From 2014 to 2024, there were five care cases reported. (No new cases in 2024.)

Hazard Notificatio

UMC Group implements daily inspections to detect potential hazards in the workplace early and strengthen safety and health to prevent accidents. Deficiencies identified during inspections are classified by severity and managed accordingly to correct them.

In case of any immediate workplace danger, workers can call the emergency contact person to notify the occupational safety personnel. Workers can stop work and retreat to a safe place on their own without endangering the safety of other workers, and will not be punished.

Prevention Measures for Impact of Occupational
Safety and Health

Through undertaking management, procurement management and change management, UMC Group further eliminates hazards and reduces health and safety risks, thereby preventing any safety and health impact on operations, products or services.

Contractor Safety Management

Policy Commitment	The Company is committed to extending the scope of its Safety and Health Policy to contracted personnel. This includes the implementation of preventive management and audit systems, ensuring workplace and operational safety, safeguarding the physical and mental health of workers, and other provisions aimed at protecting contractor safety.
Performance and Risk Pre- screening	UMC established the Sustainable Supply Chain Management and Assessment Procedure to evaluate the safety performance and potential risks of suppliers (including contractors) based on occupational safety and health-related indicators. Please refer to section 1.3.2 Supply Chain Management for the sustainability assessment of suppliers.
Operation Guideline	UMC formulated the Contractor EHS (Environmental, Health and Safety) Management Regulations to serve as safety guidelines for contractor management.
Safety Training	The Company regularly conducts UMC Contractor EHS (Environmental, Health and Safety) Education and Training to provide general safety information to contracted personnel operating on fab premises.
Contractual Agreement	Every new contractor must complete the signing of relevant agreements regarding the Contractor EHS (Environmental, Health and Safety) Guidelines and commit to comply with safety standards before engaging in transactions.
Management Objectives	The Company has established the goal of achieving ZERO accidents (including among contractors) with severity above minor levels and requires all fabs to strive toward this goal through Inter-fab Safety Performance Incentives. Those who achieve the goal each year are rewarded.
Performance Monitoring	UMC regularly convenes company-wide safety and health committee meetings to review the status of all fabs (including contractors) in achieving the goal of ZERO accidents.
Management Report	The Company regularly issues safety audit management reports to contractors.

Procurement Management

There are procurement management procedures in place that require the procurement of raw materials, machinery, projects, etc. to comply with laws and the specific safety and health regulations of the foundry, and compliance with regulations must be confirmed before acceptance and use.

Management of Change

To avoid safety and health risks and environmental impact from changes in personnel, machinery, materials, methods and environment, measures for management of changes have been established. Measures such as formal application procedure, approval procedure, implementation of necessary safety assessments, relevant personnel notification/training, and necessary technology information are in place to reduce potential risks.

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Results of the Occupational Disease Prevention Enhancement Program

Background

UMC is deeply aware of the importance of occupational disease prevention, particularly in high-tech industries such as semiconductor manufacturing. The occurrence of occupational diseases can seriously impact employee health and may also expose the Company to legal liabilities, compensation claims, and reputational damage.

Program Scope and Framework

UMC launched a three-year Occupational Disease Prevention Enhancement Program (2022–2024), covering risk assessments for more than 800 chemicals used in our operations. Based on management principles for CMR (carcinogenic, mutagenic, and reproductive toxicity) substances, these chemicals were categorized into five levels of hazard severity. The program targets both UMC employees and contractors. By proactively establishing indicators through a comprehensive exposure assessment strategy, we ensure workplace safety and health.

Key Implementation Measures

- Established a task force bringing together occupational health experts from each fab to regularly review progress.
- Selected chemical substances and conducted risk assessments, focusing on those classified as moderate to high hazard.
- Implemented the program in three phases: In 2022, we targeted CMR Category 1 occupational carcinogens; in 2023, we focused on substances included in Taiwan's List of Occupational Diseases; and in 2024, we addressed substances newly added to Taiwan's List of Occupational Diseases.
- Executed the four-pillar strategy: Inventorying substance flow and distribution; establishing a SEG (similar exposure group) database; SOP compliance verification; and awareness campaigns and training records.

Implementation Results

Year	Chemical Substances	Implementation Result			
		Identified Result		SOPs	Participants through awareness campaigns
		target substances	materials		
2022	CMR Category 1 Occupational Carcinogens	8	16	112	1,181
2023	Substances Included in Taiwan's List of Occupational Diseases	9	28	269	1,681
2024	Substances Newly Added to Taiwan's List of Occupational Diseases	12	126	189	2,373

Achievements:

- Establishing proactive indicators: Applied international methodologies and leveraged in-house occupational health expertise to implement our four-pillar exposure assessment strategy and establish proactive indicators.
- Win-win labor relations: Fostered employee cohesion and well-being, mitigated potential risks, and aligned with Sustainable Development Goal 3 (Good Health and Well-being).
- Increased visibility for occupational health efforts: Gave a keynote speech at the Taiwan-Korea Occupational Health Strategy Conference in May 2024, which received strong affirmation from attendees.

Employee Absenteeism Rate

UMC has created a safe workplace through health promotion programs and active accident prevention efforts to maintain workplace safety. The Company is committed to reducing employee absenteeism caused by injuries and illnesses. In 2024, the employee absenteeism rate at UMC was 1.11%, achieving the annual target of less than 1.2%. Starting in 2025, the management of employee absenteeism will extend across the entire UMC Group, with a target absenteeism rate of less than 1.3% in 2025. Moving forward, UMC will continue to enhance health awareness and injury prevention management, create a supportive work environment, promote employees' physical and mental well-being, and remain attentive to employee absences due to injuries and illnesses.

Year	2021	2022	2023	2024
Absenteeism Rate	0.74%	0.86%	1.06%	1.11%
Employee Data Coverage Rate	100%	100%	100%	100%

- Note: 1. Scope: UMC (fabs in Taiwan and Singapore)
2. Absence rate (%) = (Total absence hours during the fiscal year ÷ total working hours during the fiscal year) x 100%
3. The total working hours during the fiscal year are calculated based on each employee working 8 hours per day.
4. Absences include occupational injury leave, sick leave, maternity leave, and menstrual leave.

Occupational Safety and Health Statistics

UMC	Taiwan		Singapore	
2024 Relevant Index	Employees	Non-employee	Employees	Non-employee
Total No. of workers	12,695	1,810	1,764	682
Total No. of working hours	25,073,636	5,285,032	3,747,777	1,990,328
No. of people with disabling injuries	2	4	4	3
No. of lost days due to disability	33	268	23	7
No. of reportable injuries	6	5	5	3
No. of severe occupational injuries	1	1	0	0
No. of work-related deaths	0	0	0	0
Severe Occupational Injuries Rate	0.0001	0.0006	0	0
Occupational Disease Rate (ODR)	0	0	0	0
Injury Rate (IR)	0.05	0.19	0.27	0.30
Lost Day Rate (LDR)	0.26	10.14	1.23	0.70

Note: There were no third-party illnesses, injuries, disabilities or deaths caused by UMC operations.

Subsidiaries	China - HJ		China - USCXM		Taiwan - Wavetek		Japan - USJC	
2024 Relevant Index	Employees	Non-employee	Employees	Non-employee	Employees	Non-employee	Employees	Non-employee
Total No. of workers	1,751	115	1,142	615	764	36	1,261	257
Total No. of working hours	3,420,620	335,800	2,345,562	1,795,312	1,450,585	105,616	2,336,885	500,898
No. of people with disabling injuries	1	0	0	0	0	0	0	0
No. of lost days due to disability	38	0	0	0	0	0	0	0
No. of reportable injuries	1	0	0	0	0	0	0	1
No. of severe occupational injuries	0	0	0	0	0	0	0	0
No. of work related deaths	0	0	0	0	0	0	0	0
Severe Occupational Injuries Rate	0	0	0	0	0	0	0	0
Occupational Disease Rate (ODR)	0	0	0	0	0	0	0	0
Injury Rate (IR)	0.06	0	0	0	0	0	0	0.40
Lost Day Rate (LDR)	2.22	0	0	0	0	0	0	0

Note: There were no third-party illnesses, injuries, disabilities or deaths caused by USJC, HJ, USCXM, Wavetek and USJC operations.

Definition of Terms

- Non-employee workers: Refers to non-employees whose work and/or workplace is defined by the Company. The daily average number of such workers in the fabs is calculated by dividing the total number of workers in the year by 365.
- Third party: Refers to non-employees or contracted personnel.
- Working hours:
 - 1. Employees: For Taiwan fabs, working hours are calculated based on the number of workers and their respective shifts. For Singapore 12i, China HJ, China USCXM, and Japan USJC, working hours are calculated based on the actual hours worked by the employees.
 - 2. Other non-employee workers: Calculated based on 8 hours per person per day.
- Number of people with disabling injuries: Refers to the number of deaths, permanent disability, permanently total or partial disability or temporarily total disability due to occupational injuries. In 2024 UMC Group, the types of employees accident injuries are 3 walking injuries (43%), 3 chemical exposure injuries (43%), 1 other injuries (14%) : the types of non-employee workers accident injuries are 4 chemical exposure injuries (57%), 2 pinching/crushing injuries (29%), 1 collision/scraping injuries (14%).
- Number of lost days due to disability: Refers to the total number of days lost as the result of injury from a single accident. Calculated as the number of days when the injured person is temporarily (or permanently) unable to resume work, but excludes the days of injury or the days when work is

resumed. The number of days elapsed (including Sundays, holidays or company rest days) and the number of inability to work days following return to work as a result of the injury are included.

- Number of reportable injuries: Refers to the number of work-related injuries that resulted in death, job loss, impairment or transfer, emergency treatment or more, loss of consciousness, or diagnosed by a doctor as serious injuries. In 2024 UMC Group, the types of employees accident injuries are 6 walking injuries (50%), 4 chemical exposure injuries (34%), 1 puncture/cut injuries (8%), 1 other injuries (8%) : the types of non-employee workers accident injuries are 4 chemical exposure injuries (44%), 3 pinching/crushing injuries (33%), 2 collision/scraping injuries (23%).
- Number of Severe Occupational Injuries: Refers to the number of workers who, due to occupational accidents, are diagnosed by a physician and require hospitalization (excluding those under observation). In 2024 UMC Group, the types of employees accident injuries are 1 walking injuries (100%) : the types of non-employee workers accident injuries are 1 pinching/crushing injuries (100%).
- Rate of Severe Occupational Injuries = Number of Severe Occupational Injuries/Total number of workers
- ODR = Total number of occupational diseases x 200,000 / total number of working hours [per 200,000 working hours].
- IR = Number of reportable injuries x 200,000 / total number of working hours [per 200,000 working hours].
- LDR = Number of lost days due to disabling injuries x 200,000 / total number of working hours [per 200,000 working hours].

4.3.2 Creating a Reassuring Workplace



"Health, Hassle-free and LOHAS" are UMC's core values in safeguarding the physical and mental health of employees.



"Health, Hassle-free and LOHAS" are UMC's core values in safeguarding the physical and mental health of employees. Recognizing that employees are invaluable assets to the Company, UMC is committed to ensuring every staff member receives comprehensive care. By providing ongoing health support, employees can focus on their work without worry. Additionally, through the encouragement of wellness initiatives, UMC fosters healthy habits and promotes sustainable lifestyles, ultimately achieving a harmonious balance between work and family life for a true win-win outcome.



Based on the "Occupational Safety and Health Act" (later referred to as the "OSHA"), "The Labor Standards Act," "Gender Equality Act," "Sexual Harassment Prevention Act," "Guidelines for Safety and Health of Middle-aged and Elderly Workers," "Guidelines for Safety and Health of Night Shift Workers in the Workplace," UMC has formulated campaigns for maternal health protection, abnormal workload-induced disease prevention, ergonomics hazard prevention, workplace unlawful infringement prevention, and special protection of middle-aged and elderly workers.

In 2024, a series of health-promoting activities were developed to encourage employees to adopt a healthy lifestyle in their daily lives. Lucky draw prizes were also offered with the expectation that employees would continue to maintain a healthy lifestyle and take their health to the next level.

Safe Workplace Measures



» **The Construction of a Gender Equality Culture**
Gender equality measures are superior to or in accordance with the Labor Standards Act.



» **Safety Measures for Women Working at Night**
Vehicle rescue, emergency button, various emergency assistance, day and night transportation connection, and nighttime emergency response reporting mechanism.



» **Special Assistance for Pregnant Women and Employees in Need**
Priority meal pickup, reserved parking space, notification of cleaning and disinfection avoidance, night shift transfer, maternity subsidy (both male and female employees are qualified), dedicated lactation room



» **Flexible Work and Vacation System for Work and Family Balance**
Provide employees with flexible work shifts. For employees with kids under 3, the working hours can be adjusted one hour earlier or later. An extra 7-day spare leave is also provided. According to the Gender Equality at Work Act, both women and men are eligible for parental leave.



» **Sexual Harassment Prevention System**
Formulate the "Complaint and Punishment Measures for Sexual Harassment Prevention and Control in the Workplace." Provide case investigation mechanism, membership criteria, dedicated complain channels for privacy protection, education and training, and an internal police security assistance service network

Note: UMC follows various Taiwan government labor regulations, "Gender Equality in Employment Act", "Sexual Harassment Prevention Act" and other relevant norms to formulate regulations in Taiwan and promote various related measures.

Healthy Workplace ✨ Protect the Physical and Mental Health of Employees

Worker Health Promotion

UMC implements a health service management plan every year to maintain the safety, health and well-being of employees and stakeholders. In accordance with the OSHA, we review the results of the 2024 plan, then implement the 2025 health service management plan. After approval by the supervisor, the plan is announced to all employees.

Health Service

Health Service			Application						
	Description	Scope	Taiwan	Wavetek	SG	USCXM	HJTC	USJC	
 Health Check	‣ Health examination for new recruits ‣ Annual health examination ‣ Special health examinations ‣ Health examination for middle-aged and elderly workers ‣ Self-pay health examination (gynecologic examination, abdominal ultrasound, flu vaccine, etc.)	All staff and staff dependents							
 Health Management	‣ Abnormal health examination report management ‣ Care and referrals for special cases ‣ Allocate full-time nursing staff	All staff							
	‣ Maternal health protection ‣ Prevention of diseases triggered by abnormal workload								
	‣ Survey and health education for middle-aged and elderly workers ‣ Physiological assessment of respiratory protection equipment								
	‣ Selecting and matching workers/occupational sickness leave review ‣ Emergency personnel/precision/overhead operation suitability assessment								
	‣ Pandemic prevention for business travelers								
	‣ Keeping track of abnormal night shift workers								
	 Health Promotion		‣ Analysis of epidemiological and health screening data	All staff					
‣ Planning and designing self-management content through health surveys									
‣ Incentivizing participation in walking activities can effectively help improve abnormal health screening results and foster the adoption of healthier behaviors.									
 Caring for Health	‣ Through digitalized management, the health care system enables continuous monitoring, tracking, and recording of employees' health. When necessary, it also provides medical referrals and consultation assistance.	All staff							

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Health Promotion Activities in 2024

Theme	Application	Event Spine	Key Achievements
 Health Column/ Health Education	Taiwan, Wavetek, USCXM, HJTC, USJC	Diversified health education information is regularly promoted through announcements at the safety committee, various forums, emails, BBS and electronic posters.	❖ 105 pieces of health education information were sent.
 Health Lecture	UMC Group	Based on the results of health check-ups and a health needs survey, we developed a health promotion plan and invited professional doctors to give lectures at fab sites to provide health education information and knowledge.	❖ Twenty one health-related lectures were held. ❖ A total of 861 participants with an average satisfaction rate of 97 points.
 Health Examination	UMC Group	Better than regulations 1. Conduct regular health checkups annually 2. According to the advice of occupational physicians, the Employee Benefit Committee subsidizes non-regulatory items and additional items.	❖ The number of people checked reached 16,274. ❖ 100% completion rate for mandatory regulatory checks, which is higher than the 93% completion rate for voluntary regulatory checks. ❖ Participation satisfaction rate averaged 95 points. ❖ A total of 1,371 women participated in the three-in-one examination, with an average satisfaction rate of 98 points. ❖ A total of 2,885 people participated in the abdominal ultrasound examination, with an average satisfaction rate of 99 points.
 Health Promotion Activities	Taiwan	1. Planned 9 tasks under the theme of "Who is the UMC 2024 Health Points King 2.0 Challenge," including activities such as health walks, metabolic syndrome, stress relief, etc., using points collection and prize draws for achieving goals to motivate employees to value health. 2. Organizing seasonal influenza vaccination events 3. Activities to improve health check-up results and health walk classes were organized.	❖ According to the analysis of annual health check-up results of employees in the same category in the past two years, by 2024, 35% have successfully overcome metabolic syndrome. ❖ A total of 686 people received the sea sonal flu vaccine at their own expense, with a satisfaction score of 99.



UMC Group Description of Results

❖ Total number of service users: 38,543 person-times
❖ Average Satisfaction rate: 96 points

Health Autonomy Accreditation and Health promotion seal.

Site	Hsinchu headquarters building	Fab 8D	Fab 8E	Fab 8F	Fab 12A P5	Fab 8AB	Fab 8S	Fab 12A P1
Validity Period	2024-2026	2023-2025				2025-2027		

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Worry-Free Workplace Focus on Work Environment Safety

UMC's Taiwan facilities have established a health-conscious and employee-friendly workplace environment. Collaborating with the Environmental and Safety teams, the company adheres to the Occupational Safety and Health Act as a foundational framework. UMC provides comprehensive safety and health protection measures, conducts thorough hazard identification, risk assessments, and suitability evaluations, and implements preventive and corrective actions. These efforts aim to effectively prevent occupational injuries and enhance overall work efficiency.

Safety Protection Items	Safety Protection Items	Key Performance in 2024
 Occupational Disease Prevention	Regarding health protection matters such as diseases caused by abnormal work loads, maternal protection, human factor hazards, potential chronic injuries, and unlawful harm during duty execution, the health center immediately activates the care mechanism upon receiving a request. Based on the needs, it refers individuals for medical treatment, arranges occupational health consultations, conducts on-site visits, assesses and clarifies the correlation between diseases and work operations, provides individual health consultations and recommendations, and carries out subsequent assistance and follow-up care.	We conducted multiple on-site visits to understand employees' working conditions, helped identify root causes, provided early treatment, and offered necessary assistance. As a result, there was no reported case of occupational disease in 2024.
 Maternal Health Protection	UMC has established lactation rooms at each fab in Taiwan, and provides postpartum female employees with 60 minutes per workday for lactation. Additionally, a company-wide maternity protection platform has been implemented. Pregnant employees and those in postpartum or breastfeeding stages can input their information into the system. Upon receiving the notification, the health center conducts risk assessments and provides medical consultations to ensure comprehensive maternity health protection.	- Pregnant female employees receive maternal health protection. The medical staff has completed the hazard identification, risk assessment and fitness for work assessment through telephone or interviews, provided relevant health education information during pregnancy and postpartum, and gave mother gifts and informed birth subsidy and other related information. - The "Health Care System" automatically arranges care list of the employees according to their major health issues daily. The follow up and care rate was 100%.
 Special Health Protection for Middle and Old Age	Since 2019, before the announcement of "The health and safety guide for middle-aged and elderly workers," the Company already had individualized health protection plans for the middle-aged group workers. Surveys are conducted for workers over 45 years old (including 45). Work fitness survey results were assessed, and employees at risk received physician consultations.	By using the work fitness and health questionnaire, the Company can identify employees who have a low work ability. This allows the Company to arrange doctor consultations for them, provide health guidance and advice, and refer them to work safety and employee assistance programs as needed. This helps the employees cope with physical and mental problems they may face at work.
 Employee Assistance Program, EAP	Each employee can have up to 6 free psychologist consultations per year. The service is announced to every worker during the new employee training. The entire consultation process is kept confidential. Professional counselors provide employees with counseling on issues related to family, work, and interpersonal relationships in order to achieve work-life balance, and at the same time guide employees to self-adjustment of stress, alleviate different sources of stress, and enable them to overcome their challenges.	- UMC places a great deal of importance on the well-being and both mental and physical health of its employees. To support this, a wide range of professional counseling services is available 24/7 as part of a comprehensive support system - Interview or telephone consultation satisfaction rate reached 96 points
 Caring for Health	According to the injury and illness caring protocol, the health center worked together with physicians and nurses, department supervisors, human resources, legal affairs staff to perform fitness for work evaluation. For sick or injured employees, the team would help adjust the content of the work in stages according to the recovery situation, and to arrange appropriate or temporary positions without affecting the recovery process.	- By continuously raising awareness among employees about the importance of health, the Company strengthens their understanding of disease prevention and healthcare in maintaining good health. - A total of 3,620 care recipients received services for injury and illness, including ongoing follow-up care and appropriate health education.

LOHAS Workplace ✨ Emphasis on Work-Life Balance

UMC believes that healthy and happy employees are the cornerstone of a highly productive enterprise. Beyond offering a safe and health-conscious workplace, the Company fosters an employee-centered LOHAS environment that seamlessly integrates wellness benefits, vitality, and community engagement. By designing diverse and thoughtfully curated activities, UMC nurtures creativity and energy, enriching both the professional and personal lives of its employees and striking a harmonious balance between work and leisure.

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Appendix



Competition activities are organized to enhance team cohesion

To foster a vibrant workplace atmosphere and strengthen team cohesion, UMC Taiwan organizes team-building competitions in 2024 as part of the Company's anniversary celebration activities. Subsidiary USCXM holds sports tournaments, while HJ holds fun contests such as Dragon Boat Festival bag sachet toss competition, fun games, Karaoke contest, Christmas candy giveaways, and more -- all are designed to infuse the UMC team with a strong sense of unity and camaraderie. In addition to recreational events, UMC Taiwan also organizes professional sports tournaments such as U Gym Cup - badminton and basketball tournaments as part of the Happy Workplace Program. A series of parent-child courses were also organized for family members to participate.

Exploring life: Employees are encouraged to participate in social club activities, and sports facilities are offered to take care of employees' health

UMC encourages employees to not only excel in their professional roles but also to actively explore and enjoy the pleasures of life beyond work. UMC offers subsidies for employee social clubs through the Employee Welfare Committee to encourage participation in social club activities and foster connections with like-minded co-workers. In addition, in order to encourage employees to maintain good exercise habits, fitness facilities have been set up in various fab locations and some dormitories in Taiwan. Aligned with the STAY 2025 Program, UMC created an exclusive U GYM and organized large-scale celebrity group classes, attracting a total of 4,014 person-times participation.



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UMC Recreation Center and Lianyuan Activity Center

UMC Recreation Center (in the Hsinchu Science Park) and Lianyuan Sports Center (in the Southern Taiwan Science Park) offer high-quality leisure and sports facilities for employees and their families. These venues feature a variety of amenities, including swimming pools, gymnasiums, billiard rooms, basketball courts, badminton and volleyball courts, art and cultural spaces, parent-child playrooms and aerobics classrooms, providing employees with all-round sports and recreational options. Designed as exclusive, all-in-one leisure hubs, these Centers enable employees to plan fitness activities conveniently while promoting stress relief and overall well-being. Additionally, massage services performed by visually impaired therapists are available on weekends, further enhancing the care and convenience provided to UMC employees and their families.



❖ Hsinchu Science Park UMC Recreation Center ❖ Southern Taiwan Science Park Lianyuan Sports Center ❖ Five-star Fitness Equipment ❖ Multi-purpose Basketball and Badminton Court

Activity List of UMC LOHAS Workplace

Item	Content	2024 Performance
Employee Social Club Events	- There are five categories of UMC social clubs: body beauty, recreation, indoor activity, outdoor exercise, and social service. - Social clubs are evaluated annually, with top-performing clubs receiving financial subsidies. Extra funding is provided for clubs representing UMC in public competitions.	31 social clubs (Taiwan fabs) 2 rounds of evaluations were conducted, with subsidies awarded to 15 outstanding social clubs each time 2,441 club members ^{Note}
Recreational Facilities and Services (Recreation Center in Hsinchu / Lianyuan Sports Center in Tainan)	- High-quality recreation center in Hsinchu for employees and families. - Dormitory in Tainan with integrated recreational facilities. - Each site (Fab 8E, Fab 8F, Fab 8S, Fab 12A, and Fab 12A II) provides fitness equipment for employees.	- Recreation Center in Hsinchu: 74,484 person-times visitors. - Lianyuan Sports Center, located in the Southern Taiwan Science Park, optimized its fitness center, aerobics classroom and other facilities, providing a better employee experience. - Regular inspections, evaluations and replacement of equipment are conducted to ensure that employees have the best exercise environment.
Diversified Employee Activities	- Family day, team competitions and other team-building activities were organized to promote the relationships between employees and their family members, and to establish a positive corporate image. - Travel activities were also organized in each half of the year, with local revitalization as the main theme. - A total of 12 activities were held during the art season, including exhibitions, theatrical performances, musical performances, and handicraft classes, allowing employees and their families to enjoy an artistic feast together during the summer. "STAY 2025 Workplace" project was planned, centered on the theme of LOHAS and cohesion, featuring the creation of an exclusive U Gym for UMC employees and activities such as parent-child classes.	- The satisfaction score for the Family Day event reached 88.9 in Hsinchu and 85.8 in Tainan 8 sessions of light travel events were held, with a total of 366 participants. - A total of 6 events were held during the art season, with 3,539 participants. - A total of 4,014 participants in the U Gym events.

Note: Excluding temporary members and one-time participants.

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Enhancing Society Mutual-Prosperity

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Enhancing Society Mutual-Prosperity

Important Stakeholders | Employees | Suppliers | Society |

UMC has participation in diversified charity activities for a long time. With the belief of "talents are the foundation of industrial development, the environment is the foundation of human sustainability, and education is the foundation to improve the status of the disadvantaged," UMC believes that enterprises can thrive in a friendly environment, and co-prosperity with society is the only way. Therefore, UMC has joined hands with partners such as supply chains, non-profit organizations, and educational institutions to continue to exert influence through the implementation of various charity activities, inject positive energy into society and the environment, and move forward for a better society.

Performance Highlights 2024

SDGs



Note: The scope of disclosure in this chapter mainly covers information related to UMC.

Beneficiaries i

UMC's social charity participation

45,028 person-times

Semiconductor Education Programs and Science Education Project

115 Sessions conducted

UMC Love Story Group performed its seventh stage play, "The Gentle Persistence," in 2024, with a total of **5** performances.

The UMC participated for the 2024 Taiwan Science Train, this initiative guide students from **31** elementary and junior high schools across Taiwan to learn about semiconductors and water resource knowledge.

The Eco Echo Award provided a grant support to ecological conservation and green innovation projects **13** projects

Energy Saving Service Team provided services to **11** organizations, bringing energy-saving economic benefit NTD\$ **11.34** million

5.1 Promote Common-Prosperity Society

5.1.1 Core Vision and Strategy

5.1.2 Community Services Benefits Assessment

UMC upholds the corporate sustainability vision of "centered on people, in harmony with the environment, and enables communities to thrive" to integrate various internal and external resources, and actively engage in charity activities. We have diversified ways of promoting charity activities, such as "UMC Science and Culture Foundation," the "UMC LOHAS Education Foundation," and Employee Social Clubs etc., since 1996. In order to maximize the benefits of social participation, a cross-department and cross-organization, the social participation group, was established in 2021 to integrate existing organizations and make use of innovative thinking in the technology industry. Internally, it empowers employees' abilities and promotes a culture of volunteer activities. Through industry-academia cooperation, it cultivates talents and externally advocates for technology application and deepening public charity activities, with the hope that UMC can use its ubiquitous technological influence to create common good.

Diversified Ways of Promoting Charity Activities

UMC Social and Culture Foundation

Scope and Dimension: Long-term support for diversified education programs with the goal of "All-Around Education." Engage in the development of education for disadvantaged students, cultivate the future competitiveness of young talents, and collaborate with public interest groups to expand social influence

Projects

- ❖ **Diversified education**
"Spreading the Seeds of Hope Project," arts and humanities education, media literacy education
- ❖ **Reading education**
"Storytelling Volunteer," "Reading Promotion in Rural Area Program"
- ❖ **Life education**
"Love Storytelling Club," "Diversified Education Promotion in Correctional Institutions"
- ❖ **Physical education**
Supported Taiwanese sports talents and continued to sponsor the "Nantou Karate Association" and "National Tung-Shih Senior High School" baseball team.



UMC Volunteers

Scope and Dimension:

- ❖ UMC volunteers share customized semiconductor science knowledge with elementary school students in local communities.
- ❖ Initiate public charity activities to the issues of non-profit organizations in local communities at the fab and division levels.
- ❖ Charity activities organized by UMC employee social clubs.

Projects

- ❖ Semiconductor Science Education for elementary school student projects
- ❖ Fab and division volunteering projects
- ❖ Employee social club volunteering projects

UMC LOHAS Education Foundation

Scope and Dimension:

- ❖ **Sports Advocacy**
assisting disadvantaged groups in using equipment and promoting sports knowledge sharing during activities at the UMC Activity Center
- ❖ **Sports Promotion**
providing campus tour for college students.
- ❖ **Various Types of Events**
assisting the Hsinchu City and Hsinchu Science Park in providing venues or equipment for sports related events.

Projects

- ❖ Sponsoring annual events organized by the Hsinchu City Education Department every year
- ❖ Assisting in organizing sports events for the Hsinchu Science Park Cup

5.1.1 Core Vision and Strategy

Based on the belief of "take from society and give back to society," UMC has involved enormously in various charity activity projects for a long time. With the belief of "talents are the foundation of industrial development, the environment is the foundation of human sustainability, and education is the foundation to improve the status of the disadvantaged" as the starting point, UMC promotes charity activity in accordance with three major strategic principles--"Cultivating young talents," "Promoting environmental protection," and "Caring for the disadvantaged in society," which echo with the Sustainable Development Goals (SDGs) of the United Nations Through the implementation of various charity activity projects, UMC has joined hands with partners such as supply chains, social welfare groups, and educational organizations to bring positive energy to society and the environment.

Sustainability Strategy and Blueprint		2024		2025	2030
		Item	Goal	Achievement	Goal
	Social Participation	Number of beneficiarie	» 37,000	✓ 45,028	» 40,448
		Employee volunteer hours	» 6,950	✓ 9,302.5	» 42,760
	Semiconductor Education Programs	Number of semiconductor promotion events	» 60	✓ 90 場	» 62
		Annual number of participants	» 3,000	✓ 7,449 人次	» 65
		Cumulative investment in semiconductor education (base year: 2020)	» NT\$38.1 million	✓ NT\$41.65million	» 3,500
	Semiconductor Science Education Project	Number of events	» 16	✓ 25	» NT\$50 million
		Number of beneficiarie	» 4,330	✓ 4,677	» NT\$87.5 million

Note: The scope for 2024 is UMC, and it will be expanded to include UMC's foundry subsidiaries starting in 2025.

Main Theme of Social Participation

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Main Theme	Business Trend	Type	Business Key Performance	Social/Environmental Key Performance	UN SDGs
 Cultivate Young Talents	- From a forward-looking perspective to strengthen UMC's global competitiveness, UMC is committed to managing industry-academia relationships and cultivating high-tech talents to promote semiconductor research energy and technological progress by securing quality and stable manpower. - UMC volunteers share customized semiconductor science knowledge with elementary school students in local communities.	- Prospective Talent Program (PTP) - UMC Chung-Laung Liu Doctoral Scholarship - Semiconductor Science Education Project	- Expand talent pool to enhance the effectiveness of recruitment - The PTP program has been recruiting talented individuals since 2013, with a total of 6,702 talents recruited and 860 students participating. - Build high-tech talent network based on good business reputations - In 2024, UMC continuously strengthens industry-academia cooperation according to recruitment strategy and carried out projects with more than 20 colleges, including equipment operation internships, nanotechnology programs, and career planning assistance. These projects increased the recognition and support of UMC from the academia, effectively marketed UMC's corporate image, and expanded its influence among campuses and the communities, thereby laying a cornerstone for future R&D talents. - Established the UMC Chung-Laung Liu Scholarships are limited to 17 doctoral students per academic year	- Cultivate high-tech talents and enhance career prospects - Beneficiaries of special program for industry-academia cooperation: A total of over 7,000 person-times participated in these programs. - Campus relationship development and seminar management: including a series of brand image cultivation activities such as book and report discussions, career sharing, internship cooperation, company visits, and semiconductor product demonstrations. There were 90 events and over 7,000 person-times accumulated by 2024. - Enhancing investment in semiconductor education resources: a total of over NT\$40 million has been accumulated. - Semiconductor Science Education for children. - UMC custom made a set of semiconductor science knowledge video, and promoted them in near-by elementary schools 25 sessions were conducted with 4,677 person-times beneficiaries in 2024.	  
 Promote Environmental Protection	Strong economic growth leads to excessive consumption of resources. The sustainable growth of an enterprise depends on the sustainability of environmental resources. Therefore, in addition to providing customers with core products that are competitive on the market, enterprises should also bear the responsibility of environmental protection. Through energy-saving, carbon-reduction and ecological conservation initiatives, we look forward to exerting our influence and drive the value chain to jointly mitigate the impact of business operations on the environment.	- Eco Echo Award - Energy Saving Service Team	- Eco Echo Award starts to make a difference, building good corporate image - In 2024, the UMC Eco Echo Award event received more than 77 news coverages. In addition to establishing a good reputation for UMC, it also provided media exposure for the supplier partners involved. - Cost Savings Brought by the Energy Saving Service Team - Assist the beneficiaries to save on utility bills for operations. In 2024, a total of 1,152 tonnes of CO₂ was reduced, which is equivalent to an overall social/environmental economic value of NT\$11.336 million.	- Eco Echo Award support ecological conservation and green innovation - In 2024, 8 suppliers participated as partners, with a total sponsorship of NT\$4 million. 3 NGO groups' ecological conservation projects, 8 youth environmental action initiatives, and 2 green innovation projects were awarded. - Number of Service Targets of Energy-saving team 11 vendors participated as partners - Served 11 disadvantaged groups and 2,899 beneficiaries. - A total of 1,152 tonnes of CO₂ was reduced. The services provided included energy-saving and safety consultation, energy technology and engineering improvement, allowing disadvantaged groups to enjoy the wonderful energy-saving and carbon-reduction life as well.	   
 Caring for the Disadvantaged in Society	- UMC has a long-standing commitment to supporting rural communities and promoting education for disadvantaged students. By prioritizing the education of children from underprivileged families, UMC works to reduce the educational gap between urban and rural areas. - Shaping UMC's volunteer culture, encouraging clubs and departments to engage in community service for nearby communities or disadvantaged groups. By rallying colleagues to participate in volunteer activities, we aim to strengthen the Company's cohesiveness.	- Supplementary classes for disadvantaged students - Community participation and support activities for the disadvantaged - Promote on-site charity activities in fabs and divisions	- Build a positive image and maintain good corporate reputation - Total beneficiaries: 45,028 person-times Recipients of these supports include schools, non-profit organizations and environmental groups. By actively supporting disadvantaged groups, it not only builds a good image for the Company, but also earns the recognition from the public. - Build a positive image and maintain good corporate reputation 9,302.5 hours of volunteering time in total 21 fabs/divisions have participated in public charity activities, utilizing the power of each fab/division to lead colleagues in performing acts of volunteering service, and ultimately shaping a culture of volunteerism and mutual assistance.	- Number of underprivileged school children benefite 6,274 volunteers who cumulatively devoted 29,735.5 hours of their precious time - the number of beneficiaries 14,533 person-times UMC continues to promote the Spreading the Seeds of Hope project, as well as after-school tutoring service in collaborations with the Hsinchu Church of Sacred Heart of Jesus and National University of Tainan. - UMC sponsorship 90 domestic children by the Taiwan Fund for Children and Families. - An investment of NT\$ 1.08 million dollars will be made in 2024. - UMC sponsors sports education to empower underprivileged students to transform their lives through athletics. - UMC sponsors the Nantou Karate Team and National Tung-Shih Senior High School Baseball Team. - UMC aims to enhance employee engagement and boost morale by expanding opportunities for staff participation. - By focusing on social charity investment and improving the system for paid volunteer service leave, UMC aims to enhance employee willingness to engage in such activities. This helps employees understand the Company's dedication and efforts towards the community, promoting a sense of identity and encouraging them to stay with the Company.	 

Supporting Art and Cultural Activities

UMC supports domestic cultural development by empowering artistic talents through supporting domestic performing arts groups, supporting local artists, and providing performance opportunities for performing artists. In 2024, UMC invested over NT\$3.19 million to support domestic performing arts groups and cultivate artistic talents, with NT\$550,000 for empowering artistic talents through supporting domestic performing arts groups. These include: supporting the What's Young Association to organize the "Fancy Years - National Youth Drama Festival", assisting tribal schools in inheriting indigenous traditional culture, which involved 70 schools and over 400 high school students across Taiwan. Additionally, UMC sponsored the "2024 Spectacular Music and Dance - Taiwan Indigenous Music and Dance Feast" at the National Kaohsiung Center for the Arts (Weiwuying), where 7 indigenous heritage groups performed traditional songs, dances, and ceremonial rituals from the Truku, Tsou, Paiwan, Bunun, and Amis tribes. **For more details, please refer to Chapter 5.4 Commitments to Social Welfare**

UMC also allocated NT\$2.64 million to support local artists and provide performance opportunities for performing artists. Sponsorship projects include supporting the creation of works by artists such as Chang Kuang-Ho, Lin Tian-Fu, and Emmy, Pin-Shu, Chien. During the company's Family Day, UMC opened its facilities to the public and provided guided tours of the recently acquired artworks. To cultivate employees' appreciation and consumption of the arts, UMC launched a series of local art feasts during the art season under the themes "Appreciate Art", "Talk Art", and "Create Art". This included exclusive stage play performances at Hsinchu and Tainan city, allowing employees and their families to enjoy cultural activities at the Hsinchu County Cultural Affairs Bureau Performance Hall and the Tainan city Cultural Center Performance Hall. UMC also collaborated with local cultural organizations such as the Chimei Museum and the OR house to host lectures and hands-on activities with local artists, bringing art closer to their life.

Volunteer Culture Promotion

UMC has been committed to promoting and shaping a volunteer culture for a long time, from the early self-organized candlelight club to the establishment of the UMC Technology and Education Foundation. The passion for embracing society and giving back to it is deeply ingrained in the hearts of UMC employees. In order to deepen the connotation and ability of public service among employees, the UMC Technology and Education Foundation also offers diversified volunteer training courses to increase employees' vision and action methods for public service. The Employee Benefits Committee also incorporates public charity elements into various activities such as clubs, family days, light travel, art festivals, and lectures, with the goal of promoting a culture of social good. This positive influence extends to more people in need, creating a virtuous cycle from within the Company and outwards.

To demonstrate the proactive spirit of UMC employees in achieving mutual prosperity with society and to encourage colleagues to initiate volunteer service in functional divisions, fabs, or employee clubs, UMC has established a reward mechanism and also provides 16 hours of paid leave for public service activities each year to encourage more enthusiastic participation in public volunteer services. UMC employees actively participate in various forms of social charity activities, such as ecological conservation, environmental protection, supporting disadvantaged groups, promoting local development, and empowering young people. In 2024, UMC invested 9,302.5 hours, benefiting 45,028 person-times, demonstrating the importance UMC places on shaping a volunteer culture and the action, love, and selfless dedication of UMC volunteers. To expand the impact of public charity activities, UMC not only holds public charity achievement exhibitions but also establishes internal and external digital platforms to integrate internal and external public charity resources and public charity promotion results, with the goal of leveraging the ubiquitous influence of technology and creating a vision for a better society.

Volunteering Type	Content
Energy Saving	Assisting organizations for the underprivileged and large energy consumers in implementing water and energy conservation measures to improve efficiency.
Environmental Protection	Protection of mountain and beach areas, scenic spots, and public areas, as well as the preservation of wetland ecological habitats.
Education	Semiconductor science education, music instrument instruction, disaster prevention education, and chemical disaster prevention training
Caring for the Disadvantaged in Society	Donations for earthquake relief in Hualien, companionship for disadvantaged groups or elderly living alone, public benefit performances, and charity handicrafts
Regional Revitalization	Assistance for environmentally friendly farmers in planting and maintaining ecological environments.



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5.1 Promote Common-Prosperity Society

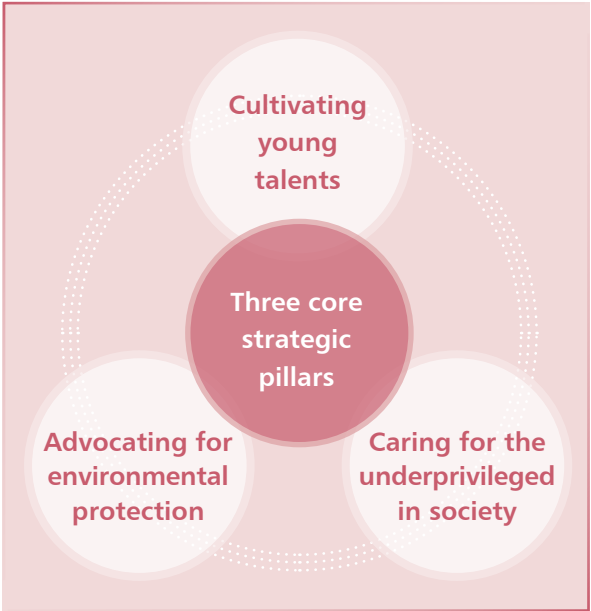
5.2 Youth Talent Cultivation

5.3 Environmental Protection Initiative

5.4 Commitments to Social Welfare

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5.1.2 Community Services Benefits Assessment

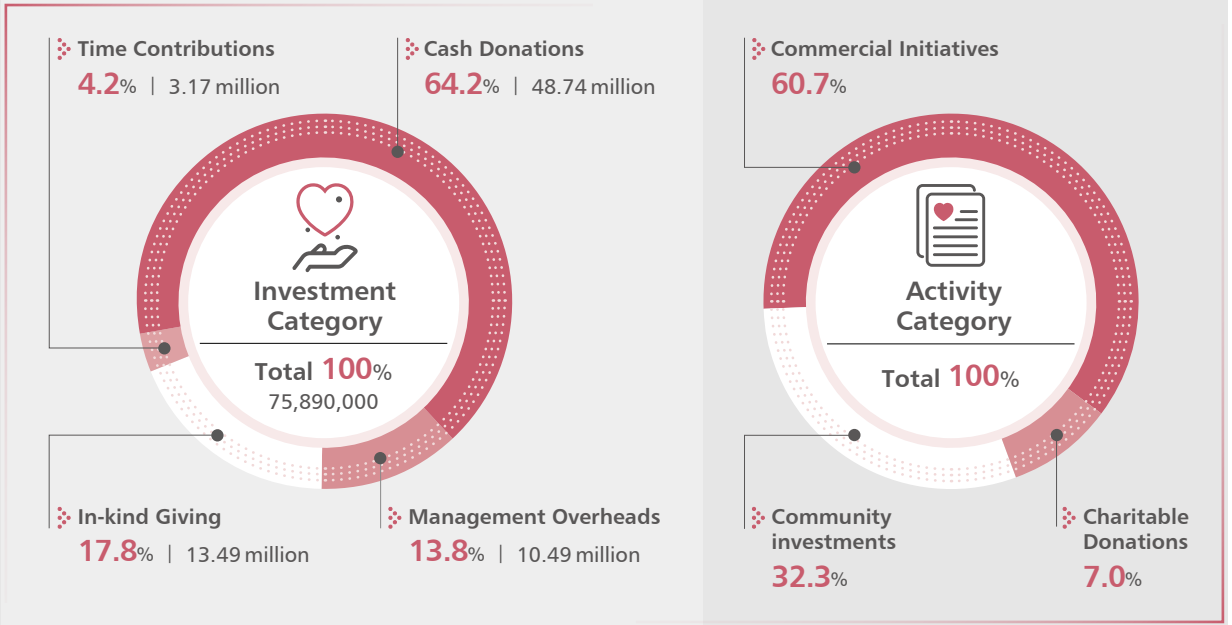


To effectively quantify the benefits brought about by community services, based on the belief of "taking from society and using it for society," UMC has long-term and deeply invested in various social activities projects. Following the business strategy direction of "Create social value" in its sustainability strategy blueprint, UMC promotes social activities through three main strategic pillars: "talent as the foundation of industrial development, environment as the foundation of human sustainability, and education as the foundation for improving the disadvantaged." UMC continuously invests resources in areas such as "empowering young talent," "advocating environmental protection," and "caring for the underprivileged." UMC referred to the community investment assessment system established by Business for Social Impact (B4SI). Investment time, cost, material donations, and management expenses were precisely recorded to evaluate the positive benefits brought about by these investments. Outputs of community services include reductions in costs, generation of benefits, and intangible influences such as positive corporate image, becoming a benchmark of corporate social responsibility (CSR), establishing positive value systems amongst school children, and helping to compensate for the inadequacy of educational resources for school children living in remote areas.

Social Charity activities Investment

UMC has dedicated a total of NT\$75.89 million to social charity activities in 2024 to ensure that the care recipients receive long-term and stable support. In addition to the investment in community participation by the UMC Science and Culture Foundation and the UMC LOHAS Education Foundation, UMC also pays more attention to the cultivation of young talents and supports a number of academic key projects through business initiatives to improve the competency of future high-tech talents. These projects include: donations to establish the UMC Dr. Chung-Laung Liu Doctoral Scholarship, UMC Management Thesis Award Fund, and "Nano Integrated Circuit Engineering" Fund of the National Cheng Kung University.

Category and Sums of Community Service Investments in 2024



Note: "Administrative expenses" include the personnel costs incurred when conducting public charity activities, such as the salary of personnel performing public charity activities.

Community Service Participation and Number of Beneficiaries in 2024

Charitable Donations ¹	NT\$ 75.89 million	Number of Beneficiaries	45,028	Total Employee Volunteer Hours ²	9,302.5
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Note: 1. This table only covers projects carried out by the Science and Culture Foundation, UMC LOHAS Education Foundation, UMC donations, and employee social clubs as well as employee donations.
2. Includes volunteer leave and holiday leave hours.

5.2 Youth Talent Cultivation

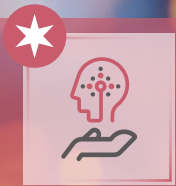
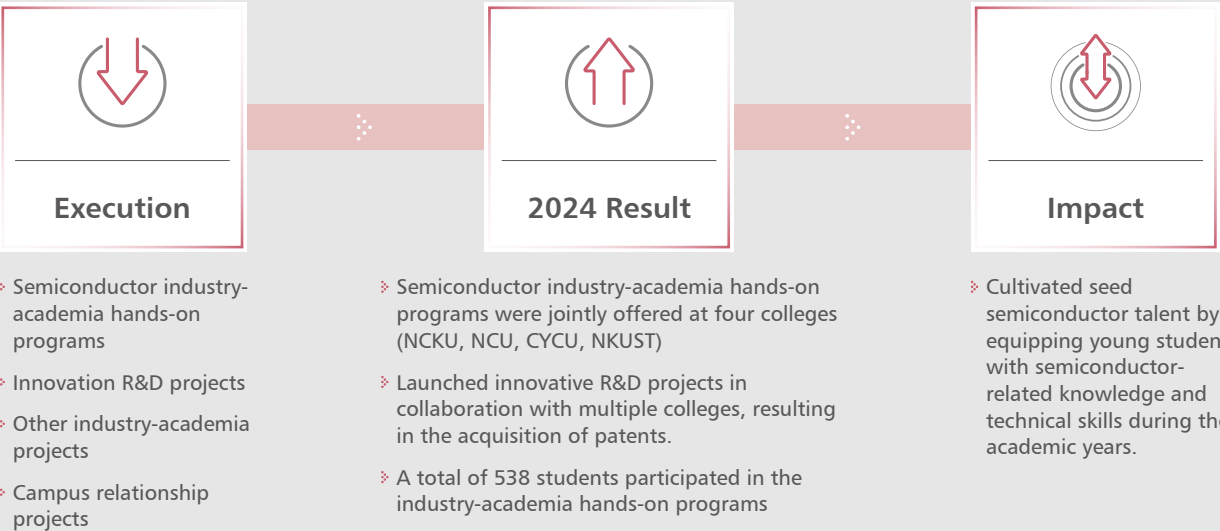
- 5.2.1 Industry-education Partnerships
- 5.2.2 Semiconductor Talent Cultivation

In order to strengthen global competitiveness, UMC is deeply engaging with campuses, dedicated to nurturing future young talent, and promoting semiconductor R&D capabilities and technological advancements. In 2024, UMC continues to focus on developing future semiconductor professionals through two key initiatives: strengthening industry-academia collaboration and fostering semiconductor talent.

5.2.1 Industry-education Partnerships

UMC continued to strengthen industry-academia partnerships in 2024. In addition to the on-going hands-on programs for semiconductor technologies in key institutions, a collaboration project was also implemented to sponsor full-time instructors for the MS Degree Program and credit courses on Nano-Integrated Circuit Engineering offered by National Cheng Kung University (NCKU). In 2024, more than 7,000 students participated in these industry-academia cooperation programs. Moreover, a series of brand image-building activities such as seminars, career sharing panels, on-site internships, and exhibits of semiconductor products were scheduled through the "UMC Campus Development Program."

Semiconductor seed - Taiwan high-tech talent cultivation



In 2024, more than 7,000 students participated in these industry-academia cooperation programs

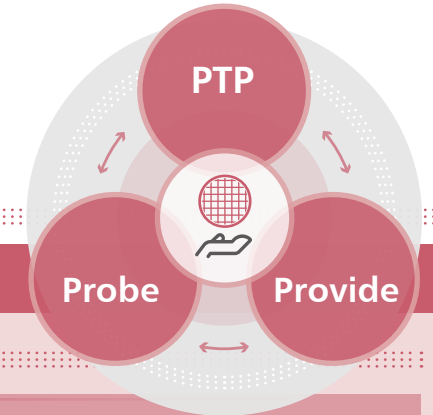


5.2.2 Semiconductor Talent Cultivation

UMC is committed to nurturing top-tier R&D talent and has established a robust 3P strategy designed to help students deepen their understanding and awareness of the semiconductor industry during their academic journey. This strategy also provides early exposure to UMC's welcoming corporate culture and its positive, healthy work environment. This effectively promotes UMC's corporate image, expands its influence on campuses and communities, and prepares future R&D talents in advance.

Planning for Outstanding R&D Talents

Semiconductor Talent Cultivation 3P Strategy



PTP

Prospective Talent Program

» PTP

UMC is committed to reducing the gap between academia and industry. Since 2013, we have been continuously promoting the Prospective Talent Program, actively offering various activities for students in science and engineering, such as lectures, internships, and mentoring programs, to help them gain industry knowledge and career planning. Through communication and interaction with enterprises, students can more quickly connect with future employment markets.

2024 Result » Since 2013, **6,702** candidates have been recruited, of which **860** are still in school.



Probe

Probing semiconductor careers

» Career Coaching Programs

UMC has conducted career planning seminars and career coaching programs in its recent collaboration with key colleges. Based on professional insights into future trends and career instructors' assessment of student characteristics and knowledge level, career plans are recommended to help students find suitable career paths.

2024 Result » Take NCKU as an example, **13** career coaching programs have already been carried out. Each batch had over **280** participants from the university and the program is still on-going to this day.



» Target Schools Visited

UMC provided on-site visits to target schools every year for students to gain early understanding of the corporate culture and healthy workplaces.

2024 Result » **800** person-times



» Internship Program

The program continues to select high-quality talent to intern at major divisions in domestic and overseas UMC facilities. The interns are guided by designated mentors so that through hands-on practice and involvement in on-going projects, interns can experience the workplace culture. In the meantime, this program effectively creates close interactions between prospective collegiate talents and UMC teams, thereby enabling mutual learning and growth through these close exchanges.

2024 Result » Over **85** students participated in the program. Following the internship, more than **45** interns (approximately **53%**) were offered full-time positions.



Provide

Providing educational resources

» Potential Management Talent Cultivation

The UMC Business Management Thesis Award was established in 2010 and started offering monetary rewards to the Award in 2011 to help further training programs for potential management professionals, encourage exchanges between academia and industry, achieve effective integration of management practice and theory, and contribute to sustainable corporate management.

2024 Result » NT\$**3.2** million was bestowed to the award.



» Scholarship for key college

UMC worked closely with key colleges to cultivate engineering and science talents for Taiwan, providing students with R&D opportunities and resources.

2024 Result » Sponsoring the Dean's Award for Graduate School of Engineering at National Taiwan University, benefiting a total of **5** students.



» Development of semiconductor schools

UMC invested in educational resources in the semiconductor colleges of key schools to promote the development of research institutes, cultivate new teaching staff, and hold technical lectures, allowing academic research and industrial technology development to progress hand in hand.

2024 Result » UMC funded **3** institutions to establish semiconductor schools.



Development of Doctoral-Level Talents

Doctoral-level research talents play an important role in advancing science and innovative research and are key to the sustainable development of the industry. In order to cultivate excellent semiconductor talents and express gratitude to Academician Liu Chung Lang for his long-term investment in electrical and electronic education, as well as his guidance and contribution to the Company's business decisions during his tenure as an independent director of UMC, the "UMC Chung-Laung Liu Doctoral Scholarship" was established in 2023. It targets the fields of semiconductor components, processes, materials, and equipment, providing a monthly scholarship of NT\$40,000 for each doctoral student for up to four years to encourage them to focus on academic research and devote themselves to technological innovation without worries. This scholarship injects more talent energy into the semiconductor industry. **17** students were awarded this scholarship in 2024.



Semiconductor Science Education for Elementary School Students

In order to strengthen the public's understanding of the semiconductor industry, understand the application of semiconductor technology in life, and take root in schoolchildren in science in terms of literacy-oriented capabilities, UMC staff formed a volunteer team to link up with the core capabilities of the industry to develop semiconductor science textbooks and popular science education courses. At the same time, volunteer employees visited elementary school campuses to deliver interactive lessons to students, demonstrating the process of IC production, manufacturing, and usage in everyday 3C products.

In 2024, the science education volunteer team went to 11 nearby elementary schools, including Tainan Municipal Yongkang District Sancun Elementary School, Hsinchu Municipal Min Fu Elementary School, Liujia elementary school, Tainan Municipal Shanhua District Dacheng Elementary School, National Nanke International Experimental High School, Hsinchu Cheng Kong Primary School, Hsinchu East District Tungmen Primary School, Hsinchu Lung Shan Elementary School, Hsinchu Guan Dong Primary School, Yung Shin Elementary School, Hsinchu Xi-Men Primary School, we organized science education booths as part of family-day events held at UMC facilities in Hsinchu, Tainan, and Singapore. UMC also participated in the National Science and Technology Council's "Taiwan Science Train" plan to conduct semiconductor science education at the plazas in front of Hsinchu and Tainan train stations. UMC volunteers also set up science education booths during UMC family days serving colleagues' children and visitors, with a total of **25** events and **4,677** beneficiaries throughout the year.



Message from the ESG Steering Committee Chairman

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5.3 Environmental Protection Initiative

5.3.1 Energy Saving Service

To enhance energy efficiency and expand the benefits of our core business functions, UMC has established the "UMC Energy Saving Service Team" with "energy saving and safety" as the core. By stepping out the foundry fabs, the Team allows UMC colleagues to bring their experiences and expertise to bear on diagnosing and improving energy conservation and fire safety for underprivileged social welfare organizations, assisting in building renovation, and enhancing safety and comfort in residential areas, so as to reduce unnecessary energy consumption of these organizations and help them to use their precious resources more effectively.

In addition to calling for the participation of colleagues, UMC also exerts its influence to integrate resources with the parent company as its core platform, in order to promote value chain collaboration, and invite subcontractors and subsidiaries to join the ranks of energy saving services for social welfare organizations, thus creating social value together. This year, the subcontractors such as Star Qiao, Chen Sheng, Cheng Tai, and Kai Fa, Ming yang, Shi Fuh, Kaiser have all acknowledged and responded to UMC's initiative by providing not only manpower but also sharing the costs for equipment improvements.

In 2024, the Energy Saving Service Teams of Hsinchu Science Park and Southern Taiwan Science Park served 11 organizations, including 7 underprivileged organizations and 4 community units (schools and hospitals), accumulating to a total of 81 units since inception of the Team. A further calculation of the public benefit resulted from the energy saving service in 2024 showed that the relevant staff input and cost brought economic benefits to underprivileged organizations and community units, such as electricity conservation amounting to NT\$11.336 million, which benefited 3,205 people and reduced CO₂ emissions by 1,152 tonnes.

Looking forward, UMC will continue to evaluate the effectiveness of its social welfare services to enable the participating partners to understand the benefits and actual contributions of energy saving services, and additionally use it as a reference for continuous internal review to help identify more opportunities for social improvement, and make future investment planning and management more efficiently. Apart from that, through the promotion and substantial implementation of energy saving services, UMC aims to promote the exchange of energy saving knowledge and education, introduce green technologies, and enhance the resilience of social welfare organizations to environmental changes, collectively creating more sustainable communities.

2024 Public Benefit Evaluation

Unit: NT\$

object	Underprivileged Organizations			Community Units		
	Labor Cost	Construction Cost	Total Cost	Labor Cost	Construction Cost	Total Cost
Input	888,000	1,371,363	2,259,363	139,500	0	139,500
Output	Economic Benefit	Environmental and Social Benefit	Total Benefi	Economic Benefit	Environmental and Social Benefit	Total Benefi
	272,261.9	19.3 tCO ₂ e	272,261	11,063,353	1,132.4 tCO ₂ e	11,063,353

- Note: 1. Scope of Statistics: Input and output of all Energy Saving Service Team's service in 2024
2. Labor Cost: Volunteer attendance cost (NT\$500/hour)
3. Construction Cost: Facility and equipment improvement cost invested
4. Economic Benefits: Operating costs that was saved, such as water and electricity bills
5. Environmental and Social Benefits: To measure carbon emissions reduction performance, the calculation is estimated by saving electricity for operating the equipment for a whole year. Electricity emission factor of 0.494 kg CO₂e/kWh in 2024 was adopted for the calculation of 2023

5.3.1 Energy Saving Service 5.3.2 UMC Eco Echo Award Initiative

★ Performance Highlights 2024

UMC Energy Saving Service Team has long been dedicated to local services and has provided energy efficiency improvement support to 81 organizations since 2016. Their service is not a one-time effort—when issues arise in previously assisted organizations, the Task Force is often the first they turn to for help. One such example is Mennonite Christian Hospital in Hualien. UMC has maintained a long-term, positive relationship with the hospital, and following the recent Hualien earthquake on April 3, the Energy-Saving Task Force and UMC Science and Culture Foundation quickly mobilized to provide support.

the Southern Taiwan Science Park UMC Energy Saving Service Team provided services to the Brightness Early Intervention Center, Tobias Social Welfare Foundation. In 2023, the team conducted an electricity safety inspection for the facility, which had been in use for over 20 years. In 2024, they continued assisting Tobias in repairing spaces long affected by water leakage and wall damage, including additional support to resolve mold issues in the third-floor corridor.

Thanks from Tobias



❖ Taipei City

- Department of Chemical Engineering, NTU

❖ Taoyuan County

- Bade Home for the Disabled
- Dun Pin High School
- Kangfu Intelligent Center

❖ Hsinchu City

- Hsinchu Church of the Sacred Heart of Jesus
- St. Joseph Social Welfare Foundation
- Renai Children and Youth House
- Jesuit Hsinchu Social Service Center
- Life Line Association
- Ai Heng Social Welfare Foundation
- Intellectual Disability Welfare Association
- Good Shepherd Social Welfare Services
- Saint Joseph Social Welfare Foundation Xi-da Branch
- Miracle Home
- Good Shepherd Faith Home
- St. Joseph Workshop – Sheng-mu Park Area
- Hsinchu Support Center, Taiwan Fund for Children and Families
- National Tsing Hua University
- Chung Hua University
- Children Are Us Foundation - Jianzhong Branch
- Chuhsin Life Education and Caring Association.
- Hsinchu Veterans Home
- Silicon Integrated Systems Corporation

❖ Hsinchu County

- Shih Guang Educational and Nursing Institution
- Grown-up with Hands Family
- Chengjheng High School
- Hsih Guang Hospice Care
- St. Joseph Home
- Hsiang Yuan Education and Nursing Institution (Hukou Township/Xinfeng Township)
- Hua Kuang Social Welfare Foundation
- National Hsinchu Special School
- Good Shepherd Dexin Home
- NTUH Hsinchu Branch
- Saint Francis Children-Youth Center
- The Garden of Hope Foundation - Hsinchu Branch Office

❖ Miaoli County

- Holy Family Intelligence Center
- Yu An Youth Development Centerr
- Yu An Youth Development Center - Day Care Center

❖ Taichung City

- Good Shepherd Taichung De-You Home
- Aggregate Corporation Ember Social Welfare Association

❖ Hualien County

- Tzu Chi
- Mennonite Christian Hospital
- Detention Center
- Prison
- Mennonite Christian Hospital - Shoufeng Branch

❖ Chiayi County

- Tung-Shih Senior High School

❖ Tainan City

- Nan Jeon University of Science and Technology
- Deguang Catholic Senior High School
- An-Nan Hospital
- Delan Social Welfare Foundation
- St. Raphael Opportunity Center
- Bethlehem Foundation
- Tainan National University of the Arts
- Kaohsiung Veterans General Hospital Tainan Branch
- Chia Nan University of Pharmacy & Science
- National Cheng Kung University
- Ren-Sing Hospital
- Tainan Babies' Home of Good Shepherd Social Welfare Foundation
- Tainan Municipal Sinhua District Koupi Experimental Elementary School
- Tainan Home of Philanthropy - Herb Garden Nursing Home
- Crowne Plaza Tainan
- A Kernel of Wheat Foundation - Guanmiao Long-term Care Center
- Brightness Early Intervention Center, Tobias Social Welfare Foundation
- Jishi Studio, Tobias Social Welfare Foundation
- Yellow Ribbon Organization Service Center in Tainan
- Taiwan Great Heart Social Welfare Association
- Kuo Pont Chemical CO., LTD.
- Jianshanpi Resort
- The Garden of Hope Foundation - Tainan Branch Office

❖ Taitung County

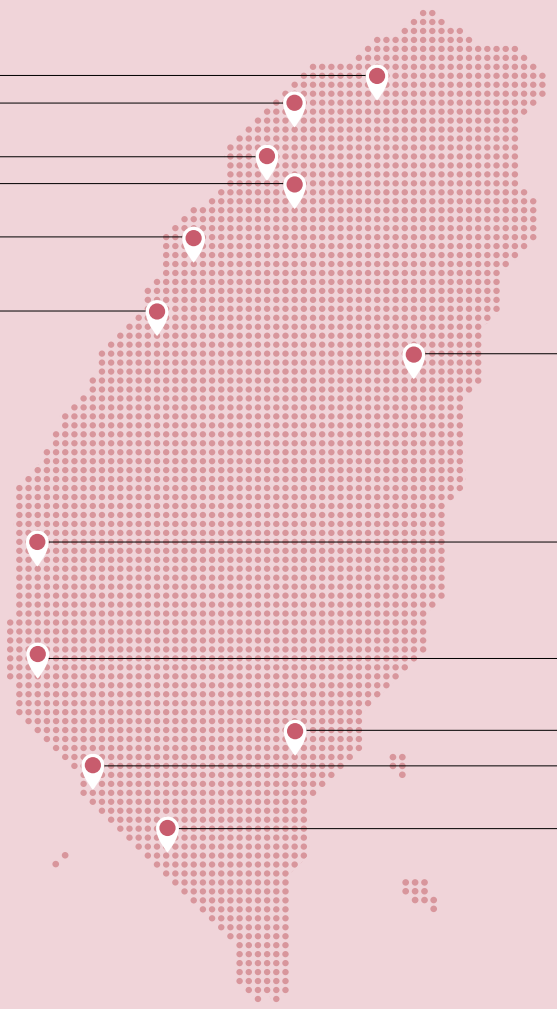
- Good Shepherd Taitung

❖ Kaohsiung City

- The Southern region office, Prader-Willi Syndrome (PWS) Foundation
- Hope Foundation for Cancer Care - Kaohsiung Branch
- Yellow Ribbon Organization Service Center in Kaohsiung
- Huayuan Children's Service Center

❖ Pingtung County

- Yanjou elementary school
- Pingtung Bethany Home



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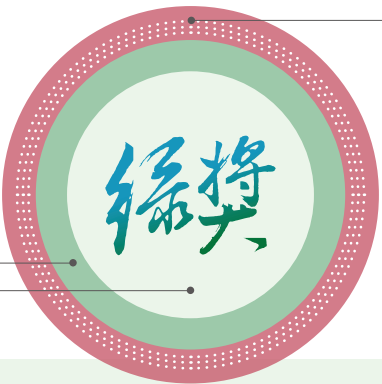
Improvement Procedure



5.3.2 UMC Eco Echo Award Initiative

UMC is well aware of the huge impact it can have on society and the environment before the climate change issue was widely recognized. After completing the first EPA (Environmental Protection Administration)-approved carbon emissions trading in 2014 in Taiwan, UMC donated the trading yield to launch the "Eco Echo Ecological Conservation Hope Project" in order to promote environmental and biodiversity conservation, restore species and ecosystems, and enhance the public awareness of environmental protection.

In 2016, the UMC Eco Echo Award was established to solicit high-quality ecological conservation projects. Since the 3rd UMC Eco Echo Award, the total prize money has grown to NT\$3 million per year, making it the highest prize money in the field of ecological conservation initiated by enterprises in Taiwan. Apart from that, in order to stimulate young students' awareness and understanding about protecting the environment and ecology, the "Youth Environmental Action Award" has also been established since the 4th UMC Eco Echo Award, with the aim to encourage young students to turn their enthusiasm and creativity for environmental protection into practical actions and further bring changes to the environment. In response to global attention on climate change and resource shortages, the 7th Eco Echo Awards added the Green Innovation Award. This award encourages start-up teams to propose solutions for net zero/ carbon reduction and resource reclamation. Total prizes were raised to NT\$4 million.



UMC used carbon trading proceeds to create a special environmental protection fund, and launched the Eco Echo Awards to spur on joint initiatives with business partners.

The Eco Echo Awards encourage ecological conservation groups, start-up teams, and students to commit to environmental protection; benefits are seen all over Taiwan.

In 2024, UMC awarded and supported three NGO ecological conservation projects, eight youth environmental initiatives, and two green innovation projects. The winning project of the "Ecological Conservation Award" support for the giant clam restoration project, the Huang River Community Environmental Education Program, and so on.

Two companies won the "Green Innovation Award." FunCrowd Inc. uses AI to connect various operational roles, conduct research, collect data for optimization, and enhance user experience, aiming to reduce the manufacturing, repeated purchasing, and disposal of cheap umbrellas at source. AddLess focuses on the circular economy in the fashion industry, promoting RAAS (Recycle as a Service). By integrating children's and second-hand clothing trading platforms, addLess achieves zero waste in clothing and provides reverse recycling solutions for retail brands.

For the "Youth Environmental Action Award," the winning students proposed diverse environmental solutions, including the conservation of species such as fish, crab-eating mongoose, and red-clawed crabs, reducing cigarette butts, and improving the soil in school flower beds. Their efforts not only influence their peers but also extend to the broader society, demonstrating their commitment to environmental protection through creative and practical actions in daily life.

In the nine years since the inception of the UMC Eco Echo Awards, 81 ecological conservation and green innovation projects have been implemented across Taiwan. In addition to our own effort, UMC has also linked resources from various parties and combined the power of supply chain partners to realize more outstanding ecological conservation projects. In 2024, UMC joined hands with partners such as 3M Taiwan, Tokyo Electron, Hermes-Epitek, Wholetech System, Faraday, Taiyo Nippon Sanso, Chiyo, and Unimicron. UMC expects the UMC Eco Echo Award to achieve the co-prosperity of enterprises and the environment, and to be the powerful guardian of Taiwan's ecological environment.

UMC Eco Echo Award Official Website



The 9th UMC Eco Echo Awards was held at UMC, with enthusiastic participation from employees.



Helping ecological conservation groups' efforts get public visibility; stimulating students to pay attention to environmental issues; raising public awareness of environmental protection; creating a better environment in Taiwan.



5.4 Commitments to Public Welfare

Over the past 29 years, the UMC Science and Culture Foundation has focused on social needs, supporting schools, communities, and public welfare teams. It emphasizes education for disadvantaged groups, youth development, and uniting corporate resources to promote social responsibility, empowering children to shape their futures through education.

★ Performance Highlights 2024

The UMC Science and Culture Foundation participated for the first time in the **2024 Taiwan Science Train**, collaborating with public sectors and other enterprises while uniting high school students from four schools to serve as gatekeepers. Through volunteer training, we aim to deepen high school students' understanding of the semiconductor industry and help them transform their knowledge into scientific challenges. This initiative guided students from **31** elementary and junior high schools across Taiwan to learn about semiconductors and water resource knowledge. In addition to laying a foundation for science education, it also assists high school students gain hands-on experience and develop their skills through these activities.

UMC Love Story Group, formed by The UMC Science and Culture Foundation, performed its 7th stage play, "The Gentle Persistence," in 2024, with a total of five performances. This script incorporates a focus on the United Nations Sustainable Development Goals (SDGs), aiming to help students understand the importance of sustainability through theatrical performances and encouraging them to care for the needs of those around them. Such activities not only enrich students' cultural lives but also foster their sensitivity to social issues and strengthen their sense of responsibility.

Seeds of Hope A Diverse Educational Program for Youth Empowerment

✦ Building a Charitable Tutorial Platform

Since 2005, the UMC Science and Culture Foundation has established tutoring centers in Hsinchu and Tainan, aiming to serve students from disadvantaged families, including single-parent families, grandparent-led families, and new residents, by providing free diverse companionship and academic tutoring. The Hsinchu tutoring center adjusted its operational model in 2013 and has since been deeply rooted in the community for 12 years. It collaborated with local universities such as National Tsing Hua University and National Yang Ming Chiao Tung University to offer service-learning courses. These courses offer academic assistance while engaging corporate and community volunteers as tutors and mentors, providing holistic support to children from disadvantaged families.

The Tainan tutoring center has long-term cooperation with National University of Tainan, effectively utilizing university resources to deepen local education and extend services to rural areas in Tainan. Through the participation of university teacher training students, tutoring resources are extended to elementary schools outside the urban area, allowing children in resource-scarce regions to receive guidance.

In addition to academic tutoring, both tutoring centers design diverse learning programs to broaden students' horizons. The tutoring classes emphasize the cultivation of life skills, fostering children's self-management abilities, and helping them establish a positive learning attitude. University students, through their participation in tutoring services, develop empathy and caring abilities, achieving personal growth while leading creative activities and courses. Furthermore, the UMC Tutorial Center of National University of Tainan assists teacher training students in preparing for teacher certification and selection exams, realizing the goal of mutual learning and growth between teachers and students.



Supporting the Nantou Karate Association

The Nantou Karate Association, led by instructors Huang Tai-chi and Liao Te-lan, is dedicated to nurturing children from disadvantaged families in Taiwan. To date, the Association has assisted over 60 students in learning karate and provides basic living support, including three meals a day and accommodation. The Association aims to help children build self-confidence and learn to take control of their life direction through karate training. Since 2005, the Foundation has continued to provide long-term support through periodic funding and care, as well as organizing course exchanges and sharing activities, helping these students grow and thrive in their lives and studies, paving the way for a better future.



Supporting National Tung-Shih Senior High School Baseball Team

The Foundation's five-year collaboration plan with Tung-Shih Senior High School baseball team has entered its fourth year. In addition to continuing financial support, the Foundation is also committed to strengthening connections and care. In September 2024, as the team traveled north to participate in the national youth baseball competition, the Foundation arranged for the team to visit the UMC headquarters in the Hsinchu Science Park and held a BaseBALL Meet & Greet event, adding more diverse elements to the collaboration plan. Through this cheerleading gathering, UMC colleagues gained a deeper understanding of the Foundation's beneficiaries, while the young players learned about the daily work and life of engineers in the Science Park, creating a valuable and meaningful learning experience.



Supporting TECO Technology Foundation's "Exclamation Mark" Plan

The UMC Science and Culture Foundation collaborates with TECO Technology Foundation to jointly support the preservation of traditional indigenous dances and rituals. The "2024 Musical and Dancing Exclamation Mark! – Taiwan Indigenous Music and Dancing Festival" was held at the National Kaohsiung Center for the Arts, bringing together 7 performance groups from the Paiwan, Amis, Bunun, Tsou, and Truku tribes to showcase their traditional music, dance, and ceremonies. This performance featured the One Song Orchestra and a professional technical team, with over 300 participants, providing the audience with a stunning cultural feast. The event was available for both live viewing and online streaming, and the Foundation, by sponsoring the live broadcast, aims to convey the cultural charm of indigenous peoples to a wider audience, injecting new energy into the continuation and innovation of traditional culture.



Supporting the Home from Above Association

UMC supports the production plan for the "Home from Above" program initiated by the Home from Above Association, guiding students from 7 elementary and junior high schools to create documentaries about their hometowns. Through camp programs and instruction from media professionals, the children learn to operate cameras and drones for filming, as well as interview local residents to deeply document the people and events of their hometowns. The project not only allows students to recognize and record the characteristics of their hometowns, but also helps them establish a strong emotional connection with their communities, fostering local cultural identity and cohesion. The project also emphasizes the cultivation of technological information and media literacy, aligning with future trends, and advocates for children to pay attention to the environment, conveying the value of their hometowns to a broader society, allowing small actions to create real and far-reaching impacts.

Supporting the What's Young Association

Since 2011, the Foundation has supported the What's Young Association to organize the "Fancy Years" National Youth Drama Festival, which is the largest high school drama creation competition in the country. Through a six-month-long competition and professional courses, the drama festival accompanies young students in exploring themselves through theaters, while learning communication, collaboration, and mutual support. The theme of 2024, "Alive," attracted participation from 70 schools and over 400 students across Taiwan. It brought together local theater groups and theater professionals to collaboratively foster creativity among high school students, allowing them to express their thoughts on existence and their hopes for the future through drama.

The drama festival not only enhances students' creativity, expressiveness, and initiative but also encourages them to reflect on themselves, understand society, and learn to empathize with others and accept themselves through teamwork. The drama festival serves as a platform for young people to find their life goals and dreams through creation, leaving a profound and meaningful mark on their youth.



Participation in the 2024 Taiwan Science Train

The Foundation participated for the first time in the Taiwan Science Train event, collaborating with public sectors such as the National Science Council and the Ministry of Economic Affairs, as well as other enterprises. High school students from 4 schools—National Taiwan Normal University Affiliated Senior High School, Hsinchu Girls' Senior High School, National Chung Hsing University Affiliated Senior High School, and Nanke International Experimental High School served as game masters. Through volunteer training, these high school students gained in-depth understanding of the semiconductor industry and apply their knowledge to design scientific challenge games. Over the course of a week-long train journey, they lead students from 31 elementary and junior high schools across Taiwan in exploring knowledge about semiconductors and water resources through interactive games. This initiative not only lays a solid foundation for science education but also provides high school students with valuable practical experience, enhancing their own capabilities.

Seeds of Caring the "Pushing the Strong to the Front and Supporting the Weak from the Back" Sponsorship Program

Sponsoring the Taiwan Fund for Children and Families

The Foundation has been continuously sponsoring 90 students from the Taiwan Fund for Children and Families (TFCF) for over 20 years, supporting TFCF in utilizing social work expertise and assistance funds to provide required assistance to vulnerable families, ensuring that children in these families receive appropriate care. In addition to providing financial supports, the Foundation encourages its employee volunteers to connect with sponsored children through heartfelt letters. By maintaining regular communication, volunteers can better understand the children's current situations and offer meaningful emotional support. Additionally, small tokens of care, such as thoughtful gifts during special occasions or everyday moments, are sent to bring warmth and joy to the children. This approach to charitable involvement not only magnifies the impact of each act of kindness but also offers busy employees diverse and flexible ways to participate in volunteer activities, fostering a deeper sense of community and compassion.

Seeds of Reading National Reading Culture Promotion

Newspaper Reading Program

Since 2003, the Foundation has partnered with Mandarin Daily News to sponsor the "Newspaper Reading Program," collaborating to identify and support "Reading Education Experimental Classes." This initiative aims to assist rural schools and teachers who lack sufficient educational resources by providing students with access to more reading materials. Through engaging with news articles, students can expand their horizons, stay informed about current events, and develop critical media literacy skills from an early age.

Storytelling Volunteers

The Foundation has been deeply engaged in reading education at Baoshi Elementary School in Hsinchu County for 14 years. In 2024, the Foundation continued to invite employee to volunteer and participate by designing diverse challenge activities focused on science education and artistic creation. These interesting activities promote communication and collaboration between UMC employees and students, igniting the staff's passion for public welfare. Through the challenge games, students get to expand their scientific knowledge while enhancing creative thinking and artistic perception.

Feedback from
participating
students

"This activity was really fun! I learned about friction, centrifugal force, and it also made me realize that we need to use 'force' in everything we do. I hope we can play science games again next time." — Baoshi Elementary School student



Seeds of Volunteering Life Education Promotion



UMC Love Story Group

Established in 2009, UMC Love Story Group is the first technologist-based theater group in Taiwan, comprised of colleagues from different fabs and divisions. Initially, the members performed existing scripts, but over time, they gradually shifted towards self-written, self-directed, and self-performed creations. Through theatrical performances, the team not only conveys life education to children but also provides a platform for technology professionals to express themselves and share their inner warmth.

The Group's latest work, the seventh production titled "The Gentle Persistence," has been meticulously prepared over two years, integrating the United Nations Sustainable Development Goals (SDGs) and delving into current social issues. In 2024, the members dedicated their lunch breaks and after-work hours to rehearsals and successfully delivered five performances that reached audiences of 480 individuals. Their performance not only deepened students' understanding of local revitalization and mental health issues but also inspired them to embrace the philosophy of "only when you are okay and I am okay, everyone will be okay." This message inspired students to care for those around them, fostering a spirit of compassion and mutual support that brings warmth and positivity to society.

Public Welfare Projects

In 2024, a total of 7,275 person-times volunteers participated in activities organized by the Foundation, accumulating 33,769.5 service hours and benefiting 46,857 person-times (including volunteers). The following is a list of the different public welfare projects:



Item	Method	Content	2024 Achievement
Seeds of Hope A Diverse Educational Program for Youth Empowerment	After-school tutoring and care Encouraging athletes Promoting arts, humanities, and cultural education Supporting popular science education	Providing after-school tutoring sessions in Hsinchu and Tainan	✦ Hsinchu and Tainan tutoring class volunteer service: 6,274 person-times; total service hours: 29,735.5 hours
		Sponsoring the Nantou Karate Association	✦ The sponsorship funds were allocated to support the training, accommodation, and meal expenses of over 60 young karate athletes.
		Sponsoring National Tung-Shih Senior High School Baseball Team	✦ The sponsorship funds were used to support the training of over 40 baseball players from Tung-Shih Senior High School and to renovate the gym.
		Supporting the What's Young Association	✦ The 24 th "Fancy Years" National Youth Drama Festival attracted more than 412 high school students participating.
		Supporting TECO Technology Foundation's "Exclamation Mark!" Plan	✦ Sponsored event: 2024 Musical and Dancing "Exclamation Mark!" – Taiwan Indigenous Music and Dancing Festival
		Supporting the Home from Above Association	✦ Sponsored the 2024 Home from Above project, with Professional teachers guiding students from 7 schools and six groups of returning student directors to complete 13 hometown documentaries.
		Participation in the 2024 Taiwan Science Train	✦ Adopted a train cart, with a total of 50 colleagues participating as volunteers, 145 high school students from 4 schools serving as gatekeepers, and students from 31 elementary and middle schools actively engaging in the activities.
Seeds of Caring "Pushing the Strong to the Front and Supporting the Weak from the Back" Sponsorship Program	Sponsoring public welfare agencies Volunteering	Sponsoring the Taiwan Fund for Children and Families and helping foster children	✦ Sponsoring the Taiwan Fund for Children and Families – NT\$1.08 million per year for a total of 90 foster children, living across various regions of Taiwan.
		Inviting UMC employees to serve as letter-writing volunteers to provide care and encouragement to foster children.	✦ Exchange of 243 letters between volunteers and students. ✦ Occasionally gifting students with birthday presents and various small tokens of care.
Seeds of Reading National Reading Culture Promotion	Promoting a Culture of Reading Volunteering	Sponsoring Mandarin Daily News to promote newspaper reading in education	✦ 2023 academic year: Newspaper reading project implemented in 24 classrooms across 16 elementary schools. ✦ 2024 academic year: Newspaper reading project implemented in 20 classrooms across 15 elementary schools.
		Leading reading activities for Baoshih Elementary School in Hsinchu County	✦ In the first half of the year, one science popularization challenge activity was held, with a total of 19 volunteers participating and 80 students benefiting; in the second half of the year, one storytelling event and one art challenge activity were conducted, with a total of 11 volunteers participating and 168 students benefiting.
Seeds of Volunteering Life	Acting for UMC Love Story Group Organizing volunteer training sessions	UMC Love Story Group Tour acting New Drama Rehearsal	✦ UMC Love Story Group performed 5 shows this year, reaching 480 audience members. Since 2009, the Group has performed a total of 74 shows, reaching nearly 23,000 audience members. ✦ UMC colleagues dedicated their lunch breaks and after-work time to prepare for the event, totaling 885.5 hours and 317 person-times participants.
		Providing volunteer training sessions for staff and social volunteers	✦ Organized 5 training courses and seminars with a total of 147 person-times participants



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2024 Highlights of Foundation Volunteer Activities

Delivering Love to Boyo's Pingtung Center, Safeguarding the Path Home for Rural Students

The Boyo Foundation's Pingtung tutoring center originally used a public service vehicle to transport students to their tutoring classes every day. After 15 years, the vehicle has traveled 240,000 kilometers and has recently been frequently breaking down for repairs. Considering the safety of driving in mountainous areas, Boyo has proposed a fundraising campaign for a new vehicle. After understanding the situation on-site, UMC donated a new public service vehicle. Chairman Tang Chuan-yi and Executive Director Chen Liang-zhi of the Boyo Foundation, along with partners from the Pingtung center, paid a visit to UMC before the Lunar New Year to express their gratitude and exchange ideas. They stated that for the Pingtung community center, the transportation vehicle is an important resource for stabilizing students' learning habits. The new public service vehicle will continue the mission of transporting students to and from tutoring classes and support other business needs of the tutoring partners. Together, we safeguard the path home for rural students in Pingtung, using education to transform children's futures and keep their dreams moving forward!



"The Gentle Persistence" Warm Premiere at Cheng Cheng High School

"The Gentle Persistence" is the seventh play by the UMC Love Story Group, and the third work that is self-written, self-directed, and self-performed. The troupe members dedicated their lunch breaks and after-work hours to fully engage in script writing, rehearsals, and prop production, accumulating over 2,000 hours of preparation. This play focuses on the United Nations Sustainable Development Goals (SDGs), specifically Goal 3 "Good Health and Well-being" and Goal 11 "Sustainable Cities and Communities," delving into issues of rural revitalization and mental and physical well-being. On Youth Day, the troupe premiered the play dedicated to the youth of Cheng Cheng High School, using the power of drama to inspire them to understand the sustainable concept of "Only when you are okay and I am okay, everyone will be okay." It is hoped that the performance could encourage young students to care for those around them and work together to create a better future.

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Baoshih Elementary School Science Exploration Activity



The Foundation has collaborated with Baoshih Elementary School for 14 years, with colleagues visiting the school each semester to lead reading courses. In 2024, the Foundation has meticulously designed a series of fun science exploration activities, allowing students from Baoshih Elementary School to explore the mysteries of science together. Through the use of easily accessible materials paired with thoughtfully crafted challenges, and guided by clear explanations from volunteers, students uncover the close connection between science and daily life. These exploration activities not only enable children to experience the joy of learning through play but also enhance their understanding and retention of knowledge. Meanwhile, the easy-to-handle volunteer tasks provide a sense of achievement for participating colleagues and enhance their willingness to engage in future charitable activities.



National Tsing Hua University International Volunteers' Charity Sales Activity



The Foundation has had over 17 years of collaboration with the National Tsing Hua University International Volunteer Team. In 2024, three teams embarked on journeys to Kenya-Swaziland, Malaysia, and Indonesia. To fund their trips, university students organized lectures and charity sales at UMC prior to departure. During lunch breaks, UMC colleagues had the opportunity to learn about foreign cultures and the volunteer teams' service initiatives, while also supporting the students and exchanging fresh perspectives.



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Engineer's Day Volunteer Fun



On Engineer's Day, UMC welcomed a total of 167 teachers and students from Baoshi Elementary School, Baoshan Elementary School, and the Juguang Branch of Baoshan Junior High School to participate in a rich learning journey that combined technology and drama. With the enthusiastic reception of nearly 40 volunteer colleagues, the children listened attentively to the introduction of the Company's art gallery and semiconductor technology, their eyes filled with a thirst for knowledge.

In addition to the guided tour, the teachers and students also watched the UMC Love Story Group's new play, "The Gentle Persistence." Through the drama, the students deeply understood the core of "sustainability"—caring for the needs around them and protecting the existence of every life with empathy. After the performance, the teachers and students eagerly shared their thoughts, stimulating diverse perspectives through dialogue and further understanding the sustainable spirit conveyed in the drama. This event not only enriched the children's learning experience but also allowed the technology professionals to feel the enthusiasm of the children.

We express our heartfelt gratitude to the volunteer partners for their commitment and to the teachers and students from the three schools for their active participation, making this day a memorable time filled with love and inspiration!



Making Friends Across Borders, Bridging Hearts

This summer vacation, UMC's Sacred Heart Tutorial Center collaborated with the National Tsing Hua University International Volunteer Team to arrange a video interaction between students and the children at the Swaziland Amitabha Care Center. The children on both sides eagerly exchanged their daily lives with curiosity. Although they grew up in different environments, they discovered many similarities: everyone loves to sing, dance, and draw, enjoys chatting and playing ball with friends after class, and even shares the amusing experience of accidentally dozing off during lessons! The students also sang together online, using a shared subcultural language to bridge the distance between them.

This activity gave the children a joyful taste of cultural exchange, sparking curiosity about the world. We hope they will step out of their comfort zones and embrace the broader world ahead.

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Dongshi High School Baseball Team Holds "Ball" Meeting

In September, during the national youth baseball competition, the Foundation specially arranged for the Dongshi High School Baseball Team to visit the UMC headquarters in Hsinchu Science Park and held a baseBALL gathering. UMC colleagues got to know UMC as a long-term sponsor through cheering activities, participating in quiz games, performing cheerleading dances, and prize drawing of baseballs with student players' autographs, showcasing their full support. Meanwhile, the young players gained a unique and inspiring experience during the visit, where they learned about the work and daily lives of engineers, combining education with motivation.

Me to We Lecture "Art Solutions - Communication Techniques with Adolescents"



The Foundation has supported the What's Young Association for over 13 years, accompanying adolescents in finding themselves through drama education. This year, we invited the founder, Yu Haowei, to share with colleagues how to help adolescents express themselves and explore their potential through drama, as well as to discuss effective communication methods with adolescents. This lunchtime discussion allowed colleagues to gain a deeper understanding of the Foundation's sponsored organization while actively supporting diverse educational initiatives, fostering greater opportunities for the growth and development of adolescents!



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Find out more about us: The Foundation's website



UMC Sacred Heart Club Tutorial Center's Facebook page



Baoshih Elementary School Art Activity

On Christmas eve, the Foundation once again invited colleagues to enter Baoshih Elementary School to serve as challenge masters, creating creative Christmas trees, unique New Year red envelopes, and Christmas decorations with students using various materials. Through station-based creation, students were able to observe works from different grades, which sparked their creativity. Every kid and adult enjoyed a wonderful afternoon in a joyful holiday atmosphere.



UMC Sacred Heart Tutorial Center Christmas Party

Winter is a season full of warmth and gratitude. The UMC Sacred Heart Tutorial Center held a Christmas Gratitude Evening event, inviting UMC colleagues, social volunteers, and students from National Tsing Hua University and National Yang Ming Chiao Tung University to participate. The students' wonderful performances lit up the evening, and the guests offered their blessings, while the students expressed their gratitude towards the tutorial center and their teachers. Everyone worked together to create Christmas ornaments and enjoyed cakes and desserts. The atmosphere reached its peak when the volunteers presented carefully prepared gifts from UMC colleagues, and everyone had a pleasant and warm night. We sincerely thank everyone for their unwavering support, which has transformed the tutorial center into a nurturing haven for the children's growth and development.

Appendix 1. GRI Content Index

Statement of Use: United Microelectronics Corporation has reported in accordance with the GRI Standards for the period 2024.1.1-2024.12.31

GRI 1 Used: GRI 1: Foundation 2021 Applicable GRI Sector Standard(s): NA

GRI 2: General Disclosures 2021		Location	Page	Note/Omission	External Assurance
1. The organization and its reporting practices					
2-1	Organizational details	About UMC	5		●
2-2	Entities included in the organization’s sustainability reporting	About This Report	4		●
2-3	Reporting period, frequency and contact point	About This Report	4		●
2-4	Restatements of information	About This Report	4		●
2-5	External assurance	About This Report Appendix External Assurance	4 212		●
2. Activities and workers					
2-6	Activities, value chain and other business relationships	About UMC Determination of Material Topics	5 17		●
2-7	Employees	4.2.1 Human Resource Development	141		●
2-8	Workers who are not employees	4.2.1 Human Resource Development 1.3.1 Localization of Supply Chain and Procurement	141 63		●
3. Governance					
2-9	Governance structure and composition	1.1.1 Board of Directors Operation Sustainable Governance	30 8		●
2-10	Nomination and selection of the highest governance body	1.1.1 Board of Directors Operation	30		●
2-11	Chair of the highest governance body	1.1.1 Board of Directors Operation	30		●
2-12	Role of the highest governance body in overseeing the management of impacts	1.1.1 Board of Directors Operation Sustainable Governance	32 8		●
2-13	Delegation of responsibility for managing impacts	Sustainable Governance	8		●
2-14	Role of the highest governance body in sustainability reporting	About This Report Determination of Material Topics	4 16		●
2-15	Conflicts of interest	1.1.1 Board of Directors Operation	33		●
2-16	Communication of critical concerns	Sustainable Governance	8		●
2-17	Collective knowledge of the highest governance body	1.1.1 Board of Directors Operation	31		●



GRI 2: General Disclosures 2021		Location	Page	Note/Omission	External Assurance
2-18	Evaluation of the performance of the highest governance body	1.1.1 Board of Directors Operation	32		●
2-19	Remuneration policies	1.1.1 Board of Directors Operation	33		●
2-20	Process to determine remuneration	1.1.1 Board of Directors Operation	33		●
2-21	Annual total compensation ratio	1.1.1 Board of Directors Operation	33		●

4. Strategy, policies and practices					
2-22	Statement on sustainable development strategy	Message from the ESG Steering Committee Chairmen	3		●
		Sustainability Strategy and Blueprint	9		
2-23	Policy commitments	4.1.1 Enforce Human Rights Management	129		●
2-24	Embedding policy commitments	4.1.1 Enforce Human Rights Management	129		●
2-25	Processes to remediate negative impacts	4.1.1 Enforce Human Rights Management	131		●
2-26	Mechanisms for seeking advice and raising concerns	4.1.2 Employer-employee Communication	134		●
2-27	Compliance with laws and regulations	1.1.3 Legal Compliance	36	There have been no significant violations resulting in fines exceeding NT\$1 million recently. However, minor violations have occurred. For detailed information on fines, please refer to the "Operations Overview" section of the company's annual reports for the years 2023 and 2024.	●
2-28	Membership associations	About UMC	6		●

5. Stakeholder engagement					
2-29	Approach to stakeholder engagement	Determination of Material Topics	19		●
2-30	Collective bargaining agreements	4.1.2 Employer-employee Communication	135	UMC and its subsidiaries USCXM and Wavetek currently do not have labor unions. Hence, no collective bargaining agreements have been signed by the employees. However, the subsidiaries HJ and USJC have established labor unions.	●

GRI 3: Material Topics 2021		Location	Page	Note/Omission	External Assurance
3-1	Process to determine material topics	Determination of Material Topics	15		●
3-2	List of material topics	Determination of Material Topics	17		●

List of Material Topics		Location	Page	Note/Omission	External Assurance
Sustainable Supply Chain Management					
GRI 3: Material Topics 2021					
3-3	Management of material topics	1.3 Building Sustainable Supply Chain	62		●
GRI 204: Procurement Practices 2016					
204-1	Proportion of spending on local suppliers	1.3.1 Localization of Supply Chain and Procurement	63		●

List of Material Topics		Location		Page	Note/Omission	External Assurance
GRI 308: Supplier Environmental Assessment 2016, GRI 414: Supplier Social Assessment 2016						
308-1	New suppliers that were screened using environmental criteria	1.3.2	Sustainable Supply Chain Management	65	UMC has established and formulated the "Supplier Sustainable Evaluation and Management Measures," incorporating sustainable management requirements into the evaluation indicators. All new suppliers must complete the ESG & RBA self-assessment questionnaire and meet the criteria before transactions can proceed.	●
308-2	Negative environmental impacts in the supply chain and actions taken			67		●
414-1	New suppliers that were screened using social criteria			65		●
414-2	Negative social impacts in the supply chain and actions taken			67		●
KPIs of Sustainability Strategy and Blueprint						
Ratio of Tier 1 suppliers who have included 'signing Employment Ethics Terms and Conditions Agreement' in their supply chain management requirements		1.3	Building Sustainable Supply Chain	62		●
Percentage of carbon intensity reduction by suppliers within the project of "low-carbon supply chain"				62		●
Ratio of audit conducted to significant high-risk suppliers				62		●
Completion rate of biodiversity survey for Tier 1 suppliers				62		●
❖ Innovation Management and Intellectual Property Rights Protection						
GRI 3: Material Topics 2021						
3-3	Management of material topics	2.1	Innovation & Technology Development	74		●
UMC Specific Topic						
R&D and patent deployment for new product, technology, and process optimization; employees must respect the company's patents and intellectual property rights.		2.1	Innovation & Technology Development	75		●
		1.1.3	Legal Compliance	36		
KPIs of Sustainability Strategy and Blueprint						
Revenue from sustainable products as % of total revenue		2.1	Innovation & Technology Development	74		●
Number of innovative energy-saving specialty process platforms		2.1	Innovation & Technology Development	74		●
Cumulative patents granted worldwide		2.1	Innovation & Technology Development	74		●
Ratio of products produced by Industry 4.0 Smart Manufacturing Systems		2.1.3	Smart Manufacturing Execution	81		●
❖ Information Security and Privacy Protection						
GRI 3: Material Topics 2021						
3-3	Management of material topics	1.2.4	Information Security Risk Management	47		●
❖ Customer Relationship Management						
GRI 3: Material Topics 2021						
3-3	Management of material topics	2.2	Customer Satisfaction	83		●

List of Material Topics		Location		Page	Note/Omission	External Assurance
UMC Specific Topic						
Establishing a smooth communication mechanism to understand customer opinions, strengthening service quality, improving customer satisfaction, and maintaining customer relationships.		2.2.1	Customer Satisfaction Improvement	83		●
		2.2.2	Continuous Quality Improvement and Innovation	85		
GRI 418 Customer Privacy 2016						
418-1	Substantiated complaints concerning breaches of customer privacy and losses of customer data	1.1.3	Legal Compliance	39	No complaints concerning breaches during the reporting period	●
KPIs of Sustainability Strategy and Blueprint						
Ratio of satisfied customers		2.2.1	Customer Satisfaction Improvement	83		●
❖ Product Responsibility and Lifecycle Assessment						
GRI 3: Material Topics 2021						
3-3	Management of material topics	2.3	Product Responsibility	86		●
UMC Specific Topic						
Management of hazardous substances and product environmental impact assessment		2.3	Product Responsibility	87		●
Management KPI						
No customer complaints, product scrap and recalls due to violations of hazardous substance management		2.3	Product Responsibility	86		●
❖ Climate Strategy and Action						
GRI 3: Material Topics 2021						
3-3	Management of material topics	3.1	Climate Action	96		●
GRI 305: Emissions 2016						
305-1	Direct (Scope 1) GHG emissions	3.1.2	GHG Emissions Management	99	1. Gases included in the calculation: CO ₂ , CH ₄ , N ₂ O, HFCs, PFCs, SF ₆ , and NF ₃ 2. There are no emissions of CO ₂ from the combustion or biodegradation of biomass in 2024.	●
305-2	Energy indirect (Scope 2) GHG emissions	3.1.2	GHG Emissions Management	99		●
305-3	Other indirect (Scope 3) GHG emissions	3.1.2	GHG Emissions Management	100		●
305-4	GHG emissions intensity	3.1.2	GHG Emissions Management	99		●
305-5	Reduction of GHG emissions	3.1.3	GHG Emissions Reduction	101		●
KPIs of Sustainability Strategy and Blueprint						
Reduction of GHG emissions per unit product		3.1	Climate Action	96		●
Reduction of GHG emissions		3.1	Climate Action	96		●

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❖ Energy Management							
GRI 3: Material Topics 2021							
3-3	Management of material topics	3.2	Energy Management	104			●
GRI 302: Energy 2016							
302-1	Energy consumption within the organization	3.2.1	Energy Mix	105			●
302-3	Energy intensity	3.2.1	Energy Mix	105			●
302-4	Reduction of energy consumption	3.2.2	Energy Efficiency Improvement	107			●
KPIs of Sustainability Strategy and Blueprint							
Reduction of electricity consumption per unit product		3.2	Energy Management	104			●
Cumulative energy savings		3.2	Energy Management	104			●
Ratio of renewable energy in total electricity used		3.2	Energy Management	104			●
❖ Water and Wastewater Management							
GRI 3: Material Topics 2021							
3-3	Management of material topics	3.3	Water Resources Management	108			●
GRI 303 Water and Effluents 2018							
303-1	Interactions with water as a shared resource	3.3.1	Water Structure	111			●
		3.3.3	Water Pollution Control	116			
303-2	Interactions with water as a shared resource	3.3.3	Water Pollution Control	115			●
303-3	Water withdrawal	3.3.1	Water Structure	117			●
303-4	Water discharge	3.3.1	Water Structure	117			●
303-5	Water consumption	3.3.1	Water Structure	117			●
KPIs of Sustainability Strategy and Blueprint							
Reduction of water withdrawal per unit product		3.3	Water Resources Management	108			●
Cumulative water savings		3.3	Water Resources Management	108			●
Key water pollution indicators better than regulated standards		3.3	Water Resources Management	108			●
Reclaimed water and desalinated water usage rate		3.3	Water Resources Management	108			●
❖ Waste Management and Circular Economy							
GRI 3: Material Topics 2021							
3-3	Management of material topics	3.4	Waste Management	118			●

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GRI 306: Waste 2020							
306-1	Waste generation and significant waste-related impacts	3.4	Waste Management	118			●
306-2	Management of significant waste-related impacts	3.4.2	Source Reduction	120			●
		3.4.3	Circular Economy Promotion	121			
		3.4.4	Contractor Management	123			
306-3	Waste generated	3.4.1	Waste Output Structure	119			●
306-4	Waste diverted from disposal	3.4.1	Waste Output Structure	119			●
306-5	Waste directed to disposal	3.4.1	Waste Output Structure	119			●
KPIs of Sustainability Strategy and Blueprint							
Reduction of waste generated per unit product		3.4	Waste Management	118			●
Cumulated waste reduction		3.4	Waste Management	118			●
Develop circular economy solutions/products (cumulative)		3.4	Waste Management	118			●
Ratio of in-fab resource conversion		3.4	Waste Management	118			●
🔗 Talent Attraction and Retention							
GRI 3: Material Topics 2021							
3-3	Management of material topics	4.2	Diversity, Equity and Inclusion in the Workplace	138			●
GRI 401: Employment 2016							
401-1	New employee hires and employee turnover	4.2.1	Human Resource Development	142, 145			●
401-2	Benefits provided to full-time employees that are not provided to temporary or part-time employees	4.2.2	Compensation and Benefits	147			●
401-3	Parental leave	4.2.1	Human Resource Development	146			●
KPIs of Sustainability Strategy and Blueprint							
Mid- and high-performance retention rate		4.2.1	Human Resource Development	138			●
🔗 Diversity and Inclusion							
GRI 3: Material Topics 2021							
3-3	Management of material topics	4.2	Diversity, Equity and Inclusion in the Workplace	138			●
GRI 405: Diversity and Equal Opportunity 2016							
405-2	Ratio of basic salary and remuneration of women to men	4.2.2	Compensation and Benefits	148			●
KPIs of Sustainability Strategy and Blueprint							
Proportion of women among managerial personnel		4.2.1	Human Resource Development	138			●

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Occupational Health and Safety						
GRI 3: Material Topics 2021						
3-3	Management of material topics	4.3	Protecting the Employees and the Work Environment	159		
GRI 403: Occupational Health and Safety 2018						
403-1	Occupational health and safety management system	4.3.1	Maintain Workplace Safety	160		
403-2	Hazard identification, risk assessment, and incident investigation	4.3.1	Maintain Workplace Safety	160		
403-3	Occupational health services	4.3.2	Creating a Reassuring Workplace	167		
403-4	Worker participation, consultation, and communication on occupational health and safety	4.3.1	Maintain Workplace Safety	161		
403-5	Worker training on occupational health and safety	4.3.1	Maintain Workplace Safety	161		
403-6	Promotion of worker health	4.3.2	Creating a Reassuring Workplace	167		
403-7	Prevention and mitigation of occupational health and safety impacts directly linked by business relationships	4.3.1	Maintain Workplace Safety	163		
403-8	Workers covered by an occupational health and safety management system	4.3.1	Maintain Workplace Safety	160		
403-9	Work-related injuries	4.3.1	Maintain Workplace Safety	165		
403-10	Work-related ill health	4.3	Protecting the Employees and the Work Environment	163	There is no case of legally defined occupational disease during the reporting period. For the number of cases of mental and physical health related problems in employees, please refer to Chapter 4.3.	
KPIs of Sustainability Strategy and Blueprint						
Disabling injury frequency rate (FR) and disabling injury severity rate (SR)		4.3.1	Maintain Workplace Safety	159		
Human Rights						
GRI 3: Material Topics 2021						
3-3	Management of material topics	4.1	Enforce Human Rights Management	128		
GRI 406: Non-discrimination 2016						
406-1	Incidents of discrimination and corrective actions taken	4.1.1	Enforce Human Rights Management	131		
GRI 407: Freedom of Association and Collective Bargaining 2016						
407-1	Operations and suppliers in which the right to freedom of association and collective bargaining may be at risk	4.1.1	Enforce Human Rights Management	131		
GRI 408 Child Labor 2016						
408-1	Operations and suppliers at significant risk for incidents of child labor	4.1.1	Enforce Human Rights Management	131		
GRI 409: Forced or Compulsory Labor 2016						
409-1	Operations and suppliers at significant risk for incidents of forced or compulsory labor	4.1.1	Enforce Human Rights Management	131		
KPIs of Sustainability Strategy and Blueprint						
The coverage rate and score of work engagement questionnaire		4.1	Respect for Human Rights	128		

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Appendix 2. SASB Index - Semiconductors Sustainability Accounting Standard 2018

Industry: TECHNOLOGY & COMMUNICATIONS SECTOR-SEMICONDUCTORS Version 2018-10

Scope: UMC Group (the parent company and its subsidiaries HJ, USCXM, Wavetek and USJC which covers revenue accounting for over 99% of UMC’s consolidated financial statements.

Topic/Code	Accounting Metric	Category	Location	Page	2024 Performance Summary	External Assurance
Greenhouse Gas Emissions TC-SC-110a.1	(1) Gross global Scope 1 emissions	Quantitative	3.1.2 GHG Emissions Management	99	(1) Amount of total Scope 1 direct emissions: 463,670 tonnes CO ₂ e.	●
	(2) Amount of total emissions from perfluorinated compounds		3.1.3 GHG Emissions Reduction	101	(2) Amount of total F-GHGs (Fluorinated greenhouse gas) emissions: 356,526 tonnes CO ₂ e.	
Greenhouse Gas Emissions TC-SC-110a.2	Discussion of long-term and short-term strategy or plan to manage Scope 1 emissions, emissions reduction targets, and an analysis of performance against those targets	Discussion and Analysis	3.1.3 GHG Emissions Reduction	101	Scope 1 GHG emissions is mainly due to the F-GHGs used in the manufacturing process. UMC has promoted reduction measures including source reduction with gas replacement, and the installation of high efficiency local scrubbers. UMC has set goals through its Sustainability Strategy and Blueprint to track the progress of action plans. The goals of GHG reduction were all achieved in this year.	●
Energy Management in Manufacturing TC-SC-130a.1	(1) Total energy consumed	Quantitative	3.2.1 Energy Mix	105	(1) Total energy consumption: 4,129,707MWh (14,866,944 GJ)	●
	(2) Percentage grid electricity				(2) Percentage of purchased electricity in total energy consumed: 77.9% (renewable energy is excluded from the purchased electricity)	
Water Management TC-SC-140a.1	(3) Percentage renewable energy	Quantitative	3.3 Water Resources Management	111 117	(3) Percentage of renewable energy in total energy consumption: 16.1%	●
	(1) Total water withdrawal				(1) Total water withdrawal: 27,562 thousand cubic meters.	
Waste Management TC-SC-150a.1	(2) Total water consumed, percentage of each in regions with High or Extremely High Baseline Water Stress	Quantitative	3.4.1 Waste Output Structure	119	(2) Total water consumption: 8,218 thousand cubic meters.	●
	Amount of hazardous waste from manufacturing and the percentage that is recycled				(3) The subsidiary HJ is located at Suzhou identified as water-stress area by WRI. The water withdrawal and water consumption of HJ were accounted for 7.8% and 3.8% respectively. In addition to the ample water resources provided by the nearby Taihu Lake, which has a storage capacity of 4.5 billion tonnes, and a dual water source and dual water plant safety supply model, the dual water sources (Taihu Lake and Jiuyang Lake) have a combined supply capacity of 650,000 tonnes/day. HJ also implements water-saving measures to improve water use efficiency and reduce the impact of water risks on operations.	
Employee Health & Safety TC-SC-320a.1	Description of efforts to assess, monitor, and reduce exposure of employees to human health hazards	Discussion and Analysis	4.3.1 Maintain Workplace Safety	160	Total hazardous waste output was 40,868 tonnes, accounting for 65.3% of total waste output. The percentage of hazardous waste recycled was 93.7%.	●
Employee Health & Safety TC-SC-320a.2	Total amount of monetary losses as a result of legal proceedings associated with employee health and safety violations	Quantitative	4.3.1 Maintain Workplace Safety	162	UMC performs hazard identification and risk assessment for routine and non-routine operations, including: 1. Workplace Hazard Assessment; 2. Safety and health risk assessment for manufacturing processes and activities; 3. Risk identification for engineering by FMEA.	●
					There were no incidents of such violations and the total related monetary loss was NT\$0.	●

Topic/Code	Accounting Metric	Category	Location	Page	2024 Performance Summary	External Assurance
Recruiting & Managing a Global & Skilled Workforce TC-SC-330a.1	Percentage of employees that are (1) Foreign nationals (2) Located offshore	Quantitative	4.2.1 Human Resource Development	143	The percentage of local employees (full-time) in each country and region as follows: 96.9% for Taiwan (including its subsidiary Wavetek); 26.0% for Singapore; 95.7% for its Chinese subsidiaries HJ and USCXM; and 96.2% for its Japanese subsidiary USJC.	●
Product Lifecycle Management TC-SC-410a.1	Percentage of products by revenue that contain IEC 62474 declarable substances	Quantitative	2.3.1 Hazardous Substance Management	88	In response to domestic and foreign regulations, UMC regularly updates its hazardous substance management regulations, which cover the substances listed in IEC62474 that require reporting. The percentage of materials not complying with IEC62474 is 0%.	●
Product Lifecycle Management TC-SC-410a.2	Processor energy efficiency at a systemlevel for: (1) servers, (2) desktops, and (3) laptops	Quantitative	-	-	Considering foudry firm is not an end product manufacturer, and thus there is no applicable content.	●
Materials Sourcing TC-SC-440a.1	Description of the management of risks associated with the use of critical materials	Discussion and Analysis	1.3.3 Sustainable Raw Materials Management	70	UMC completed its supply chain risk database and the management system of earthquake spare parts (quartz, fragile parts) for over 2,000 key raw materials supplied by approximately 100 vendors. A production area risk warning automatic is established, sending and return monitoring and risk identification system. Besides, 95% of UMC’s 8-inch fabs and 99% of 12-inch fabs have completed earthquake spare parts preparation.	●
Intellectual Property Protection & Competitive Behavior TC-SC-520a.1	Total amount of monetary losses as a result of legal proceedings associated with anticompetitive behavior regulations	Quantitative	1.1.3 Legal Compliance	36	There were no incidents of such violations and the total related monetary loss was NT\$0.	●
Activity Metrics TC-SC-000.A	Total production	Quantitative	About UMC	5	UMC Group manufactured and integrated approximately 3.451 million pieces of 12-inch equivalent wafers.	●
Activity Metrics TC-SC-000.B	Percentage of production from owned facilities	Quantitative	About UMC	5	100% production came from UMC Group's own fabs.	●

Appendix

TCFD Disclosure Comparison Table

Appendix 3. TCFD Disclosure Comparison Table

Dimension/Recommended Disclosures	Location	Page
Governance		
The board’s oversight of climate-related risks and opportunities.	1.1.1 Board of Directors Operation	32
	1.2.5 Managing Climate Risks and Opportunities	50
Management’s role in assessing and managing climate-related risks and opportunities.	Sustainable Governance	8
	1.2.1 Risk Management System	40
	1.2.5 Managing Climate Risks and Opportunities	50
Strategy		
The climate-related risks and opportunities the organization has identified over the short, medium, and long term.	1.2.5 Managing Climate Risks and Opportunities	55
The impact of climaterelated risks and opportunities on the organization’s businesses, strategy, and financial planning.	1.2.5 Managing Climate Risks and Opportunities	55
The resilience of the organization’s strategy, taking into consideration different climate-related scenarios, including a 2°C or lower scenario.	1.2.5 Managing Climate Risks and Opportunities	52
Risk Management		
The organization’s processes for identifying and assessing climate-related risks.	1.2.5 Managing Climate Risks and Opportunities	51
The organization’s processes for managing climate-related risks.	1.2.1 Risk Management System	40
	1.2.3 Improving Operational Risk Management	45
	1.2.5 Managing Climate Risks and Opportunities	52
How processes for identifying, assessing, and managing climate-related risks are integrated into the organization’s overall risk management.	1.2.1 Risk Management System	40
	1.2.3 Improving Operational Risk Management	45
	1.2.5 Managing Climate Risks and Opportunities	52
Metrics and Targets		
Metrics used by the organization to assess climaterelated risks and opportunities in line with its strategy and risk management process.	1.2.5 Managing Climate Risks and Opportunities	55
Disclose Scope 1, Scope 2, and, if appropriate, Scope 3 greenhouse gas (GHG) emissions, and the related risks.	1.2.5 Managing Climate Risks and Opportunities	55
	3.1.2 GHG Emissions Management	99
Targets used by the organization to manage climate-related risks and opportunities and performance against targets.	Sustainable Governance	8
	2.1.1 Robust Wafer Manufacturing Services	74
	3.1.3 GHG Emissions Reduction	101
	3.2.1 Energy Mix	105
	3.3.1 Water Structure	109
	3.3.2 Water Saving	112

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Appendix 4. United Nation Global Compact Index

10 Principles	Location	Page
Human Rights		
Businesses should support and respect the protection of internationally proclaimed human rights.	4.1.1 Enforce Human Rights Management	129
Make sure that they are not complicit in human rights abuses.	1.3.2 Sustainable Supply Chain Management	64
	4.1.1 Enforce Human Rights Management	129
Labor		
Businesses should uphold the freedom of association and the effective recognition of the right to collective bargaining.	4.1.2 Employer-employee Communication	134
The elimination of all forms of forced and compulsory labor.	4.1.1 Enforce Human Rights Management	129
The effective abolition of child labor.	4.1.1 Enforce Human Rights Management	129
The elimination of discrimination in respect of employment and occupation.	4.1.1 Enforce Human Rights Management	129
Environment		
Businesses should support a precautionary approach to environmental challenges.	Chapter 3 Implementing Green Operations	92
Undertake initiatives to promote greater environmental responsibility.	Chapter 3 Implementing Green Operations	92
Encourage the development and diffusion of environmentally friendly technologies.	2.1 Innovation & Technology Development	74
	2.3 Product Responsibility	86
Anti-Corruption		
Businesses should work against corruption in all its forms, including extortion and bribery.	1.1.2 Integrity Management	35

Appendix 5. ISO 26000 Index

Core Subjects and Issues	Location	Page
Organizational Governance		
Decision-making processes and structures	1.1 Excelling Corporate Governance	30
Human Rights		
Due diligence	1.3.2 Sustainable Supply Chain Management	64
	4.1.1 Enforce Human Rights Management	129
Human rights risk situations	1.3.2 Sustainable Supply Chain Management	64
	4.1.1 Enforce Human Rights Management	129
Avoidance of complicity	1.1 Excelling Corporate Governance	30
	4.1.1 Enforce Human Rights Management	129
Resolving grievances	1.1.2 Integrity Management	35
	4.1.1 Enforce Human Rights Management	129
	4.1.2 Employer-employee Communication	134
Discrimination and vulnerable groups	4.1.1 Enforce Human Rights Management	129
	5.4 Commitments to Social Welfare	186
Civil and political rights	4.1.1 Enforce Human Rights Management	129
Economic, social and cultural rights	5.1 Promote Common-Prosperity Society	174
Fundamental principles and rights at work	4.1.1 Enforce Human Rights Management	129
Labor Practices		
Employment and employment relationships	4.2 Diversity, Equity and Inclusion in the Workplace	138
	4.1.2 Employer-employee Communication	134
Conditions of work and social protection	4.1.1 Enforce Human Rights Management	129
	4.3 Protecting the Employees and the Work Environment	159
Social dialogue	Determination of Material Topics	19
	4.1.2 Employer-employee Communication	134
Health and safety at work	4.3 Protecting the Employees and the Work Environment	159

Appendix 6. TWSE-required Disclosure of Indicators

According to TWSE's "Rules Governing the Preparation and Filing of Sustainability Reports by TWSE Listed Companies", relevant attachments have been disclosed belows.

Disclosure of sustainability indicators as required by Attachment 1-8: Semiconductor Industry

Scope: UMC Group (the parent company and its subsidiaries HJ, USCXM, Wavetek and USJC which covers revenue accounting for over 99% of UMC’s consolidated financia statements.

Item	Accounting Metric	Category	2024 Performance Summary	Unit	Remarks
1	Total energy consumption	Quantitative	14,866,944	GJ	
	Percentage of purchased electricity		77.9	percentage (%)	Consumption of purchased grid electricity ÷ total energy consumption
	Percentage of renewable energy usage		16.1	percentage (%)	Renewable energy ÷ total energy consumption
2	Total water withdrawal	Quantitative	27,562	1,000 cubic meters (m³)	
	Total water consumption		8,218	1,000 cubic meters (m³)	
3	Total hazardous waste output	Quantitative	40,868	tonne (t)	
	The hazardous waste recycling rate		93.7	percentage (%)	
4	Explain the occupational accident categories	Quantitative	There were three minor or above accidents reported. The accidents can be categorized: Arm injury due to pinching: 1, Finger injury due to pinching: 1, fire alarm: 1.		quantity
	Number of people with disabling injuries		7	person	
	Percentage of people with disabling injuries		0.04	percentage (%)	No. of people with disabling injuries ÷ total employees
5	Disclosure of product life cycle management	Quantitative	Weight of scrap products and electronic waste	55.9	tonne (t)
			Percentage of recycling	90.1	percentage (%)
6	Description of the risk management associated with the use of key materials	Discussion and Analysis	The Company completed its "Supply Chain Risk Database" and the management system of earthquake spare parts (quartz, fragile parts) for over 2,000 key raw materials supplied by approximately 100 vendors. In addition, 95% of UMC's 8-inch fabs and 99% of 12-inch fabs have completed earthquake spare parts preparation. 200 major suppliers of raw materials completed risk assessment surveys, all of which met UMC's requirements.		Not applicable
7	Total amount of monetary losses as a result of legal proceedings associated with anticompetitive behavior regulations	Quantitative	0	Currency	There were no incidents of such violations in 2024 and the total related monetary loss was NT\$0
8	Total production	Quantitative	UMC Group manufactured and integrated approximately 3.451 million pieces of 12-inch equivalent wafers.		-

Appendix 2: Climate-related Response Measures of List Company

1. Climate-related Response Measures: Please refer to Chapter 1.2.5 Managing Climate Risks and Opportunities of this report.

1-1 Results of Greenhouse Gas Emissions Inventory and Verificatio

UMC has completed the inventory of 27 subsidiaries included in the consolidated financial statements since 2023.

Year	Scope	Total Emissions (tonne CO ₂ e)			Intensity (tonne CO ₂ e/million NT\$)		Verificatio			
		Scope 1	Scope 2	Scope 3	Scope 1	Scope 2	Party ^{Note 3}	percentage of the inventory information	Status	Opinion
2023	Parent company	356,911	1,376,960	1,733,871	2.29	7.36	DNV	100%	ISO 14064-3 Reasonable Assurance	Unmodified
	Subsidiary ^{Note1}	152,644	259,248	411,892			SGS/DNV ^{Note 2}	100%		
2024	Parent company	287,393	1,334,802	1,622,195	1.99	6.56	DNV	100%	ISO 14064-3 Reasonable Assurance	Unmodified
	Subsidiary ^{Note1}	176,277	189,053	365,330			SGS/DNV ^{Note 2}	100%		

Note: 1. The statistics cover wafer foundry subsidiaries HJ, USCXM, Wavetek, and USJC. The other subsidiaries with a consolidated financial report have emissions that are less than 5% of the total emissions of the consolidated entity and are not disclosed based on the materiality criteria.
2. The verification party of UMC parent company, Wavetek and USJC is DNV Business Assurance Co., Ltd.; the verification party of HJ and USCXM is SGS-CSTC Standards Technical ervices CO., Ltd.

1-2 Goals and Action Plans of Greenhouse Gas Emissions Reduction

In 2021, UMC announced its pledge to reach net zero emissions by 2050 as the first semiconductor foundry globally. The 2030 GHG emissions reduction target set by UMC was approved by the Science-Based Targets Initiative (SBTi) in 2022, and the Company continues to drive further reductions by three resolutions. In 2024, in order to actively implement UMC’s low-carbon commitments and take actions to speed up emission reduction, UMC has further signed the SBTi net-zero target commitment and raised the 2030 GHG emissions reduction target in accordance with the SBTi Net-Zero Standard to a 42% reduction of Scope 1 and Scope 2 emissions with 2020 as the base year. The GHG emissions of UMC and its foundry subsidiaries reduced by 31% in 2024 compared to the base year emissions of 2.883 million tCO₂e. For more details about UMC’s climate action, [please refer to Chapter 3.1 Climate Action of this report.](#)

Appendix 7. ESG Data Summary

Scope: UMC

Indicator		Unit	2021	2022	2023	2024	Remarks
Governance & Economic							
Total board size		seat	9	9	9	9	
Composition of the board of directors	Independent directors	seat	5	5	5	6	
	Other non-executive directors	seat	1	1	1	0	
	Executive directors	seat	3	3	3	3	
Female directo	Seat	seat	2	2	2	3	
	Proportion	%	22.2	22.2	22.2	33.3	
Average attendance rate of Board meeting		%	98.7	100.0	100.0	100.0	
Ratio of satisfied customers		%	89	91	92	92	Ratio of satisfied customers is conducted by a third-party agency since 2021.
Product Recalls		number	0	0	0	0	
R&D Expense	Amount	10 million NT\$	1,002	1,024	1,072	1,315	
	Percentage of operation income	%	6.3	4.9	6.2	7.3	
Revenue from sustainable products of total revenue		%	68.7	74.9	82.2	82.3	82.3% of total revenue derived from sustainable products in 2024, with 76.8% from low power consumption, low leakage, and power management products, 11% from automotive chips which used in advanced driver assistance systems (ADAS) and powertrain, etc., and 7.9% from microcontroller unit (MCU) and epidemic prevention & medical related products.
Accumulated granted patents worldwide		number	14,423	14,771	15,196	15,560	
Environment							
GHG	F-GHG emission per unit product (wafer weight)	kg F-GHG/wafer-tonne	133	107	80	68	
	Direct greenhouse gas emissions (Scope 1)	tCO ₂ e	712,639	591,781	356,911	287,393	
	Indirect greenhouse gas emissions (Scope 2)	tCO ₂ e	1,298,439	1,373,914	1,376,960	1,334,802	
	GHG emission per unit product (Scope 1 & 2)	tCO ₂ e/wafer-m ²	8.25	7.91	9.36	8.25	
	Othes indirect greenhouse gas emissions (Scope 3)	tCO ₂ e	2,834,727	2,064,284	1,893,167	1,416,926	
Energy	Total energy consumption (including renewable energy)	MWh	2,883,399	3,000,645	3,050,481	3,085,414	
	Purchased Electricity	MWh	2,676,665	2,796,662	2,856,931	2,781,600	
	Total renewable energy consumption	MWh	2,146	4,175	6,310	122,407	

	Indicator	Unit	2021	2022	2023	2024	Remarks
Water	Total water withdrawal	1,000 m ³	16,213	17,410	17,258	18,412	
	Total water discharge	1,000 m ³	10,676	11,475	11,389	11,734	
	Total water consumption	1,000 m ³	5,537	5,935	5,869	6,678	
	Third-party water: Municipal Water	1,000 m ³	12,024	13,053	12,591	13,058	
	Third-party water: Reclaimed Water	1,000 m ³	3,324	3,566	3,921	4,586	
	Condensate water and rainwater	1,000 m ³	733	791	746	768	
	Surface freshwater, underground water, salt water	1,000 m ³	132	0	0	0	The groundwater withdrawal in 2021 indicated the groundwater supply (withdrawn from renewable groundwater formations) by water tankers activated for Hsinchu plant to cope with a severe drought in the year.
	Total water withdrawal from third party	1,000 m ³	15,348	16,619	16,512	17,644	Including municipal water and reclaimed water
	Percentage of water withdrawal from alternative water sources	%	25	25	27	29	Including reclaimed water, condensate water and rainwater
	Water withdrawal per unit product	m ³ /wafer-m ²	66.54	70.08	93.08	93.65	
	Water consumption per unit product	m ³ /wafer-m ²	22.72	23.89	31.65	33.97	
	Manufacturing-process recycling rate	%	81.9	82.8	84.3	83.3	(All fabs' manufacturing-process water withdrawal ÷ UMC's total manufacturing process water withdrawal) x Sum of all fabs' manufacturing-process water recycling rate
General Waste	Ultra-pure water usage	1,000 m ³	18,538	19,529	19,528	19,923	
	Total waste recycled/ reused	tonne	8,557	10,387	9,813	7,919	
	Total waste disposed	tonne	1,887	1,881	1,253	734	
	Waste landfilled	tonne	1,280	1,179	669	112	
	Waste incinerated with energy recovery	tonne	376	390	305	247	
	Waste incinerated without energy recovery	tonne	231	312	279	375	
	Waste otherwise disposed	tonne	0	0	0	0	Including solidification, and chemical treatment
	Total waste disposed(UMC Group)	tonne	2,278	2,287	1,643	1,049	
Hazardous Waste	Total hazardous waste recycled/reused	tonne	21,976	22,560	18,766	18,474	
	Total hazardous waste disposed	tonne	3,116	2,586	2,482	2,223	
	Hazardous waste landfilled	tonne	0	0	0	0	
	Hazardous waste incinerated with energy recovery	tonne	2,153	1,620	1,775	1,989	
	Hazardous waste incinerated without energy recovery	tonne	298	328	231	207	
	Hazardous waste otherwise disposed	tonne	664	638	476	28	Including solidification, and chemical treatment

Note: The calculation of statistic above is based on the definition of Dow Jones Sustainability Index (DJSI). Liquid waste and waste recycled inside the fab are excluded from general waste. And waste recycled inside the fab are also excluded from hazardous waste.

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	Indicator	Unit	2021	2022	2023	2024	Remarks
Hazardous Waste	Total hazardous waste disposed	tonne	3,721	3,331	2,861	2,556	
	Direct VOC emissions	tonne	35.65	29.01	24.86	23.95	
Costs and Benefits	Amount of fines related to environmental violations	million NT\$	0	0	0	0	0 environmental violation and fine in all years.
	Cost saving	million NT\$	477	226	549	904	Cost saving mainly calculates the benefits generated through the implementation of management measures for water, electricity, waste, and GHGs.
	Environmental capital expenditures investment	million NT\$	1,670	2,920	3,317	1,849	
	Environmental operating expenses	million NT\$	1,463	1,688	1,803	1,846	
Social							
Human Resource Development	Total number of new employee hires	%	14.9	15.9	5.3	5.2	Total number of newly hired employees ÷ the average number of existing employees. The number of new employees has been adjusted to exclude employees returning from unpaid leave and corporate transfer employees. The reported information of 2021~2023 has also been adjusted.
	Total employee turnover rate	%	10.5	9.4	5.5	5.4	Total number of employees who resigned ÷ the average number of existing employees
	Voluntary employee turnover rate	%	10.4	9.3	5.4	5.4	
	Internal transfer rate	%	57.1	58.6	65.1	77.8	Internal transfer rate = number of internal transfers ÷ number of job vacancies. Including direct employees, the reported information of 2020-2022 has also been adjusted.
	Female management position rate	%	20.8	20.7	22.5	22.3	The management positions include junior, middle and top management positions
	Average amount spent per FTE on training and development	NT\$	2,882	3,743	4,074	4,053	
	Average hours per FTE of training and development	hour	20.5	26.9	44.7	77.5	
Employee Health & Safety	No. of people with disabling injuries	person	11	13	8	6	
	People with disabling injuries rate	%	0.06	0.09	0.05	0.04	No. of people with disabling injuries ÷ total employees
Total amount of charitable donations		million NT\$	241.2	104.0	75.7	75.9	

Appendix 8. External Assurance

Third-Party Independent Assurance Statement



ASSURANCE STATEMENT

SGS TAIWAN LTD.'S REPORT ON SUSTAINABILITY ACTIVITIES IN THE UNITED MICROELECTRONICS CORPORATION'S SUSTAINABILITY REPORT FOR 2024

NATURE AND SCOPE OF THE ASSURANCE

SGS Taiwan Ltd. (hereinafter referred to as SGS) was commissioned by UNITED MICROELECTRONICS CORPORATION (hereinafter referred to as the Company) to conduct an independent assurance of the Sustainability Report for 2024 (hereinafter referred to as the Report). The assurance is based on the SGS Sustainability Report Assurance methodology and AA1000 Assurance Standardv3 Type 2 high level during 2025/02/21 to 2025/06/02. The boundary of the Report includes the Company's main sites of operation located in Taiwan (Hsinchu and Tainan) and Singapore, as well as the fabs managed by UMC's subsidiaries in Taiwan (Hsinchu), China (Suzhou and Xiamen), and Japan (Mie), which only cover a partial entity of UMC's consolidated financial statements.

SGS reserves the right to update the assurance statement from time to time depending on the level of report content discrepancy of the published version from the agreed standards requirements.

INTENDED USERS OF THIS ASSURANCE STATEMENT

This Assurance Statement is provided with the intention of informing all the Company's Stakeholders.

RESPONSIBILITIES

The information in the Report and its presentation are the responsibility of the directors or governing body (as applicable) and the management of the Company. SGS has no conflict of interest in providing assurance on the Company's environmental, social and governance data, and has not been involved in the preparation of any material included in the Report.

Our responsibility is to express an opinion on the text, data, graphs and statements within the scope of assurance based upon sufficient and appropriate objective evidence.

ASSURANCE STANDARDS, TYPE AND LEVEL OF ASSURANCE

The assurance of this Report has been conducted according to the AA1000 Assurance Standard (AA1000AS v3), a standard used globally to provide assurance on sustainability-related information across organizations of all types, including the evaluation of the nature and extent to which an organization adheres to the AccountAbility Principles (AA1000AP,2018).

Assurance has been conducted at a type 2 high level of scrutiny.

SCOPE OF ASSURANCE AND REPORTING CRITERIA

The scope of the assurance included evaluation of quality, accuracy and reliability of specified performance information as detailed below and evaluation of adherence to the following reporting criteria:

Reporting Criteria Options	
1	AA1000 Accountability Principles (2018)
2	GRI (In Accordance with)
3	SASB SEMICONDUCTORS Sustainability Accounting Standard (Version 2018-10)

- The evaluation of the reliability and quality of specified sustainability performance information in th Report is limited to determined material topics or those clearly marked in the Report as conducted in accordance with type 2 of AA1000AS v3 sustainability assurance engagement at a high level of scrutiny for the Company
- The evaluation of the Report against the requirements of GRI Standards, includes GRI 1, GRI 2, GRI 3, 200, 300 and 400 series claimed in the GRI content index as material and is conducted in accordance with the standards.

SPECIFIED PERFORMANCE INFORMATION AND DISCLOSURES INCLUDED IN SCOPE

- Information related to the materiality assessment process.
- Information on the corresponding GRI topic standards for material topics and general disclosures in accordance with the GRI Standards.
- The goals and 2024 achievements in the Sustainability Strategy and Blueprint, including economic, social, and environmental KPIs which were remarked in the "GRI Content Index" that appendix of the Report.
- Information disclosed with reference to the SASB standards.

ASSURANCE METHODOLOGY

The assurance comprised a combination of desktop research, interviews with relevant employees, superintendents, Sustainability committee members and the senior management in Taiwan; documentation and record review and validation with external bodies and/or stakeholders where relevant.

LIMITATIONS

Financial data drawn directly from independently audited financial accounts, Task Force on Climate-related Financial Disclosures (TCFD) and greenhouse gas emissions had been checked by other assurance providers or engagements has been checked the transcription from these sources as part of this assurance process.

INDEPENDENCE AND COMPETENCE

The SGS Group of companies is the world leader in inspection, testing and verification, operating in more than 140 countries and providing services including management systems and service certification; quality, environmental, social and ethical auditing and training; environmental, social and sustainability report assurance. SGS affirm our independence from the Company, being free from bias and conflicts of interest with the organisation, its subsidiaries and stakeholders.

The assurance team was assembled based on their knowledge, experience and qualifications for this assignment, and comprised auditors registered with professional qualifications such as ISO 26000, ISO 20121, ISO 50001, RBA, QMS, EMS, SMS, GPMS, CFP, WFP, GHG Verification and GHG Validation Lead Auditors and experience on the SRA Assurance service provisions.

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FINDINGS AND CONCLUSIONS

ASSURANCE OPINION

On the basis of the methodology described and the assurance work performed, we are satisfied that the specified performance information included in the scope of assurance is accurate, reliable, has been fairly stated and has been prepared, in all material respects, in accordance with the AA1000 AccountAbility Principles (2018).

We believe that the organisation has chosen an appropriate level of assurance (High Level) for this stage in their reporting.

ADHERENCE TO AA1000 ACCOUNTABILITY PRINCIPLES (2018)

INCLUSIVITY

The Company has demonstrated a good commitment to stakeholder inclusivity and stakeholder engagement. A variety of engagement efforts such as survey and communication to employees, customers, investors, suppliers and other stakeholders are implemented to underpin the organization's understanding of stakeholder concerns. The concerns of stakeholders are addressed and deliberated at the highest level of governance.

MATERIALITY

The Company has established effective processes for determining issues that are material to the business. The process includes impact analysis, stakeholder concerns, and expert perspectives. Formal review has identified stakeholders and those issues that are material to each group and the Report addresses these at an appropriate level to reflect their importance and priority to these stakeholders.

RESPONSIVENESS

The Company has established principles for stakeholder engagement and regularly reports to relevant governance bodies. The Report includes coverage given to stakeholder engagement and channels for stakeholder feedback.

IMPACT

The Company has demonstrated a process on identify and fairly represented impacts that encompass a range of environmental, social and governance topics from wide range of sources, such as activities, policies, programs, products and services. Measurement and evaluation of its impacts related to material topic were in place at target setting.

QUALITY AND RELIABILITY OF SPECIFIED PERFORMANCE INFORMATION

On the basis of the verification work performed, we checked internal management records, management review meeting minutes, minutes of committee meeting, relevant ISO certificates and opinion, and publicly available information. We have confidence that the specified performance information included in the scope of assurance is reliable at a high level of scrutiny for the Company. It is encouraged to establish a regular internal review process to ensure the completeness and accuracy of disclosed sustainability information.

ADHERENCE TO GRI

The report, the Company's Sustainability Report of 2024, is reporting in accordance with the GRI Universal Standards 2021. The significant impacts were assessed and disclosed in accordance with the guidance defined in GRI 3: Material Topic 2021 and the relevant 200/300/400 series Topic Standard related to the material topics claimed in the GRI content index. The Report has properly disclosed information related to the Company's contributions to sustainability development.

For future reporting, it is recommended the Company to conduct comprehensive internal communication regarding the management and disclosure of sustainability information to enhance the precision and consistency.

ADHERENCE TO SASB

The Company has referenced with SASB's Standard, SEMICONDUCTORS, VERSION 2018-10 to disclose information of material topics that are vital for enterprise value creation. The disclosed SASB information follows the reporting boundaries of the Company's 2024 Sustainability Report. Process to identify, assess, and manage SASB-topic-related risks and opportunities were integrated into the Company's overall management process. It is recommended to adopt the latest version of the SASB Standards for future disclosures.

Signed:
For and on behalf of SGS Taiwan Ltd.

Stephen Pao
Business Assurance Director
Taipei, Taiwan
25 June, 2025
WWW.SGS.COM



AA1000
Licensed Report
000-8/V3-52L45

Message from the ESG Steering
Committee Chairman

About This Report

About UMC

Sustainable Governance

Sustainability Strategy and Blueprint

Enterprise Risk Management

Sustainable Performance Highlights

Determination of Material Topics

Special Feature of the Year

1 Leading in Corporate
Governance

2 Innovative Products and
Services

3 Implementing Green
Operations

4 Providing a Friendly
Workplace

5 Enhancing Society
Mutual-Prosperity

Appendix

GRI Content Index

SASB Index - Semiconductors
Sustainability Accounting Standard 2018

TCFD Disclosure Comparison Table

United Nation Global Compact Index

ISO 26000 Index

TWSE-required Disclosure of Indicators

ESG Data Summary

External Assurance

TCFD Assurance Statement



Independent Limited Assurance Statement

Statement No.:
C756060-2024-AG-TWN-DNV

Date of Issue:
15 April, 2025

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DNV Business Assurance Co., Ltd. Taiwan (hereafter "DNV") has been commissioned by the management of United Microelectronics Corp. (hereafter "UMC" or "the Company") to perform an assurance engagement on the Company's reporting against the Recommendations of the Task Force on Climate-related Financial Disclosures (version June 2017, hereafter "TCFD Recommendations")

Scope of Verification

Subject to our assurance engagement was the Company's reporting against the four overarching recommendations presented in the TCFD Recommendations, which has been incorporated in the Company's Sustainability Report section 1.2.5 (hereafter "the Report") for the year ended 31 December 2024 which includes relevant disclosures on climate-related management and performance. The current assurance engagement does not explicitly cover the disclosures on Scope 1, Scope 2, and Scope 3 greenhouse gas emissions, as the data has been checked in another engagement based on the verification results from multiple assurance providers.

United Microelectronics Corp.'s Responsibilities

The management of UMC has the sole responsibility for the collection, assessment, analysis, and consolidation of information within the Report. DNV's assurance engagements are based on the assumption that the data and information provided by the Company to us as part of our review have been complete, accurate, and provided in good faith.

DNV's Responsibilities and Independence

In performing the current assurance engagement, our responsibility and independent opinion are to the management of UMC. Provided the consent that UMC may refer to this Assurance Statement under the terms we agreed with the Company, DNV expressly disclaims any liability or co-responsibility for any decision a person or an entity having potential access to this Assurance Statement or our Assurance Report, may make based on our opinion.

DNV has not been involved in the preparation of any statements or data included in the Report except this Assurance Statement, and we have no other contractual relationship with UMC that constitutes a conflict of interest against the current assurance engagement under the established policies and procedures.

Conclusions

It is DNV's opinion that based on the Company's reporting against the TCFD Recommendations, nothing has come to our attention that causes us to believe that relevant disclosures in the Report is not reliable in accordance with the TCFD Recommendations. We can conclude that the Company's management of the climate-related risks and opportunities and information consolidated accordingly is reasonable and balanced, as presented in the Report.



Statement No.:
C756060-2024-AG-TWN-DNV

Date of Issue:
15 April, 2025

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Basis of Our Opinion

A multi-disciplinary team of sustainability and assurance specialists performed work at the Company's Headquarters and site level. We undertook the following activities to assess the Company's disclosures as per the four overarching recommendations:

- Review of the Company's governance structure and management procedures on climate-related risks and opportunities.
- Review of the Company's approach in identifying and assessing the climate-related risks and opportunities, which takes into consideration the following climate scenarios: IEA NZE, IEA APS (incl. NDC), IEA STEPS for the transition risks; and IPCC AR6 SSP1-2.6, and IPCC AR6 SSP5-8.5 for the physical risks.
- Review of the Company's sustainable strategy and blueprint, which states the UMC's metrics and targets of climate-related risks and opportunities.
- Review of the processes for collecting and consolidating the specified assessment data and, for a sample, checking the data consolidation.
- The verification was conducted based only on the Chinese version Report.

Limitations

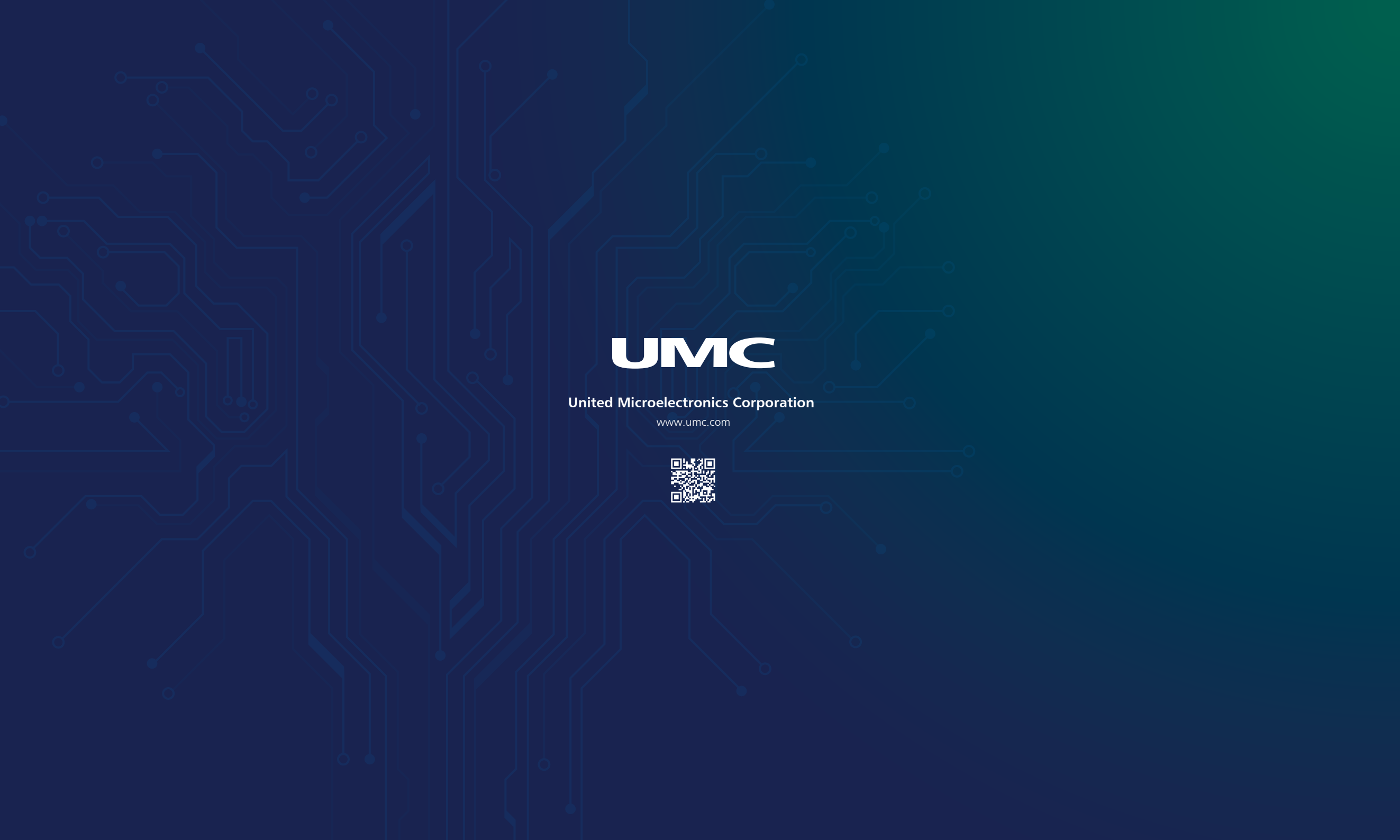
This Assurance Statement is based upon the application of sample principles and professional judgment on certain facts and assumptions, with resulting subjective interpretations. Professional judgments expressed herein are based upon the data made available to us at the moment of assurance engagement performed under the agreed scope of work and pertinent time and resource constraints.

For and on behalf of DNV Business Assurance Co., Ltd. Taiwan

Lead Verifier
Johnny Wu

Place and date:
Taipei, 15 April, 2025

Reviewer / District Manager
David Hsieh

The background of the entire image is a dark blue gradient with a complex, glowing circuit board pattern. The pattern consists of numerous thin, light blue lines that form a dense network of paths, with small circular nodes at various points along these paths. The pattern is most prominent on the left side and fades slightly towards the right.

UMC

United Microelectronics Corporation

www.umc.com

